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Functional composition and structural diversity enhance mangrove forest resilience in the Sundarbans



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Mangrove forests offer nature-based solutions for climate change mitigation by storing atmospheric CO₂, while also supporting biodiversity, livelihoods, and coastal protection. Yet, they face increasing threats from deforestation, salinity, and extreme weather events. Here, we assessed and explored the resilience of Sundarbans mangrove forests and associated drivers using satellite-derived vegetation indices, biodiversity, and environmental data, within a structural equation modeling framework. We found 1–8 major perturbations at 250-meter spatial resolution, with the lowest resilience to disturbance or stress in the central and southeastern zones of the Sundarbans. Approximately 610 to 990 km² (~10–15% of the total Sundarbans area) of mangrove forests exhibited declining resilience. The functional composition of maximum canopy height (MCH) was the strongest driver of mangrove forest resilience ($\beta = 0.61$), followed by specific leaf area (SLA; $\beta = 0.56$) and precipitation ($\beta = 0.44$). Structural diversity ($\beta = 0.22$), though weakly associated, mediated the positive association of species richness with resilience. Perturbation frequency had a significant negative direct ($\beta = -0.29$) and total ($\beta = -0.33$) association on resilience, whereas temperature exhibited only a total negative association ($\beta = -0.20$). Based on these findings, we hold that for enhancing mangrove forest resilience against disturbances and stressors, (re-)establishment and conservation efforts should focus on maintaining site-specific dominant species with tall canopy height and specific leaf area — supplemented by a few complementary species — to increase structural diversity and facilitate recovery rates.

Mangrove forests provide a wide range of ecosystem services, including the support of livelihoods of coastal communities, the protection of lives and properties from tropical cyclones and tidal surges (valued at more than \$US \$65 billion annually), the provision of habitat for biodiversity, and high carbon sequestration and storage rates^{1–6}. With these invaluable ecosystem services, mangrove forests can serve as nature-based solutions for climate change mitigation and adaptation^{2,3,7}. Despite these benefits, mangrove forests face multiple threats including land cover-change, rising temperatures, sea level-rise, disrupted freshwater flows, increasing salinity, and more frequent cyclones and other extreme events^{2,6,8–10}. These stressors collectively contribute to the degradation of mangrove habitats, affecting species

richness, composition, and abundance, as well as diversity and composition of functional traits, thus, plant ecophysiological processes, physicochemical processes in sediments, and blue carbon stocks both in the biomass and the sediment^{2,7,10,11}. Such changes potentially undermine the resilience of mangrove forests—their capacity to recover from disturbances¹²—thereby restricting their ability to provide multiple ecosystem services^{5,13–16} sustainably under changing environmental and climatic conditions.

Ecological resilience refers to the capacity of an ecosystem to recover from natural or anthropogenic disturbances^{17,18}. Along this line, mangrove forest resilience encompasses the ability of mangrove ecosystems to recover from diverse environmental challenges, spanning natural factors like fires,

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windstorms, cyclones, and pathogens, as well as anthropogenic pressures, such as aquaculture or forestry and climate change^{9,10,19}. This resilience is critical for maintaining the persistence and equilibrium of mangrove forests in rapidly changing conditions.

Theoretical investigations propose that complex systems may exhibit tipping points—critical thresholds beyond which abrupt and irreversible changes will occur^{17,18}. When these points are reached, the system collapses, exhibiting diminished resilience and increased vulnerability to shifts upon external perturbations. This phenomenon is often associated with *Critical Slowing Down* (CSD), a key indicator that the system is losing its ability to recover from disturbances²⁰. CSD is characterized by three primary signals: slow recovery from perturbations, increased autocorrelation (AC, e.g., high correlation of an observed vegetation parameter with its neighboring time), and increased fluctuations of ecosystem parameters^{18,21–23}. The detection of CSD requires long-term monitoring.

Monitoring CSD through 1-lag autocorrelation (AC1) using remote sensing imagery may offer advantages over field-based approaches to monitoring species demographics, ecophysiological traits, or productivity, especially in landscape-to-regional and global contexts²⁴. In remote sensing-based methods, temporal changes in the Normalized Difference Vegetation Index (NDVI) serve as a common proxy for ecosystem resilience²⁵ due to its high correlation with productivity. However, a prevalent limitation in prior research stems from the use of NDVI, known for its saturation effects, particularly in areas with high vegetation cover or complex canopy structure²⁶. To overcome these limitations, a more advanced and flexible approach involves using Vegetation Optical Depth (VOD)²⁷ derived from radar imagery or kernel Normalized Difference Vegetation Index¹⁵ to mitigate saturation issues. Nevertheless, a challenge lies in the limited availability of high-resolution historical VOD datasets, with publicly accessible data typically having a resolution of $\sim 9 \times 9 \text{ km}^2$ (ref. 28). Effectively addressing these challenges, the recent introduction and application of the nonlinear kernel Normalized Difference Vegetation Index (kNDVI) in terrestrial ecosystems²⁹ offers enhanced reliability and applicability. This advancement proves valuable in capturing mangrove forest dynamics over both short- and long-term periods, facilitating the assessment of mangrove forest resilience against natural and anthropogenic disturbances through remote sensing-based approaches.

The Sundarbans is one of the world's largest contiguous mangrove ecosystems, spanning over 10,000 km² across Bangladesh and India, and serves as a vital hub for both marine and terrestrial biodiversity, home to numerous endangered species, including *Heritiera fomes* (also known as the Sundari tree) and iconic fauna, such as the Royal Bengal Tiger, the saltwater crocodile, and the Irrawaddy dolphin^{30–32}. Recognized as a UNESCO World Heritage Site and a Ramsar Wetland, the Sundarbans remains one of the last essentially intact mangrove ecosystems globally. This unique environment provides significant support to the livelihoods of millions of coastal inhabitants, both directly and indirectly³³. It plays a critical role in protecting lives and property from tidal surges and tropical cyclones, thus providing disaster risk reduction and fostering coastal resilience. Additionally, the Sundarbans serves as a huge carbon sink^{34,35}, contributing to climate change mitigation through sequestration of atmospheric CO₂ and carbon storage in mangrove biomass and sediments.

Despite the multiple ecosystem services the Sundarbans provides, they are confronting numerous challenges. Alongside the global environmental stressors, such as increasing cyclone frequency, rising temperatures, and shifting precipitation patterns, the localized anthropogenic pressures, like upstream dam construction and shrimp farming, are collectively undermining the growth, integrity, and productivity of the Sundarbans^{36–43}. This degradation not only diminishes their capacity to sequester and store carbon but also threatens the very foundation of these ecosystems. For instance, the dominant tree species, *H. fomes*, has been plagued by a phenomenon known as “top dying” since the early 1960s⁴⁴. This issue has led to an annual loss of approximately \$16.72 million in timber value, while simultaneously becoming a significant source of carbon emissions^{7,35,44}. Additionally, forest composition is shifting notably; diverse stands are increasingly becoming

homogeneous, resulting in a decline in taxonomic and functional diversity. Key traits, such as wood density, tree morphometry (including diameter and height), and leaf litter quality, are being compromised, which, in turn, alters forest dynamics, growth patterns, and carbon storage potential^{40,45}. Empirical investigations indicate that the primary drivers of these changes are a reduction in freshwater flow and corresponding fluctuations in salinity levels^{7,40,45}. The changing climate—altered temperature and rainfall patterns—in the region may be causing a critical reduction in the resilience of this mangrove ecosystem. This phenomenon is reminiscent of trends observed in other tropical forests, such as the Amazon rainforest²⁷, which have recently exhibited declining resilience. Cyclone impacts^{41–43,46,47}, associated risks⁴⁸, driver of dieback⁴⁹ and empirical recovery rates¹² in mangrove forests have been documented across various regions and globally. Most existing studies focus on historical cyclone damage assessments using MODIS- and Landsat imagery-derived vegetation indices with thresholding, machine learning, and deep learning approaches, or case studies of individual cyclone events^{10,12,41,42,46–49}. However, most previous studies have concentrated on tracking mangrove forest condition and empirical recovery and future risk, while paying less attention to understanding how mangrove forests recover, what drives their post-disturbance recovery, or what role multidimensional diversity plays in shaping their recovery. Notably, this issue remains largely unexplored in the Sundarbans. Addressing this knowledge gap could provide valuable insights into the effects of climate change on mangrove forest resilience and offer guidance for future conservation and restoration efforts.

The resilience of forest ecosystems to disturbances is influenced by a complex interplay of biotic and abiotic factors^{29,50–52}. Recent studies that utilize remote sensing to assess resilience and its potential drivers have predominantly focused on environmental variables, such as temperature, drought, precipitation, water deficit, water availability, and soil nutrients, e.g., magnesium, potassium, and calcium^{20,22,27,29,51–53}. Additionally, they examine biotic factors, such as biomass and remotely sensed productivity (gross primary productivity), and how they impact resilience metrics. However, these studies often overlooked important aspects of biodiversity, including taxonomic diversity, functional diversity, and functional trait composition. They also tend to neglect forest stand structural complexity as well as crucial physio-chemical properties (such as salinity, pH, texture) and key nutrients (e.g., phosphorus, sulfide, iron and nitrogen)^{7,54}. These biogeochemical factors interact with changing temperatures and precipitation patterns, playing a pivotal role in the overall resilience of forests following disturbances. Moreover, most studies have employed simple linear regression or machine-learning approaches to explore the relationships between environmental variables and resilience^{17,20,24,27,29}. While these methods can identify direct correlations, they often fail to capture indirect or total associations among variables such as temperature, precipitation, or taxonomic and functional diversity, particularly in terms of their mediating effects on resilience. For instance, temperature might show a weaker direct correlation with resilience metrics than precipitation. Yet, temperature can significantly reduce species richness and affect key functional traits^{55,56}—such as leaf area, wood density, or maximum canopy height—that are fundamental to functional composition and diversity⁵⁷. Conversely, precipitation may exhibit a stronger positive association with these elements, suggesting that, when considering indirect effects, the overall impact of temperature on resilience could be more substantial than that of precipitation or vice versa.

Similarly, beyond direct associations, taxonomic diversity may enhance resilience by improving functional diversity or functional trait composition. Analyzing the comparative strength of these associations can help elucidate key ecological theories and hypotheses⁵⁷ that govern resilience. For example, the niche complementary effect theory posits that a positive association between taxonomic or functional diversity indicates that a greater number of species is essential for recovery from disturbance or stress. In contrast, the selection theory suggests that recovery may rely on a few dominant species rather than a broader diversity of species. Taxonomic diversity could also positively influence resilience by enhancing sediment nutrient quality through increased organic matter input. Integrating these multifaceted biotic and abiotic properties—while considering their

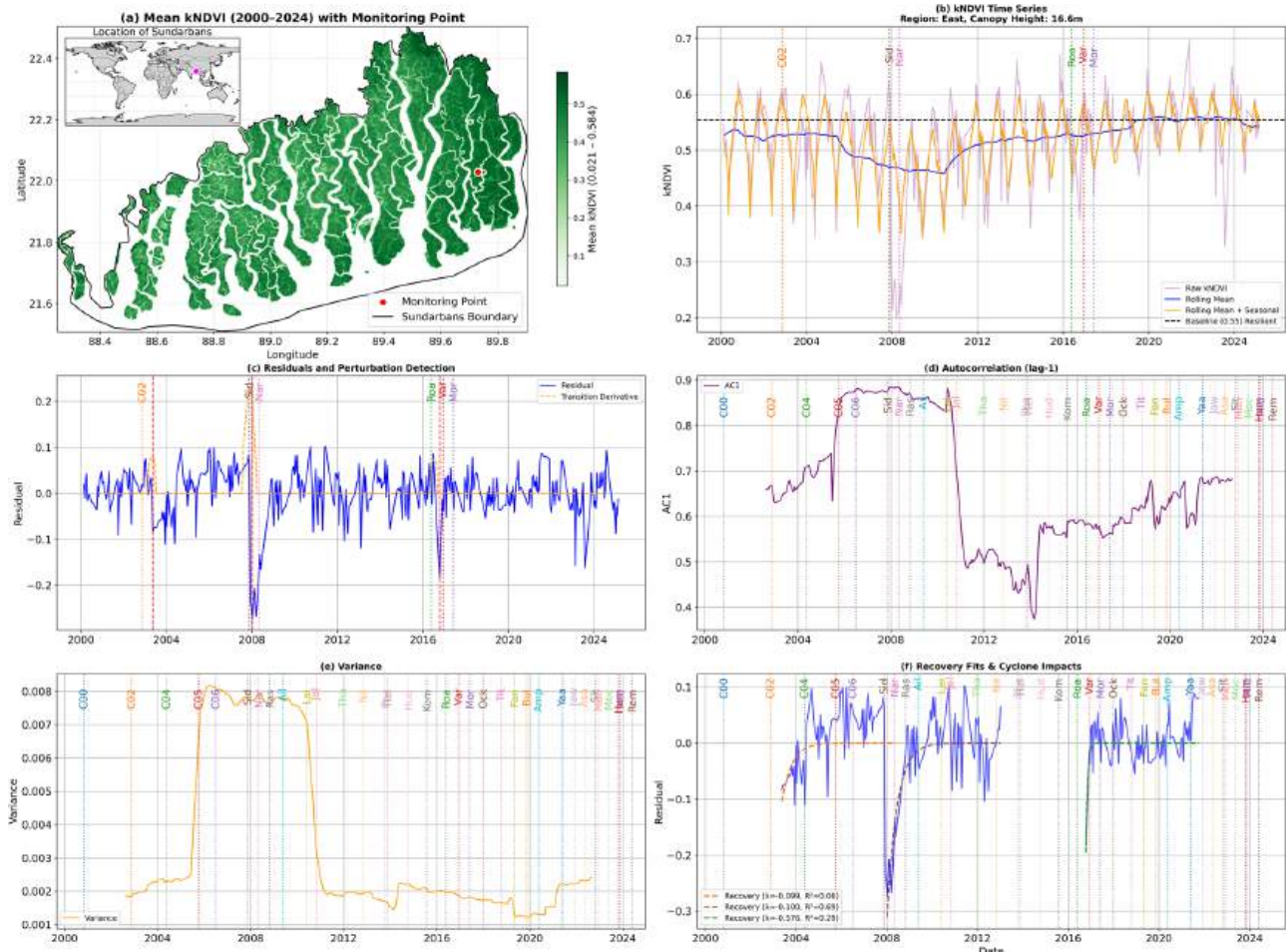


Fig. 1 | Vegetation dynamics and resilience indicators in the Sundarbans. a Long-term Mean NDVI (2000–2020) in the Sundarbans with monitoring point (Longitude: 89.7314, Latitude: 22.0295) and boundary; **b** kNDVI time series of particular monitoring sites in the eastern region, showing raw and smoothed kNDVI values, with cyclone impact periods marked; **c** Perturbation detection through 12-months moving windows and 95% thresholding on residual time series, highlighting perturbation periods; **d** Temporal 1-lag autocorrelation of residual kNDVI, signaling the resilience loss or critical slowing down; **e** Variance in residuals captures fluctuations in the kNDVI time series, with detected perturbations flagged, showcasing

critical slowing down; **f** Empirical post-disturbance recovery rate trajectories from local minima, following cyclone impacts, indicating vegetation response and resilience over time. The magenta circle on the global map (**a**) shows the position of the Sundarbans on the Globe. C00 (Unnamed), C02 (Unnamed), C04 (Unnamed), C05 (Unnamed), C06 (Unnamed), Sid Sidr, Nar Nargis, Ras Rashmi, Ail Aila, Lai Laila, Jal Jal, Tha Thane, Nil Nilam, Pha Phailin, Hel Helen, Hud Hudhud, Kom Komen, Roa Roanu, Var Vardah, Mor Mora, Ock Ockhi, Tit Titli, Fan Fani, Bul Bulbul, Amp Amphan, Yaa Yaas, Jaw Jawad, Asa Asani, Sit Sitrang, Man Mandous, Moc Mocha, Ham Hamoon, Mid Midhili, Rem Remal.

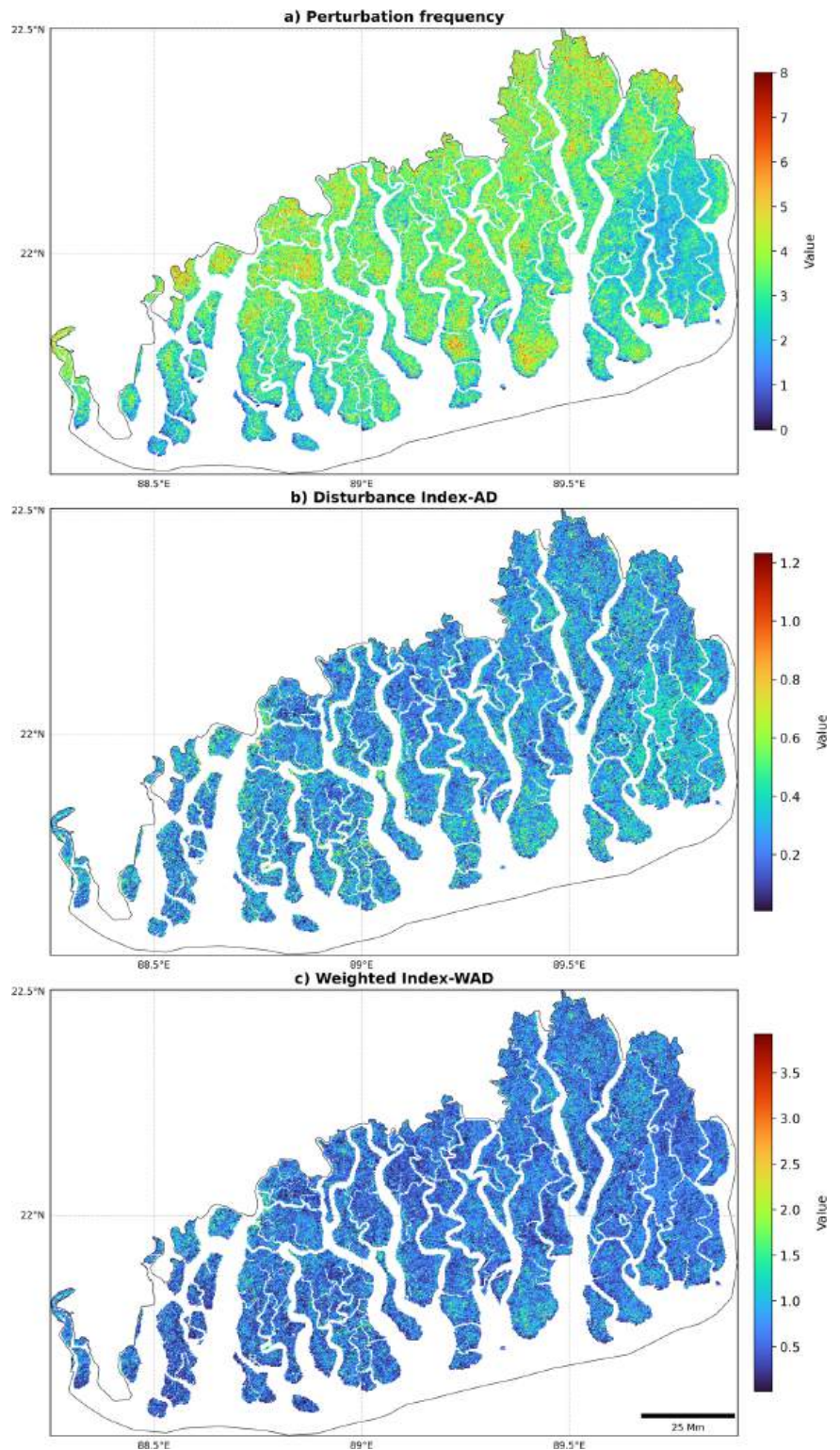
associations with precipitation and temperature as proxies for climatic conditions and extreme events—into advanced and complex modeling approaches, such as structural equation modeling, could yield new scientific insights. This integration can enhance our understanding of the intricate interactions among biodiversity, the environment, and resilience, contributing significantly to our understanding of community ecology and climate change and their effects on improving (re-)establishment measures⁴⁵ and the sustainable management of existing mangrove ecosystems.

Here, we aim to (i) assess resilience (log of 1-lag autocorrelation, as an indicator of resilience) of mangrove forests using Moderate Resolution Imaging Spectroradiometer (MODIS) monthly imagery and (ii) explore the drivers of resilience and the mechanisms that link biodiversity, environmental factors, disturbance (perturbation frequency), and resilience in mangrove forests³⁵. To investigate the drivers and mechanisms of mangrove forest resilience, we consider both biotic and abiotic factors relevant to resilience. We use taxonomic diversity, functional diversity, and functional composition, including wood density, maximum canopy height, leaf dry matter content, leaf photosynthesis rate, specific leaf area, and leaf succulence as biodiversity variables. For climatic and sediment physicochemical properties, we examine mean annual precipitation, mean annual

temperature, sediment salinity, sediment pH, sediment sulfide, phosphorus, and iron concentration. We investigate objective ii by testing three hypotheses to address the ecological theories that regulate mangrove forest resilience in the Sundarbans: H1) temperature and precipitation exhibit both direct and indirect associations with resilience by modifying biotic factors and physical and nutrient properties of the sediment and disturbance; H2) taxonomic diversity, functional diversity and functional composition of traits exhibit both direct and indirect associations with resilience metrics through their influence on the physical and nutrient properties of the sediment where the functional composition of traits has a higher association with resilience than taxonomic or functional diversity; H3) disturbance exhibits both direct and indirect associations with resilience biotic and sediment physicochemical variables.

We found that approximately 610–990 km² (~10–15% of the total Sundarbans area) of mangrove forests exhibited declining resilience. The structural equation model revealed that the functional composition of maximum canopy height was the strongest driver of mangrove forest resilience, followed by specific leaf area and precipitation. Perturbation frequency had a significant negative direct and total association with resilience, whereas temperature exhibited only a total negative association. These findings underscore the urgent need for trait-based conservation and

Fig. 2 | Spatial pattern of mangrove forest perturbation frequency and damage severity in the Sundarbans. **a** Perturbation frequency map showing the number of disturbance events, with warmer colors (yellow to red) indicating higher frequency; **b** Disturbance Index (AD)-based on the average drop- showing the intensity of disturbance, where red areas represent higher disturbance levels and blue areas indicate lower disturbance; **c** Weighted Index (WAD)-based on weighted average drop, decay factor and time-reflecting cumulative disturbance impact, with red indicating higher weighted values and blue indicating lower values. White lines denote water channels and geographic boundaries. The consistent layout across all panels allows for comparative analysis of disturbance patterns and resilience gradients within the region.



restoration strategies that prioritize site-specific dominant species with tall canopy height and high specific leaf area—supplemented by a few complementary species—to enhance structural diversity and facilitate mangrove recovery against disturbances and stressors.

Results

Perturbation count and disturbance index

We estimated the perturbation frequency using residual kNDVI after removing long-term and seasonal trend (see the example along with

resilience indicators and empirical recovery rate at five locations in Fig. 1a–f, Figs. S1–4) which showed that every location across the Sundarbans experienced at least one significant perturbation in the past 25 years (Fig. 2a). A high number of perturbations spanned scattily across the Sundarbans with more homogeneous patches were observed in the seaward zone of Khulna range, northern boundary, and northeastern boundary and central part of Sundarbans. Interestingly, the forest stands between the southeastern zone of Chandpai and the northwestern zone of the Sharankhola range (blue-colored homogeneous patches, Fig. 2a) had 1–2

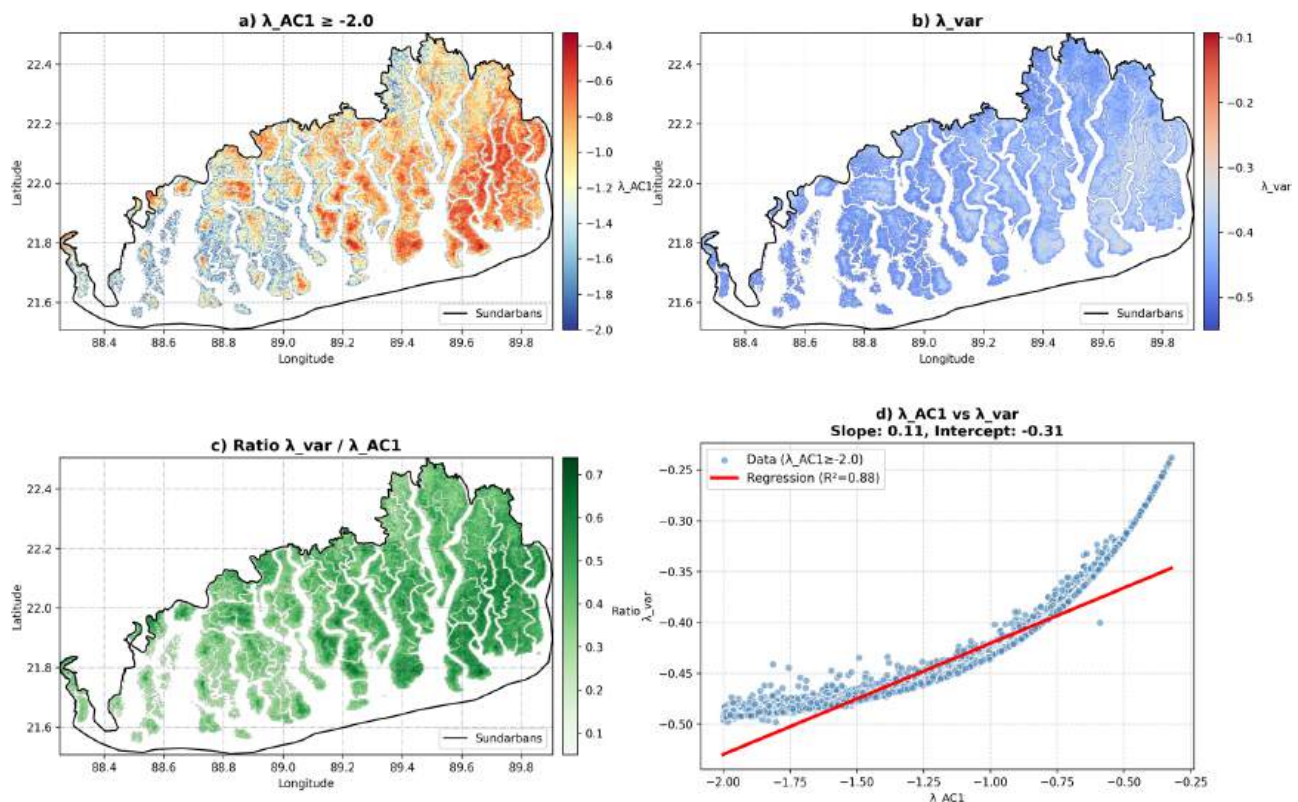


Fig. 3 | Spatial patterns of mangrove forest resilience indicators (λ_{AC1}) in the Sundarbans. a Recovery rate λ_{AC1} , estimated from the 1-lag autocorrelation of pixel-level kNDVI residuals; **b** Recovery rate λ_{VAR} , derived from the temporal variance

decay of kNDVI residuals; **c** The ratio $\lambda_{VAR}/\lambda_{AC1}$; **d** Scatter plot, indicating a substantial spatial similarity between λ_{VAR} and λ_{AC1} . Forest areas with λ values close to zero exhibit low resilience.

perturbations. However, those forest stands, the disturbance indices (Disturbance Index-AD and Disturbance Index-WAD; Fig. 2b, c, see “Method”), reveal a similar intensity of damage where the perturbation count is more than 4 (Fig. 2a–c).

Resilience pattern and trends in the Sundarbans

We estimated the two-recovery rates, denoted as lambda (λ) in the Sundarbans mangrove forests, using 1-lag autocorrelation (AC1) and variance (VAR)—the two widely used resilience indicators—by, analyzing MODIS 16-days images from 2000 to 2024, showing diverse patterns of resilience (Fig. 3). For example, from the central (Satkhira range) to the eastern region (southern parts of the Chandpai Range, as well as the northeastern, central and southern parts of the Sharankhola Range) seaward forest stands showed lower resilience (red color gradient) than stands along the northern boundary (Fig. 3). However, in the western areas (mostly Indian) of the Sundarbans, mangrove stands near the northern boundary (dark red, in Fig. 3a) exhibited lower resilience than the Bangladeshi Sundarbans (Fig. 3a). In most of the mangrove stands both resilience indicators (λ_{AC1} , and λ_{VAR}) showed a strong similarity at their spatial pattern in ratio metrics (Fig. 3a–c; higher value, darker green indicates higher level of similarity), and simple regression showed a strong linear association ($R^2 = 0.88$; Fig. 3d). To observe the resilience status, we followed two complementary methods. First, we used quantile threshold-based resilient classes (Fig. 4)—Low Resilient (Q1: kNDVI ≤ 0.39), Moderately Resilient (Q2: $0.39 < \text{kNDVI} \leq 0.49$) and Resilient (Q3: kNDVI > 0.49)—and evaluated their transitions relative to the 2000 baseline, Fig. 5). Second, we quantified temporal trends using Kendall’s tau rank correlation with 3-years, 5-years, and 7-years overlapping windows with 1-year shift (see method; Fig. 5, Figs. S5, S6). This analysis is crucial for understanding how resilience in mangrove stands changed over time and for identifying areas at risk (Fig. 4, 5). The spatio-temporal maps of resilience class show that over the past 25 years, a

significant number of forest stands had shifted in their resilience classes from their baseline years and with cyclone severity, mostly towards moderate resilience (Fig. 4, Table 1). For example, from 2007 to 2009 (cyclones Sidr, Rashmi, and Aila), areas of 820.97, 732.54, and 1138.74 km², respectively, shifted from resilient to moderately resilient, whereas areas of 356.89, 784.68, and 945.04 km², respectively, shifted from moderately resilient to weakly resilient (Table 1). When we considered resilient status maps relative to the baseline year 2000, we found that the lowest resilience loss occurred in 2022 (Fig. 4, Table 1), while maximum resilience loss was recorded in 2009 (2110.52 km²; Fig. 4, Table 1). Significant losses were also observed in 2007 (1189.21 km²), coinciding with Cyclone Sidr on 7 December 2007, and in 2008 (2056.79 km²; Fig. 4, Table 1). The average resilience loss from 2001 to 2024 was 606.83 km², based on average kNDVI (Fig. 5, Table 1). Notably, in 2004 and in the years after 2009–2012, more than 1000 km² of mangrove forests exhibited reduced resilience every year relative to the baselines of 2000 (Table S1). The Kendall’s tau values for other window sizes (3, 5, and 7 years), which were assessed to investigate the robustness of our resilience assessment, are provided as Supplementary materials (Fig. 5; Figs. S5, S6). Our analysis revealed that 61.4% (4030.88 km²) of the studied areas show non-significant Kendall’s tau values, suggesting stable conditions. Conversely, 23.6% (988.81 km²) of mangrove stands exhibited negative Kendall’s tau ($P < 0.05$), indicating increased resilience (Fig. 5). However, 15.1% (988.81 km²) of the areas displayed significant positive Kendall’s tau values, indicating resilience loss (Fig. 5). The other two windows of trend analysis also exhibited similar spatial patterns of resilience trends, with slight variations in the percentage and area coverage under neutral, resilience gain and resilience loss categories (Figs. S5, S6). Trend-based resilience analysis depicted a steep resilience decrease from the central to western regions and near terrestrial or northern boundaries, along the coastline. However, the southeastern part of the Sundarbans (the southern part of the Chandpai and Sharonkhola range) exhibited lower resilience losses (Fig. 4).



Fig. 4 | Threshold-Based Vegetation Resilience Classification, 2000–2024. Maps categorize vegetation into three resilience classes relative to the baseline (2000) using a quantile approach: Low Resilient (red; Q1: $kNDVI \leq 0.39$), Moderately Resilient (yellow; Q2: $0.39 < kNDVI \leq 0.49$), and Resilient (green; Q3: $kNDVI > 0.49$).

Individual years (2000–2024) are shown with bold titles, while the bottom-right map illustrates spatial patterns of the mean resilience over 2001–2024 relative to 2000. The black boundary outlines the Sundarbans region.

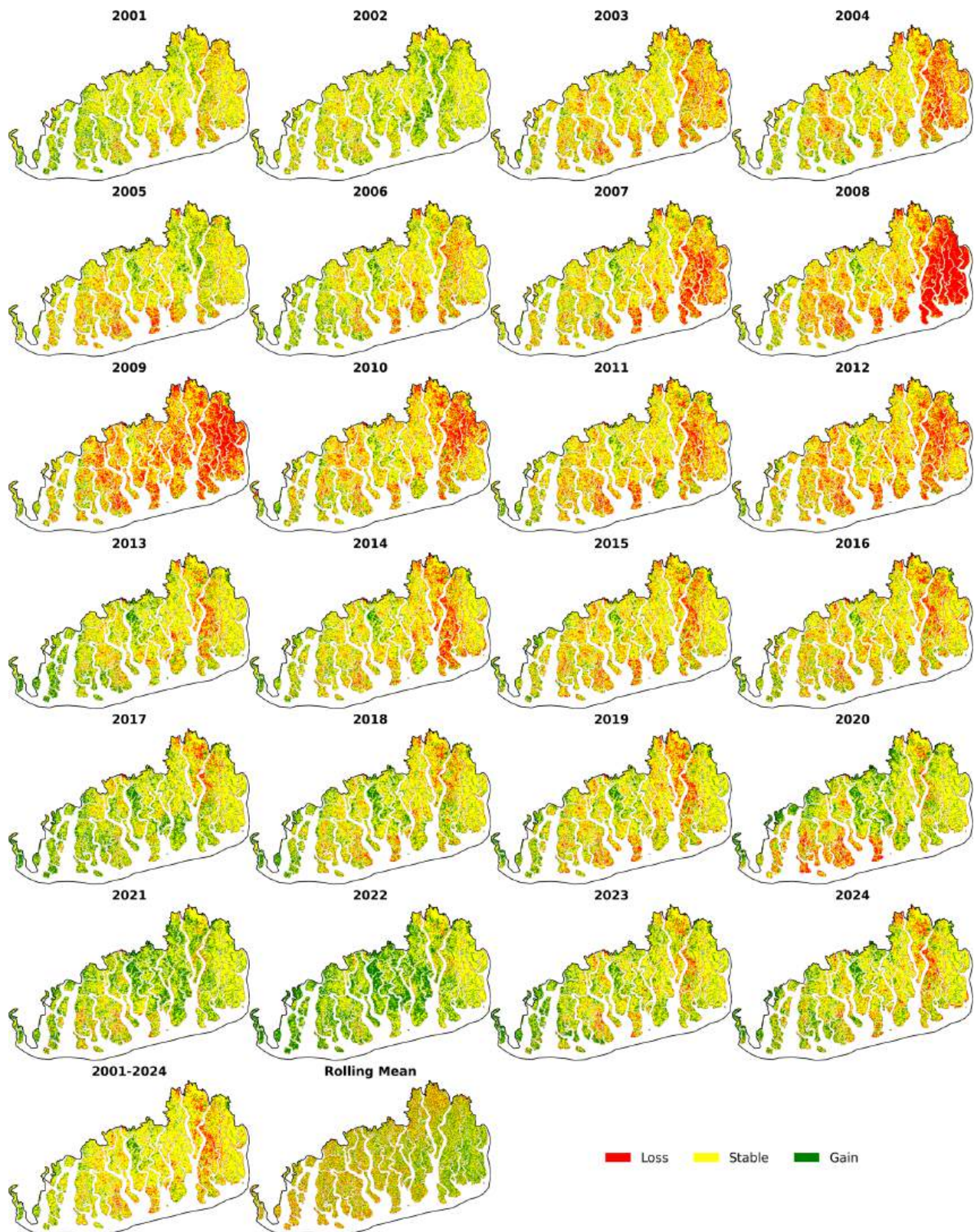


Fig. 5 | Annual, mean, and rolling mean (2002–2022) resilient status from 2001 to 2024 relative to 2000. Resilience status (stable: yellow; loss: red; and gain: green) is based on the transition of resilient classes in shown in Fig. 4 where classes were defined as Weakly Resilient (red; Q1: $kNDVI \leq 0.39$), Moderately Resilient (yellow;

Q2 $0.39 < kNDVI \leq 0.49$), and Resilient (green; Q3: $kNDVI > 0.49$). The 5-year rolling mean was quantified for 2002–2022: as our study period covers 2000–2024, we were unable to apply a full 5-year rolling window for the boundary years 2000, 2001, 2023, and 2024 due to insufficient adjacent data.

Table 1 | Summary of resilient class transition, total gain and loss from 2001 to 2024 (annual and mean) relative to baseline year 2000

Year	R1 to R2	R1 to R3	R2 to R3	Total Gain	R3 to R2	R3 to R1	R2 to R1	Total Loss	Stable
2001	445.28	11.61	416.74	873.63	281.62	2.49	116.12	400.23	4380.41
2002	424.39	6.01	607.11	1037.51	124.11	2.17	121.47	247.75	4368.4
2003	269.35	6.02	178.37	453.74	592.06	4.93	337.09	934.08	4265.53
2004	275.09	5.96	179.42	460.47	754.64	4.92	336.35	1095.91	4097.34
2005	317.07	3.08	334.86	655.01	324.05	5.69	269.94	599.68	4398.54
2006	396.17	8.89	300.3	705.36	402.76	3.57	205.55	611.88	4336.42
2007	291.43	4.12	145.24	440.79	820.97	11.35	356.89	1189.21	4023.79
2008	211.56	3.25	58.5	273.31	732.54	539.56	784.68	2056.78	3322.86
2009	149.48	5.15	81.99	236.62	1138.74	26.74	945.04	2110.52	3304.63
2010	261.64	9.22	173.44	444.3	753.54	10.05	463.76	1227.35	3981.14
2011	265.55	6.88	168.99	441.42	642.49	11.64	422.49	1076.62	4132.86
2012	212.83	4.66	91.62	309.11	825.01	15.09	610.98	1451.08	3892.27
2013	470.94	31.33	424.74	927.01	336.25	6.49	203.72	546.46	4179.7
2014	369.59	10.23	227.75	607.57	537.82	11.04	342.76	891.62	4154.62
2015	334.4	10.4	209.03	553.83	505.19	8.51	347.7	861.4	4236.87
2016	324.81	10.02	246.47	581.3	489.38	13.85	394.26	897.49	4170.15
2017	573.48	42	711.98	1327.46	242.79	6.17	147.1	396.06	3927.33
2018	504.1	25.94	477.86	1007.9	312.57	8.5	214.61	535.68	4106.51
2019	428.87	16.17	319.55	764.59	397.57	13.7	332.36	743.63	4140.07
2020	514.82	29.06	573.79	1117.67	278.16	27.87	344.43	650.46	3881.75
2021	637.09	42.43	1007.98	1687.5	127.38	10.35	136.14	273.87	3686.49
2022	739.56	98.62	1330.63	2168.81	82.62	7.09	78.61	168.32	3313.72
2023	561.05	47	615.93	1223.98	251.8	11.64	187.34	450.78	3971.95
2024	547.21	31.33	482.4	1060.94	340.06	16.02	233.17	589.25	3997.51
Mean	379.11	1.03	160.43	540.57	430.18	4.55	172.09	606.82	4508.17

R1 Low Resilient (Q1: kNDVI $\leq$ 0.39), R2 = Moderately Resilient (Q2: 0.39 < kNDVI $\leq$ 0.49), and R3 = Resilient (Q3: kNDVI > 0.49). These resilience classes are based on the quantile threshold baseline year (2000).

Drivers and mechanisms of mangrove forest resilience

Using the perturbation frequency and the two-disturbance metrics along with climatic, physicochemical sediment properties and taxonomic, structural, and trait diversity, we developed six structural equation models (SEM): 3 for λ_{ACI} and 3 for λ_{VAR} (by adding separately three disturbance-related variables). Based on the fit indices (Chi Square = 2.464, p = 0.292, Goodness of Fit Index of 0.999, Comparative Fit Index of 0.999, and a standardized root mean square residual of 0.009), our fitted SEM models did not show any significant deviation from our dataset (Fig. 6; Table S1). However, according to the Akaike Information Criterion (AIC) and the Coefficient of determination (R^2), SEM λ_{VAR} exhibited (AIC = 3776.63 and R^2 = 0.49) a better performance than λ_{ACI} (AIC = 3786.92 and R^2 = 0.44) with disturbances as perturbation frequency. Thus, we report the variance-based SEM. The model fitted well with 14 predictors, including mean annual temperature (hereafter, temperature), total annual precipitation (hereafter, precipitation), sediment salinity, sediment pH, sediment concentrations of sulfide, phosphorus and iron, sediment texture (silt content), perturbation frequency (count of perturbations), species richness, structural diversity, functional compositions of maximum canopy height and specific leaf area and the functional diversity of succulence (other functional traits had a variance inflation factor over 10; Tables S2, 3).

Biotic and abiotic variables displayed distinct types of associations with resilience, biotic variables exerting the dominant role (Fig. 6). For instance, perturbation frequency exhibited a significantly moderate negative association with resilience (β = -0.29, P < 0.05; Fig. 6, Table 2; according to the conventional benchmarks for standardized regression coefficients in

ecological SEMs: β < 0.2 indicates a weak association, 0.2 < β < 0.5 a moderate association and $\beta \geq 0.5$ strong association^{58,59}), whereas neither precipitation nor temperature demonstrated any significant direct association (P > 0.05; Table 2). Both sediment pH and phosphorus content had a direct, significant negative association with resilience (P < 0.05, Table 2). However, sediment salinity and the sediment contents of iron and sulfide did not have any significant direct association (Table 2).

Structural diversity showed a significant positive direct association with resilience, while species richness and functional diversity did not exhibit any significant direct association (Fig. 6; Table 2). The functional compositions of the maximum canopy height (β = 0.62, P < 0.05; Fig. 6, Table 2) and specific leaf area (β = 0.62, P < 0.05; Fig. 6, Table 1) showed a strong significant positive association with resilience. Thus, functional composition diversity metrics were the dominant driver of mangrove forest resilience in the Sundarbans. We did not find any direct association of sediment salinity with resilience (P > 0.05). However, we observed an indirect, weaker negative association of sediment salinity with resilience through species richness to structural diversity pathways (β = -0.25; P < 0.05; Fig. 6).

Species richness showed a significant positive association with structural diversity (β = 0.35; P < 0.05; Fig. 6). Consequently, through this specific pathway, species richness showed significantly weaker positive associations with resilience (β = 0.08 0; P < 0.05; Fig. 6; Table 2). However, the total indirect and total association of species richness with resilience were not significant (P > 0.05; Table 2). In contrast, the total association of functional diversity with resilience was negative and moderate (β = -0.22, P < 0.05; Table 2). Although precipitation showed no direct association with

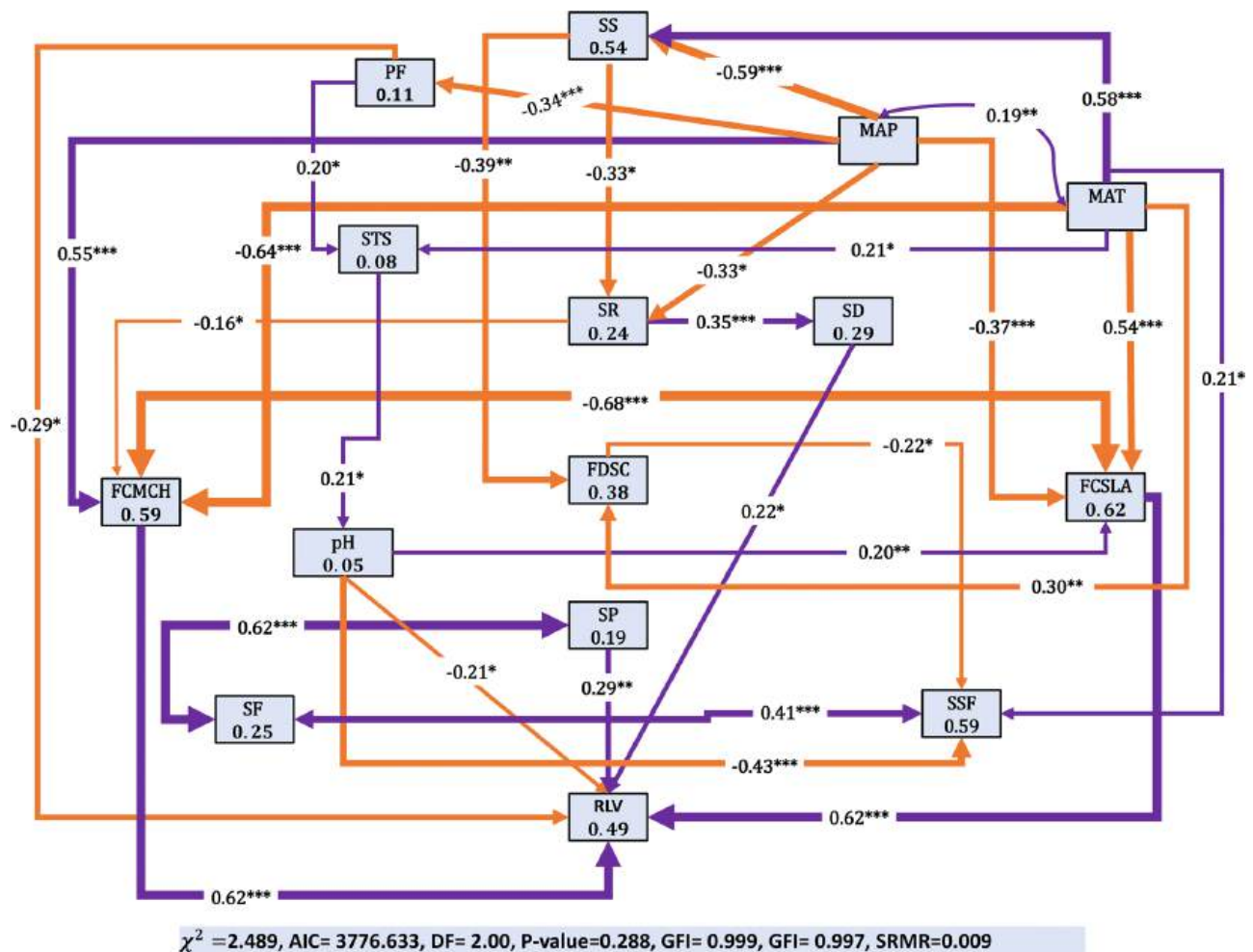


Fig. 6 | Structural Equation Model illustrating the mechanisms connecting biodiversity measures, abiotic factors, and resilience (λ_{VAR}). Hypothesized paths with asterisks (***) $P < 0.001$, ** $P < 0.01$, * $P < 0.05$ indicate significant associations between variables; non-significant associations ($P > 0.05$) are not shown for the sake of clarity (please see Table 2). The non-significant χ^2 (Chi-Square) of 2.489 ($P = 0.288$), with a goodness-of-fit index (GFI) of 0.997, a comparative fit index (CFI) of 0.999, and a standardized root mean square residual of 0.009, suggests no significant deviation from the model and datasets at 2 degrees of freedom. Orange lines indicate negative associations, while purple lines indicate positive associations between covariables. Numbers on the arrows represent standardized regression weights (β) rather than correlation coefficients between predictors and dependent variables. The association was estimated using maximum

likelihood and validated with 1000 bootstrap samples. For interpretation, we follow conventional benchmarks for standardized regression coefficients in ecological SEMs: $\beta < 0.2$ indicates a weak association, $0.2 < \beta < 0.5$ a moderate association and $\beta \geq 0.5$ strong association^{58,59}. R^2 Coefficient of Determination within the endogenous variables represents their explained variance by all predictors. MAP mean annual precipitation, MAT mean annual temperature, SS sediment salinity, SR Species richness, FCMCH functional composition of maximum canopy height, SD Structural diversity (variation of diameter at breast height), FCSLA functional composition of specific leaf area, FDLS functional diversity leaf succulence, P sediment phosphorus concentration, SF sediment iron concentration, SSF sediment sulfide concentration, pH sediment pH level, STS sediment texture silt, PF perturbation frequency.

resilience, it exhibited an indirect association with resilience through specific functional trait pathways. Notably, precipitation showed an indirect positive association through the functional composition of maximum canopy height path ($\beta = 0.34$, $P < 0.05$; Fig. 6, Table 2). Conversely, precipitation also showed an indirect negative association with resilience mediated by functional composition of specific leaf area ($\beta = -0.22$, $P < 0.05$; Fig. 6). However, the positive association through canopy height was higher than the negative association through specific leaf area ($\beta = 0.12$; Fig. 6). Precipitation also showed a negative association with resilience through the sediment salinity to functional composition of maximum canopy height pathway (Table 2). More importantly, precipitation also exhibited a positive indirect association with resilience through reducing the perturbation frequency ($\beta = 0.10$, $P < 0.05$; Fig. 6 and Table 2). These indirect positive and negative associations, combined with a weaker positive direct association, resulted in the moderate total positive association of precipitation with resilience ($\beta = 0.44$, $P < 0.05$; Fig. 6; Table 2). Temperature exhibited an indirect negative association with resilience through functional composition of maximum canopy

height ($\beta = -0.40$, $P < 0.05$; Fig. 6; Table 2) and through the sediment salinity to species richness to structural diversity pathway (Fig. 6; Table 2). Conversely, temperature also showed an indirect positive association with resilience through the functional composition of specific leaf area ($\beta = 0.33$, $P < 0.05$; Fig. 6; Table 2). However, the total association of temperature with resilience was negative ($\beta = -0.20$, $P < 0.05$; Table 2). The total association of perturbation frequency with resilience was significant and negative ($\beta = -0.29$, $P < 0.05$; Table 2). Although both maximum canopy height and specific leaf area with resilience, the total association of the functional composition of maximum canopy height ($\beta = 0.61$, $P < 0.05$; Table 2) was slightly stronger than that of specific leaf area ($\beta = 0.56$, $P < 0.05$; Table 2).

Discussion

Mangrove ecosystems worldwide are experiencing frequent external pressures—such as warming, rising sea levels, changing precipitation patterns, or fluctuating salinity levels—that could affect the resilience of mangrove forests against disturbances, such as tropical cyclones or stressors such as

Table 2 | Summary of the mean resilient class transition and total gain and loss from 2002 to 2022 (5-year rolling mean and mean) relative to 2000

Year (Rolling to)	R1 to R2	R1 to R3	R2 to R3	Total Gain	R3 to R2	R3 to R1	R2 to R1	Total Loss	Stable
2002	283.13	0.27	156.18	439.58	284.14	1.14	113.22	398.49	4817.5
2003	323.28	1.14	200.04	524.45	317.15	1.84	143.66	462.65	4668.46
2004	310.4	1.3	181.37	493.07	348.55	2	166.59	517.14	4645.36
2005	283.76	0.92	114.42	399.1	559.23	2.65	214.93	776.82	4479.65
2006	267.16	0.65	80.28	348.09	939.17	4.82	272.81	1216.8	4090.68
2007	231.88	0.87	62.89	295.63	1039.59	6.65	353.51	1399.75	3960.18
2008	226.24	0.98	60.97	288.18	1100.21	9.78	405.99	1515.98	3851.4
2009	199.53	1.14	47.05	247.71	1142.23	13.56	472.5	1628.29	3779.57
2010	186.68	1.3	44.34	232.33	1144.07	13.46	533.13	1690.66	3732.58
2011	243.48	2.39	91.95	337.81	769.64	7.35	403.14	1180.13	4137.62
2012	296.78	3.03	125.96	425.78	596.09	6.98	308.69	911.76	4318.03
2013	313.13	3.74	134.49	451.35	546.95	6.33	292.96	846.24	4357.97
2014	324.75	3.79	144.52	473.06	518.73	5.74	288.5	812.97	4369.48
2015	406.7	6.44	233.57	646.7	398.42	4.76	205.29	608.48	4400.33
2016	414.33	6.11	240.71	661.15	390.61	5.36	207.24	603.21	4390.93
2017	430.71	7.46	267.05	705.22	355.89	6.39	207.26	569.53	4380.44
2018	474.65	8.43	324.16	807.24	317.62	7.69	188.46	513.77	4334.4
2019	547.23	11.94	455.7	1014.86	244.08	7.58	147.62	399.28	4241.15
2020	598.39	15.77	548.28	1162.44	203.66	7.8	130.56	342.02	4151.11
2021	616.78	17.56	570.39	1204.72	197.41	8.4	125.1	330.91	4119.88
2022	644.92	19.34	618.92	1283.18	182.16	8.94	114.94	306.04	4066.18
Mean	351.66	0.92	135.94	488.52	482.03	4.39	192.24	678.66	4488.38

R1 = Low Resilient (Q1: kNDVI $\leq$ 0.39), R2 = Moderately Resilient (Q2 0.39 < kNDVI $\leq$ 0.49), and R3 = Resilient (Q3: kNDVI > 0.49). These resilience classes are based on the quantile threshold baseline year 2000.

salinity^{48,60,61}. In this study, we assessed mangrove resilience against disturbance in the Sundarbans using 25 years of MODIS kNDVI data, applying empirical quantile-based thresholding and theoretical recovery rates (λ_{AC1} and λ_{VAR}) alongside biotic and abiotic drivers. Resilience varied spatially and temporally, with the lowest values in seaward zones (from Satkhira to Sharankhola, Bangladesh) and higher resilience near the northern boundary—although Indian Sundarbans mangrove stands showed lower resilience than their Bangladeshi counterparts (Fig. 3; as indicated by red color gradients). Quantiles-based thresholding and a rolling window-based trend analysis revealed significant resilience loss over time, potentially upon increasing environmental pressure, across 11% (607 km²) and 15% (989 km²) of the region, respectively, with a peak decline in 2009 (37%, or 2110 km²) following three consecutive cyclones (Sidr, Rashmi, and Aila). With SEM analysis, we found that mangrove forest resilience in the Sundarbans is primarily driven by the functional composition of maximum canopy height (MCH) and specific leaf area (SLA), while structural diversity and species richness play minor roles (hypothesis H2). Mean annual precipitation was the dominant abiotic driver of resilience, mainly mediated by MCH and SLA, whereas perturbation frequency and sediment phosphorus content showed negative associations (hypothesis H1 and partly H3). Thus, mangrove forest resilience in the Sundarbans is shaped by a dynamic interplay of functional trait dominance, structural complexity and precipitation. These findings highlight the urgent need to integrate trait-based approaches and disturbance regimes into conservation and (re-)establishment strategies under changing environmental conditions.

The observed spatial pattern of resilience metrics, spatiotemporal patterns of threshold-based resilience class, indicates that resilience varies widely across the Sundarbans, followed the pattern of cyclone impact zones, and was driven by tree height, hydrological conditions, species composition, stream proximity and the increase in natural disturbance frequency^{31,55}. For

example, southeastern, northern and southwestern areas showed higher recovery rates than the seaward zones (from Satkhira to Sharankhola) in Bangladesh. The northern boundary typically experiences cyclones with lesser intensity, particularly in the Khulna and Chandpai Range, as these regions are located 75–90 km inland off the coastline. Forests in these boundary regions are highly diverse stands with a mixture of fast-growing (*Avicennia officinalis*, *Excoecaria agallocha*), and slow-growing (*Heritiera fomes*, *Bruguiera sexangular*, *Xylocarpus mekongensis*) species^{31,55}. These species are characterized by great height, with *Excoecaria agallocha* in the northern boundary area developing greater canopy height than those in the seaward and western forest stand typically associated with higher salinity levels^{31,55}. Additionally, southeastern and northern boundary regions receive more freshwater from the Passur and Baleshwar rivers than the western and central parts of the Sundarbans⁶². One notable exception is the low-resilience hotspot located between the southeastern Chandpai range and the Sharankhola range along both the east and west sides of the Shala River (dark red, Fig. 3a; light blue, Fig. 3b). Despite receiving more freshwater input than the southern and central western zones of the Sundarbans, being dominated by *H. fomes* and *E. agallocha* with greater canopy height and exposed to moderate perturbation frequency, the area unexpectedly exhibits low resilience. The likely reason is exposure to severe cyclones along their central track—most notably Sidr—followed by successive moderate cyclones (Rashmi, Aila, Mora, Bulbul, Amphan; Fig. 2a, b). The spatial map of resilience classes (2007–2013) further reveals that it took six years for these regions to regain its baseline resilient class, like 2000, underscoring the long-lasting impact of extreme events like Sidr and the compounded stress from tropical cyclones in the lower upstream reaches compared to the Passur and Baleshwar rivers.

Mangrove forest resilience or recovery from disturbances in the Sundarbans is strongly driven by the functional composition of whole-plant

traits and the leaf economic spectrum. Our findings reveal that the functional composition of MCH and SLA qualify as the strongest biotic predictors of mangrove forest resilience in the Sundarbans. This result suggests that the selection effect theory^{63,64}—dominant species with their key functional traits drive ecosystem process—is primarily governing mangrove forest resilience in the Sundarbans. Mangrove stands with tall-growing species capitalize on vertical light gradients⁶³, enabling rapid canopy reformation and biomass accumulation after disturbances. Concurrently, high SLA—reflecting an acquisitive leaf strategy along the leaf economic spectrum⁶⁵—supports faster photosynthesis, nutrient turnover and growth rates⁶³. Together, these traits synergistically accelerate post-disturbance recovery, reinforcing the critical role of functional trait composition for mangrove forest resilience, like other ecosystem processes and properties⁶³ (e.g., biomass, productivity and carbon storage).

Studies in mangrove and terrestrial ecosystems, consistently link species diversity and structural diversity (variation in DBH and tree height) to key ecosystem properties and processes, such as productivity, aboveground biomass, carbon storages, and these descriptors of community composition can also promote ecosystem resilience^{7,66–68}. Our findings of a direct positive association of structural diversity with resilience, and an indirect positive association of species richness through increasing structural diversity, aligns with this growing body of work, indicating the minor relevance of a niche complementarity effect^{57,63} on mangrove forest resilience. Forest stands with high DBH variation reflect both horizontal and vertical heterogeneity^{67,69,70}, characterized by the spatial co-existence of trees with different sizes and growth forms. This complexity promotes niche partitioning, reduces competitive exclusion, and enhances microhabitat diversity, leading to more efficient light capture and resource use⁷¹. In this way, structural diversity buffers mangrove forests against disturbance and enhances their recovery potential. Additionally, species-rich forest stands with high structural diversity may prevent catastrophic canopy collapse during “super cyclones”. For example, if dominant tall trees are damaged, smaller or intermediate-sized individuals maintain partial canopy cover, stabilize sediments through diverse rooting architectures and facilitate faster regrowth^{72,73}. Thus, while functional trait composition primarily drives post-disturbance recovery in the Sundarbans, species diversity enhances resilience by promoting structural heterogeneity, increasing the range of growth forms and life-history strategies and fostering functional redundancy^{63,64,70}.

The rapid change in global temperatures is causing more frequent extreme weather events, such as tropical cyclones, as well as rising sea levels and, in conjunction with changing precipitation patterns, fluctuating salinity levels, which could influence the resilience of mangrove forests^{36,60,61}. The moderate total positive association of mean annual precipitation with resilience highlights their pivotal role in mangrove forest resilience. Our SEM analysis also showed how precipitation patterns exerted this positive influence mainly through three different pathways: functional composition of MCH, salinity and perturbation frequency. Higher rainfall plays a crucial role in alleviating water stress, increasing freshwater flow and diluting porewater salinity, which, in turn, reduces the energetic cost of osmoregulation and improves overall physiological performance. This allows mangrove trees to allocate more energy into vertical growth^{72–74}. Moreover, consistent precipitation helps maintain a hydrological balance and stabilizes sediment conditions⁷². In this way, rainfall can support enhanced growth capacity of dominant tall species and likely contributes to faster post-disturbance recovery, reinforcing the mass-ratio effect observed in our study.

Our results on the negative direct, indirect and total associations of perturbation frequency (PF), suggest that mangrove forest stands exposed to frequent cyclones, tidal surges and anthropogenic pressures exhibit lower resilience. Severe cyclones typically cause direct physical damage of mangrove trees, including canopy top breakage, uprooting and defoliation^{10,12}. Indirectly, these disturbances alter biogeochemical processes by creating hypersaline condition, increasing the sediment silt content, and elevating sediment pH—all of which are detrimental to mangrove health, productivity and, ultimately, mangrove forest resilience^{12,48,72}. Increased salinity

can render nutrients inaccessible to plant roots or force plants to spend more energy in coping with osmotic stress, thereby weakening their physiological performance^{72–74}. On the other hand, increased sediment pH, resulting from silt accumulation (Fig. 6) and reduced tidal flushing, can further reduce the availability of essential micronutrients, such as iron and manganese⁷². This alteration in physicochemical mangrove sediment properties disrupts microbial activity that is critical for nutrient-cycling and organic matter decay, compounding the stress on mangrove ecosystems and diminishing their capacity to recover from disturbance^{54,75}.

Phosphorus, although a key essential nutrient for mangroves, playing a crucial role in photosynthesis, growth and sediment microbial activity^{72,74}, can become ecologically disruptive when present in excessive amount. The direct negative association of the sediment phosphorus content with resilience suggests that excessive phosphorus-enrichment in mangrove forest sediments may disrupt the nutrient stoichiometric balance and interfere with microbial processes^{72,74}. In mangrove forest sediments, a balanced ratio of carbon (C), nitrogen (N) and phosphorus (P) is essential for sustaining microbial metabolic processes and ecological interactions. This stoichiometric balance supports efficient microbial activity, organic matter decay and nutrient-cycling^{54,74,75}, all of which are critical for mangrove forest health and resilience. High phosphorus levels can skew this ratio, leading to disproportionate growth of microbes that rapidly decay organic matter^{72,74} which, in turn, accelerates microbial respiration, depleting oxygen in the sediment and creating hypoxic conditions⁷². Under low oxygen availability, nutrient-cycling is impaired, and root respiration is restricted, ultimately weakening mangrove health and reducing ecosystem resilience.

Our findings have key implications for community ecology, policy-making for mangrove (re-)establishment, protection and management, as well as biodiversity stewardship⁷⁶. For example, for the first time, we detected hotspots of mangrove forest resilience loss in the Sundarbans, particularly in the northeastern part of the region. Thus, our resilience metrics (λ_{ACI} and λ_{VAR}) can be used as an indicator of forest integrity in reporting global conservation and climate stewardship such as it aligns with the Sustainable Development Goals (Goal 15: Life on Land) and the Kunming-Montreal Biodiversity Framework (Target 8)^{77,78}. As rising temperatures decrease mangrove forest resilience, mangrove-hosting countries can address the Kunming-Montreal biodiversity framework—specifically target 8—by indicating hotspot areas (8.5) that have experienced impacts of climate change⁷⁸. Furthermore, improvement of forest management by preserving adult individuals of dominant species⁷⁹, as encroachers primarily target large trees (positive association with resilience), increasing patrolling, raising awareness about the significance of protecting mangrove forests in adjacent communities and incorporating the effects of rising temperatures into forest policy, the Bangladesh forest department can address the Kunming-Montreal biodiversity framework targets 8.2 and 8.7, respectively⁷⁸. Similarly, in line with SDG15.2: end deforestation and restore degraded forests, SDG15.5: protecting biodiversity and natural habitat; SDG15.8: improve forest management through co-management approach⁸⁰, we found adult individuals of the dominant species and structural diversity to enhance mangrove forests resilience.

Beyond management implications, our results also highlight both the selection effect and, to a lesser degree, the niche complementarity hypotheses⁵⁷. Specifically, strong positive associations of functional composition of maximum canopy height and specific leaf area with resilience, but a weaker positive indirect association of species richness and a weaker positive direct association of structural diversity with resilience, demonstrate a coexistence of selection and niche complementarity effects in determining mangrove forest resilience. These findings underscore that the selection effect is a dominant driver of mangrove forest resilience, even when considering the direct associations of temperature, rainfall and sediment salinity with biodiversity^{54,60}. For future mangrove (re-)establishment projects⁸¹, it is essential to plant site-specific dominant species, with special attention to those species with great canopy height or growth form, along with few taxonomically and functionally diverse and distinct species to maintain a functionally rich and structurally diverse forest stand. On the

other hand, functional distinctiveness of mangrove species had been proven the major driver of multiple ecosystem properties and processes^{7,82}. With functional diversity having no significant effect on mangrove forest resilience, while functional composition of maximum canopy height and specific leaf area, and structural and taxonomic diversity increase resilience, these findings suggest a balanced approach towards (re-)establishing mangrove forests with diverse and functionally distinct, but partly functionally redundant, species. Ultimately, adopting such a strategy will foster multiple ecosystem processes and enhance the resilience of mangrove forests against the growing challenges posed by climate change and human pressures.

Method and materials

Remote sensing data source

We used the MODIS (Moderate Resolution Imaging Spectroradiometer) 16-day composite Normalized Difference Vegetation Index (NDVI) dataset (products MOD13Q1 (250 m) product, from February 2000 to December 2024. We ensured high-quality pixels in our analysis by using the QC_250 m quality assurance band to exclude pixels affected by cloud cover, cloud shadows, and other atmospheric effects. For getting stable forest, we intersected between 2000 and 2024 MODIS annual tree cover percentage (>2). As we used MODIS 250m-resolution pixels adjacent to or over small rivers or canals, the mixed pixels often reflect both vegetation and water bodies. To remove those pixels from our analysis, we further intersected the Landsat of 2000 and 2024 after masking the waterbody using a Modified Normalized Water Index with a threshold of 0.3⁸³. Our analysis was conducted on the Google Earth Engine (GEE) platform⁸⁴, mounted on Google Colaboratory, providing access to extensive geospatial datasets. We used the most up-to-date vegetation index, the Kernel Normalized Difference Vegetation Index (kNDVI; Eq. 1), which is less sensitive to the saturation effect compared to the Normalized Difference Vegetation Index²⁶.

In time series analysis, potential trends derived from seasonal variation and temporal differences can affect the resilience metric²⁵. Therefore, before calculating the perturbation detection, disturbance index, empirical recovery rate, theoretical recovery rate, and trend analysis, we employed two principal methods to detrend the kNDVI dataset to satisfy the assumption of stationarity^{25,27}—the rolling mean, a smoothing technique using a five-year rolling window to remove temporal trends; and harmonic regression, a statistical technique that fits a harmonic model to remove seasonal trends²⁵ (Fig. 1).

The Kernel Normalized Difference Vegetation Index (kNDVI) was calculated as

$$kNDVI = \tanh\left(\frac{NIR - RED}{NIR + RED}\right)^2 \quad (1)$$

Perturbation detection, disturbance metric, recovery rate

We detected a perturbation in the residual time series using a moving window approach²³. We employed a window size of 24 (12 months). We used 95% thresholding to detect the perturbation frequency on the residual time series²³. Furthermore, the method may flag non-significant fluctuations, especially in time series lacking strong disturbances. To mitigate this, we applied a minimum amplitude threshold: the raw time series must decrease by at least 0.01 to qualify as a valid perturbation²³. Further, we employed a Savitsky-Golay filter with a 7-point window to reduce high-frequency noise in the resultant derivative time series^{23,85}.

To assess whether the recovery rate was significant—and thus confirm that the system returned to its original state rather than shifting to an alternative state—we performed a two-sample Kolmogorov–Smirnov test ($P < 0.05$) comparing the residual kNDVI values. The baseline was defined as the dynamic mean of residual kNDVI during the year preceding perturbation detection⁴⁸ and the post-perturbation period spanned the subsequent 5 years. For each valid perturbation, we identified a local minimum

in the residual kNDVI time series within a 4-month window around the detected event to account for potential lags in response, then fitted an exponential equation to estimate the recovery rate. For pixels experiencing multiple perturbations, we used the average recovery rate²³. We also incorporated the potential cyclones occurrence using a 90-days window, there were 34 cyclones formed in the Bay of Bengal from 2000 to 2024 (Table S6). Based on the total count of valid perturbations, we develop two disturbance metrics: (1) average drop in kNDVI, and (2) weighted average drop in kNDVI (combine count, decay factor, and severity; Eqs. 2, 3). In both disturbance metrics, higher values indicate greater impact of perturbations.

$$\text{Average drop disturbance index (AD)} = \sum_{i=1}^N d_i \quad (2)$$

$$\text{Weighted Disturbance index (WD)} = N \left(\frac{\sum_{i=1}^N d_i e^{-\frac{t_i}{365}} \frac{d_i}{\max(d)}}{\sum_{i=1}^N e^{-\frac{t_i}{365}} \frac{d_i}{\max(d)}} \right) \quad (3)$$

Where N is the number of valid perturbations, d_i is the drop magnitude of residual kNDVI induced by the i -th perturbation, t_i is the number of days, and $\max(d)$ is the maximum drop magnitude observed for the pixel.

Resilience

Theoretical studies suggest that complex systems can reach tipping points—critical thresholds that lead to sudden and irreversible changes^{17,18,24}. As systems approach these points, reduced resilience makes them more vulnerable to shifts from minor, continuous disturbances. These tipping points can be measured through direct observations of slower recovery from perturbations, increased autocorrelation (AC1), and increased variance in the time series kNDVI^{17,18,24}. We measured resilience as the recovery rate (λ)—the rate at which a system returns to equilibrium following a small perturbation. Assuming that local dynamics near equilibrium can be approximated by an Ornstein-Uhlenbeck process, the continuous-time dynamics of deviations X_t from equilibrium is described by Eq. 4.

$$\frac{dX_t}{dt} = \lambda X_t + \sigma \xi_t \quad (4)$$

where $\lambda < 0$ denotes the recovery rate (higher negative values indicate faster recovery to equilibrium and thus higher resilience), σ is the noise intensity, and ξ_t is a Wiener process (i.e., integrated Gaussian white noise).

For empirical application, this process is discretized at a regular time interval Δt (set to 1 for simplicity), yielding a first-order autoregressive (AR1) model

$$X_{n+1} = aX_n + \tilde{\sigma}\eta_n \quad (5)$$

where X_n is the system state (e.g. kNDVI anomaly) at discrete time step n , $a = e^{\lambda\Delta t}$ is the lag-1 autocorrelation coefficient, $\tilde{\sigma}$ is the standard deviation of the discrete driving noise, and $\eta_n \sim N(0,1)$ is independent Gaussian white noise.

Under this formulation, two independent estimates of recovery rate λ can be derived directly from the observed kNDVI time series (Eqs. 6, 7):

1. From lag-1 autocorrelation⁴ (AC1)

$$\lambda_{AC1} = \frac{1}{\Delta t} \log(a) \quad (6)$$

2. From the variance of the kNDVI time series

$$\lambda_{var} = \frac{1}{2\Delta t} \log\left(1 - \frac{\tilde{\sigma}^2}{\text{Var}[X]}\right) \quad (7)$$

Equation 7 can be solved for λ as below:

$$\text{Var}[X] = \tilde{\sigma}^2 \frac{\sigma^2}{1 - e^{2\lambda\Delta t}} = -\frac{\sigma^2}{2\lambda} \quad (8)$$

Both λ_{ACI} and λ_{VAR} depend only on quantities directly quantifiable from the observed kNDVI time series: the slope α and the residual variance $\tilde{\sigma}^2$ from the linear regression in Eq. (5), and the total variance $\text{Var}[X]$. Under the ideal critical slowing down (CSD) conditions—namely, stationarity, linearity near equilibrium, and the additive Gaussian noise—the two estimates should approximately equal. Therefore, deviation from one-to-one relationship between λ_{ACI} and λ_{VAR} can serve as a diagnostic for the validity of CSD-based resilience estimate derived from a given time series²⁵.

Baseline and forest condition change. We analyzed mangrove vegetation dynamics from 2000 to 2024, using the mean kNDVI relative to the baseline condition in 2000. Based on the empirical distribution of kNDVI in 2000, we defined three resilience-based health classes using quantile thresholds: Resilient (Q3; kNDVI > 0.49), Moderately Resilient (Q2; 0.39 < kNDVI ≤ 0.49), and Low Resilient (Q1; kNDVI ≤ 0.39). Applying this fixed threshold, we classified annual mean kNDVI value for 2000–2024, average for 2001–2024, 5-year rolling mean for each year from 2002 to 2022 and the mean of those rolling windows (Fig. 4). Using this classification, we tracked interannual transitions among resilience classes and developed spatial maps depicting stable, gain, and loss in mangrove health status for all the periods mentioned above (Fig. 5, Tables 1, 2).

Forest stand structure and diversity, environmental data. We used ground-measured plot-level live tree species DBH (diameter at breast Height) ≥ 10 cm data from the Sundarbans Forest carbon inventory 2010³⁵. A total of 150 plots were surveyed from 5 December 2009 to 30 May 2010³⁵. Each plot comprised of five nested subplots with a 10-m radius: one central subplot and four additional subplots located 50 m away in the cardinal directions (Fig. S7)³⁵. In our current study, we used 100 plots; the remaining 50 plots were excluded because they were either fully underwater in the river due to riverbank erosion or subplots with water bodies more than 20% due to the presence of a small canal, or on the bank of the river or a big canal^{7,35}. To estimate the taxonomic diversity, we counted the total number of species in each plot. We used publicly available plant trait data specific to the Sundarbans, acquired in recent years^{45,86}, for calculating functional trait-related diversity.

The seven functional traits that were used in this study (Table S5) fall into three categories: leaf traits (specific leaf area, leaf litter nitrogen, leaf dry matter content, leaf photosynthesis rate, and leaf succulence), stem traits (wood density), and whole plant traits (maximum canopy height)^{7,45,86–88}. We averaged the leaf litter nitrogen values across the three distinct seasons in the Sundarbans: pre-monsoon (February to May), monsoon (Jun to September), and post-monsoon periods (October to January)^{35,86}. We used the community mean trait for species whose trait value was not available (relative abundance species < 2%)⁷.

To quantify functional trait diversity, we used two diversity metrics: functional diversity (functional dispersion; Eq. 9) and functional composition (community-weighted mean; Eq. 10). We used the FD package in R⁸⁹ to calculate both functional dispersion and community-weighted mean, where the species-wise sum of basal area in a plot was used as a weighting factor, provided that basal area represents the best contribution of a species in a plot or community⁶³. The data was log-transformed and standardized to a zero mean and one standard deviation.

$$\text{FDIs} = \frac{\sum a_j Z_j}{\sum a_j} \quad (9)$$

where a_j is the basal area proxy for abundance of species j , and z_j is the distance of species j to the weighted centroid c , which signifies the centroid of

the n species within the trait space.

$$\text{CWM}_{ij} = \sum_{i=1}^S P_{ij} t_i \quad (10)$$

where CWM_{ij} is the community-weighted mean for particular trait t for the j th plot, S refers to the number of species, P_{ij} is the relative contribution to the total basal area of a specific species i for a plot j , and t_i represents the average trait value of species i .

We used high-resolution (1-km) climatic data from CHELSA climatology dataset⁹⁰ (Dataset: CHELSA-climatologies•CHELSA; https://www.chelsa-climate.org/datasets/chelsa_climatologies) maintained by the Swiss Federal Institute for Forest, Snow, and Landscape Research. Long-term monthly temperature and precipitation data spanning historical (1980–2010) and future projections (2011–2040), were employed to calculate pixel-wise mean annual temperature (derived from monthly average) and annual precipitation (sum across months). We extrapolated the soil physiochemical properties—including sediment silt content, sediment phosphorus (P), sediment ammonium (NH₄), sediment potassium (K), and sediment sulfide—using field data collected by Saker et al.⁴⁵ from 110 permanent sample plots (plot size: 120 × 20 m; Fig. S7) in the Sundarbans, between January and June 2014. These plots encompass nearly all forest types in the Sundarbans, as the permanent sample plots were stratified by vegetation type and salinity zone. We used the Google Earth Engine (GEE) platform to generate spatial maps of soil physiochemical properties using Inverse Distance Weighted Interpolation⁹¹. By using the temporary carbon plot shapefile (100 plots, each comprising five circular subplots with 10-m radii), we extracted mean soil properties from the extrapolated maps, MODIS-derived resilience layers and Chelsha climatic data^{35,82}. For each plot geometry, we applied GEE's, `reduceRegions()` function with `ee.Reducer.mean()` at a 30-m scale to summarize values⁸⁴.

Statistical analysis

We assessed the trends in recovery rate (λ) using Kendall's τ with overlapping 5-year windows spanning 2002–2022²⁵. For example, the window centered on 2002 included data from 2000–2004, the window for 2003 used data from 2001 to 2005, and so on, up to 2022, which incorporated data from 2020–2024. As our study period covers 2000–2024, we could not compute full 5-year rolling window for the boundary years 2000, 2001, 2023, and 2024 due to insufficient adjacent data. A significant positive Kendall's τ means an increase in λ ($P < 0.05$), which indicates a loss of resilience, while a negative Kendall's τ ($P < 0.05$) implies a decrease in λ and a gain in resilience. Insignificant trends indicate a stable condition. We employed structural equation modeling⁵⁸ to investigate the interactive association among covariables. To satisfy the linearity assumption⁹², we applied Box-Cox transformation and standardized all variables to a mean of zero and one standard deviation. We performed a multicollinearity test by calculating variance inflation factors (VIFs) to identify and eliminate highly correlated variables among trait-specific functional composition and functional diversity metrics. Following a widely used threshold in the literature ($\text{VIF} < 10$)^{7,66,82,93}, we retained only those variables with VIF value below 10. This resulted in four traits of functional diversity (Table S2) and three traits of functional compositions (Table S3). The traits included in the functional diversity group were specific leaf area, maximum canopy height, leaf succulence, and leaf litter nitrogen, while those in the functional composition group were specific leaf area, maximum canopy height, and leaf litter nitrogen.

Using perturbation frequency and the two-disturbance metrics—along with climatic, sediment physicochemical properties, and taxonomic, structural, and functional trait diversity—we constructed six structural equation models (SEM): three for λ_{ACI} and three for λ_{VAR} (Table S1). Each pair model incorporated one of the three disturbance-related variables: perturbation frequency, average drop disturbance index (AD), and weighted average drop disturbance Index (WD). The biotic variables included seven

Table 3 | Standardized direct, indirect, and total associations of abiotic and biotic drivers with resilience (λ_{VAR}) in the Sundarbans mangrove forest, based on Structural Equation Model (SEM) shown in Fig. 6

Covariables	MAT	MAP	PF	STS	SS	pH	SR	SD	FCSLA	FDSC	FCMCH	SF	SP	SSF
Standardized Direct Effects														
STS	0.21	-0.04	0.20**	0	0	0	0	0	0	0	0	0	0	0
SS	0.58***	-0.59***	0.02	-0.10**	0	0	0	0	0	0	0	0	0	0
pH	0.15	-0.13	-0.01	0.19*	-0.20**	0	0	0	0	0	0	0	0	0
SR	-0.06	-0.40**	0.15**	0.11	-0.33***	-0.09	0	0	0	0	0	0	0	0
SD	-0.11	-0.20*	0.07	0.01	-0.10*	-0.01	0.35***	0	0	0	0	0	0	0
FCSLA	0.53**	-0.37***	0.02	-0.01	0.12*	0.20**	0.01	-0.27***	0	0	0	0	0	0
FDSC	0.30*	0.12	0.15	0.07	-0.39***	-0.12	0.17	0.20*	0	0	0	0	0	0
FCMCH	-0.64***	0.55***	0.02	-0.03	0.05	-0.06	-0.16*	-0.1	0	0	0	0	0	0
SF	-0.21*	0.21*	0.06	-0.07	-0.06	-0.13	-0.05	-0.04	0.01	0	0	0	0	0
SP	-0.26*	0.14	0.06	-0.03	0.23*	-0.16	0.03	-0.21*	0.25**	0	0	0	0	0
SSF	-0.43***	0.54***	-0.13*	-0.09	-0.25*	0.11	0.13	0.05	0.09	-0.22***	0	0	0	0
RLV	-0.09	0.21*	-0.29**	0.09	-0.01	-0.21**	0.09	0.22*	0.62***	-0.21*	0.61***	0.17	-0.29***	0.11
Standardized Indirect Effects														
STS	0	-0.07*	0	0	0	0	0	0	0	0	0	0	0	0
SS	-0.02	0.01	-0.02	0	0	0	0	0	0	0	0	0	0	0
pH	-0.07	0.1	0.04	0.02	0	0	0	0	0	0	0	0	0	0
SR	-0.17*	0.13	0.02	0.02	0.02	0	0	0	0	0	0	0	0	0
SD	-0.14	-0.06	0.06	0.05	-0.11*	-0.03	0	0	0	0	0	0	0	0
FCSLA	0.15**	-0.01	-0.03	0.02	0.02	0.01	-0.1	0	0	0	0	0	0	0
FDSC	-0.30**	0.07	0.07	0.05	-0.07	-0.02	0.07	0	0	0	0	0	0	0
FCMCH	0.08	0.04	-0.05	-0.04	0.08*	0.02	-0.03	0	0	0	0	0	0	0
SF	-0.07	0.07	-0.05	-0.05	0.1	0.02	-0.05	-0.03	0	0	0	0	0	0
SP	0.30*	-0.2	-0.06	-0.08	0.15*	0.07	-0.13	-0.09	0	0	0	0	0	0
SSF	-0.08	0.02	-0.03	0.05	0.03	0.04	-0.02	-0.06	0	0	0	0	0	0
RLV	-0.12	0.23*	-0.04	-0.08	0.1	0.14*	-0.13	-0.19*	-0.06	-0.01	-0.01	0	0	0
Standardized Total Effects														
STS	0.21**	-0.11	0.20**	0	0	0	0	0	0	0	0	0	0	0
SS	0.56***	-0.58***	0	-0.10**	0	0	0	0	0	0	0	0	0	0
pH	0.08	-0.03	0.03	0.21**	-0.20**	0	0	0	0	0	0	0	0	0
SR	-0.23**	-0.27***	0.17**	0.12**	-0.32***	-0.09	0	0	0	0	0	0	0	0
SD	-0.25**	-0.26**	0.13	0.06	-0.21*	-0.04	0.35***	0	0	0	0	0	0	0
FCSLA	0.68***	-0.39***	-0.01	0.01	0.14**	0.21**	-0.09	-0.27***	0	0	0	0	0	0
FDSC	0	0.2	0.22**	0.12**	-0.46***	-0.14*	0.24**	0.20*	0	0	0	0	0	0

Table 3 (continued) | Standardized direct, indirect, and total associations of abiotic and biotic drivers with resilience (λ_{VAR}) in the Sundarbans mangrove forest, based on Structural Equation Model (SEM) shown in Fig. 6

Covariates	MAT	MAP	PF	STS	SS	pH	SR	SD	FCSLA	FDSC	FCMCH	SF	SP	SSF
FCMCH	-0.56***	0.59***	-0.03	-0.08	0.13	-0.05	-0.19**	-0.1	0	0	0	0	0	0
SF	-0.28**	0.28**	0	-0.12	0.04	-0.11	-0.1	-0.06	0.01	0	0	0	0	0
SP	0.05	-0.05	0.01	-0.11	0.38***	-0.09	-0.1	-0.30***	0.25**	0	0	0	0	0
SSF	-0.51***	0.56***	-0.16	-0.04	-0.22*	0.15*	0.1	-0.01	0.09	-0.22***	0	0	0	0
RLV	-0.20*	0.44***	-0.33***	0.01	0.1	-0.07	-0.03	0.02	0.56***	-0.22*	0.61***	0.17	-0.29***	0.11

All the values are standardized regression weights (β), not correlation coefficients. The association was estimated using maximum likelihood and validated with 1000 bootstrap samples. For interpretation, we follow conventional benchmarks for standardized regression coefficients in ecological SEMs: $\beta < 0.2$ indicates a weak association, $0.2 < \beta < 0.5$ a moderate association and $\beta \geq 0.5$ strong association^{83,9}. Significance levels: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. MAP mean annual precipitation, MAT mean annual temperature, SS sediment salinity, SR species richness, SD structural diversity (variation of stem diameter at breast height; 1.37 m), FC functional composition, SLA specific leaf area, MCH maximum canopy height, FDis functional dispersion, LS leaf succulence, P sediment phosphorus concentration, SF sediment sulfide concentration, SSF sediment sulfide concentration, pH sediment pH level, STS sediment texture (silt content), PF perturbation frequency.

functional trait-based diversity metrics, species richness (as a measure of taxonomic diversity), and co-efficient of variation of DBH (as a measure of structural diversity)^{56,94}. Environmental covariables comprised eight sediment and climate variables—sediment sulfide, sediment phosphorus, sediment iron concentration, sediment pH, sediment texture silt, precipitation, temperature, and sediment salinity. We gradually removed the hypothesized path with the highest p-value and high correlation until the model satisfied the model-fitting measures⁷. In this approach, from the functional diversity group, specific leaf area, leaf litter nitrogen, and maximum canopy height were dropped, while from the functional composition group, leaf litter nitrogen was dropped. The final model with 14 covariates with a target variable (λ_{ACI} and three for λ_{VAR}) satisfied the model-fitting statistics (Fig. 6, Table 3; Table S1), with final VIF values for all variables below 10 (Table S4). The retained predictor covariables were species richness, functional diversity of leaf succulence, functional composition of specific leaf area, maximum canopy height, structural diversity, sediment salinity, sediment sulfide, sediment phosphorus, sediment iron concentration, sediment pH, sediment texture silt, perturbation frequency, precipitation, and temperature.

We estimated the total association of a predictor by summing its direct and indirect associations through different mediators⁵⁸. A specific indirect association was estimated by multiplying all the hypothesized paths from that predictor to that mediator⁵⁸ (to its mediator if the hypothesized paths are available). We constructed the SEM³¹ in R studio using Lavaan, semPlot, and semTools packages. Additionally, we estimated the indirect and total associations of predictors with endogenous variables in Amos 22. We defined the association impact size of the predictor on the target variables according to the conventional benchmarks for standardized regression coefficients in ecological SEMs: $\beta < 0.2$ indicates a weak association, $0.2 < \beta < 0.5$ a moderate association, and $\beta \geq 0.5$ a strong association^{58,59}. For raster image visualization, we used different R packages⁸⁹ (sf, raster, ggplot2, dplyr, cowplot, gridExtra, rasterVis).

We evaluated the SEM⁹⁵ models using four commonly used goodness-of-fit statistics: an insignificant chi-squared (χ^2 ; Eq. 11) value; a goodness-of-fit index (GFI; Eq. 12) closes to 1; a comparative fit index (CFI; Eq. 13) closes to 1; and a standardized root mean square residual (SRMR; Eq. 14) nearly zero. These indicate that the model does not significantly deviate from our dataset^{58,95}.

$$\chi^2 = (N - 1)f \quad (11)$$

$$GFI = \frac{\text{trace}(S - \sum(\hat{\theta}))^2}{\text{trace}(S^2)} \quad (12)$$

$$CFI = \frac{\max(\chi_0^2 - df_0) - \max(\chi_k^2 - df_k)}{\max(\chi_0^2 - df_0)} \quad (13)$$

$$SRMR = \sqrt{[\rho^{*-1}(e'W_s e)]} \quad (14)$$

where χ^2 represents the Chi-square test statistic; CFI stands for Comparative Fit Index; SRMR is the Standardized Root Mean Square Residual; N denotes the sample size; f refers to the minimized discrepancy function; 0 indicates the baseline model; df stands for degrees of freedom; k is the tested or hypothesized model; S denotes sample covariance matrix; $\sum(\hat{\theta})$ is referred as the model-implied covariance matrix; trace is denoted as the trace of a matrix, which is the sum of the diagonal elements; ρ^* indicates the number of non-duplicated elements in the covariance matrix; e is a vector of residuals from the covariance matrix; s refers to the vector of the ρ^* non-redundant elements in the observed covariance matrix; W denotes the weight matrix; and W_s is a diagonal weight matrix used to standardize the elements of the covariance matrix.

Data availability

The satellite data (MODIS/061/MOD13Q1; <https://doi.org/10.5067/modis/mod13q1.061>) used in this study is publicly available⁹⁶ and can be accessed through Google Earth Engine (https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1)⁸⁴. The climatic data (long-term monthly precipitation and temperature) is publicly available at Dataset: CHELSA-climatologies • CHELSA⁹⁰. Plant functional traits, salinity, and sediment nutrients are publicly available on the University of Glasgow data archive (Enlighten Research Data System) at <https://doi.org/10.5525/gla.researchdata.693> via Sarker et al.⁴⁵. Plot-level biodiversity, remote sensing-based resilience indicators, perturbation frequency, environment, and sediment properties data that support the findings of this study are publicly available in Zenodo⁹⁷ repository, which can be accessed at <https://zenodo.org/records/18470105>. The raw forest inventory data (tree-level DBH, height, and geographic location) from the Bangladesh Forest Department has not been made public and is subject to confidentiality clauses under open-access policies. However, these data can be accessed through a formal request to the Chief Conservator of Forest, the Officers in Charge of the Education and Research Wing, and the Bangladesh Forest Information System wing of the Bangladesh Forest Department (<https://bfis.bforest.gov.bd/bfis/>).

Code availability

Codes used to assess the resilience class and their status, resilience metrics, disturbance metrics, and to prepare figures are publicly available in Zenodo⁹⁷ repository, which can be accessed at <https://zenodo.org/records/18470105>.

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Author contributions

R.M.M. conceived and designed the study, analyzed and interpreted the results, and wrote the draft. R.M.M. participated in the field survey. R.M.M., M.Z., M.S.R., D.D., J.L., M.N.I.K., and Q.W. reviewed and revised the paper. Q.W. contributed to concept and methodological development, provided funding for this research, and supervised the study.

Competing interests

The authors declare no competing interests.

Additional information

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Co-benefits of protecting mangroves for biodiversity conservation and carbon storage

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The conservation of ecosystems and their biodiversity has numerous co-benefits, both for local societies and for humankind worldwide. While the co-benefit of climate change mitigation through so called blue carbon storage in coastal ecosystems has raised increasing interest in mangroves, the relevance of multifaceted biodiversity as a driver of carbon storage remains unclear. Sediment salinity, taxonomic diversity, functional diversity and functional distinctiveness together explain 69%, 69%, 27% and 61% of the variation in above- and belowground plant biomass carbon, sediment organic carbon and total ecosystem carbon storage, respectively, in the Sundarbans Reserved Forest. Functional distinctiveness had the strongest explanatory power for carbon storage, indicating that blue carbon in mangroves is driven by the functional composition of diverse tree assemblages. Protecting and restoring mangrove biodiversity with site-specific dominant species and other species of contrasting functional traits would have the co-benefit of maximizing their capacity for climate change mitigation through increased carbon storage.

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The role of forests in carbon sequestration renders their protection a cost-effective nature-based solution for reducing atmospheric concentration of carbon dioxide^{1,2}. How this carbon sequestration property is influenced by human-induced changes in forest biodiversity has led to hundreds of studies, mainly in terrestrial ecosystems^{3–7} (temperate, boreal, tropical, subtropical, agroforestry and plantation forests) and a few mangrove studies^{8–13}. Biodiversity-carbon research^{3,4} in terrestrial ecosystems has spanned trait-based (morphological, phenological, physiological and chemical characteristics of plants) biodiversity (e.g., functional composition, community-level mean value of plant traits and functional diversity, variation of species traits within a community) along with conventional diversity (species and taxonomic diversity) approaches. However, biodiversity-carbon studies in mangroves, among the most carbon-rich ecosystems in the tropics^{14–16}, are still essentially restricted to conventional diversity^{8–10,14,17,18}. This reveals a crucial knowledge gap for understanding mangroves' contribution to mitigating climate change, as functional diversity studies can elucidate the mechanisms behind species-diversity association with carbon storage^{4,19}. Filling this knowledge gap will inform mangrove conservation policy, such as REDD+ (Reducing Emissions from Deforestation and forest Degradation plus), the Post-2020 biodiversity agenda of the Convention of Biological Diversity, and mangrove rehabilitation and afforestation in general²⁰.

Previous studies have shown that species diversity has both a direct and indirect association with ecosystem processes through both functional diversity and functional composition, (i.e., the functional distinctiveness of taxonomic units and of the community), on carbon storage in terrestrial ecosystems^{4,19,21}. The association of these components of biodiversity with productivity or carbon storage have usually been explained by two hypotheses: niche complementarity and selection⁴. The niche complementarity hypothesis suggests that the capacity to store carbon in a community is primarily determined by coexisting species with greater trait variance^{4,21}. On the other hand, the selection theory assumes that the most abundant species with their key functional traits determine the ecosystem-wide carbon storage^{4,21}. Thus, if species diversity or functional diversity influences carbon storage, this would lend support to the niche complementarity hypothesis as a mechanism for the storage of blue carbon. Conversely, if functional composition influences the storage of blue carbon, this lends support to the selection hypothesis⁴. Alternatively, both mechanisms may jointly control carbon storage in some tropical forests, including blue carbon storage if both species richness or functional diversity and functional composition influence mangrove carbon storage^{4,22}.

Plant traits related to carbon storage are used to elucidate the relationship of functional diversity and functional composition with carbon storage²³. For instance, wood density, maximum canopy height, specific leaf area, leaf dry matter content, leaf photosynthesis rate, leaf carbon and nitrogen content and their ratios are commonly used plant traits in functional ecological studies^{3,4,24}. Different functional traits occupy different niche axes and determine the growth, productivity, stability and resistance of overall carbon storage in a community^{3,25}. For example, high wood density provides mechanical support, affects water and nutrient supply and hence plays a key role in plant growth, survival and forest carbon stock²⁶. Maximum canopy height, an indicator of light interception²⁷ that reflects the relative resource-use capacity of a species in a community, determines community scale biomass allocation and carbon storage^{3,4,28–30}. Plant traits such as leaf carbon, nitrogen and their ratio, leaf dry matter content or contents of phenolic compounds, as indicators of litter quality for decay and decomposition, could influence nutrient

cycling and productivity^{31–34}. For instance, higher litter nitrogen content or phenolic content could enhance microbial litter decay and drive sediment carbon content and productivity in mangroves^{34–36}.

The carbon fixed by mangrove plants, and accumulated in biomass and sediment organic matter over time, is influenced by integrated mechanisms involving both abiotic and biotic influences^{37,38}. On a wider geographical scale, climatic factors (abiotic) including temperature and precipitation have an association with carbon storage^{9,30}, while the local or regional hydrological system, such as freshwater flow or the tidal regime, controls porewater salinity in the sediment^{9,37,39,40}, which, in turn, influences plant growth, stand structure and ecosystem productivity⁴¹ and therefore blue carbon storage. Sediment salinity also controls total sediment nutrients³⁹ (nitrogen, phosphorus and total organic matter), as well as the distribution of the plant species⁴² and, hence, their functional traits. For example, low-salinity areas are rich in plant species with a higher potential for photosynthetic nitrogen use efficiency⁴³, resulting in higher dry matter content and higher growth form (maximum canopy height) compared to plant species in high-salinity areas. Salinity-driven variation of species richness and composition can also affect two other components of biodiversity that are based on plant functional traits: functional diversity and functional composition of different plant traits in a community. As a result, sediment salinity may also have indirect associations with blue carbon storage in mangroves through these three components of biodiversity: species richness, functional diversity and functional composition. The integration of the functional composition of the community, (i.e., the functional distinctiveness of species within the community), into modelling along with other traditional and modern diversity measures and sediment salinity can provide new insights into the relationship between mangrove biodiversity and blue carbon storage.

Here we explore the relationships among species richness, taxonomic diversity, functional diversity and functional composition with aboveground and belowground plant biomass carbon, sediment organic carbon and total ecosystem carbon storage after considering for the association of sediment salinity, using a structural equation model (a robust approach compared to multiple linear or bivariate relationship, as it considers direct, indirect and total association of predictors⁴⁴) in the mangroves of the Sundarbans Reserved Forest, Bangladesh. In this study, we analyzed a large-scale forest carbon inventory dataset which includes 90 forest plots distributed across the forest area (Supplementary Table 1 provides a description of field data including sediment salinity). Each plot is composed of five 10 m radius circular subplots⁴⁵. We estimated different biomass carbon pools, using both locally developed species-specific and common allometric equations for mangroves^{45–48}. For estimating sediment organic carbon, we used the organic carbon storage of the top 1 m (Supplementary Table 1; methods⁴⁵). We used maximum canopy height, wood density, photosynthesis rate, and four-leaf litter qualities²⁷ (carbon, nitrogen, C:N ratio, and dry matter content; Supplementary Table 2) as plant functional traits. For species diversity, we used species richness (count of species per plot) and Shannon diversity. We calculated the functional dispersion and community weighted mean as proxies for functional diversity and functional composition, respectively. We focus on two questions: (i) how are different carbon pools in mangroves related to species diversity, functional diversity and functional composition, after accounting for direct and indirect associations of the key gradient of sediment salinity? and (ii) how can these relationships guide conservation and rehabilitation or afforestation policies for mangroves? For answering the first question, we hypothesized that (i) sediment salinity has direct and indirect associations with

blue carbon storage through species diversity, functional diversity and functional composition; (ii) species diversity has direct and indirect associations with blue carbon storage through functional diversity and functional composition; (iii) functional composition has a greater association with blue carbon storage than the other two biodiversity components. Species richness and functional distinctiveness of wood density, maximum canopy height and leaf litter nitrogen are positively associated with blue carbon storage where functional distinctiveness, had the strongest association with blue carbon storage. Our findings indicate that blue carbon storage in mangroves can be best sustained through compositionally and functionally diverse tree assemblages.

Results

Bivariate relationships among different covariates. Bivariate analyses (to observe independent associations of all predictors with different blue carbon storage, apart from our structural equation model) revealed that aboveground and belowground plant biomass carbon, sediment organic carbon and total ecosystem carbon storage decreased with increasing sediment salinity (Fig. 1a–d, $P < 0.05$). Species richness (measured by counting observed species in each plot) and functional composition of wood density and maximum canopy height had positive associations with aboveground and belowground plant biomass carbon, sediment organic carbon (although maximum canopy height had no significant effect; $P > 0.05$) and total ecosystem carbon storage (Fig. 1i–p). However, the functional composition component of leaf litter nitrogen content and functional diversity component of leaf litter dry matter content did not have any significant association with any of the blue carbon storage (Fig. 1q–x; $P > 0.05$). Separately, sediment salinity, species richness, functional composition of wood density and maximum canopy height explained less variance in different blue carbon storage (Fig. 1a–p). All components of biodiversity (except species richness, $P > 0.05$) decreased significantly with increasing sediment salinity (Fig. 2a–e). Species richness had no significant association with functional diversity or the functional composition factor of wood density (Fig. 2f and i), but we observed a significant negative and positive association, respectively, with functional composition aspects of maximum canopy height and the nitrogen content of the leaf litter (Fig. 2e; $P < 0.05$). With increasing functional distinctiveness (depicted by functional composition) of maximum canopy height, the functional distinctiveness of wood density increased, while the functional distinctiveness of leaf litter characteristics showed a decreasing trend (Fig. 2j and k).

Drivers of blue carbon storage in mangrove. While bivariate analyses only show one-way associations of sediment salinity and multiple biodiversity measures with different blue carbon storage, structural equation models show interactive associations of these factors with different blue carbon storage (Fig. 3). Sediment salinity had no significant direct association with blue carbon storage, but had an indirect association with them through the functional composition of different traits (Fig. 3). Four structural equation models for aboveground and belowground plant biomass carbon, sediment carbon and total ecosystem carbon storage fit the datasets well, indicating no significant deviation between observed datasets and models (Fig. 3), as indicated by a Chi-square test statistic of $\chi^2 = 6.83$, $P = 0.093$, a comparative fit index close to 1 (CFI = 0.99), and a standardized root mean square residual close to 0 (SRMR = 0.03; Supplementary Table 5). Sediment salinity, species richness, functional composition of wood density, maximum canopy height and leaf litter nitrogen, as well functional diversity of leaf litter dry matter content together

explained 69%, 69%, 26 % and 61% of variation of aboveground and belowground plant biomass carbon, sediment carbon and total ecosystem carbon storage, respectively (Fig. 3). The functional diversity and functional composition of the other traits were strongly correlated and their variance inflation factors were > 10 and thus they were not retained in the structural equation models (Supplementary Figs. 1–2; Supplementary Tables 3–4; methods).

For aboveground and belowground biomass carbon, the strong negative association of sediment salinity was mediated by the functional composition of wood density, while for sediment organic carbon and total ecosystem carbon it was mediated strongly by the functional composition of maximum canopy height (Table 1). On the other hand, the greatest association of species richness was induced by increasing the functional composition of litter nitrogen on aboveground and belowground biomass carbon (Table 1). However, for sediment organic carbon and total ecosystem carbon, the direct associations of species richness were the strongest paths in the structural models (Fig. 3c and Table 1). When we considered all direct associations, including sediment salinity, species richness, functional composition and functional diversity, functional composition always had the strongest associations with different types of blue carbon storage, with the association size depending on the type of traits, while functional diversity had the lowest association (Fig. 3). For instance, the functional composition of wood density was the strongest driver in the case of aboveground plant biomass carbon, whereas the functional composition of both wood density and maximum canopy height were the strongest drivers for belowground plant biomass carbon (Fig. 2a, b).

Discussion

We quantified the interacting associations of species richness, functional diversity and functional composition of mangrove forests with different blue carbon storage in the Sundarbans Reserved Forest, one of the largest mangrove forests in the world and among the largest carbon deposits in the tropics¹⁶. We found that species richness and functional composition had a positive relationship with blue carbon storage, while functional diversity had a negative association after considering the association of sediment salinity in structural equation models. Sediment salinity mainly decreased functional diversity and distinctiveness of species but not species richness, and hence, its associations with different blue carbon storage are mediated by functional diversity and functional composition of the community. Species richness had positive indirect associations with plant above- and belowground biomass carbon through functional composition, while it had both direct and indirect (through functional composition) associations with sediment organic carbon and total ecosystem carbon. Our results also indicate that among the multiple facets of biodiversity, functional composition exerted the strongest and functional diversity the weakest -associations with different blue carbon storage. From this study, we derive several key implications for community ecology in mangroves and the conservation value of mangroves for climate change mitigation.

First, the positive association of species richness on different blue carbon storage contributes to answering the unexplored question of whether increasing species richness in relatively species-poor mangrove ecosystems (compared with tropical and subtropical terrestrial forests) can enhance blue carbon storage. Our results identify (in line with our second hypothesis) niche complementarity²¹ as one of the determinants of mangrove carbon pools in the Sundarbans Reserved Forest. Hence, stands with higher species richness exhibit higher carbon storage and vice versa. Mangroves hold their largest carbon storage in their

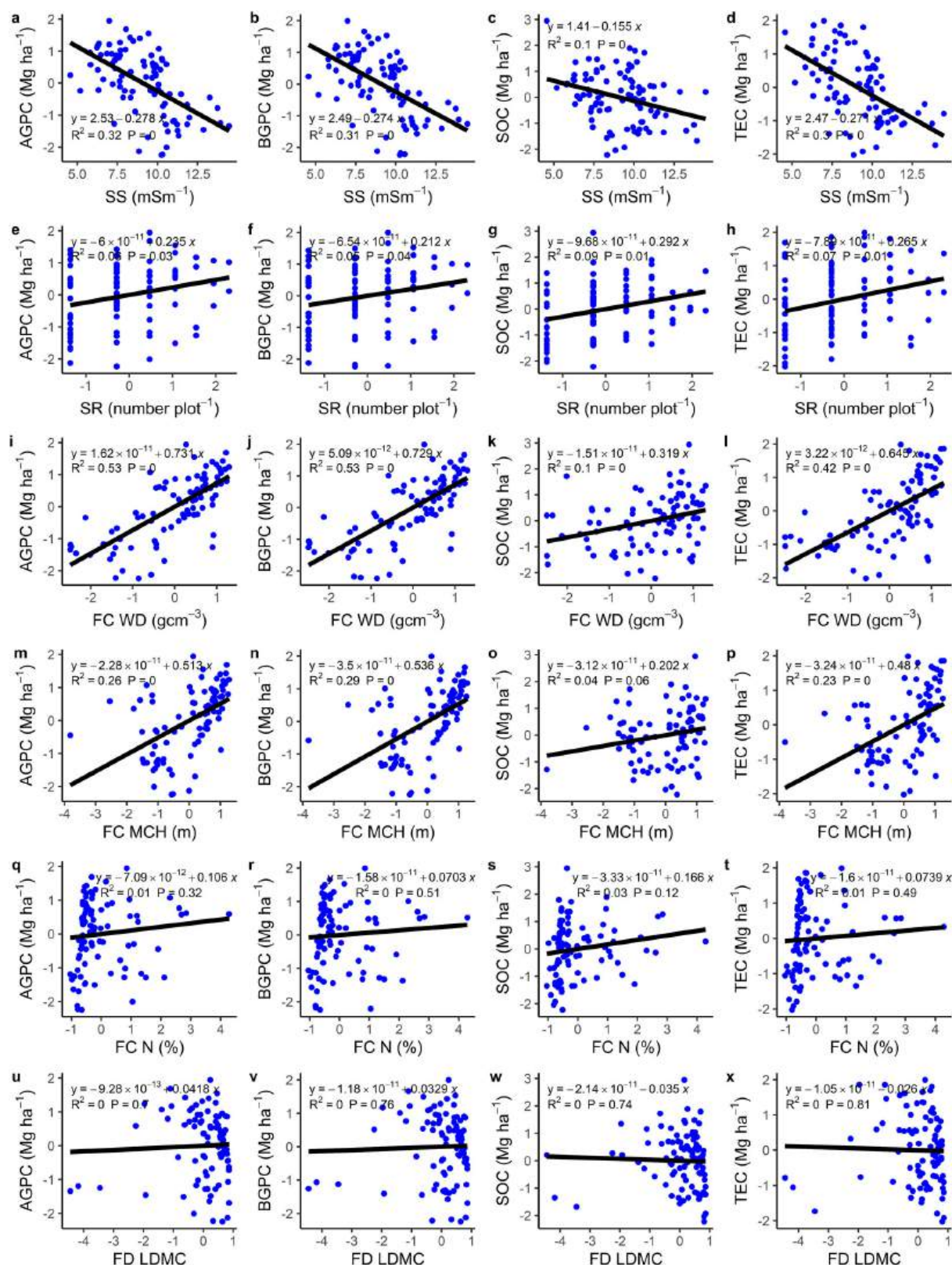


Fig. 1 Bivariate relationships between sediment salinity (SS), species richness (SR), functional composition and diversity of different traits for hypothesized causal paths in the structural equation models. a–x Sediment salinity (a–d), species richness (SR; e–h), functional composition wood density (FC WD; i–l), functional composition maximum canopy height (FC MCH; m–p), functional composition leaf litter nitrogen (FC N; q–t), functional diversity leaf litter dry matter content (FD; u–x) versus aboveground plant biomass carbon (AGPC), belowground plant biomass carbon (BGPC), sediment organic carbon (SOC) and total ecosystem carbon (TEC), respectively. Each blue point represents a sampling plot, while the black line indicates the fitted regression line. *P* value indicates the significance level.

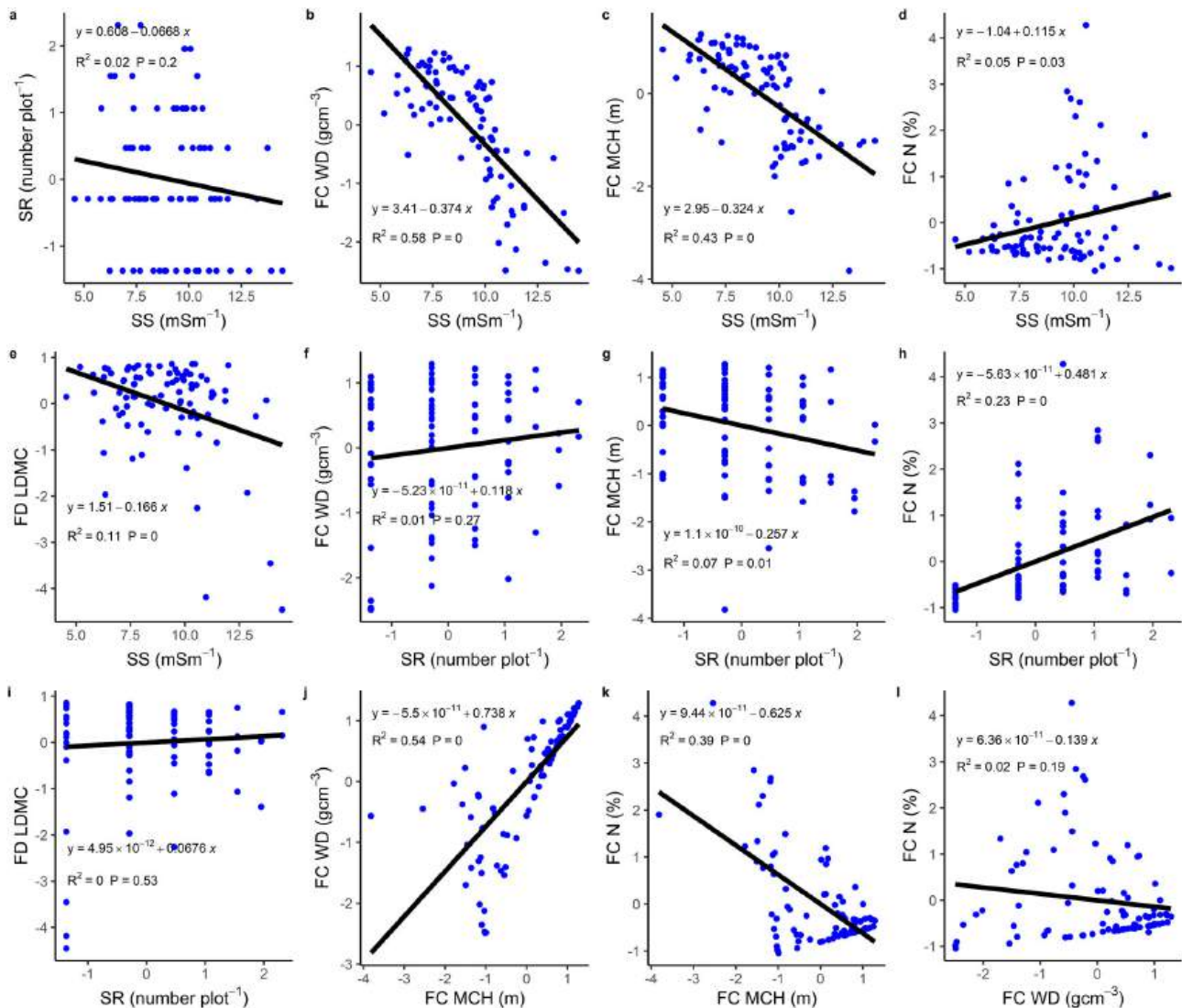


Fig. 2 Bivariate relationships between sediment salinity (SS), species richness (SR), functional composition, and functional diversity of different traits for hypothesized causal paths in the structural equation models. a–l Sediment salinity- versus species richness (a), -functional composition wood density (FC WD; b), -functional composition maximum canopy height (FC MCH; c), -functional composition litter nitrogen (FC N; d), -functional diversity leaf litter dry matter content (FD LDMC; e); species richness- versus functional composition wood density (f), -functional composition maximum canopy height (g), -functional composition litter nitrogen (h), -leaf litter dry matter content (i); functional composition maximum canopy height versus functional composition wood density (j); functional composition maximum canopy height versus functional composition litter nitrogen (k); functional composition wood density versus functional composition litter nitrogen (l). Each blue point represents a sampling plot, while the black line indicates the fitted regression line.

sediments⁹ (11–98% of total ecosystem carbon stock depending on the site), especially in mixed stands¹⁴. In the present study, species diversity had a greater impact on sediment carbon than on other carbon pools, suggesting that maintaining a mixed species stand is beneficial when management objectives include increasing carbon storage in the sediment¹³. In addition, diverse stands also contribute to sediment retention through providing diverse trunk (buttress) and root systems (pneumatophores, prop and stilt roots, knee roots).

Second, the positive and stronger standardized association of functional composition than that of species diversity and functional diversity with different blue carbon storage reveal that the selection effect²¹ (our third hypothesis) is the main driver of blue carbon storage. Thus, dominant species with high wood density and stature²² and low nutrient retention capacity promote blue carbon storage in the Sundarbans Reserved Forest. Taking into account the negative relation between species richness and functional composition factor of maximum canopy height,

complementary resource-use strategies could be confined among dominant species⁴⁹.

Third, our findings may explain how indirect associations of species diversity mediate blue carbon storage in mangroves. A recent study in a South African mistbelt tropical forest⁴ suggested that species richness exerted indirect associations with above-ground biomass carbon storage through both functional diversity and functional trait composition. However, in the present study, functional diversity proved less relevant. Hence, even relatively low functional diversity (e.g., of leaf litter dry matter content) can span the full range of traits (e.g., low and high wood density, maximum canopy height and litter nitrogen), rendering trait composition more relevant than trait diversity.

Fourth, the ecosystem carbon balance, through gains and losses of different carbon pools in above- and belowground plant biomass over time, determines whether the ecosystem is a sink¹⁶ of atmospheric CO₂, and taller plant species play a crucial role in this regard. The strong predictability of multiple facets of

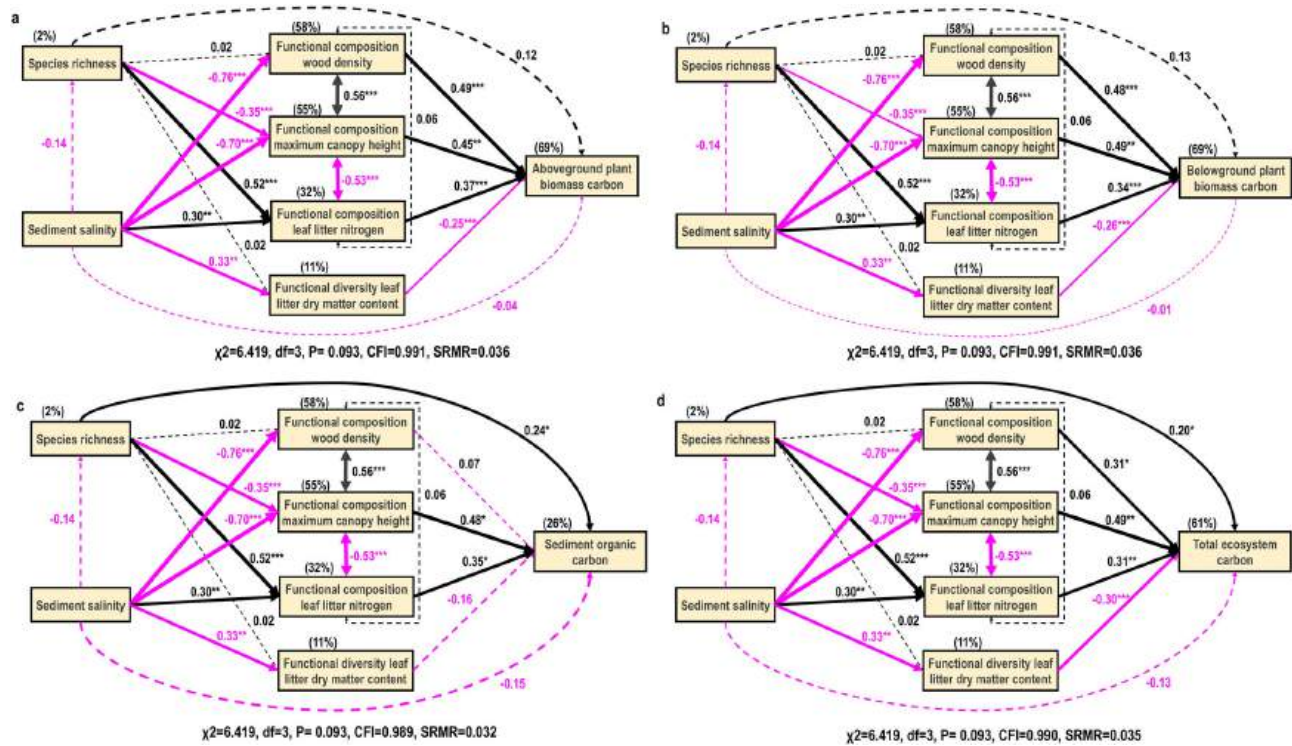


Fig. 3 Structural equation models (SEMs) for understanding the diversity association with blue carbon storage. **a–d** Aboveground plant biomass carbon (**a**), Belowground plant biomass carbon (**b**), Sediment organic carbon (**c**), and Total ecosystem carbon (**d**). All four SEMs had a similar nonsignificant χ^2 (Chi-Square) of 6.83 ($P = 0.093$) with a comparative fit index close to one (CFI; 0.99) and standardized root mean square residual close to zero (0.036) indicating no significant deviation from model and datasets at 3 degree of freedom. The lines with pink and black indicate a negative and positive association between the two covariables. Arrows with numbers indicate the standardized association of predictors with dependent variables. Numbers with percentages above boxes independent variables indicate their explained variance (Coefficient of determinant: R squared) by all the predictors. The paths value with star mark indicate their significance level (***: $P < 0.001$, **: $P < 0.01$, *: $P < 0.05$) while the insignificant paths were indicated as dotted line ($P > 0.05$).

Table 1 Indirect and total standardized association of sediment salinity and species richness on carbon pools.								
Indirect and total association pathways	Aboveground plant Biomass carbon		Belowground plant biomass carbon		Sediment organic carbon		Total ecosystem carbon	
	std.as	p value	std.as	p value	std.as	p value	std.as	p value
Indirect association of sediment salinity through species richness	−0.017	0.303	−0.017	0.298	−0.033	0.265	−0.028	0.249
Indirect association of sediment salinity through FD leaf litter dry matter content	0.082	0.011	0.086	0.010	0.055	0.128	0.100	0.008
Indirect association of sediment salinity through FC wood density	−0.371	0.000	−0.366	0.000	0.053	0.721	−0.234	0.032
Indirect association of sediment salinity through FC maximum canopy height	−0.318	0.003	−0.342	0.002	−0.339	0.035	−0.348	0.003
Indirect association of sediment salinity through FC leaf litter nitrogen	0.112	0.011	0.104	0.014	0.107	0.053	0.095	0.026
Indirect association of sediment salinity through species richness and FD leaf litter dry matter content	−0.011	0.248	−0.012	0.247	−0.007	0.324	−0.013	0.245
Indirect association of sediment salinity through species richness and FC wood density	0.050	0.222	0.050	0.223	−0.007	0.731	0.032	0.267
Indirect association of sediment salinity through species richness and FC maximum canopy height	0.043	0.235	0.046	0.231	0.046	0.270	0.047	0.236
Indirect association of sediment salinity through species richness and FC leaf litter nitrogen	−0.015	0.248	−0.014	0.252	−0.014	0.281	−0.013	0.263
Indirect association of species richness through FD leaf litter dry matter content	−0.006	0.823	−0.006	0.823	−0.004	0.824	−0.007	0.823
Indirect association of species richness through FC wood density	0.007	0.826	0.007	0.826	−0.001	0.852	0.005	0.827
Indirect association of species richness through FC maximum canopy height	−0.159	0.008	−0.171	0.006	−0.170	0.048	−0.174	0.009
Indirect association of species richness through FC leaf litter nitrogen	0.193	0.001	0.180	0.002	0.184	0.029	0.164	0.009
Total association of sediment salinity	−0.444	0.000	−0.466	0.000	−0.141	0.216	−0.362	0.000
Total association of species richness	0.159	0.068	0.137	0.119	0.254	0.009	0.191	0.028

The indirect and total associations of sediment salinity and species richness were based on structural equation models. The significant standardized associations (std.as) have p value < 0.05 . FD Functional diversity, FC Functional composition.

biodiversity of biomass carbon, ecosystem carbon, and even a substantial portion of sediment carbon in the present study suggests that plant diversity sustains carbon flows by influencing the functional composition of the community and, thereby, sediment carbon balance¹⁸ and persistence for a longer time through litter input. Our results suggest that it is mostly a loss in functional distinctiveness (trait composition of wood density, maximum canopy height and leaf litter nitrogen) of mangroves that will have negative associations with blue carbon storage and, hence, on the potential of mangroves for mitigating climate change.

Fifth, in terms of policy implications, our findings demonstrate that ecosystems with high carbon storage-capacity and biodiversity can provide the co-benefits of both climate change mitigation and biodiversity conservation⁵⁰, and thus, meet both REDD + and Post-2020 biodiversity policies. Along this line, our findings indicate that the conservation of a mangrove forest of high species richness that covers the full range of traits (such as wood density, maximum canopy height and litter nitrogen) will be more efficient in capturing and storing carbon in plant biomass and sediment than replanting mono-specific mangrove stands. Beyond this, however, our findings also have further implications for rehabilitation or afforestation efforts. Even simple mangrove communities with low functional diversity will be efficient upon implementation, if the few implemented species differ sufficiently in their functional traits, i.e., they exhibit high functional distinctiveness. Following the concept of Ecosystem Design⁵¹ for rehabilitation, selecting a combination of only few site-specific species with distinct traits, such as wood density, maximum canopy height and litter nitrogen, for instance, might yield the same results as trying to reestablish a highly diverse community, while being more feasible and more promising with respect to rehabilitation or afforestation success.

Finally, the indirect negative impact of sediment salinity on functional composition (the main driver of blue carbon storage) and, therefore, on blue carbon storage raises concerns about the adverse associations of rising sea levels^{37,38} and the disruption of natural water flow⁴⁰. Mangroves typically receive freshwater through upstream rivers, importing nutrients and lowering salinity levels and, thus, increasing the productivity of the stand and the storage of blue carbon^{37,39}. On the other hand, rising sea levels will increase the inundation frequency in currently less-saline mangrove stands that are usually dominated by tall species with high wood density but low salt-tolerance⁴⁵. Hence, sea level-rise will not only alter community composition^{39,52} toward salt-tolerant species with lower wood density and canopy height⁵³, but also reduce the functional distinctiveness of wood density and maximum canopy height, thus diminishing the blue carbon storage capacity. In the current study we did not directly measure salinity at each vegetation plot. The plot-level salinity was obtained by a spatial interpolation of salinity measurements at 32 locations, which were randomly distributed in the same area as our vegetation plots. Although the spatial interpolation gave a good accuracy the small bias due to the interpolation might slightly affect the results of structural equation models, such as the weak relationship between salinity and blue carbon pools.

Methods

Field inventory data source and carbon pools assessment. We used Sundarbans Reserved Forest inventory data, which were collected in 2009–2010 from 150 plots spanning over the whole SRF. Each plot was 4 min latitudinal and 2 longitudinal away from each other and composed of five 10 m radius nested circular subplots⁴⁵. We used 90 (90 × 5 subplots) plots out of these 150 plots in our current study. The excluded 60 plots had at least one subplot that was occupied by a canal or river for more than 10% of its total area, or was completely underwater, or was severely disturbed by cyclone or fire. For diversity assessment, we used living tree data (stem diameter at breast height (DBH) ≥ 10 cm) per plot, while for total ecosystem carbon we included live and dead carbon mass of trees and saplings, seedling carbon,

non-tree shrubs and herbs, and sediment organic carbon⁴⁵. We used a species-specific allometric equation to calculate aboveground biomass of *Excoecaria agallocha*⁴⁸ and a common allometric equation for all other species⁴⁶. We also used a common allometric equation for belowground biomass of all trees⁴⁷. Sapling (DBH ≤ 10) biomass was measured similar to trees. For non-timber vegetation and seedlings, destructive methods were followed for estimating biomass⁴⁵. We converted the dry biomass of trees, understory, and down wood to carbon mass by multiplying by 0.5, as forest biomass contains half carbon by mass²⁵. Sediment carbon was estimated at 1 m depth with two depth range: at the midpoint of 0–30 cm (15–20 cm) and 30–100 cm (60–65 cm) depth intervals from each of the five subplots. Bulk density was estimated after air-drying (in the field), and oven-dried to constant mass at 60 °C (to stop microbial decomposition) at the Khulna Integrated Protected Area Co-Management cluster office for determining bulk density. Sediment samples were further oven-dried at 105 °C Bangladesh Research Institute, Chittagong, Bangladesh. Walkley–Black's wet oxidizing method⁵⁴ was used for determination of organic carbon concentration⁴⁵. We grouped carbon as above- and below-ground of live tree, sediment organic carbon and total ecosystem carbon storage (live and dead tree and sapling, seedling, non-tree shrubs and herbs and sediment organic carbon)⁴⁵.

Functional traits and diversity metrics. We used wood density, leaf photosynthesis rate, maximum canopy height, and four leaf-litter qualities (carbon; nitrogen; carbon to nitrogen carbon ratio; and leaf dry matter content) as functional traits. Maximum canopy height for each tree was extracted from 1997 forest inventory data in the Sundarbans Reserved Forest (Supplementary Table 1) while the wood density data was collected from the Bangladesh Forest Research Institute⁵⁵. We used leaf trait values that were specific to Sundarbans^{36,56}. The values of leaf litter traits³⁶ were averaged from data representing three different seasons (monsoon, pre-monsoon and post monsoon) (Supplementary Table 1). For species (relative abundance species < 2%) with missing traits, we used the average trait value (bold value; Supplementary Table 1). We assessed species richness by counting the number of species in each plot, while plot-wide diversity was calculated by Shannon diversity index (Eq. (1))⁵⁷. Functional composition of traits was calculated using community weight mean (Eq. (2)), while for functional diversity computation we used the functional dispersion (Eq. (3)) metric in the FD package in R⁵⁸. Both metrics are abundance weighted and the species wise total basal area in each plot was used as abundance because species contribution in an ecosystem is best represented by basal area⁵⁹. As recommended, we standardized all the traits before calculating community weighted mean and functional dispersion.

$$H = \sum_{i=1}^S [P_i \times \ln(P_i)] \quad (1)$$

where H is the Shannon diversity for a plot. S is the number of species (species richness), P_i is the proportional of individuals of species i in the plot.

$$CWM_{ij} = \sum_{i=1}^S P_{ij} t_i \quad (2)$$

where CWM_{ij} is the community weighted mean of trait t for j plot, S is the species richness, P_{ij} is the proportional of relative basal area of species i for j plot, and t_i is the mean value of trait of species i .

$$FDis = \frac{\sum a_j Z_j}{\sum a_j} \quad (3)$$

where a_j is the abundance of species j in terms of basal area and Z_j is the distance of species j to the weighted centroid c which represents the centroid of the n species in trait space.

Sediment salinity and nutrients. We followed the Evidence Density Estimation interpolation method⁶⁰ for generating spatial distribution maps of surface layer (0–15 cm) sediment salinity and sediment nutrient index (measured as a function of total organic matter, total nitrogen and total phosphorus) using data from 32 field sites⁶¹. These field sites spanned the whole Sundarbans Reserved Forest and were collected from 2010 to 2012. From these spatial maps, we extracted our subplot-wise (five subplots for each plot) sediment salinity and sediment nutrient index. Then we averaged the five subplot values of sediment salinity and sediment nutrient index for getting a plot-wide mean value.

Statistical analysis. We log transformed and standardized all four carbon pools, species richness, Shannon diversity index, functional composition and functional diversity of all traits to better meet linearity assumptions⁴⁴. Functional diversity and functional composition of different traits were highly correlated (Supplementary Figs. 1 and 2). To remove the highly correlated variables from both functional diversity and functional composition of different traits, we applied multiple linear regressions for quantifying the variance inflation factors (a way to measure multicollinearity) using a threshold value of 10 (ref. 62). We employed multiple linear regression (total ecosystem carbon storage was used as the response variable), starting with including all the covariates and incrementally removing the variables with highest variance inflation factor until achieving the threshold value (all covariates with a variance inflation factor below 10). This multicollinearity test resulted in functional diversity being composed of maximum canopy height, leaf

litter nitrogen and leaf litter dry matter content; and in functional composition being composed of wood density, maximum canopy height and leaf litter nitrogen (Supplementary Table 3 and Table 4). We then employed structural equation models using the Lavaan Package⁶³ in R for assessing the direct, indirect and total associations of sediment salinity, sediment nutrient index and multiple facets of biodiversity with different blue carbon pools^{44,64}.

We constructed structural equation models that are based on theories relating diversity and ecosystem function in the literature, that have been tested in terrestrial ecosystems⁴. We particularly focused on niche complementarity and selection theories where species richness and functional diversity of traits represented the former theory while functional composition of different traits represent the latter theory. We hypothesized that (i) Sediment salinity has direct and indirect associations with different blue carbon storage sediment nutrients, species richness and both functional composition and functional diversity, (ii) species diversity also has direct and indirect associations with blue carbon storage through functional composition and functional diversity of different traits⁴, and (iii) functional composition has a greater association with different blue carbon storage than the two other biodiversity components⁴. Based on these hypotheses, we tested a total of 32 structural equation models in different combinations of species diversity, functional composition and functional diversity of different traits (variance inflation factor value below 10; Supplementary Table 3 and 4) along with sediment nutrient and salinity (Supplementary Table 5). For every carbon pool, we first started with all the covariates, then gradually removed the hypothesized path with the highest *p* value and continued until the model satisfied the model fitting measures. Of these 32 structural equation models, the combination of sediment salinity, species richness, functional composition of wood density, maximum canopy height and litter nitrogen, and functional diversity of leaf litter dry matter content exhibited the best fit (satisfied model fitting measures) with our dataset for aboveground and belowground plant biomass carbon, sediment organic carbon and total ecosystem carbon storage (Supplementary Table 5). We estimated the indirect association of sediment salinity on carbon pools by multiplying the standardized associations of all the paths from sediment salinity to species richness to functional composition and functional diversity to carbon pools⁶⁴. For estimating the indirect associations of sediment salinity and species richness, we multiplied the standardized associations of all the paths from them to carbon pools through different mediators⁶⁴. The total association of sediment salinity and species richness on carbon pools was calculated by adding the standardized direct and the standardized indirect associations⁶⁴.

We assessed the performance of structural equation model by using different fit indices that are widely applied for structural equation model. For example, an insignificant Chi-square test ($P > 0.05$; Eq. (4)), a comparative fit index (Eq. (5)) > 0.90 , and a standardized root mean square residual (Eq. (6)) close to 1 indicates no significant deviation of the structural equation model from the dataset^{24,58,64}. We used the Lavaan package to calculate all the fit indices where they were in-built⁵⁸. Finally, we also checked the multicollinearity of the final fitted models using a variance inflation factor test (Supplementary Table 6).

$$\chi^2 = (N - 1)f \quad (4)$$

$$CFI = \frac{\max(\chi_0^2 - df_0) - \max(\chi_k^2 - df_k)}{\max(\chi_0^2 - df_0)} \quad (5)$$

$$SRMR = \sqrt{[\rho^{*-1}(e' W_s e)]} \quad (6)$$

Where, χ^2 : Chi-square test statistic; CFI: Comparative fit index; SRMR: Standardized root mean square residual; *N*: sample size; *f*: minimized discrepancy function; 0: baseline model; *df*: degree of freedom; *k*: tested or hypothesized model, ρ^* : number of nonduplicated elements in the covariance matrix; *e*: a vector of residuals from a covariance matrix; *s*: a vector of the ρ^* nonredundant elements in the observed covariance matrix; *W*: a weight matrix; *W_s*: a diagonal weight matrix used to standardize the elements of covariance matrix.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The datasets analyzed during the current study are available from the corresponding authors on reasonable request.

Code availability

All the codes used to generate results, and to visualize the result during the current study are available from the corresponding authors on reasonable request.

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Author contributions

R.M.M. conceived the study, R.M.M. and I.A. participated in field survey, R.M.M. and D.D. analyzed data, R.M.M., I.A., D.D., M.Z., M.K. and M.X. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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A global map of mangrove forest soil carbon at 30 m spatial resolution

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Supplementary material for this article is available [online](#)

Abstract

With the growing recognition that effective action on climate change will require a combination of emissions reductions and carbon sequestration, protecting, enhancing and restoring natural carbon sinks have become political priorities. Mangrove forests are considered some of the most carbon-dense ecosystems in the world with most of the carbon stored in the soil. In order for mangrove forests to be included in climate mitigation efforts, knowledge of the spatial distribution of mangrove soil carbon stocks are critical. Current global estimates do not capture enough of the finer scale variability that would be required to inform local decisions on siting protection and restoration projects. To close this knowledge gap, we have compiled a large georeferenced database of mangrove soil carbon measurements and developed a novel machine-learning based statistical model of the distribution of carbon density using spatially comprehensive data at a 30 m resolution. This model, which included a prior estimate of soil carbon from the global SoilGrids 250 m model, was able to capture 63% of the vertical and horizontal variability in soil organic carbon density (RMSE of 10.9 kg m^{-3}). Of the local variables, total suspended sediment load and Landsat imagery were the most important variable explaining soil carbon density. Projecting this model across the global mangrove forest distribution for the year 2000 yielded an estimate of 6.4 Pg C for the top meter of soil with an $86\text{--}729 \text{ Mg C ha}^{-1}$ range across all pixels. By utilizing remotely-sensed mangrove forest cover change data, loss of soil carbon due to mangrove habitat loss between 2000 and 2015 was $30\text{--}122 \text{ Tg C}$ with $>75\%$ of this loss attributable to Indonesia, Malaysia and Myanmar. The resulting map products

from this work are intended to serve nations seeking to include mangrove habitats in payment-for-ecosystem services projects and in designing effective mangrove conservation strategies.

1. Introduction

Mangrove forests, occupying less than 14 million ha (Giri *et al* 2011), just 2.5% of the size of the Amazon rainforest, provide a broad array of ecosystem services (Barbier *et al* 2011). Mangroves are critical nursery habitats for fish, birds and marine mammals (Mumby *et al* 2004, Nagelkerken *et al* 2008), act as effective nutrient filters (Robertson and Phillips 1995), buffer coastal communities from storm surges (Gedan *et al* 2011) and support numerous rural economies (Spalding *et al* 2014, Temmerman *et al* 2013). These ecosystem service benefits have been valued at an average of 4200 US\$ ha⁻¹ yr⁻¹ in Southeast Asia (Brander *et al* 2012). Despite these ecosystem service benefits, mangroves are highly threatened by both urban expansion and other 'higher value' land uses because of their close proximity to major human settlements. There are no reliable estimates of original mangrove cover, but some authors have suggested that 35% or more of original cover may have been lost and wider areas have been degraded (Valiela *et al* 2001, Spalding *et al* 2010). Loss rates have slowed dramatically in the past 10–20 years in most areas, however they remain considerable, with rates up to 3.1% annually in some countries (Hamilton and Casey 2016). The major drivers of loss are conversion for aquaculture, especially shrimp farming, agriculture and urban development (Alongi 2002, Valiela *et al* 2001, Spalding *et al* 2010, Richards and Friess 2016) but loss due to extreme climatic events are also becoming more common (Duke *et al* 2017).

With the growing recognition that effective action on climate change will require a combination of emissions reductions and removals (Rockström *et al* 2017), protecting, enhancing and restoring natural carbon sinks have become political priorities (Boucher *et al* 2016, Grassi *et al* 2017). Mangrove forests can play an important role in carbon removals; in addition to being some of the most carbon-dense ecosystems in the world (Donato *et al* 2011), if kept undisturbed, mangrove forest soils act as long-term carbon sinks (Breithaupt *et al* 2012). As such, there is strong interest in developing policy tools to protect and restore mangroves through payment for ecosystem services (Friess *et al* 2016, Howard *et al* 2017).

Mangroves can store significant amounts of carbon in their biomass (Hutchison *et al* 2014); however, the vast majority of the ecosystem carbon storage is typically found in the soil (Donato *et al* 2011, Muriyarto *et al* 2015, Sanders *et al* 2016). For example, Kauffman *et al* (2014) found that within the same estuary, soil carbon contributed 78% of total ecosystem

carbon storage in tall mangroves but 96%–99% of total ecosystem carbon in medium and low stature mangrove stands. Importantly, Kauffman *et al* (2014) found that conversion of these mangrove forests to shrimp ponds resulted in the loss of 90% of this carbon from the top 3 m of soil (612–1036 Mg C ha⁻¹). In addition to avoided emissions, many mangrove forest soils are accreting as sea level rises (Krauss *et al* 2014), providing continual carbon sequestration on the order of 1.3–2.0 Mg C ha⁻¹ yr⁻¹ (Breithaupt *et al* 2012, Chmura *et al* 2003). Clearly, there can be a major climate benefit to halting or even slowing the rate of mangrove conversion, with a rough potential estimated to be 25–122 Tg C yr⁻¹ (Pendleton *et al* 2012, Siikamäki *et al* 2012). For nations with large mangrove holdings, protection and restoration can make major contributions to meeting climate mitigation targets (Herr and Landis 2016).

While many mangrove forests do accumulate large quantities of soil carbon, others do not. There can be significant variability in soil carbon stocks across different mangrove forests (Jardine and Siikamäki 2014) but also within the same mangrove forest (Adame *et al* 2015, Kauffman *et al* 2011). Understanding the distribution of soil carbon in mangrove forests will be very important in prioritizing protection and restoration efforts for climate mitigation. The controls on soil carbon stocks are diverse and are likely scale dependent; however, some generalizations can be made. Mangrove forests, no matter how productive, will struggle to have high soil carbon stocks in the upper meter of soil if they receive large annual sediment loads. Mangrove forests in river deltas, such as the Sundarbans (Banerjee *et al* 2012) and the Zambezi river delta in Mozambique (Stringer *et al* 2016), typically only contain a few percent organic carbon throughout the soil profile. These locations may still have very high carbon stocks, but the density of carbon is low due to the high allocthonous input of mineral sediments. Conversely, forests with moderately low productivity can accumulate large amounts of soil carbon if they are in an isolated hydrogeomorphic setting (Ezcurra *et al* 2016). Within the same mangrove forest there are typically steep hydrogeomorphic gradients from the seaward to landward extent of the forest which results in zonation of both vegetation (Snedaker 1982) and soil carbon storage (Kauffman *et al* 2011, Ouyang *et al* 2017, Ewers Lewis *et al* 2018) but not necessarily for the same reasons. Within a similar hydrogeomorphic position, forest productivity and soil edaphic conditions (e.g. redox potential, pH, salinity) driving decomposition rates are often the dominant controls on soil carbon density. Consideration of this nested

hierarchy of controls will be necessary to successfully capture the variability in soil carbon at both local and global scales.

Accurate estimates and an understanding of the spatial distribution of mangrove soil carbon stocks are a critical first step in understanding climatic and anthropogenic impacts on mangrove carbon storage and in realizing the climate mitigation potential of these ecosystems through various policy mechanisms (Howard *et al* 2017). Previous global estimates (Atwood *et al* 2017, Jardine and Siikamäki 2014), do not capture enough of the finer scale spatial variability that would be required to inform local decisions on siting protection and restoration projects. To close this information gap, we have: (1) compiled and published a harmonized global database of the profile distribution of soil carbon under mangroves, (2) used this database to develop a novel machine-learning based data-driven statistical model of the distribution of carbon density using spatially comprehensive data at an ~ 30 m resolution, (3) projected the model results across global mangrove habitat for the year 2000 (Giri *et al* 2011), and (4) overlaid estimates of mangrove forest change between 2000 and 2012 (Hamilton and Casey 2016) to estimate potential soil carbon emissions from recent forest conversion.

2. Methods

2.1. Mangrove soil carbon database

A harmonized globally representative database (available at: [10.7910/DVN/OCYUIT](https://doi.org/10.7910/DVN/OCYUIT)) was compiled from peer-reviewed literature, grey literature and from contributions of unpublished data from a number of researchers and organizations. Details of database development and a statistical summary of the data are given in the supplemental information available at stacks.iop.org/ERL/13/055002/mmedia.

2.2. Spatial modelling of soil organic carbon

In order to maximize the utilization of available soil carbon data, we developed a machine learning-based model of organic carbon density (OCD) which models OCD as a function of depth (d), an initial estimate of the 0–200 cm organic carbon stock (OCS) from the global SoilGrids 250 m model (Hengl *et al* 2017), and a suite of spatially explicit covariate layers (X_p):

$$\text{OCD}(xyd) = d + \text{OCS}_{\text{SG}} + X_1(xy) + X_2(xy) + \dots X_p(xy)$$

where OCS_{SG} is the aggregated organic carbon stock estimated for 0–200 cm depth using global SoilGrids 250 m approach down-sampled from 250 m–30 m resolution, and xyd are the 3D coordinates northing easting and soil depth (measured to center of a horizon). Note here that we model spatial distribution

of OCD in three dimensions (soil depth used as a predictor) using all soil horizons layers at different depths, which means that a single statistical model can be used to predict OCD at any arbitrary depth. This 3D approach to modeling OCD reduces the need for making complex assumptions about the downcore trends in OCD, and maximizes the use of collected data.

The derived spatial prediction model is then used to predict OCD at standard depths 0, 30, 100, and 200 cm, so that the organic carbon stock (OCS) can be derived as a cumulative sum of the layers down to the prediction depth for every 30 m pixel identified as having mangrove forest in the year 2000 (Giri *et al* 2011). Importantly, we found that there is a spatial mismatch between the global mangrove forest distribution (GMFD) of Giri *et al* (2011) and satellite imagery (figure S2). To best resolve this spatial mismatch, we have adjusted the GMFD by growing all vectors by one pixel (~ 30 m) and then filtering out any pixel that falls over water by using Landsat NIR band (see SI for more details).

Environmental covariates have been compiled to represent the postulated major controls on OCS in soils generally (McBratney *et al* 2003) and specifically for mangrove ecosystems (Balke and Friess 2016). Covariates included:

1. Vegetation characteristics including percent forest cover (Hansen *et al* 2013) and Landsat bands 3 (red), 4 (near infrared), 5 (shortwave infrared) and 7 (shortwave infrared) for the year 2000 (Hanson *et al* 2013) retrieved from http://earthenginepartners.appspot.com/science-2013-global-forest/download_v1.3.html;
2. Digital elevation data, which at or near sea-level approximately follows forest canopy height (Simard *et al* 2006), from the shuttle radar topography mission (SRTM GL1; NASA, 2013) was retrieved from <https://lpdaac.usgs.gov>, maintained by the NASA EOSDIS Land Processes Distributed Active Archive Center (LP DAAC) at the USGS/Earth Resources Observation and Science (EROS) Center, Sioux Falls, South Dakota;
3. Long-term averaged (1990–2010) monthly sea surface temperature (SST) averaged into four seasons were generated in Google Earth Engine from NOAA AVHRR Pathfinder Version 5.2 Level 3 Collated data (Casey *et al* 2010) and downscaled to 30 m resolution using bicubic resampling;
4. The M2 tidal elevation amplitude product (FES2012) from a global hydrodynamic tidal model which assimilates altimetry data from multiple platforms was used to represent tidal range at each location. The FES2012 product was produced by Noveltis, Legos and CLS Space Oceanography Division and distributed by Aviso, with support from Cnes (www.aviso.altimetry.fr/);

5. Averaged (2003–2011) monthly total suspended matter (TSM) averaged into four seasons estimated from MERIS imagery collected by the European Space Agency's Envisat satellite. Processed and validated TSM data was retrieved from the GlobColour project (<http://hermes.acri.fr>).
6. A mangrove typology map delineating mangroves into estuaries and then either organogenic or mineralogenic based on an analysis of TSM and tidal amplitude data (Zu Ermgassen, *unpublished data*).

Sea surface temperature, tidal amplitude and TSM are 4 km resolution ocean products and needed to be extrapolated to each pixel containing mangrove forest. Missing values in the sea surface temperature, tidal amplitude and TSM were first filled-in using spline interpolation in SAGA GIS, then down-scaled to 30 m resolution using bicubic resampling in GDAL. Including SoilGrids and depth, there were a total of 20 covariates used in building the mangrove OCD model.

The ability of the training points to represent the entire covariate space of the global mangrove domain was assessed by conducting a principal components analysis (PCA) on 15 000 randomly selected points and the 1613 points used in the spatial model. Spatial variables were detrended and centered by subtracting the mean and dividing by the standard deviation (s.d.) before entering into the PCA analysis.

Soil carbon typically varies in highly non-linear ways with depth and across the landscape and as such the ability of standard parametric models to capture this variation is limited (Jardine and Siikamäki 2014, Hengl *et al* 2017). Here we model the spatial (xyd) distribution of OCD using a machine learning random forest model implemented in the ranger package (Wright and Ziegler 2015) in the R environment for statistical computing (R Core Team 2000). Given the clustered nature of the point data, we have implemented a spatially balanced random forest model design. Model performance was assessed with a 5 fold (Leave-Location-Out) cross-validation procedure where 20% of complete locations were withheld in each model refitting (Gasch *et al* 2015). The relative importance of using SoilGrids as a covariate was assessed by implementing the cross-validation procedure with and without this variable.

Finally, prediction error of OCS for 0–1 m depth was derived for ± 1 s.d. based on the quantile regression approach of Meinshausen (2006) and implemented in R via the ranger package. This procedure is relatively computationally demanding so a random subset of approximately 15 000 points were selected to calculate prediction errors. All modeling was run on ISRIC High Performance Computing servers with 48 cores of 256 GB RAM.

2.3. Data analysis

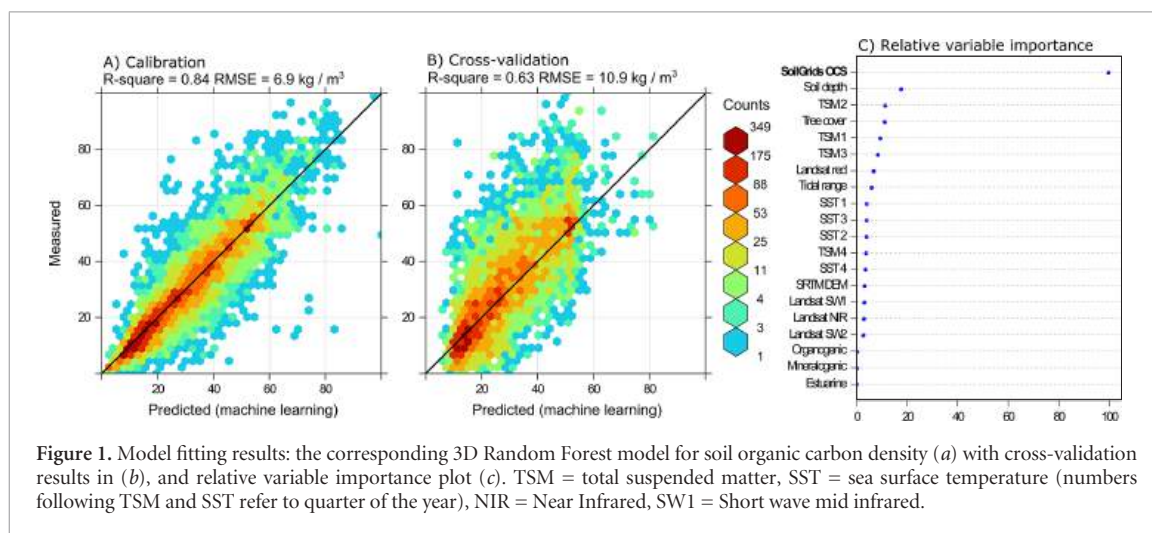
Soil carbon stocks were calculated for the global extent of mangroves for the year 2000 by summing the OCS in each pixel for 1 and 2 m depths. Country level carbon stocks were also calculated for the same depths. Given the fringing nature of mangroves, a global spatial vector data layer was built that allocated the offshore area for each country where mangrove forests can be found. It was derived from the Exclusive Economic Zone for each country. This layer was then dissolved with the onshore areas for each associated country and subsequently used to quantify mangrove OCS tonnage and areal extent.

Potential loss of OCS due to mangrove habitat conversion was calculated between 2000 and 2015 by summing the OCS in mangrove forest pixels which were identified to be deforested. While this analysis cannot distinguish between natural and anthropogenic disturbance, human-driven land use change is believed to be by far the dominant driver of deforestation in mangrove ecosystems (Alongi 2002, Murdiyarso *et al* 2015). We define deforested using the Global Forest Change dataset (Hansen *et al* 2013) available online from: <http://earthenginepartners.appspot.com/science-2013-global-forest>. We chose to use this approach for estimating deforestation instead of using the derived mangrove tree cover loss data produced by Hamilton and Casey (2016) as used by Atwood *et al* (2017) because the Hamilton and Casey (2016) analysis only considered forested area as area actually covered by trees (i.e. if a 100 ha forest has 80% tree cover then it is counted as 80 ha of forest). In our opinion, this definition mischaracterizes forest area extent. Next, an estimate of the soil carbon emissions associated with land use conversion is needed. The amount of OCS lost can be highly variable and depends on the new land use (Kauffman *et al* 2014, 2016b, Jones *et al* 2015) and probably on soil properties. Pendleton *et al* (2012) used a 25%–100% loss range. Donato *et al* (2011) used a low estimate of 25% of the OCS in top 30 cm and 75% in top 30 cm + 35% from deeper layers as a high estimate. Expanding on earlier work, Kauffman *et al* (2017) found that on average 54% of belowground carbon (soil + roots) to 3 m was lost after conversion to shrimp ponds and pastures. Given the limited number of studies comparing soil OCS change with land use change, in this work we adopt the same 25%–100% range as used by Pendleton *et al* (2012) applied to the first meter of soil. Finally, country level statistics for OCS loss were calculated as described above. All global and country level analyses were performed on the 30 m resolution dataset in Google Earth Engine (Gorelick *et al* 2017).

3. Results

3.1. Model results

The random forest model was successful in capturing the major variation in OCD across the mangrove



database (figure 1(a)) with an R^2 of 0.84 and a root mean square error (RMSE) of 6.9 kg m^{-3} compared to the mean OCD value of 29.6 kg m^{-3} . Cross-validation results yield an R^2 of 0.63 and an RMSE of 10.9 kg m^{-3} (figure 1(b)), which is the de-facto mapping accuracy to be expected on the field. Low OCD values were slightly over-predicted and high OCD values were under-predicted (figure 1(b)). The initial OCS prediction from SoilGrids 250 m was the most important variable explaining mangrove OCD. Running the 5 fold cross-validation without and with SoilGrids indicated that this single variable explained improved model performance by nearly 50% (R^2 increased from 0.42–0.63). Seasonal total suspended matter (TSM), depth of sample, mangrove tree cover, Landsat Red band, sea surface temperature (SST), and tidal range were the next ten most important variables, respectively (figure 1(c)). Quantile regression analysis indicated that the full uncertainty (± 1 s.d.) about a mean prediction of carbon stocks to 1 m depth averaged 40.4% of the mean OCS with lowest relative uncertainty in the most carbon-rich mangrove forests (figure S7).

3.2. Mangrove soil carbon storage

Projection of the mangrove OCD model to global mangrove forests revealed the distribution of soil carbon storage in these ecosystems (figure 2). The mean (± 1 s.d.) OCS to 1 m depth was $361 \pm 136 \text{ Mg C ha}^{-1}$ with a range of $86\text{--}729 \text{ Mg C ha}^{-1}$. At the national level (table S1), Bangladesh had the lowest per ha stocks, averaging just 127 Mg C ha^{-1} followed by China and the nations bordering the Persian Gulf and Red Sea with an average OCS of 214 and 233 Mg C ha^{-1} , respectively. The highest per ha stocks were found in many of the Pacific island nations, averaging 505 Mg C ha^{-1} with much of Southeast Asia ranking well above the global mean.

While the national level comparisons are revealing, by modeling at a 30 m resolution much richer details of potential within forest variation in OCS are

seen (figure 2). Mangrove forests dominated by sediment laden fluvial inputs typically have consistently low OCS as seen in the Sundarbans and Madagascar (figures 2(a) and (e)). In non-deltaic mangroves, the model appears to have captured the large zonal variation in OCS that is often observed in field studies (figures 2(b) and (c)).

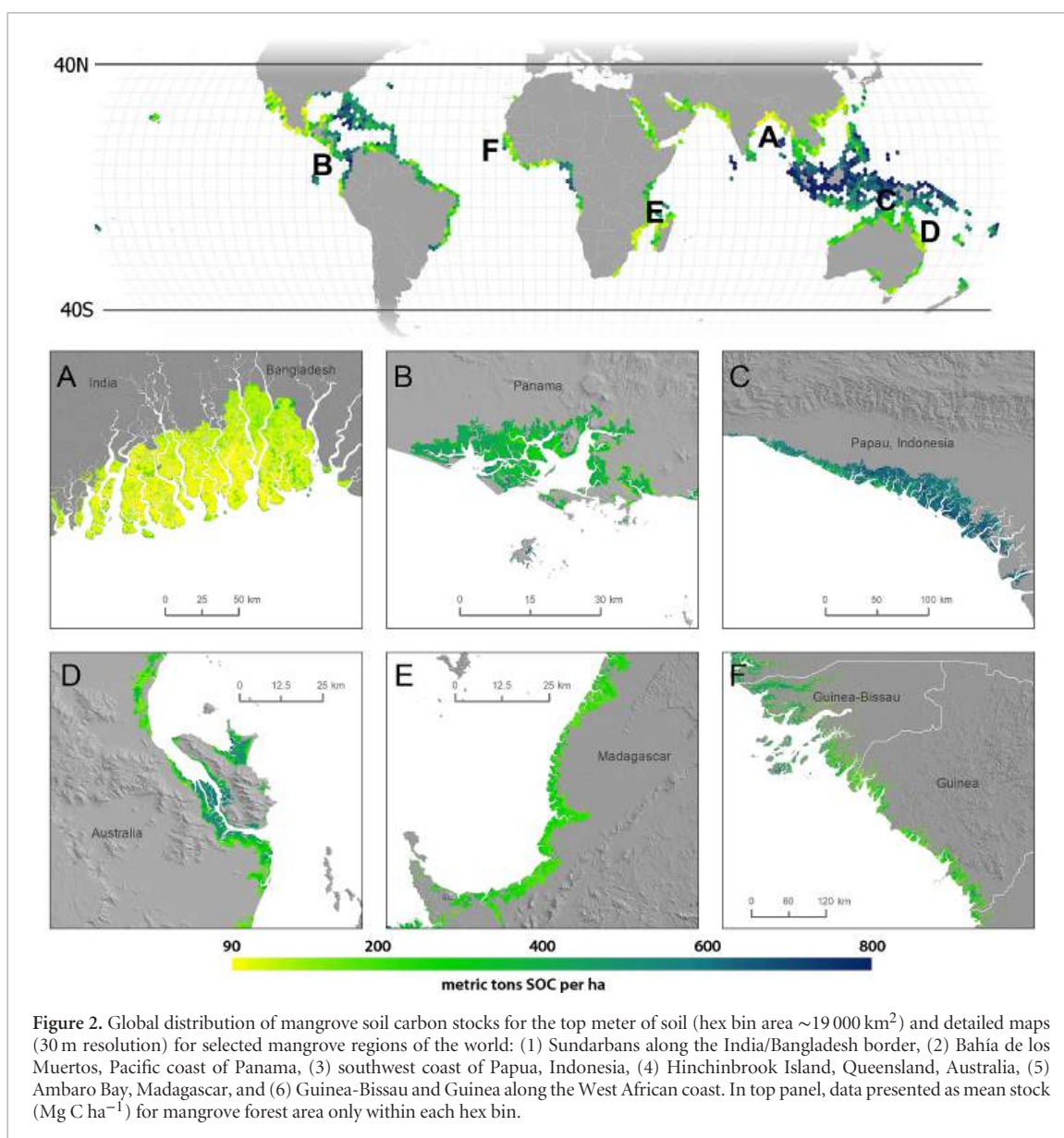
3.3. Soil carbon loss due to habitat loss

Utilizing the Hanson *et al* (2013) global deforestation analysis (figure S8), we found that 278049 ha (1.67% of total) of area identified as mangrove habitat in the year 2000 was deforested resulting in the committed emission of $30.4\text{--}122 \text{ Tg C}$ ($111\text{--}447 \text{ Tg CO}_2$) from mangrove forest soils due to land use change between 2000 and 2015 (figure 3). The relative rank of nations in terms of loss of mangrove forest area and OCS were often not the same (table S1). Indonesia alone was responsible for 52% of this global loss with Malaysia and Myanmar representing another 25% of the global total loss (figure 3(c)). When visualized as a percent loss from year 2000 stocks, a slightly different pattern emerged (figure 3(d)). Guatemala had the highest percent loss of mangrove OCS (0.9%–6.8%) followed by several southeast Asian nations, but high percent losses were also found in several Caribbean island nations as well as the United States and several west African countries.

4. Discussion

4.1. Amount and distribution of Mangrove SOC

Our new estimate of global mangrove OCS of 6.4 Pg C in the upper meter and 12.6 Pg C to 2 m is largely consistent with past efforts to calculate this value (Donato *et al* 2011, Jardine and Siikamäki 2014, Sanders *et al* 2016). However, our estimate is double that of Atwood *et al* (2017) primarily due to their use of the Hamilton and Casey (2016) estimate of mangrove extent instead of Giri *et al* (2011). Importantly, by using an environmental covariate model,

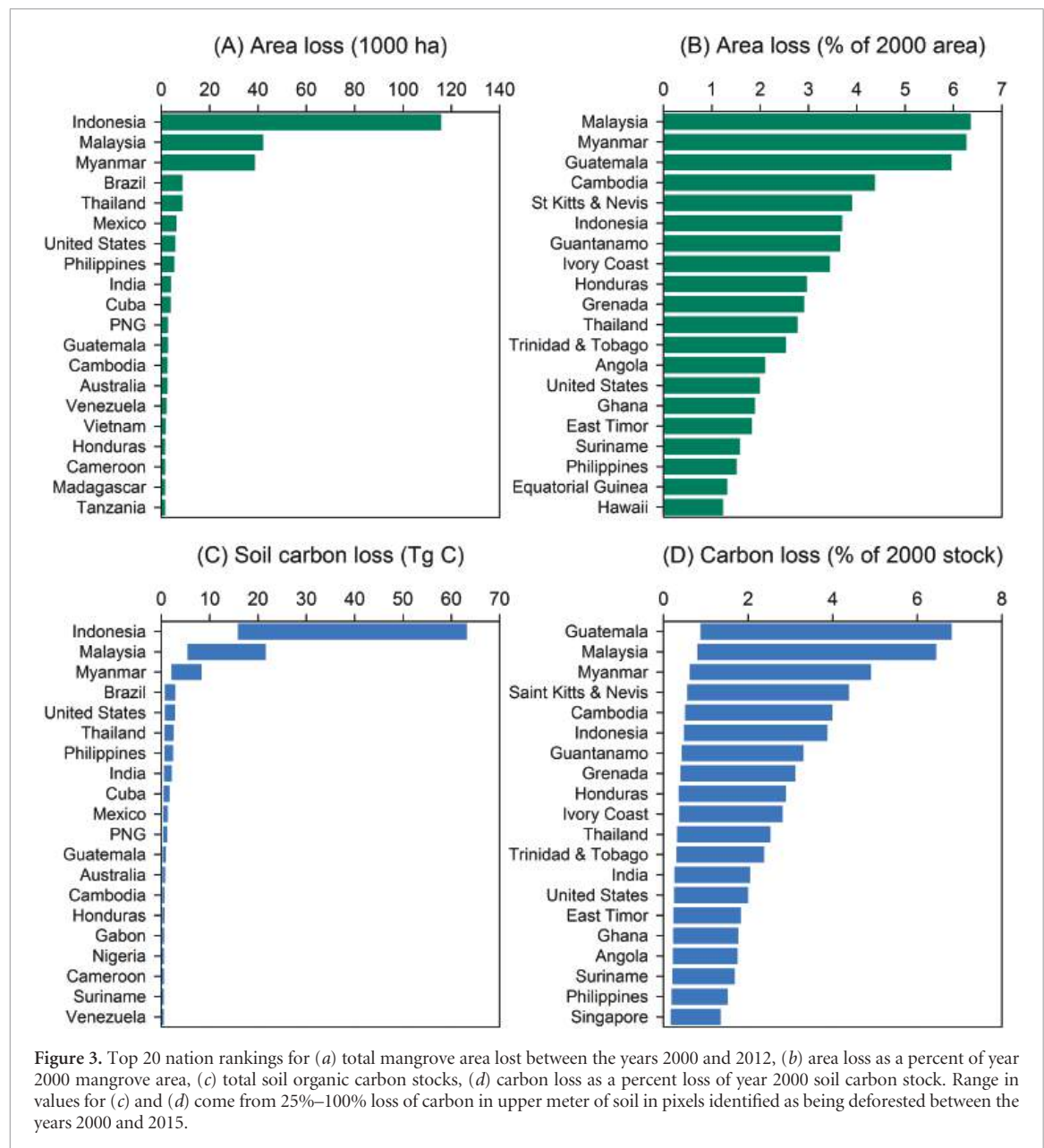


we have been able to make plausible estimates for regions where no sampling has taken place instead of relying on global mean values (i.e. Atwood *et al* 2017).

The total amount of soil carbon was similar in our analysis and the most comparable analysis, that of Jardine and Siikamäki (2014), but the spatial distribution of carbon-rich versus carbon-poor mangroves varied substantially. For example, we found much higher OCS levels in West Africa than in East African nations (figure 2 and table S1) but the reverse was found by Jardine and Siikamäki (2014). Large discrepancies were also found for Colombia, Sri Lanka and many of the countries bordering the Red Sea. These differences were most likely driven by lack of data in those regions at the time of the analysis by Jardine and Siikamäki (2014) given that nearly half the data in our database was collected after their study was published. Additionally, our analysis suggested a much larger range in OCS ($86\text{--}729\text{ Mg C ha}^{-1}$) compared to $272\text{--}703\text{ Mg C ha}^{-1}$ in the analysis of

Jardine and Siikamäki (2014). This difference in range was likely due to the inclusion of more data from sub-tropical and temperate mangroves (figure S3).

The depth trend analysis (figure S6) and random forest variable importance (figure 1(b)) both indicated that depth should be considered in calculation of OCS. For locations that were either stable peat domes or located in estuaries receiving large annual sediment loads, a stable OCD profile distribution would be expected and this was found for many sites (figures S6(a) and (e)). However, where mangroves are growing in a mineral matrix that is receiving only low sediment loads, a decline in OCD may be expected as the carbon inputs from the productive mangrove forest would be concentrated in the surface horizons (figure S6(b)). Still in other cases (figure S6(c)), changes in hydrologic/sediment regimes can lead to irregular depth patterns or even an increase in OCD with depth.



While total area of mangroves was a key determinant of total soil carbon storage, amongst the top 25 mangrove OCS holding nations, there was a nearly even split between nations with smaller area of high soil carbon density forests and those nations with lots of low soil carbon density forests (figure 4). Indonesia was the clear exception to this trend with the largest mangrove holdings which also contain rich carbon stocks resulting in Indonesia alone holding nearly 25% of the world's mangrove OCS.

Compared to terrestrial carbon pools, mangrove forests rank low due to their limited spatial extent. For example in the upper meter of soil, permafrost affected soils are estimated to store 472 ± 27 Pg C (Hugelius *et al* 2014), tropical forests contain ~ 188 Pg C, and soils under permanent cropping contain ~ 150 Pg C (table 1). However, on an equal area basis, mangrove forests on average store more soil carbon than most other ecosystems (table 1). Importantly, our

analysis has demonstrated mangrove soil carbon is highly variable and many mangroves actually store fairly modest amounts of carbon in the upper one or two meters of soil. While not the focus of this analysis, it is important to point out that while some mangrove forests store modest levels of OCS in the upper meter of soil, they can have high sequestration rates and conversely carbon-dense mangroves can have low annual sequestration rates (Lovell *et al* 2010, MacKenzie *et al* 2016).

4.2. Drivers of soil carbon storage

Our spatial modelling framework, in which global predictions were combined with local high resolution images, was successful as the general patterns of carbon variation from SoilGrids 250 m were maintained, while the spatial detail was significantly improved by moving from 250 m–30 m spatial resolution. The initial SoilGrids 250 m OCS prediction (Hengl *et al* 2017)

Table 1. Soil organic carbon stocks (mean with 5th–95th percentile in parentheses) and total storage for different terrestrial ecosystems compared to mangrove forests.

Land cover category (IGBP code) ^a	Area (10 ⁶ ha)	1 m soil organic carbon stock	
		Pg C	Mg C ha ⁻¹
Mangrove forest ^b	16.6	6.4	361 (94–628)
Gelisols (permafrost soils) ^c	1878	472.0	389 (178–691)
Evergreen Needleleaf forest (1) ^d	286	60.0	210 (121–346)
Evergreen Broadleaf forest (2)	1248	188.4	151 (85–271)
Deciduous Needleleaf forest (3)	116	29.3	253 (163–412)
Deciduous Broadleaf forest (4)	165	22.1	134 (83–223)
Mixed forest (5)	771	152.8	198 (93–343)
Closed shrublands (6)	56	6.2	110 (39–223)
Open shrublands (7)	1933	325.8	169 (49–329)
Woody savannas (8)	1179	185.8	158 (82–274)
Savannas (9)	1010	112.9	112 (52–201)
Grasslands (10)	1810	280.1	155 (56–289)
Permanent wetlands (11) ^e	104	25.1	241 (114–474)
Croplands (12)	1177	149.6	127 (60–200)
Cropland/Natural veg. mosaic (14)	868	117.7	136 (58–238)

^a data for MODIS-based IGBP land cover classes (Friedl *et al* 2010) extracted from 1 m OCS map for the year 2010 produced by Sanderman *et al* (2017).

^b mangrove area and OCS data from this study.

^c permafrost area from Tarnocai *et al* (Tarnocai *et al* 2009), OCS data from Hugelius *et al* (2014).

^d some overlap between class 1 (evergreen Needleleaf forest) and gelisols.

^e class 11 (permanent wetlands) likely has overlap with mangrove area.

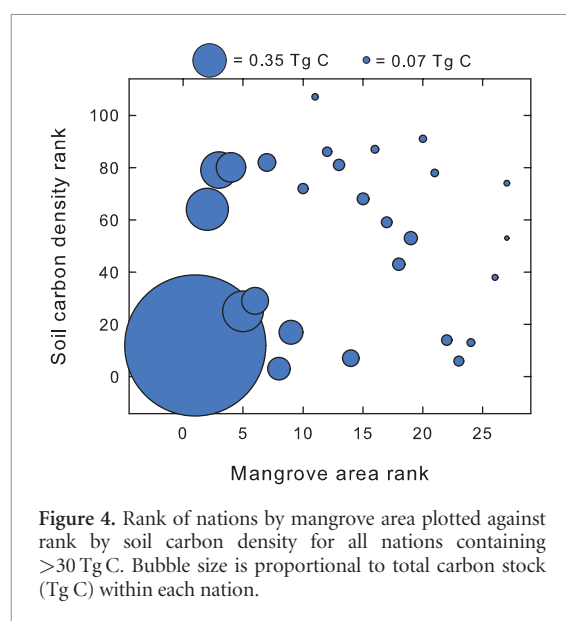


Figure 4. Rank of nations by mangrove area plotted against rank by soil carbon density for all nations containing >30 Tg C. Bubble size is proportional to total carbon stock (Tg C) within each nation.

was based upon machine learning algorithms using 237 covariates that covered the main state factors of soil formation (Jenny 1994)—climate, relief, living organisms (vegetation) and parent material—and was four times more important in predicting mangrove OCD than any of the local covariates. However, the local covariates allowed for a much more refined picture of the spatial variation in OCS within mangrove forests (figure 2) which was not captured in the 250 m resolution SoilGrids 250 m prediction. Covariates related to hydrogeomorphology (TSM and tidal range), as hypothesized, were important predictors of local variation in OCD. Both TSM and tidal range were strongly negatively correlated with OCD suggesting that locations with either high sediment loads or strong tidal flushing do not accumulate large carbon

stocks. The mangrove typology ended up being uninformative likely because most of the data that went into this classification was already captured in the model.

Covariates related to mangrove biomass (SRTM elevation and Landsat bands) were also important in explaining the local distribution of OCD (figure 1(b)). However, as pointed out by Bukoski *et al* (2017), it is unclear whether the importance of these vegetation-related data are causal drivers of differences in OCD or they just happen to co-vary in the same way as OCD. To further explore the relationship between forest biomass carbon and soil carbon storage, we extracted aboveground biomass data from Hutchison *et al* (2014) and compared it to our OCS results (figure 5). While a clear positive trend was found between biomass and soil carbon storage ($R^2 = 0.26$), there is clearly a lot of variance especially at lower biomass levels where nearly the full range in OCS can be found.

Depth below the soil surface was an important covariate in modeling OCD distribution (figure 1(b)). This finding was supported by the database depth trend analysis (figure S6) which indicated that a flat depth distribution of OCD would be an incorrect assumption 37%–64% of the time. These findings suggest that the simple scaling performed in figure S5 and in several previous assessments (Bukoski *et al* 2017, Jardine and Siikamäki 2014, Atwood *et al* 2017) may not be an accurate estimation of total OCS especially when extrapolating from a surface horizon sample alone.

4.3 Soil carbon loss due to land conversion (2000–2015)

Our analysis suggests that mangrove soils have lost or are at least committed to losing 30.4–122 Tg C due to the land use conversion that occurred between the

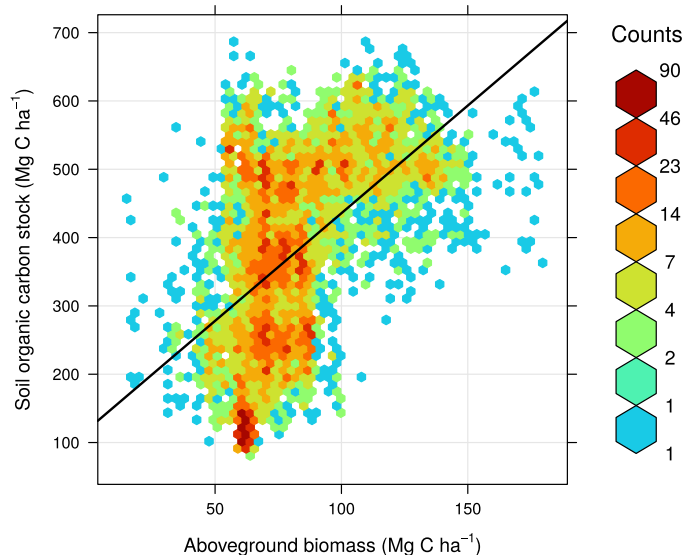


Figure 5. Comparison of aboveground biomass carbon (Hutchison *et al* 2014) with organic carbon stocks (OCS) for the top meter of soil. Points were generated by casting 10 000 random points into mangrove areas and extracting values from both maps. Linear regression $R^2 = 0.26$.

years 2000 and 2015 (figure 3). Given that at the global level the rate of mangrove forest lost was consistent over this time period (Hamilton and Casey 2016), we estimated an annual soil carbon emission of 2.0–8.1 Tg C yr⁻¹. This value is significantly lower than previous estimates (Donato *et al* 2011, Pendleton *et al* 2012) for two reasons. First, we use remote sensing-based measurements of actual mangrove loss instead of applying a large range of annual conversion rates, which are notoriously variable according to their source (Friess and Webb 2011). Second, we have summed the actual OCS values for each of the pixels where land conversion has taken place (i.e. figure S8) instead of applying a conversion rate to a mean OCS value.

The three nations of Indonesia, Malaysia and Myanmar contributed 77% of global mangrove OCS loss for this time period (figure 3). Despite similar area loss (figure 3(a)), Malaysia lost approximately twice as much soil carbon as Myanmar due to the large differences in carbon density between these two nations (mean OCS = 485 ± 57 versus 245 ± 63 Mg C ha⁻¹, respectively). This comparison highlights the importance of using local OCS values for estimating carbon emissions attributed to mangrove conversion.

Not all land use conversions result in equal loss of OCS. Conversion of mangrove forest to shrimp ponds results in a rapid and near complete loss of carbon in the upper meter of soil (Kauffman *et al* 2014), as well as losses deeper in the soil profile (Kauffman *et al* 2017). Conversion to other agricultural uses such as pasture for beef production (Kauffman *et al* 2016b) and cereal crops (Andreetta *et al* 2016) also appear to result in large soil carbon emissions. However, mangrove degradation and loss due to over harvesting for fuelwood (Jones *et al* 2015) or due to natural

disturbance (Cahoon *et al* 2003) likely leads to more moderate emissions as decomposition and erosion exceed new plant carbon inputs.

It is important to note that nearly all available data on OCS loss due to conversion to other land uses come from organogenic mangrove forests. In a mineral-dominated mangrove systems with only a few percent sediment OCC, we would not expect the same level of carbon loss as when peat deposits are drained or removed. In fact, reclamation of deltaic sediments for paddy rice cultivation can lead to increases in OCS (Kalbitz *et al* 2013), although methane emissions would be expected to increase. Additionally, if mangrove habitat is lost due to deforestation without a change in hydrologic regime, mineral-dominated mangroves can continue to accrete carbon, but at a lower rate than in a system that has additional organic matter inputs from the mangroves themselves (Pérez *et al* 2017).

4.4. Limitations and uncertainties

While we endeavored to ensure that the model input data was of the highest quality possible, there undoubtedly remain unknown errors in the database which are contributing to model error. Machine learning models are particularly sensitive to outlier values and extrapolation (Murphy 2012). Various research groups use different methods for determining the organic carbon concentration (OCC) of a sample with not all publications reporting whether or not results were corrected for occurrence of inorganic carbon or whether or not roots were excluded before further processing. Bulk density (BD) is a difficult parameter to measure accurately in many soils, and based on our analysis of BD versus OCC (figure S1) some reported data are unlikely to be accurate. We

developed a procedure to correct potential BD errors, but a pedotransfer function gives only an approximation of the true value. Given the importance of depth in our models (figure 1(b)), it was unfortunate that so many investigations only report OCS for large depth increments. We suggest that future studies using the common practice of collecting subsamples within larger horizon increments (e.g. Kauffman and Donato 2012, Kauffman *et al* 2016a) report the specific depth increment of the sample rather than that of the entire core. This additional level of transparency in the data would allow mass-preserving splines (Bishop *et al* 1999) to be fit through the distinct measurement intervals, resulting in unbiased estimates of OCS.

The largest uncertainty in the input data likely resulted from imperfect information on plot location. Whether, accidental or purposeful (i.e. not wanting to identify exact locations), spatially misplaced data in publications are of limited utility in geospatial applications. All of the covariate data for each mangrove point were selected from spatial layers resulting in the potential for a mismatch between the recorded OCS and the spatial predictors. In this study, we visually inspected all coordinates against Google Earth imagery and our adjusted mangrove domain, and then contacted many authors to seek further information and manually adjusted coordinates when we were confident that the adjustments lead to better spatial location. In the final analysis, 199 soil profiles had to be excluded from analysis because we could not confidently locate these plots within the adjusted mangrove spatial domain.

5. Conclusions

This work has produced three resources which we hope to be of significant value to the blue carbon research and management communities: 1) a large harmonized database of soil carbon data from mangrove ecosystems; 2) high-resolution (30 m) predictions with error of soil carbon stocks across all mangrove forests globally; 3) estimates of potential soil carbon losses due to mangrove habitat loss between 2000 and 2015. By using a statistical data-driven model, we have been able to produce credible estimates OCS for the numerous mangrove regions where no field data exist. We found that mangrove OCS is highly variable ($86\text{--}729\text{ Mg C ha}^{-1}$ in the top meter) but that much of the variability could be captured using spatially-comprehensive predictors in a machine-learning framework. Of the 6400 Tg C in the upper meter of soil, $30\text{--}122\text{ Tg}$ have likely been lost due to deforestation since the year 2000 with 77% of this loss attributed to Indonesia, Malaysia and Myanmar. These spatially-explicit estimates of mangrove soil carbon storage and loss will provide a practical first step for enabling nations to prioritize mangrove protection as part of their climate mitigation and adaptation plans.

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Data availability

The mangrove soil carbon database and model outputs can be downloaded from Harvard dataverse at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/OCYUIT>.

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ORIGINAL RESEARCH

Improved assessment of mangrove forests in Sundarbans East Wildlife Sanctuary using WorldView 2 and TanDEM-X high resolution imagery

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Canopy height, high resolution satellite image, mangrove, optical data, species identification, Sundarbans Reserved Forest

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Abstract

Recent developments of remote sensing techniques which can capture both the structure and function of the ecosystem provide a more representative view of the landscape. These unique Earth observations were used to help improve traditional forestry surveys by providing species-specific land cover classes for mangrove forests in the Sundarbans East Wildlife Sanctuary. By combining optical data from WorldView2 (WV2; 2 m pixel) and a canopy height model derived using radar data from TanDEM-X (TDX; 12 m pixel), we identified nine mangrove and five non-mangrove classes by following an Iterative Self-Organizing Data Analysis Algorithm. Three dominant mangrove species accounted for nearly 50% of the sanctuary. *Heritiera fomes* disproportionately covered the largest area at 43%, overturning previous field-based estimates of *Excoecaria agallocha* dominance. *E. agallocha* and *Sonneratia apetala*, covered 3% and 1.47% of the sanctuary, respectively. Four mixed species classes were also identified with clear vegetation zonation patterns that trended toward species homogeneity with increasing distance from shore. The overall land cover accuracy (WV2: 89.33%; WV2-TDX: 89.89%), the Kappa Coefficient (WV2: 0.88; WV2-TDX: 0.89) and change statistics between WV2 and WV2-TDX land cover classifications indicate that the WV2 imagery can separate mangrove community types without structural data. The combination of the land cover classifications and the canopy height model indicated that *H. fomes* were not only the most dominant forest but also, on average, the tallest (12.3 m) among the other eight mangrove types. Our large-scale mapping with high resolution optical and radar platforms can capture subtle changes in mangrove vegetation and canopy structural gradients more accurately and be used to monitor biodiversity changes and Aichi Biodiversity Targets and Indicators, which would contribute to biodiversity policy updating.

Introduction

Mangroves are the woody vegetation found in sheltered coastal transition zones of tropical and subtropical countries (Alongi 2008). These ecosystems are now a central concern for rehabilitation, protection, and climate change mitigation and adaptation due to their unique ecosystem services and anthropogenic threats they face. The livelihoods of coastal communities are supported by mangrove ecosystems and provide resources for household materials, serve as habitats for fish and endangered wildlife, and help to safeguard communities against cyclones and storm surge (Kathiresan and Bingham 2001; Kathiresan and Rajendran 2005; Alongi 2008; Thant et al. 2010; Uddin et al. 2013). However, in recent decades, there has been widespread deforestation or clearing of mangrove forests globally because of agricultural expansion, urbanization, shrimp farming and overexploitation of timber (Richards and Friess 2016; Thomas et al. 2017). In concert with human development, sea level rise, cyclones, and other natural disturbances can alter floral and faunal species compositions, vegetation structure and carbon sequestration potential (Gilman et al. 2008).

International biodiversity conservation programs such as Aichi Biodiversity Targets 2020 (O'Connor et al. 2015) are providing new strategies to limit and monitor changes in critical regions. There are 20 specific targets which were established by the Convention of Biological Diversity that aim to halt biodiversity loss (See O'Connor et al. (2015) for details) with a set of biodiversity indicators that aim to quantitatively monitor the progress of those targets (Petrou et al. 2015). Specific Aichi Biodiversity Targets; 5 – *Habitat loss, fragmentation and degradation*; 11 – *Protect areas*; and 14 – *Ecosystem services and safeguarded* can be directly or indirectly monitored using remote sensing techniques with the support of *in situ* data (see Petrou et al. 2015). Essential Biodiversity Variables such as ecosystem extent and fragmentation, ecosystem composition, ecosystem function, species traits, and land cover change are potential indicators for assessing the progress of these Aichi Biodiversity Targets (Pereira et al. 2013; O'Connor et al. 2015; Vihervaara et al. 2017). Therefore, the regular evaluation of mangrove species communities is necessary for conserving mangrove forests, informing sustainable management, and the reevaluating regional and national policies (Giri 2016; Thomas et al. 2017).

Field-based inventory monitoring is the desired method for assessing forested resources. However, mangrove forests and their remote environment settings, complex terrain, dangerous conditions and daily tidal flooding cause economic and logistic constraints that can be difficult to overcome (Kamal and Phinn 2011). Considering these

setbacks, remote sensing has played an important role in augmenting field inventories, by increasing the spatial and temporal mapping of mangrove areas (See Kamal and Phinn 2011; Viennois et al. 2016; Shapiro et al., 2015; Stringer et al. 2015). Over the last few decades, both multispectral and radar imagery have been used in mapping regional and global mangrove coverage, measuring mangrove communities and structure at the local scale, and estimating other biophysical properties (Giri et al. 2011; Heumann 2011; Kamal and Phinn 2011; Kuenzer et al. 2011; Lagomasino et al. 2016; Thomas et al. 2017; Bunting et al. 2018; Lee et al. 2018).

For remote sensing to be considered an asset for Aichi biodiversity Targets and other similar programs that traditionally rely on field inventory data, it is necessary to develop spatially explicit models that represent the variability, range, and coverage of dominant and mixed species vegetation types. Several studies have identified mangrove species groups using high resolution images. Very high resolution multispectral satellite images from WorldView, IKONOS and QuickBird have been used to map mangrove species at various sites but with a limited number of species (Wang et al. 2004; Neukermans et al. 2008). Imagery from the WorldView satellite series have proven more successful to other high resolution satellite images (e.g. IKONOS, QuickBird and SPOT5) because of their revisit cycles, geometric accuracy and multispectral bands (Aguilar et al. 2013; Zhu et al. 2015). Mangrove species have been mapped with WorldView imagery using supervised (Heenkenda et al. 2014; Wang et al. 2015; Viennois et al. 2016) and object-based (Kamal et al. 2015; Zhu et al. 2015) classification methods that utilize spectral bands, textures and differential spectral features to differentiate land cover types. Using land cover classification methods that combined spectral bands, textural information and differential spectral features (Wang et al. 2015) performed better on species discrimination than methods that incorporated only spectral or textural information alone (Heenkenda et al. 2014; Wang et al. 2015).

Optical imagery and biophysical spectral indices can result in saturation effects for dense mangrove forests and may reduce the accuracy of the overall classification result (van Ewijk et al. 2014). Combining a canopy height layer with high spatial resolution multispectral bands can provide fine-scale information than can better separate land cover types into dominant and co-dominant mangrove species classes (van Ewijk et al. 2014; Kamal et al. 2015). Although, vertical structure has been included in vegetation species classification (van Ewijk et al. 2014), this technique has been limited for mangrove ecosystems at large scales because of the scarcity of spatially explicit canopy height information. Recent advances in the large-scale mapping of mangrove canopy

height using interferometric synthetic aperture radar imagery can be a potential option to help improve mangrove species classification. Using these advanced techniques, mangrove canopy height has been estimated at a 12 m spatial resolution and an RMSE of less than 2 meters for expansive sites in Mozambique, Mexico and Bangladesh using TanDEM-X (TDX), a German Space Agency mission comprised of two satellites flying in formation (Lee and Fatoyinbo 2015; Lee et al. 2015; Lagomasino et al. 2016).

The aforementioned studies mapping mangrove species based on high-resolution imagery were applied in relatively small areas of primarily restored mangroves with a limited number of species to map. Furthermore, many of these studies used only optical imagery and had a limited discussion of the distribution of species along physical gradients, a common phenomenon in mangrove forests. Modifying the application of the high resolution optical and radar imagery to monitor large, undisturbed mangroves like the Sundarbans Reserved Forest is necessary to test the improvement of vegetation mapping and underlying ecological landscape processes. In this study, we focused on mangrove land cover classification of dominant and co-dominant species using WV2 images and TDX derived canopy height for the Sundarbans East Wildlife Sanctuary (SEWS), a section of the larger Sundarbans Reserved Forest in Bangladesh. The objectives of this study were as follows: (1) to produce a high resolution map of species functional types (monitoring indicators of Aichi Biodiversity Targets-2020) and their associated vertical canopy structure (structural biodiversity indicator) for mangroves forests in SEWS, and (2) to combine multispectral and canopy height data to test the improvements of classification methods in SEWS.

Materials and Methods

Study area

The Sundarbans Reserved Forest is one of the most diverse mangrove forests in the world (Rahman et al. 2015b; Islam et al. 2016). The total area of the Sundarbans Reserved Forest is 6017 km² and mangroves occupy about 69% of the area and the remaining 31% are covered by water (Fig. 1). The forest is home to 528 species of vascular plants where 24 species are true mangrove and 70 species are mangrove associates (Rahman et al. 2015b). The Sundarbans Reserved Forest also has a rich faunal biodiversity with a total of 1135 recorded species, including a large community of endangered species like the Royal Bengal Tiger, the Ganges and Irrawaddy dolphins, and saltwater crocodiles (Aziz and Paul 2015).

The forest supports and protects the livelihoods of the local communities. Because of the rich biodiversity, and the high socioeconomic and ecological value, the forest was declared a Ramsar Wetlands Site in 1992 and a UNESCO World Heritage Site in 1997 (Singh et al. 2010; Abdullah-Al-Mamun et al. 2017; Islam et al. 2018).

The World Heritage Site in Sundarbans Reserved Forest, is partitioned into three protected areas: Sundarbans West Wildlife Sanctuary, Sundarbans South Wildlife Sanctuary and the SEWS (Fig. 1). The three sanctuaries cover a total of 139 699 ha which is about 23% of the entire Sundarbans Reserved Forest (Rahman et al. 2017). In this study, we concentrated on SEWS within the old sanctuary boundary that covered 40 768 ha as delineated by the Bangladesh Forest Department (BFD) in 1997.

In 2010, under the financial support of United State Agency for International Development, BFD conducted a forest carbon inventory at 150 mangrove plots. Each plot was comprised of five, 10 m radius subplots where one central subplot was surrounded in the cardinal directions by four additional subplots (Rahman et al. 2015a). Of the 150 plots, nine of them occurred within the SEWS study region and were used in this study for calibration and validation (Rahman et al. 2015a). In each of the subplots, the diameter-at-breast height (DBH) of every tree (DBH ≥ 10 cm) was measured along with the height of three tallest trees. Saplings (DBH < 10 cm and height ≥ 1.37 cm; 3 m radius plot), seedlings (height < 1.37 cm; 2 m radius plot), palms (4 m radius plot), shrubs (4 m radius plot), herbs (2 m radius plot), and lianas were also measured (See Rahman et al. 2015a for detail). The detailed stand structures of the study site were given in Table 1 and Table S1.

Remote sensing data processing

For the land cover classification of mangrove species in SEWS, we used two types of Earth observation: (1) passive sensor high resolution (2 m spatial resolution) from WV2 and (2) active sensor for example, TDX Synthetic Aperture Radar (SAR) images (12 m spatial resolution). The WV2 imagery provided multispectral information about land cover types, while the TDX imagery was used to produce accurate canopy height models for mangrove forests (Lee et al. 2015; Lagomasino et al. 2016). WV2 imagery was comprised of eight multispectral bands that cover wavelengths between 430 and 1050 nm. The spectral range of these eight multispectral bands are coastal blue (400–450 nm), blue (450–510 nm), green (510–580 nm), yellow (585–625 nm), red (630–690 nm), red edge (705–745 nm), NIR1 (770–895 nm) and NIR2 (860–1040 nm) (Rapinel et al. 2014). Three images from

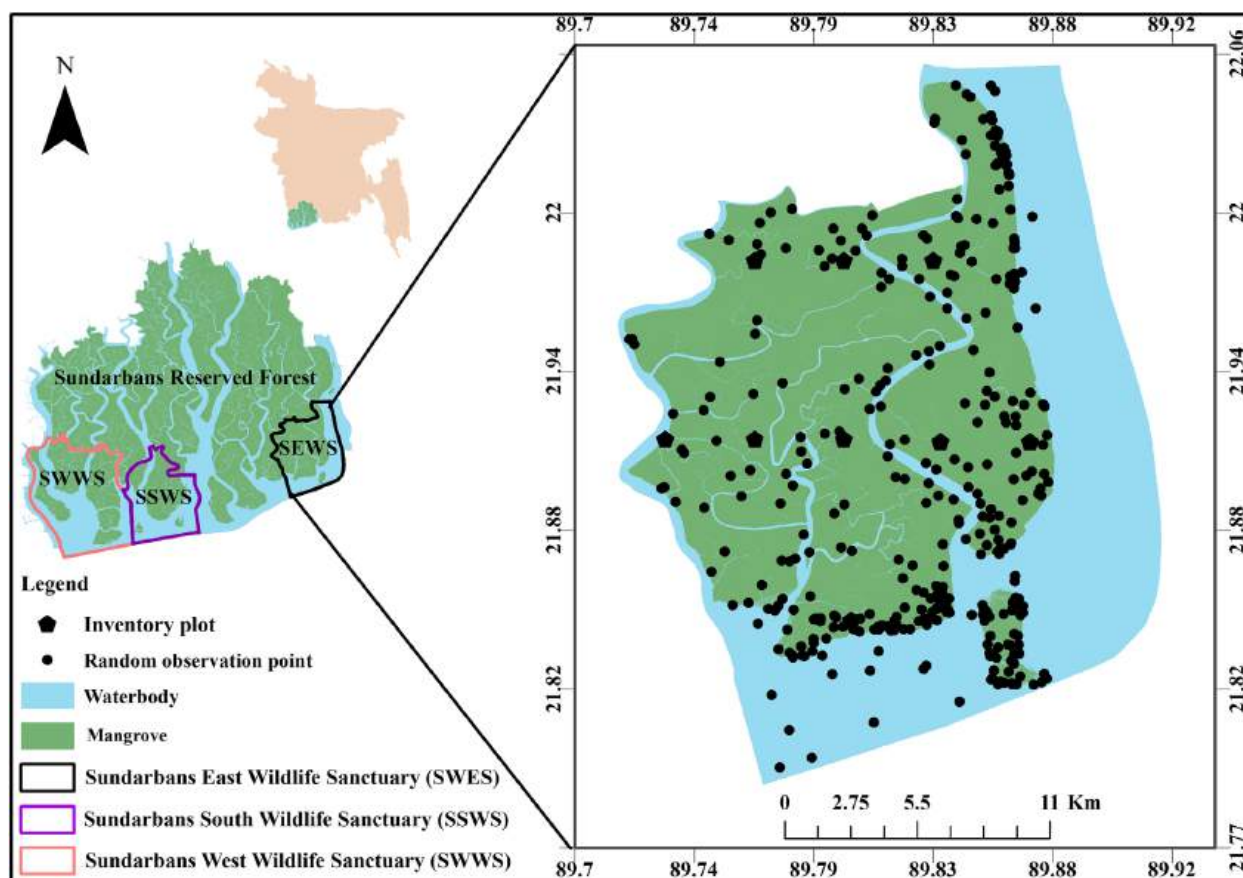


Figure 1. Position of field inventory plot (black polygon) and Google earth observation point (black circle) at Sundarbans East Wildlife Sanctuary in Sundarbans Reserved Forest.

Table 1. Structural composition and mean DBH ($\pm$ SE) of mangrove species at inventory plots (total 9 plots; each of which consist of five 10 m radius circular subplots) at Sundarbans East Wildlife Sanctuary.

Species	Number of tree	Basal area ($\text{m}^2 \text{ha}^{-1}$)	Relative abundance (%)	Relative dominance (%)	Mean DBH (cm)
<i>Heritiera fomes</i>	1105	15.75	72.18	72.06	15.30 ± 0.14
<i>Excoecaria agallocha</i>	374	4.14	24.43	18.93	13.65 ± 0.19
<i>Xylocarpus mekongensis</i>	27	0.61	1.76	2.77	18.60 ± 1.51
<i>Sonneratia apetala</i>	9	1.19	0.59	5.44	46.79 ± 4.91
<i>Amoora cucullata</i>	8	0.08	0.52	0.35	12.9 ± 60.95
<i>Xylocarpus granatum</i>	4	0.06	0.26	0.29	16.38 ± 2.26
Unknown species	1	0.01	0.07	0.04	13.00
<i>Avicennia officinalis</i>	1	0.01	0.07	0.04	12.00
<i>Cynometra ramiflora</i>	1	0.01	0.07	0.03	10.00
<i>E. indica</i>	1	0.01	0.07	0.05	14.30
Total	1531	21.86	100.00	100.00	15.12 ± 0.14
Stem density (tree/subplot)	34				

December 26, 2015 and one from January 15, 2016 were acquired from Digital Globe through cad4nasa and the NextView License Agreement (Neigh et al. 2013). Each image was radiometrically calibrated for WorldView2 imagery in ENVI to generate a Top-of-Atmosphere reflectance image (ToA). Each calibrated ToA was then passed on to Fast Line-of-sight Atmospheric Analysis of Hypercubes to account for localized atmospheric effects. After corrections, the three images representing surface reflectance were mosaicked, and several normalized band ratios were determined from the multispectral data to provide additional information into the classification (Equations 1–4). Band ratios can be helpful in reducing errors associated with land type classification modeling. The NDVI-Red Edge (NDVI_{re}) has shown to be a good measure of biophysical plant traits and has been helpful in discriminating mangroves from non-mangroves because of the rapid change in reflectance within the Red-Edge region (Ahamed et al. 2011; Heenkenda et al. 2014). The NDVI-Green (NDVI_g) has been shown to be less impacted by atmospheric effects (Gitelson et al. 1996). We calculated four normalized difference band ratios:

- (1) Normalized Difference Vegetation Index (NDVI)

$$\frac{(NIR1 - RED)}{(NIR1 + RED)} \quad (1)$$

- (2) Normalized Difference Vegetation Index (NDVI_g)

$$\frac{(NIR1 - GREEN)}{(NIR1 + GREEN)} \quad (2)$$

- (3) Normalized Difference Vegetation Index (NDVI_{re})

$$\frac{(NIR1 - REDEGE)}{(NIR1 + REDEGE)} \quad (3)$$

- (4) Normalized Difference Vegetation Index (NDVI_{re2})

$$\frac{(NIR2 - REDEGE)}{(NIR2 + REDEGE)} \quad (4)$$

For this classification scheme, we combined information about the land cover types (multispectral and band ratios) and forest structure (TDX canopy height) to help distinguish generalized mangrove species types. Using single-pass TDX data, Lee and Fatoyinbo (2015) estimated mangrove canopy height with an accuracy that was comparable to airborne lidar. By using a similar method to Lee and Fatoyinbo (2015), Lee et al. (2015) estimated large-scale mangrove canopy height across the Sundarbans using TDX images. The estimated canopy height correlated highly with the field measurement data ($R^2 = 0.85$) with a root mean square error (RMSE) of 0.77 m which resulted in a 10% estimation accuracy of the mean forest

height (Lee et al. 2015). In the current study, we used this canopy height model over our study region within the SEWS. The 12 m pixel size image was resampled to the same spatial resolution as the WV2 imagery (2 m). In total, there were 13 different map layers that were used in the land cover classification scheme; eight multispectral, four band ratios, and one canopy height layer.

Unsupervised classification

We used an Iso Cluster Unsupervised Classification algorithm in ArcMap 10.2.1 to cluster similar land cover types based on all 13 layers (multispectral, band ratios and canopy height) (Fig. 2). The ISODATA algorithm is a method of iteration that makes clusters of similar groups into one by measuring the Euclidean distance between cluster centers (Dhodhi et al. 1999). In the classification, we set the number of classes to 50, the minimum class size to 10 and the sample interval to 5. The classification result returns a one band image with a numbered set of land cover types.

We assigned a vegetation or other land cover type for each of the 50 classes from the resultant land cover classification. The assignment of land cover types was done manually using supplemental information from the field inventory data, field photos, field experience and Google Earth imagery interpretation. Once all land cover classes were identified, similar cover types were then aggregated in a single class. In order to remove pixels that were associated with shadows or canopy gaps, we applied two majority filters. To remove small shadows or canopy gaps, the first majority filter selected the highest occurrence of a land cover type within a 3-pixel $\times$ 3-pixel kernel. For land cover types associated with large shadows, we applied an 11-pixel $\times$ 11-pixel kernel. The larger kernel was only applied to the canopy shadow land cover class. The 11-pixel $\times$ 11-pixel kernel filter added the shadows or canopy gap to the neighboring land cover class. Any additional canopy gaps or shadows that were still identified were then assigned to the nearest land cover class. The class assignment and filtering were performed using ENVI (Version 5.2).

Accuracy assessment

We randomly generated 356 points. The dominant land cover class covered ~45% of the SEWS. We therefore, randomly generated 356 points, in accordance with multinomial probability theory, in order to access accuracy with a 95% confidence interval (Congalton and Green 2009). The randomly generated reference points were confirmed by visual interpretation of Google Earth imagery collected during the same time period as the WorldView acquisitions, field photos and expert field knowledge (Congalton 2001; Yu and Gong 2012; Rahman et al. 2015a, 2017).

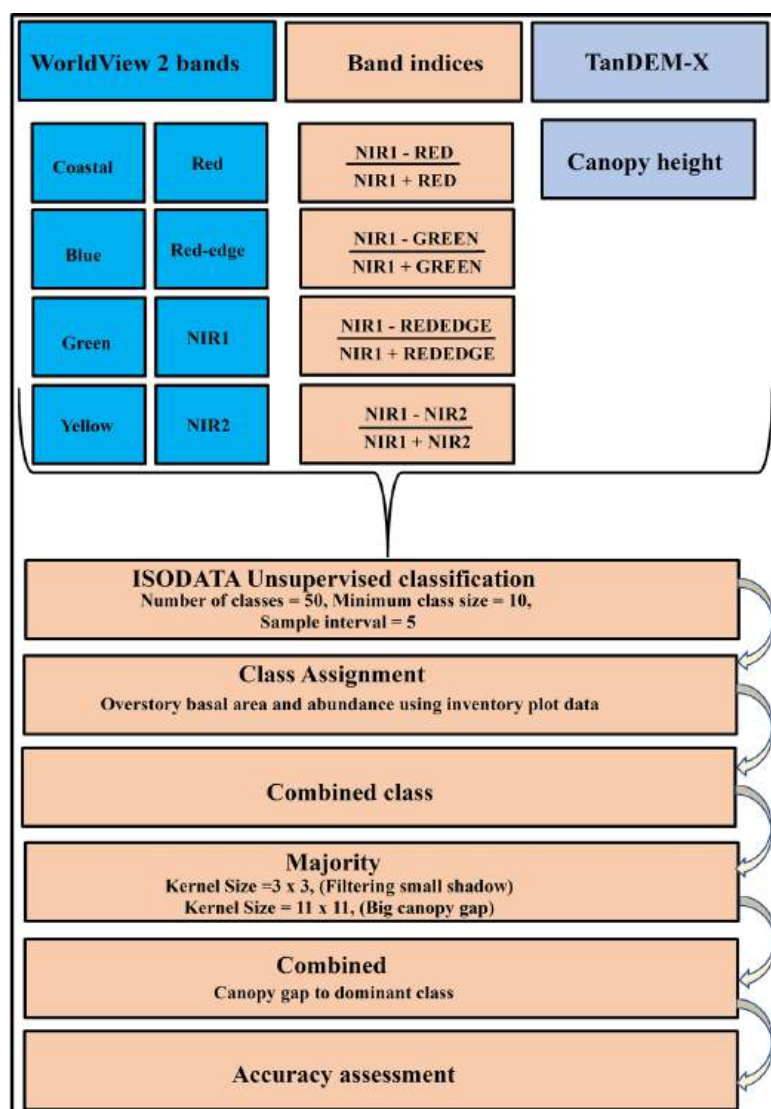


Figure 2. Flowchart of the classification method.

Results

Land cover types

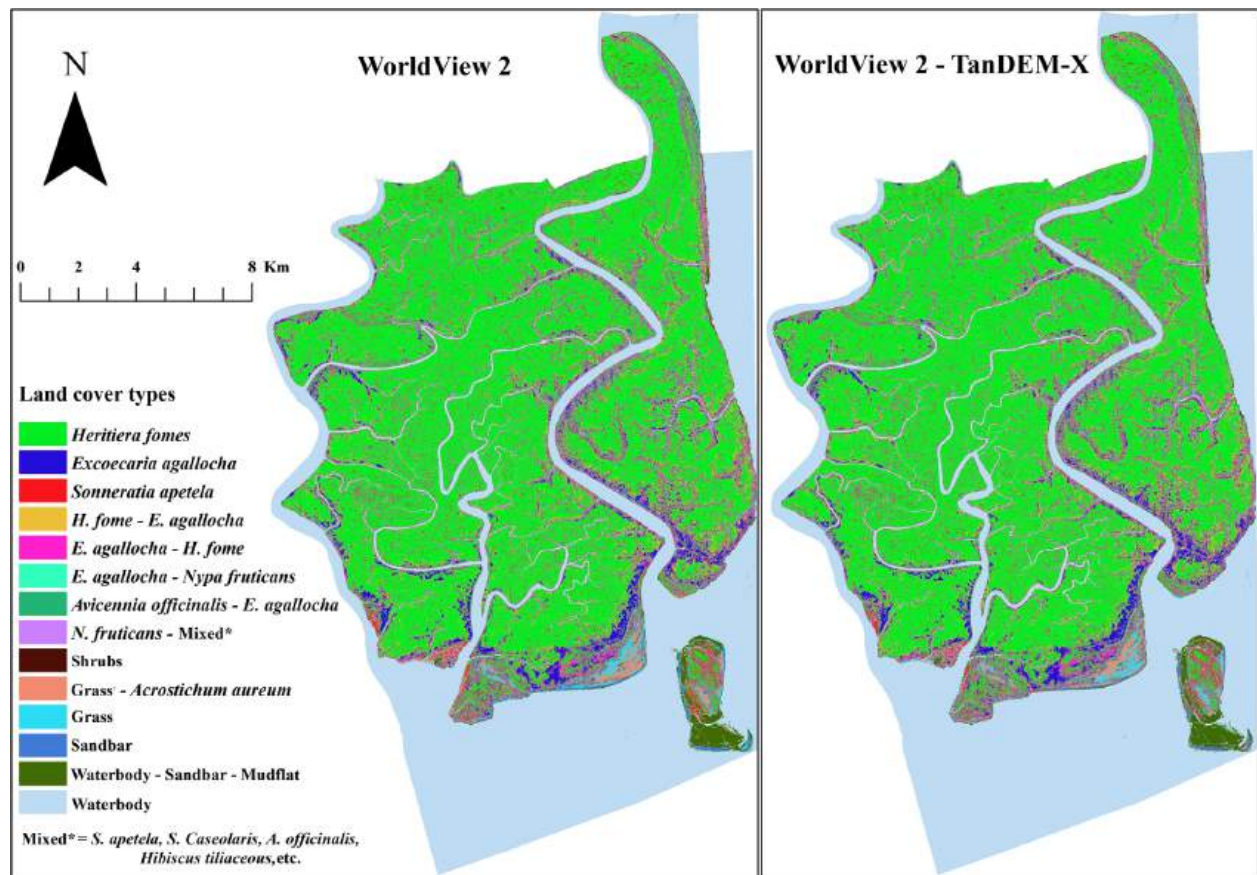
A total of 14 land cover classes were identified within the SEWS using WV2 and TDX imagery. Of these 14 land types, nine were classified as trees, shrubs and palms (65.01–65.02%) which included both mangrove and non-mangrove vegetation (Table 2). The remaining five land cover types were separated into Grass – *Acrostichum aureum* and Grass-other classes, (1.17–1.18%) as well as Sandbar, Waterbody – Sandbar – Mudflat, and Waterbody (33.80–33.80%; Table 2). Three of the nine tree/shrub classes were separated into mangrove classes that were dominated by a single species: (1) *H. fomes*, (2) *Excoecaria*.

agallocha, and (3) *Sonneratia apetala*. The other five tree/shrub classes were identified as mixed species types, (4) *H. fomes* – *E. agallocha*, *Nypa fruticans* – Mixed, (5) *E. agallocha* – *H. fomes*, (6) *E. agallocha* – *N. fruticans*, (7) *Avicennia officinalis* – *E. agallocha*, (8) *Nypa fruticans* – Mixed (*N. fruticans*, *S. caseolaris*, *S. apetala*, *Phoenix paludosa*, *Hibiscus tiliaceus* and *Avicennia officinalis*) and (9) Shrubs (Table 2). The spatial distributions of these land cover classes are shown in Fig. 3. Of the 32 930.77 ha we mapped within the SEWS, 67.7% was classified as terrestrial and the remaining 32.3% was classified as aquatic. The most dominant vegetation type in terms of area was *H. fomes* which covered 44.76–44.82% of the total area of SEWS based on WV2 and WV2-TDX classification maps, while the next three largest area classes *E. agallocha* –

Table 2. Comparison of land cover areas between WV2 and WV2-TDX at Sundarbans East Wildlife Sanctuary.

Land cover type	WorldView 2 (E1)		WorldView 2 – TanDEM-X (E2)		(E1 – E2)/E1 (%)
	Area (ha)	Percentage	Area (ha)	Percentage	
<i>Heritiera fomes</i>	14761.14	44.82	14741.15	44.76	0.00
<i>Excoecaria agallocha</i>	993.09	3.02	996.02	3.02	0.00
<i>Sonneratia apetala</i>	465.92	1.41	479.50	1.46	–0.03
<i>H. fomes</i> – <i>E. agallocha</i>	2086.76	6.34	2090.71	6.35	0.00
<i>E. agallocha</i> – <i>H. fomes</i>	2251.32	6.84	2251.38	6.84	0.00
<i>E. agallocha</i> – <i>N. fruticans</i>	200.26	0.61	200.63	0.61	0.00
<i>Avicennia officinalis</i> – <i>E. agallocha</i>	483.61	1.47	485.52	1.47	0.00
<i>Nypa fruticans</i> – Mixed*	145.84	0.44	146.25	0.44	0.00
Shrubs	19.81	0.06	20.40	0.06	–0.03
Grass – <i>Acrostichum aureum</i>	180.20	0.55	181.70	0.55	–0.01
Grass	203.52	0.62	205.90	0.63	–0.01
Sandbar	54.77	0.17	54.58	0.17	0.00
Waterbody – Sandbar – Mudflat	433.21	1.32	439.28	1.33	–0.01
Waterbody	10651.32	32.34	10637.73	32.30	0.00
Total	32930.76	100.00	32930.77	100.00	–0.10

Mixed* = *S. apetala*, *S. caseolaris*, *A. officinalis*, *Hibiscus tiliaceus* etc.

**Figure 3.** Comparison of land cover types between WorldView 2 and WorldView 2 – TanDEM-X for the Sundarbans East Wildlife Sanctuary.

H. fomes, *H. fomes* – *E. agallocha* and *E. agallocha* covered, 6.84%, 6.34% and 3.02%, respectively (Table 2). We found no significant difference in the area or distribution of the individual land cover types between the two classification approaches; WV2 and WV2-TDX. The total difference between the two maps was 0.10% (Table 2; Fig. 3).

There were clearly noted vegetation community zonations along the shore-perpendicular gradients. In most cases, stands became more monospecific with increasing distance from the shore (Fig. 4). For example, from the canal bank to inland the presence of vegetation types was, *N. fruticans* – Mixed followed by *E. agallocha* – *N. fruticans*, then either followed by *E. agallocha* – *H. fomes* or *H. fomes* – *E. agallocha*, followed by *H. fomes* or *E. agallocha* (Fig. 4). Several other zones were also found such as *S. apetala* or *E. agallocha* or *E. agallocha* – *H. fomes* followed by *H. fomes* or *E. agallocha* – *H. fomes* and then followed by *E. agallocha* or *H. fomes* (Fig. 4).

From one-way analysis of variance test, we found that mean canopy height varied significantly among the nine vegetation classes ($P < 0.05$; Fig. S1). According to Tukey Honestly Significant Difference *post-hoc* test, the canopy height

was significantly different between vegetation classes ($P < 0.05$). The canopy height of *H. fomes* was, on average (12.30 ± 2.93 m) and significantly taller than the other eight vegetation classes ($P < 0.05$). Conversely, the Shrubs class had the shortest canopy height (7.37 ± 3.95 m; $P < 0.05$; Fig. S1). The canopy height profile exhibited an increasing trend, shorter to taller, with increasing distance from the shoreline (Fig. S2). The plot level mean canopy height varied from 6.58 ± 0.56 m to 16.17 ± 0.21 m (Table S1).

Accuracy assessment

The overall land cover classification showed a strong agreement between the WV2 and WV2-TDX classification maps and the reference points, yielding an overall accuracy and the Kappa Coefficient of 89.33% and 0.88%, and 89.89% and 0.89%, respectively for WV2 and WV2-TDX based classifications (Tables S2 and S3). The specific land cover accuracy, also revealed that there was strong agreement between most of the land cover types of WV2 and WV2-TDX and random reference points (Tables S2 and S3). The Sandbar class had the highest producer's and user's

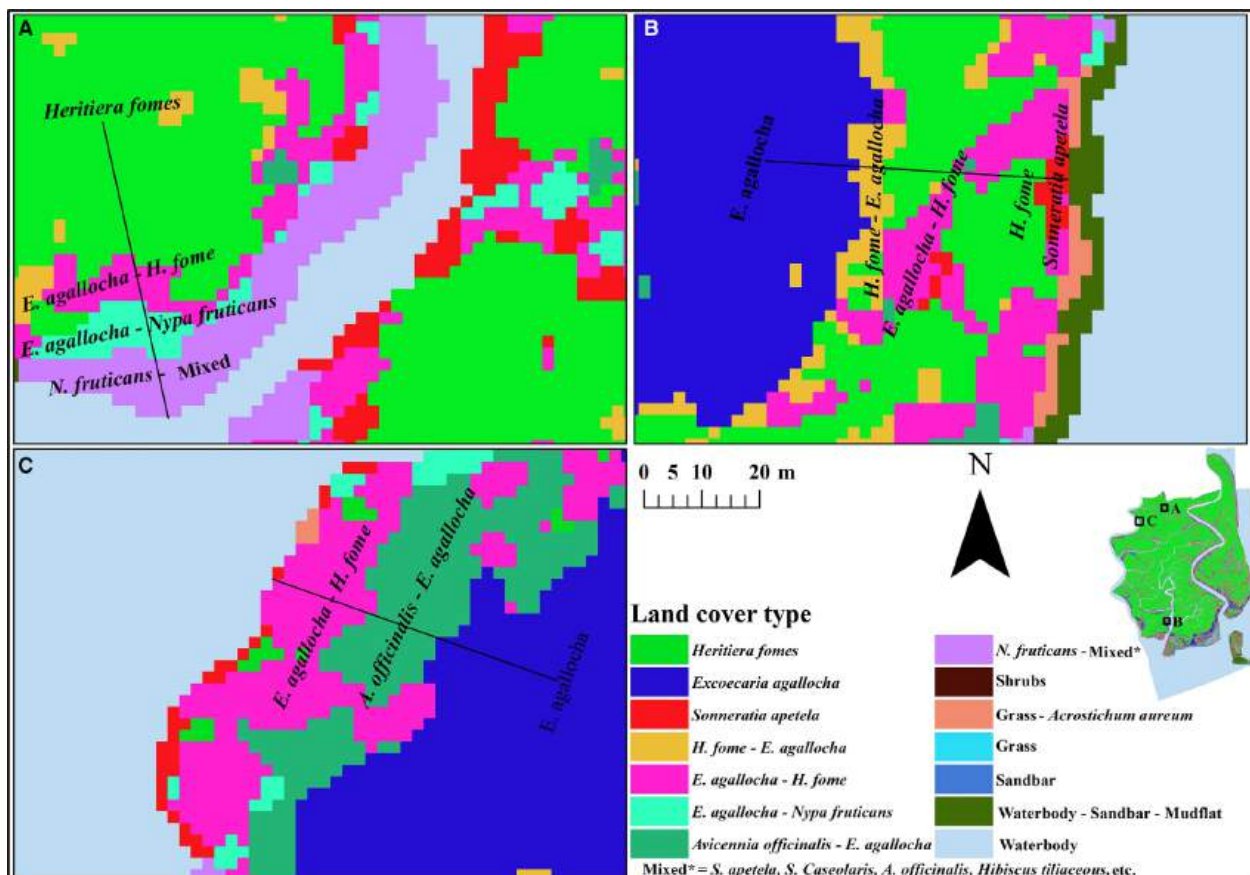


Figure 4. Species zonation at Sundarbans East Wildlife Sanctuary within 50 m transect: (A) from canal bank to inland side; (B) from big river bank to inland side; (C) from big river bank to inland side.

accuracy. The most dominant tree species *H. fomes*, had similar producer's accuracy in both WV2 and WV2-TDX-based classification but slightly lower user's accuracy in WV2 than WV2-TDX (Tables S2 and S3). While *E. agallocha* showed the best user's and producer's accuracy in both WV2 and WV2-TDX classification among the tree species (Tables S2 and S3). In both WV2 and WV2-TDX classification, *S. apetala* and *N. fruticans* – Mixed classes had lower producer accuracy (Tables S2 and S3).

Discussion

In this study, we applied a hybrid classification schema for separating mangrove vegetation into individual, co-dominant, and mixed species classes by combining high resolution optical (WV2) and radar (TDX) imagery. The land cover classification was performed using an ISO-DATA Algorithm and showed a strong agreement with Google Earth visual interpretations and field inventory data. Both land cover classifications using either spectral data only and combining spectral data with canopy height were able to separate mangrove from non-mangrove, where together the terrestrial classes covered more than two-third of the total areas of SEWS (Table 2).

There was little difference between the land cover area and distribution between the two classifications approaches (WV2 vs. WV2-TDX). High resolution WV2 bands along with the four vegetation indices (used in this study), can separate dominant mangrove species with high accuracy. This could suggest that the spectral data alone provided enough information to discriminate between mangrove land cover classes or that the coarse spatial resolution of the TDX data cannot resolve differences in vegetation at the individual species scale within the area of study (Table 2). A potential reason for the lack of classification improvement may be a result of TDX canopy model spatial resolution (12 m). As a rule-of-thumb the potential spatial resolution of imagery needs to be two-to-five times finer than the monitored objects (in this case tree canopy; Petrou et al. 2015). The average number of trees per subplot (314.59 m²) in our study was 34 which means one species occupied about 9 m² areas (Table 1). WV2 can capture the individual tree canopy but TDX canopy height was aggregated over multiple tree canopies. Despite the lack of improvement to the classification, the magnitude and the variability in the canopy structure for each of the vegetation types was estimated over a large area.

Canopy height imagery from lidar can have a similar spatial resolution to WV2 imagery and may have a better potential to improve species classification at fine-scales (Holmgren et al. 2008). However, the acquisition of airborne lidar can be costly for regular acquisitions. High resolution satellite stereo-imagery from platforms like

WV2, IKONOS, or Cartosat-1 that have been applied to other mangrove, boreal and temperate forests could be an efficient approach for regular monitoring (Straub et al. 2013; Neigh et al. 2014; Lagomasino et al. 2016; Persson and Perko 2016). Despite, the coarser resolution canopy height that was estimated in the current study, it can still be useful in biomass and carbon assessments when coupled with field data (Shapiro et al. 2015; Aslan et al. 2016). Using the TDX canopy height, we identified that when *H. fomes* occurred in dominate stands the canopy was taller than when it was found in mixed stands, for example in case of *H. fomes* – *E. agallocha* and *E. agallocha* – *H. fomes* classes. The reason behind this difference may be due to the variation of site conditions between monospecific and mixed stand (Fig. 4; Ewel et al. 1998).

Species dominance in SEWS

The dominant class in SEWS was identified as *H. fomes*, which were also reported by Rahman et al. (2017). This differs from previous reports that indicate that *E. agallocha* was more dominant than *H. fomes* in SEWS (Islam et al. 2014). The previous inventory was based on a field assessment using 12 transects of 200 m long with two plots (20 m × 50 m): one at 0–50 m (stream side) and another at 150–200 m (inland or “forest proper”) from shore. Over this transect distance, our results indicate similar vegetation communities, *N. fruticans* – Mixed class (stream side of canal) and *E. agallocha*, *H. fomes* or *E. agallocha* – *H. fomes* or *H. fomes* – *E. agallocha* (forest proper). The spatial distribution of species or groups of species in our study reveals that *H. fomes* stands are primarily found in the inner-most parts of almost all islands, far from most transect based inventories (Fig. 3). Islam et al. (2014) reported that the near shore site was the most diverse which we also found in our study. However, their report suggested that the protected areas were dominated by *E. agallocha*. Our findings suggest that ~45% of the total area in the SEWS was covered by *H. fomes* while *E. agallocha* covered only 3% (Table 2 and Fig. 3). Thus, our current approach can overcome some of the spatial limitations and potential biases of field plot-based studies and more accurately represent the spatial distribution of dominant species (overall accuracy; 89.33%) with the use of high resolution WV2 satellite imagery.

Comparison with other studies

Both the error matrix and visual interpretation reveal that the classified vegetation type in the present matched well with randomly sampled observation points and inventory plots (Tables S2 and S3; Heenkenda et al. 2014). The overall accuracy of both datasets (WV2: 89.33% and WV2-TDX:89.89%) in our study, were similar to that of

Zhu et al. (2015) that report an overall accuracy of 89% in mapping mangrove species in Chinese mangrove forests using WV2 images. Mapping mangrove at the species level in Northern Territory, Australia using a supervised classification also resulted in a high accuracy (89%) (Heenkenda et al. 2014). Our overall accuracy was also within the range of overall accuracy (83.78% to 94.40%) reported by Wang et al. (2015), who combined original bands, textural features and differential spectral features of WorldView3 images to identify several mangrove species. These studies covered relatively small areas of natural or restored sites with clear homogeneous patches and only a limited number of species delineated. However, in our study, we covered a large complex natural mangrove ecosystem (SEWS 32 930.76 ha) within the Sundarbans Reserved Forest and produced maps with representative dominant and mixed species types.

Zonation in SEWS

The spatial distribution of species in our study suggested that there were clear vegetation community zonation patterns that occurred with increasing distance from the shoreline. For example, from the bank of small river or canal, we found either *Nypa fruticans* or mixed stand of *N. fruticans* – Mixed, followed by mixed stand of *E. agallocha* – *N. fruticans*, or *E. agallocha* – *H. fomes* or *H. fomes* – *E. agallocha*, which ended up with a monospecific zone of either *E. agallocha* or *H. fomes* (Fig. 4). While the seaward or large river bank forest margins transitioned from either *S. apetala* or *E. agallocha* into monospecific stands of either *E. agallocha* or *H. fomes*, then followed by mixed stands of *E. agallocha* – *H. fomes*, or *A. officinalis* – *E. agallocha* or *H. fomes* – *E. agallocha* which ended up with monospecific zone of *H. fomes* (Fig. 4). Our findings contrasted with the field-based study on zonation patterns in Sundarbans Reserved Forest (Ellison et al. 2000). There they examined 11 blocks by randomly laying out three random 200-m transects at each block throughout the whole Sundarbans Reserved Forest, which also included our study site. They concluded that there were no specific vegetation zonation patterns in the Sundarbans Reserved Forest (Ellison et al. 2000). The limited number of transects at two blocks (six transects within 32 930.76 ha) in SEWS, may not be enough for observing species zonation or the traditional sampling method of sequential quadrats may not be in line with the natural distribution or species zoning. The results presented here highlight that further study is needed to combine remote sensing with representative transects and plots in order to provide a more accurate assessment of protected areas and will be necessary to explore the zonation patterns along with other ecosystem information (Shapiro et al. 2015; Stringer et al. 2015).

Confusing classes

Visual interpretation of Google Earth imagery showed that there were minor misclassifications between Waterbody – Sandbar – Mudflat, *S. apetala*, *E. agallocha*, and Grass – *Acrostichum aureum*, and *N. fruticans* – Mixed. These misclassifications may be a result of the complicated mangrove environment, where the spectral reflectance values measured by WV2 over the mangrove vegetation can be affected by wet soils, water and atmospheric vapor (Chauvaud et al. 1998; Adam et al. 2010; Heenkenda et al. 2014). *S. apetala* was confused primarily by *H. fomes* (12%), followed by Grass – *Acrostichum aureum*, *E. agallocha* – *H. fomes*, *N. fruticans* – Mixed, Waterbody – Sandbar – Mudflat (Tables S2 and S3). *S. apetala* is a pioneer species in Sundarbans and primarily found in newly colonized mudflats and along the banks of canals and creeks, whereas *H. fomes* is the climax successional species and was primarily located in the more stable and interior portions of the islands. However, in some of these newly formed areas, *S. apetala* was misclassified as *H. fomes* (Fig. 3). This may have been a result from shadow effects, because of similar height structures, or similar spectral signatures (Adam et al. 2010). In its mature state, *S. apetala* is taller than *H. fomes*, but at earlier life stages its height could be similar to *H. fomes*. Because there were no distinctions between these two classes using both WV2 and WV2-TDX, new methods or further research is needed to improve the separability between these two classes, *S. apetala* and *H. fomes*. For example, methods that combine both spectral and textural properties of WV images may help to improve this classification (Wang et al. 2015).

In some cases, we were able to differentiate *A. officinalis* (very large tree with wide spread canopy) from the *A. officinalis* – *E. agallocha* class and *N. fruticans* (usually grows along the canal bank as large patches) from the *N. fruticans* – Mixed class. However, because these species, *A. officinalis* and *N. fruticans*, were unique in some locations but mixed in other locations, as confirmed by visual interpretation and expert knowledge, we combined each of these species into two different mixed classes, *A. officinalis* – *E. agallocha* and *N. fruticans* – Mixed, respectively. In the case of *N. fruticans* – Mixed, there was some confusion with *E. agallocha* – *H. fomes* (13.04–17.39%) and *E. agallocha* – *N. fruticans* (7.41–8.70%; Tables S2 and S3). The misclassification between *N. fruticans* – Mixed and *E. agallocha* – *H. fomes* may have been complicated because of the presence of common liana or vines over the canopy of these species. A finer scale resolution canopy height in addition to textural properties may improve the separability of mangrove species in future studies.

Implication for Aichi Biodiversity Targets-2020 progress assessment

Remote sensing-based monitoring of biodiversity indicators has been recognized at global biodiversity forums like the Convention of Biological Diversity (O'Connor et al. 2015; Petrou et al. 2015). Our study results could be used to directly monitor some of the major indicators for specific Aichi Biodiversity Targets-2020. The main indicators for Target 5 are trends in the extent, condition and vulnerability of ecosystems, biomes, and habitats (Petrou et al. 2015). Using WV2, we measured 14 land cover types of SEWS. These results would provide the information of tracking the progress and change in key biomes and habitats and monitoring management effectiveness for this protected area (Petrou et al. 2015).

The mangrove species *H. fomes* is a major component of the Sundarbans Reserved Forest but it is also an endangered species according to the IUCN Red List. Aichi Biodiversity Target 12 aims to prevent the extinction and to improve the conservation status of threatened species (CBD, 2010). An improvement of the conservation status means that a species increases in population and moves into a lower threat status (CBD, 2010). In our study, we estimated that *H. fomes* covered 45% of the SEWS, a much larger area than previously measured. Using similar mapping techniques, we can not only estimate the location of endangered plant species, like *H. fomes*, but can also monitor changes in their extent. Both aspects of this monitoring will help to better inform the status of IUCN Red List species and the management strategies needed to improve conservation.

The canopy height maps with a spatial resolution of 12 m that we generated can be coupled with *in situ* data (Rahman et al. 2015a; Aslan et al. 2016) for assessing carbon storage. In this way, we can also relate our findings to the Essential Biodiversity Variables; trends in distribution, condition, and sustainable ecosystem services for equitable human well-being (Petrou et al. 2015). These Essential Biodiversity Variables tie back in to Aichi Biodiversity Target 14 – *Ecosystem Services Safeguard* and 15 – *Ecosystem Resilience enhanced* which we show can be monitored by WV2 and TDX imagery by following our approach. Thus, similar remote sensing based approaches that we demonstrated with this study can be instrumental in the future and continual monitoring, reporting, and verification of mangrove biodiversity indicators and fine-scale forest cover changes.

Conclusions

In this study, we combined both high resolution optical WV2 and TDX canopy height imagery to classify dominant mangrove species and land cover types following an

ISODATA unsupervised approach. Our results showed a strong agreement with field referenced data and photography, visual interpretation with Google Earth, and expert knowledge. The overall classification accuracy was 89.89–89.33% with a kappa coefficient range of 0.89–0.88. There was little improvement in the land cover classifications by including the coarser scaled height layer. Despite the lack of improvement in classification accuracy using the height information, the variability and magnitude of canopy structure was measured across each of the vegetation types. The finding of *H. fomes* as the dominant species in the SEWS challenged the previous conclusion that *E. agallocha* was the dominant species. The canopy height under each dominant mangrove type will ultimately be useful in determining aboveground biomass and forest volume. The method presented here has the potential to be applied to the entire the Sundarbans Reserved Forest as well as other mangrove forests. By using WV2 imagery and spaceborne derived canopy information, we can map dominant species types, canopy structure and other mangrove biodiversity indicators over time which will directly address Aichi Biodiversity Targets-2020.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1. Plot level species composition and forest characteristics across the Sundarbans East Wildlife Sanctuary.

Table S2. Error matrix between WV2 derived land cover class and Google Earth observation point.

Table S3. Error matrix between WV2-TDX derived land cover class and Google Earth observation point.

Figure S1. TanDEM-X derived mean canopy height ($\pm$ SD) across the nine mangrove classes at Sundarbans East Wildlife Sanctuary. The different letters indicated on bars are significantly different ($P < 0.05$) as tested with Tukey Honestly Significant Difference.

Figure S2. Mangrove canopy height map at 12-m spatial resolution for the Sundarbans East Wildlife Sanctuary mangrove forest.



Original research article

High carbon stocks in roadside plantations under participatory management in Bangladesh

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ABSTRACT

Plantations are important REDD+ strategies for increasing carbon sequestration while enhancing local livelihoods. Reforestation along roads and highways under participatory forest management in southwestern Bangladesh could contribute to REDD+. This study assessed the diversity and structure of roadside plantations in order to develop a basal area based generalized allometric model for estimating above- and below-ground tree biomass carbon in Southwestern Bangladesh. All woody plants with d.b.h. ≥ 2 cm were identified and their diameters measured in 108 systematically selected zigzag plots of equal size (2×10 m). A total of 36 species in 17 families were recorded. Leguminosae accounted for 28% of species and 94% of the total estimated biomass carbon. We estimated a mean stem density of 4528 ha^{-1} , basal area of $52.6 \text{ m}^2 \text{ ha}^{-1}$ and biomass carbon of $192.80 \text{ Mg ha}^{-1}$. *Samanea saman*, *Dalbergia sissoo*, *Acacia nilotica*, and *Leucaena leucocephala* accounted for most density, basal area, and carbon. We developed and validated three allometric models with equal strength (R^2 0.94–0.98) using generalized linear regression. Roadside plantations in Bangladesh can now surely participate in the UNFCCC's carbon mitigation and adaptation mechanism, but challenges to their long-term sustainability must be addressed.

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1. Introduction

Global warming and biodiversity loss are the two important currently debated issues among the world's scientists and policy makers (Zhang et al., 2011), caused mainly by fossil fuel burning and deforestation during the last few decades (van der Werf et al., 2009). The last century finished with an increase in global temperature by 0.74°C and the atmospheric CO_2 concentration of 379 ppm (UNFCCC, 2007; IPCC, 2013). Furthermore, atmospheric carbon dioxide would be doubled by 2050 if the current rate of increase continues and will lead to the global temperature rise of up to $2\text{--}4^\circ\text{C}$ (IPCC, 2013). A projection by IPCC (2013) revealed that by the end of 21st century the global sea level will rise by 28–98 cm due to melting of polar ice, which would badly alter low-lying coastal countries (e.g. Bangladesh, Maldives, The Netherlands) existence and livelihoods pattern. Forests retention, coupled with various reforestation and afforestation programmes, tropical in particular, can play

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an important role in mitigating global climate change through sequestering atmospheric carbon (see [Dixon et al., 1994a,b](#); [Jose, 2009](#); [Kumar, 2011](#)).

Forests in Africa, Latin America, South and Southeast Asia, and the Pacific region have been experiencing the highest carbon emissions due to deforestation ([FAO, 2010](#)). In the least developed countries like Bangladesh, 62% of total carbon emissions originate from deforestation ([IPCC, 2007](#)). The Kyoto Protocol, the main instrument of the UNFCCC, has introduced the CDM concept among the low-income people who can store carbon through changes in land use patterns ([Roshetko et al., 2007](#); [Takimoto et al., 2008](#)). Under the current arrangements, reduced emissions from deforestation and forest degradation, and enhancing forest carbon stocks in developing countries (REDD+) does not consider smallholder trees, but only large scale forests. However, small patches of trees outside forests, if not cut, can contribute towards reducing emissions of carbon to the atmosphere. The only reason why smallholder trees are not a major focus under the REDD+ arrangement is because their contribution towards carbon sequestration is not well documented ([Nair, 2012](#)). To improve country negotiations for REDD+ and other carbon market mechanisms, it is necessary to provide evidence about the potential contribution of smallholder trees outside forests to biodiversity conservation, livelihood options and carbon sequestration ([Nair, 2012](#); [Jashimuddin and Inoue, 2012](#)).

Yet, globally attention has been given to what extent managed landscapes, such as agroforests, community forests, village woodlots and roadside plantations under participatory management, could hold carbon and contribute to climate change mitigation (see [Albrecht and Kandji, 2003](#); [Roshetko et al., 2007](#); [Saha et al., 2010](#); [Kumar and Nair, 2011](#); [Nair, 2012](#)). The global coverage of agroforests is 1023 million ha which represents a carbon sequestration potential of 1.9 Pg of carbon over 50 years at a rate of 94 Mg ha⁻¹ in managed landscapes ([Dixon et al., 1994a,b](#); [Nair et al., 2009](#)).

Participatory forest management has been practiced in Bangladesh since 1976. Several pilot projects from 1982 to 1987 provided the experience for launching the countrywide participatory forest management project in the forms of restored forests, agroforests, village woodlots, and road and highway plantations in 1989 ([Kabir and Webb, 2005](#)). Increasing the supply of forest products, especially fuel wood, to improve rural socioeconomic conditions and reversing the process of environmental degradation through proper soil and water conservation are the main objectives of participatory forest management in Bangladesh. As an outcome, 48,420 ha of roadside plantations, 30,666 ha of woodlots and 8778 ha of agroforestry plantations have been raised during the last 30 years in Bangladesh ([Jashimuddin and Inoue, 2012](#)). Members of the poorer sections, with special preference to landless, land-poor and rural women of the surrounding rural communities, are targeted as participants in the roadside plantations. The participants who protect the plantations have the right to harvest and consume or market all intermediate products in the forms of leaves, twigs and dead branches for household fuel. According to Forest Department policy, the participants have a right to receive a pre-determined share of 40% of the receipts from auction of the trees in a local market after the rotation (usually 10–12 years). Yet, commonly practiced roadside plantation under participatory management in Bangladesh has so far received no research attention in estimating its potential contribution to livelihood supplementation and carbon sequestration.

Species-level tree biomass carbon estimation using diameter at breast height (dbh) with a tree density based allometric model is becoming popular ([Pandey et al., 2014](#); [Rahman et al., 2014](#)). However, for quick calculation of biomass carbon, a basal area based allometric model could be another important option as both basal area and biomass carbon have a strong relation to dbh ([Torres and Lovett, 2012](#); [Rahman et al., 2014](#)). Studies have made significant contributions in estimating ecosystem level aboveground carbon stocks using basal biomass (see [Torres and Lovett, 2012](#); [Rahman et al., 2014](#)). The present focus of REDD+ is examining to what extent carbon sequestration through forest restoration and plantation establishment is related to biodiversity conservation, poverty reduction, and carbon sequestration. Therefore, this study aimed to develop generalized allometric equations based on basal area to estimate tree biomass carbon content of the roadside plantations under participatory management in southwestern Bangladesh.

2. Methods

2.1. Study area

Southwestern Bangladesh is primarily a floodplain landmass lying between 21.50° and 23.91°N latitude and 88.55° and 90.35°E longitude ([Fig. 1](#)). The region is bounded by India in the west and the Bay of Bengal in the south. Excessive siltation and sedimentation in the upper stream has resulted in phenomenal floods in the plain land areas of the region during the monsoon and severe drought during the dry season. Regular floods and droughts cause enormous damage to the lives and property in the region. The deltaic landscape of this region is a primarily low (<10 m above a.s.l.), flat, and fertile plain. The coastal plain is partly sandy and saline, and varies from 1 to 15 km in width ([Kabir and Webb, 2008](#)). Calcareous to non-calcareous soils and peat are the basic soil types extending over the study area. Coastal regions have some areas with acid sulphate and peat soils ([Kabir and Webb, 2008](#)).

Thirty-six percent of the study area is cultivable, 9% uncultivable, 37% littoral mangrove forestland, and 18% un-surveyed ([BBS, 2013](#)). Rice, wheat, jute, sugarcane, pulses, and potatoes are the principal agricultural crops from the cultivable lands ([Kabir and Webb, 2008](#)). Various types of vegetables, spices, fruits, and nuts are also important cultivated crops. There are no primary forests in the study area except the inaccessible littoral Sundarbans mangrove forest. Intensive shrimp (tiger prawn) culture is a newly emerging economic activity along the coastal regions of southwest Bangladesh ([Kabir and Webb, 2008](#)).

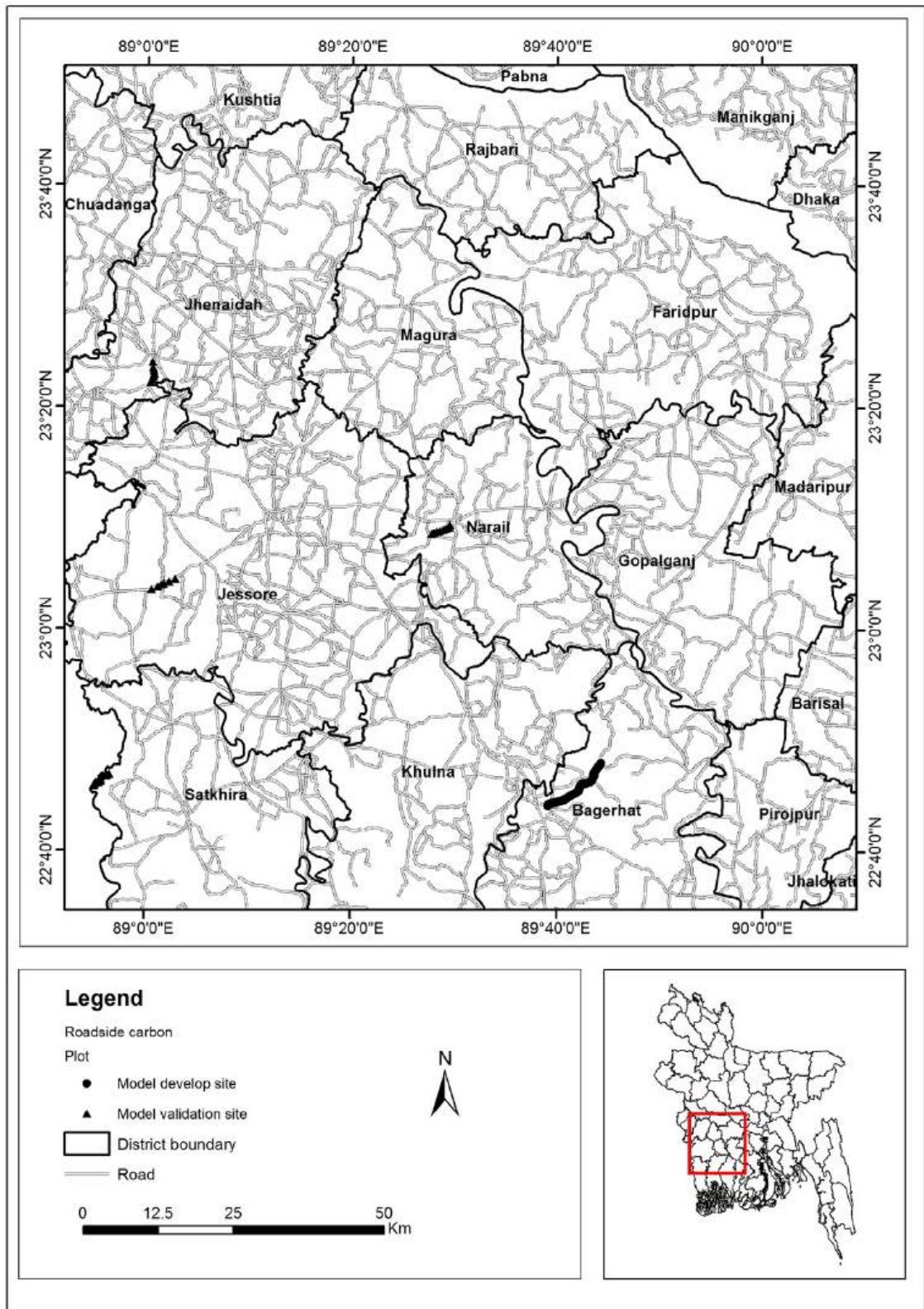


Fig. 1. Study area. Bagerhat, Satkhira, Jessore, Narail and Jhenaidah districts in southwestern Bangladesh.



Fig. 2. Photographs of roadside plantation in southwestern Bangladesh. Tender plantation along both the mud (upper left) and paved (lower left) roads. Beneficiary of the plantation (upper right) and the harvested products (lower right) after the rotation for benefit sharing among stakeholders.

All the districts in southwestern Bangladesh adjoin the capital of the country with a relatively good communication network. Highway and railway lines connect all the districts of the region. Paved roads connect each sub-district with the adjoining district headquarters. The majority of the roads within sub-district boundaries are mud with slowly developing paved roads. The majority of rural roads are muddy and become unusable during the rainy season. The Local Government and Engineering Department (LGED) have been converting rural mud roads into paved roads, but the progress is slow. In rural areas, communication facilities are developing rapidly under the rural public works programme. Almost all roads and highways in southwestern Bangladesh have been planted on both sides with the active participation of the local community and local government under the social forestry programme of the Bangladesh Forest Department (see Fig. 2).

A tropical to subtropical monsoon climate characterizes the region, marked with seasonal variations, moderately warm temperatures, heavy rainfall, and excessive humidity (Kabir and Webb, 2008). Three distinct seasons—summer (March–May), rainy (June–October), and winter (November–February) – are characteristic of the region (Kabir and Webb, 2008). The mean annual temperature is 26 °C (range: 19–32 °C). In some places temperatures go down to 7 °C during the winter and reach up to 40 °C or more during the summer (Kabir and Webb, 2008). Total rainfall during the monsoon accounts for 80% of the total annual rainfall. The annual average rainfall of the study area is 1800 ± 268 mm, ranging from 1400 to 2600 mm (Kabir and Webb, 2008). The potential evapotranspiration is 65–129 mm. The annual average relative humidity of the region is 78% (Kabir and Webb, 2008). March is the least humid month (65%) while humidity during the monsoon (June–September) is 75% (Kabir and Webb, 2008).

2.2. Sampling design

A total of 108 plots of equal size (2×10 m) were selected following systematic sampling in a zigzag manner (Fig. 3) on both the sides of road, in order to reduce microsite variation. Successive plots were 200 m apart for model development (Fig. 3(a) and 500 m apart for model validation (Fig. 3(b)). Plots were laid out in two slots in same year plantations (2005–2006). At first in Bagerhat district, 77 rectangular plots were laid out along the roadside plantation to collect data to develop a basal area based allometric model. Secondly, 31 plots were laid out along the roadside plantation in four other districts in the region, 10 from Satkhira, 8 from Jessore, 7 from Narail and 6 from Jhenaiddah districts (Fig. 1), in order to validate the basal area based allometric model. The data from all 108 plots were later used to assess the floristic composition, stand structure, species diversity and biomass carbon.

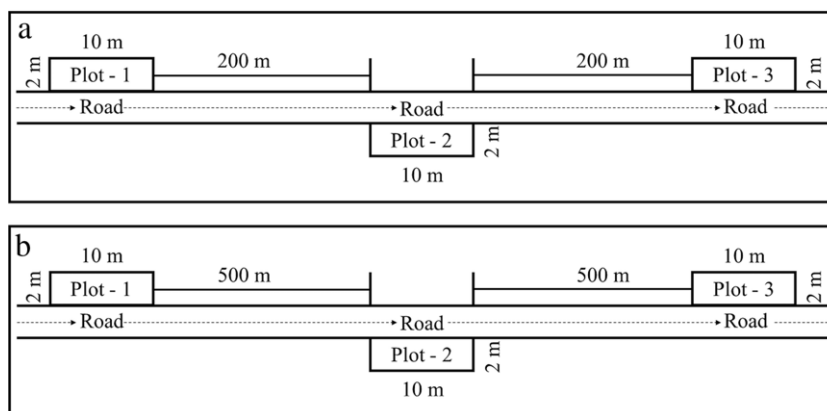


Fig. 3. Schematic diagram of the zigzag plot layout (a) for model development and (b) for model validation along the roadside plantations in southwestern Bangladesh.

2.3. Data collection

All woody plants with dbh ≥ 2 cm, were censused from all sample plots of the roadside plantations. In each sample plot, every individual tree was identified and recorded to species level. The diameters of all identified trees were measured at breast height (1.3 m above ground) using a diameter tapeTM and recorded accordingly.

2.4. Data analysis

2.4.1. Species diversity and structure

Species diversity H' (Shannon and Weaver, 1949), evenness J' (Pielou, 1977) and richness R (Margalef, 1958) indices were computed using BASIC program SPDIVERS.BAS (Ludwig and Reynolds, 1988). The importance value index was computed using the derivatives of density, basal area, and frequency of each recorded species in characterizing the stand structure (see Zhang et al., 2005). ANOVA was used to test the differences between species diversity, richness and evenness, biomass carbon, basal area and tree density of the roadside plantations across study sites. The Least Significant Difference (LSD) test was performed for multiple comparisons if any significant difference was found between species diversity, richness and evenness, biomass carbon, basal area and tree density of the roadside plantations across the study sites.

2.4.2. Tree biomass and carbon

Tree biomass estimation from species- and site-specific allometric models is tedious using the destructive harvest method, particularly in tropical and subtropical regions, because of the presence of numerous species and individuals in multiple layers. Common allometric equations that have been developed by felling trees in different tropical and subtropical regions of the world are an easier alternative. Taking this point into account, the allometric equation for aboveground carbon estimation of Chave et al. (2005) was employed, as it covers a wide geographical and diameter range of vegetation of all types. Below ground biomass and carbon were estimated using the regression model suggested by Cairns et al. (1997) as the most cost effective and practical method of determining root biomass.

The wood density data were obtained from the World Agroforestry Database (Carsan et al., 2012) and the Global Wood Density Database (Chave et al., 2009; Zanne et al., 2009). The relatively poor physical condition of the roadside plantations and their isolation from natural systems usually provides approximately 20% less total biomass than what natural systems provide (Aguaron and McPherson, 2012). Estimated biomass using Chave et al. (2005) and Cairns et al. (1997) allometric equation was therefore multiplied by 0.80 to calculate total biomass as suggested by Aguaron and McPherson (2012). Finally, the total biomass was multiplied by 0.5 to compute actual tree carbon content as 50% of wood's total biomass is considered to be carbon. A Generalized Linear Regression Model (GLRM) was employed to develop and validate the basal area based carbon estimation model.

3. Results

3.1. Species diversity and structure

A total of 36 species in 17 families were recorded from 108 sample plots (Table 1). Of the 36 species, *Samanea saman*, *Dalbergia sissoo*, *Acacia nilotica*, and *Leucaena leucocephala* were the most important species considering stem density, basal area, and biomass carbon content (Table 1). Leguminosae was the most dominant family, accounting for 28% of all the recorded species and 94% of the total estimated biomass carbon (Table 2). A total of 978 individuals were recorded in the

Table 1

Relative abundance, relative frequency, relative coverage, importance value, and biomass carbon content of trees in roadside plantations in southwestern Bangladesh. List according to the biomass carbon content (from high to low).

Species	RD	RF	RC	IVI%	BCC%
<i>Samanea saman</i> (Jacq.) Merr.	16.56	30.12	17.77	21.48	27.01
<i>Acacia nilotica</i> Karst.	32.52	17.58	13.55	21.22	24.37
<i>Dalbergia sissoo</i> Roxb.	18.61	22.71	19.58	20.30	23.82
<i>Leucaena leucocephala</i> (Lam.) de Wit	12.07	14.71	9.64	12.14	12.68
<i>Melia azedarach</i> L.	1.84	2.31	4.22	2.79	2.28
<i>Gmelina arborea</i> Roxb.	1.84	1.41	3.61	2.29	0.78
<i>Swietenia macrophylla</i> King	1.64	0.48	3.92	2.01	0.20
<i>Acacia auriculiformis</i> A. Cunn. ex Benth.	1.33	1.60	3.01	1.98	1.28
<i>Albizia procera</i> (Roxb.) Benth.	0.72	2.05	1.81	1.52	2.56
<i>Spondias pinnata</i> (L.f.) Kurz	1.33	0.30	2.71	1.45	0.11
<i>Trewia polycarpa</i> Benth. & Hook.f.	1.33	1.50	1.20	1.34	0.97
<i>Azadirachta indica</i> A.Juss.	1.74	0.32	1.51	1.19	0.25
<i>Artocarpus heterophyllus</i> Lam.	1.02	0.20	1.81	1.01	0.08
<i>Terminalia arjuna</i> Wight & Arn.	0.72	0.36	1.81	0.96	0.30
<i>Senna siamea</i> (Lam.) Irw. & Barneby	0.92	1.15	0.60	0.89	0.77
<i>Albizia richardiana</i> King & Prain	0.61	0.45	1.51	0.86	0.35
<i>Phyllanthus emblica</i> L.	0.51	0.26	1.20	0.66	0.25
<i>Bombax ceiba</i> L.	0.41	0.49	0.90	0.60	0.31
<i>Mangifera indica</i> L.	0.92	0.24	0.60	0.59	0.12
<i>Tectona grandis</i> L.f.	0.41	0.10	1.20	0.57	0.06
<i>Pithecellobium dulce</i> (Roxb.) Benth.	0.41	0.44	0.60	0.48	0.60
<i>Psidium guajava</i> L.	0.31	0.03	0.90	0.41	0.02
<i>Polyalthia longifolia</i> (Sonn.) Hook.f. & Thomson	0.31	0.03	0.90	0.41	0.02
<i>Aegle marmelos</i> (L.) Correa	0.20	0.23	0.60	0.35	0.27
<i>Ziziphus nummularia</i> (Burm.f.) W. & A.	0.20	0.18	0.60	0.33	0.19
<i>Morus alba</i> L.	0.20	0.14	0.60	0.32	0.06
<i>Tamarindus indica</i> L.	0.20	0.03	0.60	0.28	0.02
<i>Lagerstroemia speciosa</i> (L.) Pers.	0.20	0.03	0.60	0.28	0.01
<i>Anthocephalus chinensis</i> (Lmk.) A. Rich. ex Walp.	0.20	0.18	0.30	0.23	0.08
<i>Ficus religiosa</i> L.	0.10	0.21	0.30	0.20	0.12
<i>Moringa oleifera</i> Lam.	0.10	0.08	0.30	0.16	0.01
<i>Trema orientalis</i> (L.) Blume	0.10	0.05	0.30	0.15	0.02
<i>Alstonia scholaris</i> (L.) R. Br.	0.10	0.04	0.30	0.15	0.02
<i>Ficus benghalensis</i> L.	0.10	0.01	0.30	0.14	0.00
<i>Annona reticulata</i> L.	0.10	0.00	0.30	0.14	0.00
<i>Syzygium cumini</i> (L.) Skeels	0.10	0.00	0.30	0.14	0.00

Table 2

Relative abundance, relative frequency, relative coverage, importance value, and biomass carbon content of plant families in roadside plantations in southwestern Bangladesh. List according to the biomass carbon content (from high to low).

SL. no	Family	RD	RF	RC	IVI	BCC%
1	Leguminosae	83.95	52.94	90.84	75.91	93.47
2	Meliaceae	5.21	12.75	3.10	7.02	2.73
3	Verbenaceae	2.25	7.35	1.51	3.70	0.84
4	Anacardiaceae	2.25	4.90	0.54	2.56	0.23
5	Euphorbiaceae	1.84	3.92	1.75	2.51	1.21
6	Moraceae	1.43	4.90	0.56	2.30	0.27
7	Combretaceae	0.72	2.94	0.36	1.34	0.30
8	Myrtaceae	0.41	1.96	0.04	0.80	0.02
9	Annonaceae	0.41	1.96	0.03	0.80	0.02
10	Bombacaceae	0.41	1.47	0.49	0.79	0.31
11	Rutaceae	0.20	0.98	0.23	0.47	0.27
12	Rhamnaceae	0.20	0.98	0.18	0.46	0.19
13	Lythraceae	0.20	0.98	0.03	0.40	0.01
14	Rubiaceae	0.20	0.49	0.18	0.29	0.08
15	Moringaceae	0.10	0.49	0.08	0.23	0.01
16	Ulmaceae	0.10	0.49	0.05	0.21	0.02
17	Apocynaceae	0.10	0.49	0.04	0.21	0.02

sample plots. No significant difference was observed in species richness, diversity, and evenness across the five study sites ($P < 0.05$; Table 3).

An average density of 4528 stems ha^{-1} (range: 3000–7950), dbh of 12.45 cm (range: 8.02–12.67) and basal area of 52.6 $\text{m}^2 \text{ha}^{-1}$ (range: 20.7–95.3) were recorded from roadside plantations across five study sites (Table 4). Multiple comparisons with LSD found no significant differences in stem density and diameter across the study sites ($P < 0.05$), while

Table 3

Mean species diversity, evenness and richness indices of roadside plantations across five study sites in southwestern Bangladesh. Figures in parenthesis are standard errors.

Species parameters	Study sites					Average
	Bagerhat	Jessore	Jhenaidah	Narail	Satkhira	
Diversity (H')	1.26 (0.08)	1.28 (0.42)	1.25 (0.35)	1.01 (0.30)	0.98 (0.22)	1.22 (0.07)
Evenness (J')	0.75 (0.04)	0.81 (0.26)	0.79 (0.22)	0.64 (0.19)	0.62 (0.14)	0.74 (0.04)
Richness (R)	1.00 (0.07)	1.22 (0.41)	1.13 (0.34)	0.88 (0.26)	0.68 (0.19)	0.98 (0.07)

Table 4

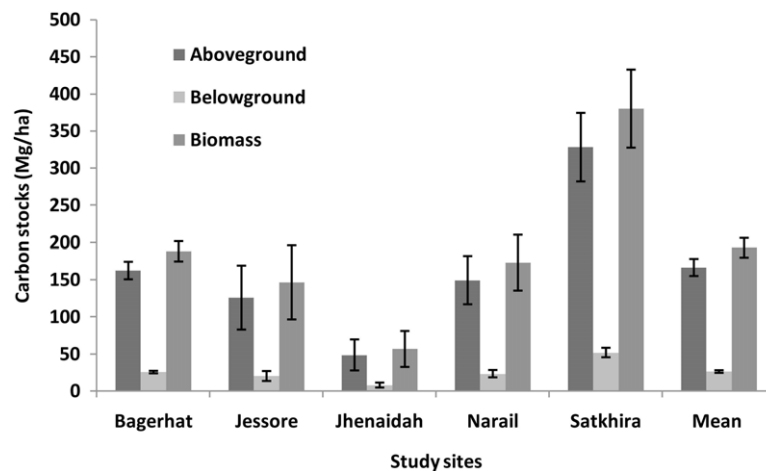
Stem density, diameter and basal area of roadside plantations across five study sites in southwestern Bangladesh. Figures in parenthesis are standard errors.

Parameter	Study sites					Average
	Bagerhat	Jessore	Jhenaidah	Narail	Satkhira	
Density (ha^{-1})	4487 (362)	3000 (532)	3000 (675)	3143 (705)	7950 (626)	4528 (297)
Diameter (cm)	11.59 (0.43)	12.67 (1.21)	8.02 (1.30)	12.51 (1.35)	11.67 (0.36)	11.45 (0.35)
Basal area ($\text{m}^2 \text{ha}^{-1}$)	51.9 (3.2)	44.5 (13.8)	20.7 (8.6)	43.2 (10.1)	95.3 (6.1)	52.6 (3.0)

Table 5

Pearson correlation analysis showing relationship of aboveground, belowground, and biomass carbon with species diversity, evenness, and richness of roadside plantations across five study sites in southwestern Bangladesh. Figures in parenthesis are P values.

Carbon level	Pearson correlation (r)		
	Species diversity (H')	Species evenness (J')	Species richness (R)
Aboveground	0.03 (0.77)	0.03 (0.75)	−0.06 (0.52)
Belowground	0.03 (0.72)	0.03 (0.74)	−0.07 (0.50)
Total biomass	0.03 (0.76)	0.03 (0.75)	−0.06 (0.52)

**Fig. 4.** Carbon and biomass content of the roadside plantations across five study sites in southwestern Bangladesh.

basal area was significantly different ($P > 0.05$). Satkhira district accounted for the highest density and basal area while Jhenaidah district accounted for the least (Table 4).

3.2. Tree biomass and carbon content

An average biomass carbon of $192.80 \text{ Mg ha}^{-1}$ (range: 56.75–380.11) was calculated from the roadside plantations (Fig. 4); 86% above ground and 14% below ground. Above- and below-ground biomass carbon content was significantly different across the five study sites ($P < 0.05$). The LSD results for multiple comparisons revealed that the Satkhira site has the highest biomass carbon content and Jhenaidah has the lowest. However, no significant difference was found in biomass carbon between Bagerhat, Jessore and Narail districts ($P > 0.05$). Sites with higher basal area and stem density tend to store more carbon. Pearson's correlation analyses of plot level aboveground, belowground, and biomass carbon content with species diversity, evenness and richness indices showed no significant relationships ($P > 0.05$; Table 5).

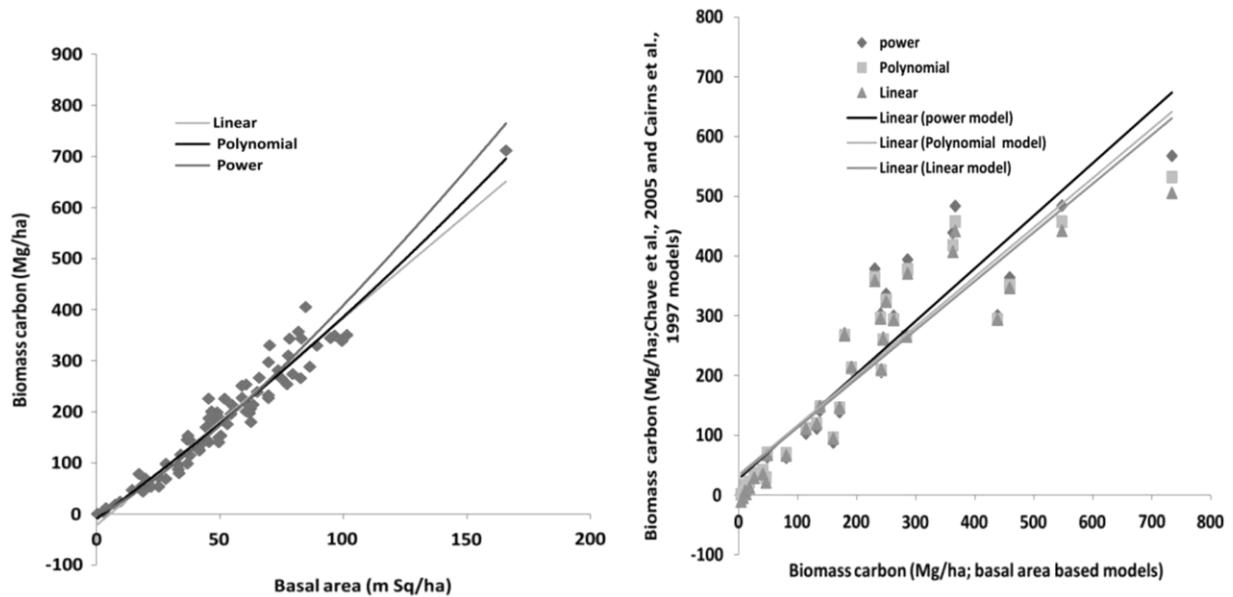


Fig. 5. Basal area based biomass models at the left side of this study validated with models established by Chave et al. (2005) and Cairns et al. (1997) at the right side.

3.3. Basal area based allometric models and their validation

Three types of models were developed for carbon assessment from the plot level mean basal area (Eqs. (1)–(3)). We found a strong (mean $R^2 = 0.96$; for Linear 0.94, Polynomial 0.94 and Power 0.98 models) and significant ($P < 0.05$) relationship between mean biomass carbon and mean basal area for roadside plantations (Fig. 5). All three models were tested against 31 plots with Chave et al. (2005) and Cairns et al. (1997) for validation. Results of the GLRM revealed that all three models showed significant ($P < 0.05$) and strong relationships ($R^2 = 0.84$) with established models (Chave et al., 2005; Cairns et al., 1997) based on biomass carbon content (Fig. 5). Given the high regression R^2 (range: 0.94–0.98), our three models are equally strong in calculating the tree biomass carbon content. Therefore our basal area based allometric models are equally suitable for calculating biomass carbon content from the trees in open stands.

$$\text{Biomass C} = 4.061 \times \text{BA} - 22.516 \quad (1)$$

$$\text{Biomass C} = -9.188 + 3.501 \times \text{BA} + 0.005 \times \text{BA}^2 \quad (2)$$

$$\text{Biomass C} = 1.239 \times \text{BA}^{1.360} \quad (3)$$

4. Discussion

Controlling the present level of atmospheric carbon dioxide through reducing deforestation, increasing afforestation or reforestation, and preventing biodiversity loss is a significant concern among scientists and policy makers (Kanowski et al., 2011; Pandey et al., 2014). The importance of engaging in meaningful action to combat deforestation is recognized in the United Nations Framework Convention on Climate Change (UNFCCC) and parties are discussing policies and approaches to reduce CO_2 emissions from deforestation in a post-2012 international agreement on climate change. The UNFCCC recognizes various mitigation and adaptation options: firstly, the Clean Development Mechanism (CDM); secondly, Reduced Emissions from Deforestation and Forest Degradation REDD; and most recently the new strategy—reducing emissions from deforestation and forest degradation, and enhancing forest carbon stocks in developing countries (REDD+). These are intended to engage multi-scale stakeholders in conservation and sustainable management of forest resources for enhancing carbon sequestration in developing countries with incentives as a reward for mitigating global climate change (Gardner et al., 2012).

Afforestation and reforestation are the integral parts of REDD+ as an effective mechanism for reducing global climate change (Bonan, 2008; Wang et al., 2011; Pandey et al., 2014). The parties involved need accurate information on carbon stocks, biodiversity and the socioeconomic status of the communities in developing countries participating in the REDD+ financial mechanism (Pandey et al., 2014). Thus, protecting forest under REDD+ could be an effective measure to mitigate greenhouse gas emissions and will provide other ecosystem services, such as watershed management and biodiversity conservation (Pandey et al., 2014).

Our recorded 36 tree species in 34 genera and 17 families from roadside plantations was higher than recorded (19 species) from roadside plantations in Kanchanpur District, Nepal (Baral et al., 2013), but fewer than recorded (62 species) from a national highway plantation in Taiwan (Wang, 2011), or in homestead plantations in Southwestern Bangladesh (146 species; Kabir and Webb, 2008; 58 species; Motiur et al., 2006). These differences may reflect geographic and physiographic coverage, environmental gradients and the purpose of plantation management. For example, in roadside plantations fast-growing timber species are planted with the objective of getting a quick return, while in homegardens, multipurpose tree species, including fruit, timber, and ornamental species are planted.

Repeated disturbances such as grazing, soil work and others, favour species that grow best in degraded conditions, such as legumes (Brakenhielm and Liu, 1998). Indeed, the tree communities were dominated by several species of legumes in our study. *Samanea saman* (Rain tree), *Dalbergia sissoo* (Sissoo), *Acacia nilotica* (Babul), and *Leucaena leucocephala* (Ipil-ipil) had similar contributions to stem density, basal area, and biomass carbon (Table 1). Legumes accounted for 28% of the tree species and 94% of the biomass carbon in the five plantations (Table 2). Legumes also dominate in other forests in South Asia, including an urban forest of Chennai, India (Muthulingam and Thangavel, 2012) and natural forests in Tamil Nadu, India (Gamble and Fischer, 1934).

The stem density (3668 tree ha⁻¹) in our study was much higher than recorded from other Asian countries. For example, 705 tree ha⁻¹ was recorded from Taiwanese highway plantations of the same age (Wang, 2011) and 279 tree ha⁻¹ from urban roadside forests in Shenyang, China (Liu and Li, 2012). Compared with some natural and restored forest systems in Bangladesh, our result for stem density and basal area (52.6 m² ha⁻¹) was higher than 381 trees ha⁻¹ and 53.5 m² ha⁻¹ in Chittagong Hill Tracts (South) Forest Division (Nath et al., 1998), 459 trees ha⁻¹ and 16.88 m² ha⁻¹ in Chunati Wildlife Sanctuary, Cox's Bazar (Rahman and Hossain, 2003), 464 trees ha⁻¹ and 27.07 m² ha⁻¹ in Dudpukuria-Dhopachori Wildlife Sanctuary of Chittagong South Forest Division (Hossain et al., 2013), 257 tree ha⁻¹ in Ukhiya natural forests of Cox's Bazar Forest Division (Ahmed and Haque, 1993) and 369 stem ha⁻¹ in Bamu reserve forests of Cox's Bazar. This very high density of trees in roadside plantations compared to many other natural and restored forest systems in Bangladesh may be because of maintaining a certain tree spacing (1.5 m × 1.5 m) along with the higher survival rate from confirmed protection. Planting hedgerow crops, mainly *Cajanus cajan* on the inner side (close to the road) and *Acacia nilotica* on the outer side (away from the road), of roadside plantation is another reason for having higher stem density. Roadside plantations in southwestern Bangladesh may therefore play an important role in producing more timber and revenue for the local surrounding community as livelihood – and more importantly sequester carbon – compared to other natural and restored ecosystems in Bangladesh.

The niche complementary effect states that ecosystem functions like productivity and thereby carbon storage capacity are regulated by species richness and diversity (Tilman et al., 1996, 2006; Hooper et al., 2005). However, we found a very weak insignificant positive relation between aboveground, belowground, and biomass carbon content with species diversity and evenness. Our finding is analogous to the study conducted in forest, agroforest and pasture land use system in tropical Panama (Kirby and Potvin, 2007), in Mexican tropical forest (Martinez-Sanchez and Cabrales, 2012), and in a mixed-species plantation in Sardinilla, Panama (Ruiz-Jaen and Potvin, 2011).

Our established three basal area based allometric models can be useful for carbon calculation from the plot level mean basal area as all three models showed strong relationships in the GLRM analysis ($R^2 = 0.84$). The mean biomass carbon (192.80 Mg ha⁻¹) was higher than the reported range (65–158 Mg ha⁻¹) of tree biomass carbon for Bangladesh (Gibbs et al., 2007). Biomass carbon content was also higher than the reported average tree biomass carbon content of 83.72 Mg ha⁻¹ (Shin et al., 2007) and 110.94 ton ha⁻¹ (Ullah and Al-Amin, 2012) in hill forest of Bangladesh. Studies documented 11.71 ± 3.57 Mg ha⁻¹ in Eastern Australia roadsides (Eldridge and Wilson, 2002), 45.49 Mg ha⁻¹ in Buter Street and 22.29 Mg ha⁻¹ in Penn Street, USA (Keating et al., 2005), 22.83 Mg ha⁻¹ in USA national average urban forest carbon storage (Nowak and Crane, 2002), and 34.95 Mg ha⁻¹ in roadsides of Shenyang, China (Liu and Li, 2012). We thus estimated a much higher amount of carbon in roadside plantations compared to other studies from tropical and subtropical regions. This may be due to the higher stem density and basal area (Table 2) in our study. This way roadside plantations in Bangladesh can play an important role in atmospheric carbon sequestration.

Despite their relatively poor physical condition and isolation, roadside plantations are still important for both livelihoods and carbon in Bangladesh. However, it is important to consider the role of the local village community in plantation establishment, protection, management, and conservation as they sustainably harvest non-timber products from the plantation to supplement livelihood. The villages would take the initiative to protect the plantation from further degradation, establish management rules, and enforce them with the active involvement of local forest officials. It is now widely acknowledged by social scientists that collective action and locally-initiated resources conservation can lead to successful conservation outcomes (Ostrom, 1990; Webb and Shivakoti, 2007). In addition, self-initiated protection, given rights to use, and strong community leadership contributed to safeguarding the plantations. Although natural and protected forests are usually more effective repositories of plant diversity than plantations managed for extractive use, participatory-managed plantations can be vital repositories of livelihoods and carbon alongside the protected area system in Bangladesh. This should be considered as a near-term participatory-managed conservation success, even if the level of diversity, livelihood supplement and amount of carbon stored estimated are low compared to other forest systems in Bangladesh under more complete protection.

Roadside plantations in southwestern Bangladesh still retain enough structural complexity to give a good start to livelihood supplementation and carbon sequestration. Given the challenges to long-term repositories to livelihood and carbon stocking discussed above, action should be taken to assist in the sustainable retention of roadside plantations in Bangladesh.

We recommend two investments that should increase the probability of long-term roadside plantation retention in Bangladesh. First, roadside plantations need to be facilitated with manual enrichment planting. Well-planned enrichment planting of species would maintain maximum density, improve soil conditions and create microenvironments favourable for other taxa such as wildlife. Enrichment planting would require a major human resource investment by the communities as well as generous technical backstopping and distribution of appropriate seedlings by the Forest Department, but could potentially have significant and positive impacts on forest conditions. Second, there needs to be strict implementation of operational rules, strengthening of institutions for regular monitoring, and increased authority to implement sanctions against violators. Vested interest in the plantation and its benefits, combined with the capacity and authority to protect the plantation, should improve the outcomes of plantation management by the communities. If both plantation establishment and vigilant protection is initiated promptly, roadside plantations in Bangladesh could be on a trajectory towards significant livelihood supplementation and carbon sequestration in addressing the long lasting poverty and environmental issues in Bangladesh respectively. This was the REDD+ mechanism of poverty alleviation and climate change addressed simultaneously.

Care must be taken when interpreting our results. Although indeed in this case study the participatory management did result in the retention of diverse plants in roadside plantations, the future trajectory of biodiversity conservation, livelihood supplementation and carbon sequestration potential is not known. Moreover, this single case study is not necessarily representative of all participatory managed roadside plantations in Bangladesh. Accumulation of longitudinal case studies will be required before we may generalize results like ours. Participatory managed roadside plantation may vary greatly in biodiversity, intensity of use, and trends of environmental services over time and geographical space. Thus, more research is required on the impacts of community managed roadside plantation for livelihood options and carbon sequestration potential in Bangladesh, and in Asia. The results of such research would have important implications in generalizing the impacts of managed landscapes, with particular reference to roadside plantations, on global poverty alleviation and climate change adaptation and mitigation strategies.

5. Conclusion

The participatory managed roadside plantations in southwestern Bangladesh contribute to livelihood supplementation from the pre-determined share (40%) of the receipts from the final harvest. Despite the relatively poor physical condition of the roadside plantations and their isolation, they are still important for livelihoods and for carbon sequestration. Although the plantations are harvested every 10–12 years, new plantations are established immediately after harvest, starting another cycle of carbon sequestration. The large area (4.65 million ha) of these plantations in Bangladesh suggests that their participation in the UNFCCC's financial based carbon mitigation strategies (e.g. CDM) could provide additional benefits to the local communities in a comanagement system such as Payment for Environmental Services (PES). Roadside plantations should be considered as a near-term participatory management conservation success, even if the contribution to livelihoods and the carbon pool estimated are low compared to other protected forest systems in Bangladesh. Their ability to contribute to livelihoods and carbon sequestration could be increased by management actions (enrichment planting and strict implementation of rules in use) to assist their sustainable retention. Despite this pioneering study, more research is needed into both the livelihood options and carbon sequestration potential of roadside plantations in Bangladesh and elsewhere in Asia.

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Article

Remote Sensing-Based Mapping of Senescent Leaf C:N Ratio in the Sundarbans Reserved Forest Using Machine Learning Techniques

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Abstract: Carbon to nitrogen ratio (C:N) of senescent leaf is a crucial functional trait and indicator of litter quality that affects belowground carbon and nitrogen cycles, especially soil decomposition. Although mapping the C:N ratio of fresh mature canopies has been attempted, few studies have attempted to map the C:N ratio of senescent leaves, particularly in mangroves. In this study, four machine learning models (Stochastic Gradient Boosting, SGB; Random Forest, RF; Support Vector Machine, SVM; and Partial Least Square Regression, PLSR) were compared for testing the predictability of using the Landsat TM 5 (LTM5) and Landsat 8 to map spatial and temporal distribution of C:N ratio of senescent leaves in Sundarbans Reserved Forest (SRF), Bangladesh. Surface reflectance of bands, texture metrics of bands and vegetation indices of LTM5 and Landsat 8 yearly composite images were extracted using Google Earth Engine for 2009–2010 and 2019. We found SGB, RF and SVM were significant different from PLSR based on MAE, RMSE, and R^2 ($p < 0.05$). Our results indicate that remote sensing data, such as Landsat TM data, can be used to map the C:N ratio of senescent leaves in mangroves with reasonable accuracy. We also found that the mangroves had a high spatial variation of C:N ratio and the C:N ratio map developed in the current study can be used for improving the biogeochemical and ecosystem models in the mangroves.

Keywords: functional trait; litter quality; machine learning; spatial modeling; remote sensing; mangrove

1. Introduction

Mangroves grow in the intertidal zone of tropical and subtropical countries, are highly productive and have huge carbon sequestration capacity, especially into sediments [1,2]. Thus, mangroves play a tremendous role in climate regulation. For sustaining productivity and carbon sequestration, the nutrient cycling needs to be maintained [3,4]. Through litter decomposition, a keystone ecosystem process, the nutrient supply is maintained in mangrove forests, which is essential for plant growth [3,5–8]. Leaf litter quality such as carbon to nitrogen (C:N) ratio is often regarded as the key determinant of litter decomposition [3,5–8]. Litter quality also influences the activities of soil microorganisms that play the direct role in litter decomposition [7,9–11]. For example, poor litter quality (higher C:N ratio) with lower amount of labile C with higher amount of tannin content causes microbial immobilization [8,9,12]. In a forest ecosystem, litter mostly comes from the falling of senescent leaves [7,13]. The variability of C:N ratio in senescent leaves largely depends on species at community and ecosystem scale and

on temperature and rainfall at large scale [9,12,14,15]. In mangrove, to a large extent, for instance, in Sundarbans Reserved Forest, species distribution follows hydrological gradients in terms of salinity and fresh water flow that might be influenced by the balance of C:N ratio and thereby nutrient cycling and primary productivity [7,16–18].

Furthermore, mangrove forest faces tremendous disturbances both human mediated and natural such as upstream fresh water harvest, deforestation, sea level rise, salt intrusion, temperature change and tropical cyclone [3,19–21]. Due to those disturbances the mangrove is altered by reducing sedimentation, increasing salinity, and changes in species composition. For example, the fast-growing salt-intolerant species are replaced by the slowly growing salt-tolerant species [3,19–21]. Consequently, these changes in environmental factors and species composition have altered leaf C:N ratio and thereby affected nutrient cycling and primary productivity of the ecosystem [3,16,22]. Therefore, mapping the spatial and temporal variation of C:N ratio is necessary for monitoring nutrient dynamics, forest productivity, and soil carbon cycles in the mangrove forests. In spatial and temporal context, remote sensing-based upscaling of C:N with in situ data is feasible in terms of logistic supports, time, inaccessibility, and periodic monitoring [22,23].

Of the different dimensions of plant diversity, the trait based diversity (functional diversity) has been regarded as the main driver of ecosystem services and function [24] because it has a mechanistic relationship with plant productivity and other ecosystem functions such as microbial litter decomposition and nutrient cycling [25]. Functional diversity is defined as the variation, range and value of functional traits in a given community [24,26,27]. It has been suggested that the stand level biogeochemistry is largely driven by the community weighted mean of trait value of dominant species (usually weighted by species relative basal area or abundance at plot or stand) [24,25,28]. For example, community weighted mean of leaf C:N, C, N determines forest biomass [29], productivity [25], forest growth [30], and litter decomposition [31,32]. Thus, scientists have used this plot level community weighted mean of trait value as a bridge between field based trait value with remote sensing properties for scaling up ecosystem level from regional to global scale in terrestrial ecosystems [23,25,33].

The optical remote sensing-based approach has been proven a practical, cost-effective and accurate way to upscale the plot based in situ foliar trait value to forest landscape level in terrestrial forest ecosystems [34–36]. For example, Walis et al. [34] used Landsat 8 for leaf toughness, specific leaf area, foliar N concentration, and foliar P concentration in the tropical montane forest. Another open-access remote satellite imagery is Sentinel 2, which has been used recently in foliar chemical and morphological properties such as N [37] mass per area, chlorophyll (Cab), N and carbon content [38]. Those studies mainly focused on different traits of green leaf in forest ecosystems, but application of open source satellite imagery in assessing the senescent leaf C:N as a litter quality is limited. The reason may be the inability of remote sensing to capture the senescent leaf spectral properties as old leaves are not present in upper canopy [35,39]. However, the stoichiometric studies showed that there is high correlation between green and senescent leaves C:N in forest, which may allow us to map C:N ratio of senescence by means of multispectral remote sensing [40–42].

Furthermore, the above mention studies used mainly spectral reflectance and texture (Grey Level Co-occurrence Matrices, GLCM) properties of different bands and normalized vegetation indices. The drawback of normalized vegetation indices is that it has a saturation effect over dense canopy that leads to poor accuracy in model validation and mapping of biophysical and biogeochemical canopy properties [35,43]. Along with texture and vegetation indices, the canopy moisture, water (Modified Normalized Difference water index; MNDWI), Disease Water Stress Indices (DWSI), thermal band (B6 of Landsat) can be used to improve model accuracy as functional traits vary with environmental factors and community composition [23,33,35,39,44]. In addition, other simple ratios between Visible, NIR (Near Infra-Red) and SWIR (Short-wave infra-red) bands, such as B3/B1, B5/B7, and B7/B2, that are sensitive to moisture content and forest vigor may also contribute to improving the accuracy of C:N ratio mapping [33,35,39,44].

Remote sensing-based modeling of functional traits in forest ecosystems has been followed by both parametric and non-parametric advance machine learning regression approaches [23,34,37,38,45]. The advantage of non-parametric machine learning over parametric modeling approaches such as multiple linear modeling is that it can handle both multicollinearity between predictors and noisy data with efficiency in selecting the key predictor from a large amount of predictors [46–49]. Moreno-Martínez et al. [23] reported the outperformance of Random Forest (RF) modeling technique over multiple linear model, neural networks, Kernel Ridge Regression, and Gaussian Process Regression in mapping leaf functional trait in terrestrial forests at the global scale. Similarly, Gara et al. [38] and Chumura et al. [37] mapped leaf N and C concentration accurately with RF. On the other hand, the application of two other machine learning modeling techniques, support vector machines (SVM) and artificial neural networks, in modeling N and C concentrations over intact and fragmented forest ecosystems have been found to have equal performance [45]. Another ensemble of learning regression method is Stochastic Gradient Boosting (SGB), which can deal with plenty of data and combine important predictors along with average predictors for strong prediction of dependent variable. It found an outperformance of RF in other forestry field (biomass mapping) [50]. However, application of SGB in functional trait such as C:N modeling and mapping is not well known.

In this study, to our current knowledge, for the first time in the mangrove ecosystem, we tested four machine learning models to map the spatial and temporal distribution of the community weighted mean of C:N of senescent in the Sundarbans Reserved Forest in Bangladesh using Landsat TM 5 and Landsat 8. Advanced remote sensing-based modeling approach (machine learning) can contribute in assessing the essential biodiversity variables for biodiversity target achievement (e.g., National Biodiversity Strategies and Action Plans related to Aichi Biodiversity Target 2020) [51]. Particularly, in remote sensing-based forest biochemical properties (C:N as an indicator of decomposition), monitoring and information systems can improve the process of measurement, reporting and verification for achieving global targets and commitments [52]. Therefore, the specific objectives of current study were (i) to select the best machine learning model to predict C:N ratio of senescence leaf using spectral and textural properties of bands and vegetation indices of Landsat TM 5, (ii) to develop a spatial map for observing the spatial variation of C:N ratio of senescent leaf as litter quality trait that controls ecosystem function (decomposition and nitrogen using strategy) in the Sundarbans Reserved Forest, and (iii) to assess the temporal change of C:N ratio of senescent leaf in Sundarbans Reserved Forest (2009–2010 to 2019) using Landsat TM5 and Landsat 8.

2. Materials and Method

2.1. Description of Study Area

The Sundarbans is the world's largest contiguous mangrove ecosystem (10,017 km²) shared between Bangladesh (62%) and India (38%) and was declared a Reserve Forest in 1878 [53,54]. The Bangladesh part of Sundarbans is called Sundarbans Reserved Forest (SRF). The SRF is situated in the southwestern part of Bangladesh and is one of the most diverse mangrove forests in the world [16,55] (Figure 1). The total area of SRF is 6017 km², in which mangrove forests occupy about 69%, and the rest is water bodies such as rivers, small streams and canals. The mangrove extent in SRF accounts for 5% of global mangrove forests (83,495 km²) [56]. It accounts for 4.07% of the total area of Bangladesh and 40% of the total forest coverage managed by the Forest Department [57]. Because of its multiple ecosystem services and biodiversity value, it is likely to influence regional and global environments under the changing climate [16,18]. This forest was declared as Ramsar Wetlands Sites in 1992 and the UNESCO (United Nations Educational, Scientific and Cultural Organization) World Heritage Site in 1997 [53].

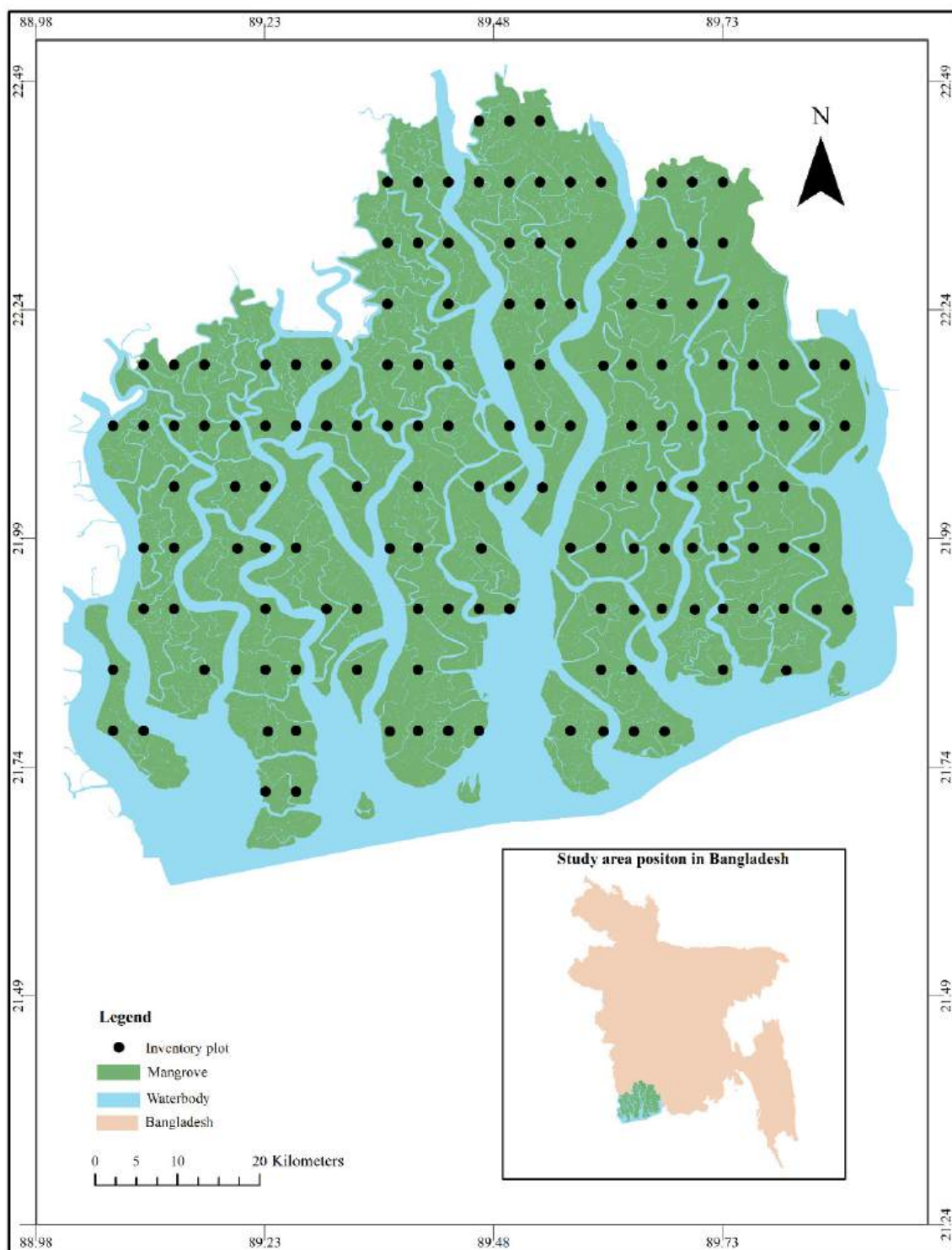


Figure 1. Distribution of sampling plots in Sundarbans Reserved Forest.

A total of 528 species of vascular plants where 24 species are true mangroves and 70 species are mangrove associates have been reported so far [54]. This forest is also rich in faunal biodiversity with 1135 recorded species, including a large community of endangered species like the Royal Bengal Tiger, the Ganges and Irrawaddy dolphins and saltwater crocodiles [58]. The forest supports and protects the livelihoods of the local communities [59]. This forest is heterogamous in terms of species, hydrological system, and salinity gradient and also faces anthropogenic and natural disturbances similar to other mangroves, which affects the litter quality, growth, productivity and nutrient cycling [7,16–19].

2.2. Field Data

In this study, Sundarbans Reserved Forest inventory data were used, which were collected in 2009–2010 from 150 plots spanning over the whole Sundarbans Reserved Forest for calculating community weighted mean of C:N. The field survey at the 150 plots was conducted from 5 December 2009 to 30 April 2010 with two field inventory groups, each led by an Assistant Forest Conservator. Each group consisted of one Assistant Forest Conservator, one Forest Ranger/Deputy Ranger, two foresters, two students, two laborers and two armed guards. Each group was assigned a small engine boat with boatman. Each trip was seven to ten days long (an interval between two surveys) depending on stored food and availability of fresh water. Before starting each journey, both groups discussed together with detailed maps and GPS (Global Positioning System) units to plan for the next plots. Local knowledge of laborers, guards, Forest Department district staff and even fishermen aided the crews' efforts to find suitable routes to plots and minimize walking distance and time. Generally, each group completed one plot per day, but often this pre-planning activity helped the groups to complete more than one plot a day. This was due to the developed skill of our crew members and easy access to the plot. Sometimes, we even completed a third plot by working both groups together. All the documentation had been reserved. Doing forest inventory in mangrove forest is always challenging. Sometimes, we had to walk more than 600 m across the dense forest and make a pool to cross a small canal (if it was unavoidable). One day, we even had to swim a canal with a width of around 10 m because after finishing the survey, when we returned to the canal, it was full of water due to the high tide.

The plots were 4 min latitude and 2 longitude away from each other and were composed of five 10-m radii nested circular subplots [16]. We also took four pictures at four directions at the center of each subplot. All the documentation had been reserved. For detailed field inventory methods see Rahman et al. [16]. This study used 93 of those 150 plots, because at least one subplot (out of five subplots) of the remaining 57 plots contained more than 10% water bodies or was underwater completely. Living tree species that had DBH (diameter at breast height) ≥ 10 cm (18 species including one unknown species) were used in this study, and species wise basal area (cross sectional area at 1.30 m height) per plot was calculated for use as the weighted value in estimating community weighted mean of C:N (Equation (1)) [25].

This study used the C:N data of Chanda et al. [13]. C:N data were averaged because they analyzed the data in three different seasons (monsoon, pre-monsoon and post-monsoon). They plucked 30 senescent leaves for each species and made a composite sample for estimating C and N. Then, they oven dried each composite sample at 75 °C for approximately 7 days until a constant weight was achieved [13]. After that, they grounded the dried samples to a fine consistency using a mortar and pestle and then analyzed for carbon and nitrogen content (Perkin Elmer 2400 Series II CHNS/O Elemental Analyzer) [13]. We used the initial C:N ratio of litter. The average C:N value (29.22; Table 1) of all the species estimated by Chanda et al. [13] was used for the species in which C:N was missing. The community weighted mean was calculated using the FD package in R [60]. Plot wise community weighted mean was estimated as the mean trait value of each weighted by the relative basal area of the species in a given plot by following the Equation (1) [61,62]. The community weighted mean of C:N varied from 18.40 to 39.99 with a mean value of 35.16 across the whole Sundarbans Reserved Forest.

$$CWM = \sum_{i=1}^S P_i \times C : N_i \quad (1)$$

where CWM is the community weighted mean of C:N ratio, S is the species richness, P_i is the proportional of relative basal area of species i , and $C:N_i$ is the C:N ratio of species i .

Table 1. Species-wise mean leaf carbon to nitrogen (C:N) ratio ($\pm$ SD) with the relative abundance of species. Bold-type C:N values (for missing trait value of species) were mean trait value for all species measured by Chanda et al. [13], and SD were calculated from the three seasonal values. The relative abundance of species was based on live trees have DBH $\geq$ 10 cm (2009–2010) forest inventory data of 150 plots Rahman et al. [16].

Species	Frequency (Number of Plots Species Found)	Total Number of Individuals	C:N $\pm$ SD
<i>Heritiera fomes</i>	127	9438	38.70 $\pm$ 1.84
<i>Excoecaria agallocha</i>	148	7241	35.73 $\pm$ 1.30
<i>Xylocarpus mekongensis</i>	85	434	16.07 $\pm$ 0.76
<i>Bruguiera sexangula</i>	23	253	44.73 $\pm$ 2.15
<i>Avicennia officinalis</i>	35	127	24.47 $\pm$ 0.35
<i>Xylocarpus granatum</i>	20	75	19.43 $\pm$ 0.68
<i>Amoora cuculata</i>	31	74	29.22 $\pm$ 0.75
<i>Sonneratia apetala</i>	7	36	23.73 $\pm$ 0.91
<i>Initia bijuga</i>	6	12	29.22 $\pm$ 0.75
<i>Sonneratia caseolaris</i>	2	12	19.73 $\pm$ 0.83
<i>Cynometra ramiflora</i>	8	10	29.22 $\pm$ 0.75
<i>Aegiceras corniculatum</i>	6	9	49.67 $\pm$ 2.06
Unknown species	5	9	29.22 $\pm$ 0.75
<i>Excoecaria. indica</i>	5	8	29.22 $\pm$ 0.75
<i>Pongamia pinnata</i>	3	7	29.22 $\pm$ 0.75
<i>Rhizophora mucronata</i>	3	7	56.93 $\pm$ 5.42
<i>Lumnitzera racemosa</i>	1	4	22.43 $\pm$ 2.55
<i>Hibiscus tiliaceus</i>	1	2	29.22 $\pm$ 0.75

2.3. Remote Sensing Data

The Landsat surface reflectance data from Thematic Mapper 5 (Landsat TM 5) were used in this study, which have been corrected for atmospheric, reflectance and topographic effects [63]. Both Landsat TM 5 and Landsat 8 have a 16-day repeat cycle, and their band specification is shown in Table 2 [64]. Cloud and cloud shadow pixels in all images were masked based on the pixel quality attributes provided in the products. Composite images of Sundarbans Reserved Forest for each sensor, Landsat TM 5 (2009–2010) and Landsat 8 (2019), used in this study, were aggregated from 89 and 75 images, respectively [63]. The median value of seven multispectral bands was used for modeling purposes because it was robust against extreme values and produced a good quality image which was representative of the time period [63]. One hundred and fifty-seven predictors were extracted from seven bands of Landsat TM 5 that comprised surface reflectance and 18 Gray Level Co-occurrence Metrics (GLCM) for each of the seven bands and Vegetation indices (VI). Of those 150 predictors, 48 ($r \geq 0.20$ with C:N; Table 2; Supplementary Materials) were used in modeling C:N. All the image processing was done using the Google Earth Engine Platform and R environment (Version 3.5.3).

Table 2. List of bands (Landsat TM 5 and Landsat 8), Grey Level Co-occurrence Metrics (GLCM) texture and vegetation indices with code used in this study.

Predictors	Source
Landsat TM 5 (Band name, wavelength, resolution)	[64]
B1 (Visible Blue, 0.45–0.52 μ m, 30 m)	
B2 (Visible Green, 0.52–0.60 μ m, 30 m)	
B3 (Visible Red, 0.63–0.69 μ m, 30 m)	
B4 (NIR, 0.76–0.90 μ m, 30 m)	
B5 (SWIR 1, 1.55–1.75 μ m, 30 m)	

Table 2. Cont.

Predictors	Source
B6 (Thermal, 10.40–12.50 μm , 120 m)	
B7 (SWIR 2, 2.08–2.35 μm , 30 m)	
Landsat 8 (Band name, wavelength, resolution)	
B2 (Visible Blue, 0.450–0.51 μm , 30 m)	
B3 (Visible Green, 0.53–0.59 μm , 30 m)	
B4 (Red, 0.64–0.67 μm , 30 m)	
B5 (Near-Infrared, 0.85–0.88 μm , 30 m)	
B6 (SWIR 1, 1.57–1.65 μm , 30 m)	
B7 (SWIR 2, 2.11–2.29 μm , 30 m)	
B10 (TIRS 1, 10.6–11.19 μm , 100 m)	
B11 (TIRS 2, 11.5–12.51 μm , 100 m)	
GLCM (Grey Level Co-occurrence Metrics) texture	
asm (Angular Second Moment)	[65]
contrast (Contrast)	[65]
corr (Correlation)	[65]
variance (Variance)	[65]
idm (Inverse Difference Moment)	[65]
savg (Sum Average)	[65]
svar (Sum Variance)	[65]
sent (Sum Entropy)	[65]
ent (Entropy)	[65]
dvar (Difference variance)	[65]
dent (Difference entropy)	[65]
imcorr1 (Information Measure of Corr. 1)	[65]
imcorr2 (Information Measure of Corr. 2)	[65]
maxcorr (Max Corr. Coefficient)	[65]
diss (Dissimilarity)	[66]
inertia (Inertia)	[66]
shade (Cluster Shade)	[66]
prom (Cluster prominence)	[66]
Vegetation index	
MSI (Moisture stress Index)	[67]
PSSR (Pigment Specific Simple Ratio)	[68]
PBI (Plant biochemical index)	[69]
SI (Salinity index)	[70]
DWSI (Disease-Water Stress Index)	[71]
SRRNI (Simple Ratio Red and NIR)	[72]
IO (Simple Ratio Red and Blue)	[73]
SRSN1B (Simple index SWIR 1 and blue)	This study
SRSN2B (Simple index SWIR 2 and blue)	This study
SRSN2G (Simple index SWIR2 and green)	This study
SRNB (Simple index NIR and blue)	This study
LWC (Leaf water content)	[74]
CVI (Chlorophyll Vegetation Index)	[67]
SIPI (Structure Intensive Pigment Index 3)	[68]
MNDWI (Modified Normalized Difference Water Index)	[75]
SCI (Soil Composition Index)	[76]

3. Modeling Approaches

3.1. Stochastic Gradient Boosting (SGB)

Stochastic Gradient Boosting (SGB) is a hybrid approach combining both boosting and bagging techniques [77,78]. In SGB, a small classification or regression trees are sequentially built from the residuals of the preceding tree(s). Instead of focusing on the full training set, the SGB carries out a boosting by selecting (without replacement) at each step a random sample of the data, leading to a

gradual improvement of the model. Thus, increasing the chance that incorrectly classified observations will be correctly classified in the next tree. The weighted majority of classification determines the final classification of each observation across the sequence of trees [47]. Ridgeway [77] provided a comprehensive background and mathematical functions of SGB.

3.2. Random Forest (RF)

Random forest (RF) is an ensemble classifier or regression technique [79]. It is based on the well-known classification and regression tree approach (CART) [79], which consists of many decisions or regression trees, in which each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the data [80]. Random forest with a large number of trees is robust against overfitting, noise, as well as non-informative and correlated features [79]. RF is a widely used modeling technique in remote sensing-based studies in forestry such as biomass [50] and functional traits mapping [23,38,45].

3.3. Support Vector Machine (SVM)

Support vector machine (SVM) belongs to the regression model family and is a kernel-based learning method from statistical learning theory [81]. SVMs mainly involve a projection of the data into a high-dimensional feature space using a valid kernel function and then application of a simple linear regression within this enhanced space [82]. The resulting linear regression function in the high-dimensional feature space corresponds to a non-linear regression in the original input space [83].

3.4. Partial Least Square Regression (PLSR)

One of the most common methods to build a multi-independent model is Partial Least Square Regression (PLSR), and its advantage is that PLSR can handle data with strong collinearity and noise, as well as situations in which the number of variables considerably exceeds the number of available samples [84]. In modeling of functional traits, this modeling technique has also been used [85].

Data preparation, implementation of all modeling methods (SG B, RF, SVM and PLSR), accuracy assessment, mapping process and variable importance were performed using the “caret” package in RStudio (version 3.5.3) [86]. The “gbm”, “cforest”, “svmRadial” and “kernelpls” packages were used for SGB, RF, SVM and PLSR modeling techniques in RStudio (version 3.5.3) [86], respectively.

3.5. Model Development

In this study, we constructed repeated cross-validation (fold = 10) with ten repetitions for model fitting and model validation with the R “caret” package. In the 10-fold cross validation, at each repetition, the sample data were split into 10 subsets, where nine subsets of samples were used to train models, and one was treated as validation samples. We set a seed value of seven for each modeling method, providing the same dataset for each modeling method. To select the simple and optimal model as suggested by Breiman et al. [79], this study used the “one standard error rule”. By using the cross validated models, we predicted C:N ratio for 93 plots for each modeling method. Then we also fitted a simple linear regression between field C:N ratio and predicted C:N ratio for testing models fitness in terms of slope and intercept parameters against the 1:1 line. We did not perform any collinearity test or feature selection, since all the modeling methods are non-parametric, and they have their own resistance against collinearity.

3.6. Model Performance

In this study, three common performance indices, namely Mean Absolute Error (MAE), root mean squared error (RMSE) and coefficient of determination (R^2) were used. MAE % and RMSE % were also used as model performances which were calculated by dividing MAE and RMSE by mean value of field C:N ratio. MAE summarizes and assesses the quality of a machine learning model. The MAE

of a model refers to the mean of the absolute values of each prediction error in all the test dataset. Therefore, it refers to the results of measuring the difference between all predictions and observations. RMSE (an absolute measure of fit) is a measure of the average deviation of the estimates from the observed values. It is not scaled to any particular values. RMSE shows how much the predictions deviate, on average, from the actual values. Lower values of RMSE indicate better fit. RMSE is a good measure of how accurately the model predicts the response, and it is the most important criterion for fit if the main purpose of the model is prediction. R^2 (relative measure of fit) varies between 0 and 1. R^2 is the proportion of the variance in the dependent variable that is predictable from the independent variable(s). Therefore, the value near to 1 means that the two variables are perfectly correlated and vice versa.

$$MAE = \frac{1}{n} \sum_{i=1}^n |P_i - O_i| \quad (2)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n [O_i - P_i]^2}{n}} \quad (3)$$

$$R^2 = \left[\frac{\sum_{i=1}^n (O_i - O_{avg})(P_i - P_{avg})}{\sqrt{\sum_{i=1}^n (O_i - O_{avg})^2} \sqrt{\sum_{i=1}^n (P_i - P_{avg})^2}} \right]^2 \quad (4)$$

where P_i , O_i , O_{avg} , P_{avg} , n and p are the predicted and observed values, average of observed and predicted soil properties values at the i th point, number of data and the total number of explanatory variables in the model, respectively.

Moreover, the relative improvement (RI) was calculated to compare the improvement of models and the accuracy of the prediction for C:N. The RI are defined as Equation (5):

$$RI_x = 100 \times \frac{\text{Model fitting parameter}_x - \text{model fitting parameter}_{ref}}{\text{Model fitting parameter}_{ref}} \quad (5)$$

where model fitting parameters are MAE, RMSE and R^2 ; subscript x is the model whose model fitting parameter is higher (R^2) and lower (MAE and RMSE) than that of the reference model (least performing model).

We also performed a one-way ANOVA (Analysis of Variance) for testing the significance of the differences among different modeling techniques. For multiple comparison between modeling techniques, Least Significant Difference (LSD) was used. Furthermore, we employed simple linear regression model among different modeling techniques to see the pixel to pixel association of prediction of C:N ratio by them. After selecting the best modeling method based on Landsat TM 5 composite image for 2009–2010, we applied it to Landsat 8 composite image for 2019 for observing the temporal change of C:N.

4. Results

4.1. Models Comparison

The mean MAE, RMSE and R^2 were varied from 2.06 ± 0.62 to 2.43 ± 0.68 , 2.88 ± 0.85 to 24 ± 1.15 , and 0.37 ± 0.23 b to 0.51 ± 0.22 a, respectively (Table 2). A significant difference in model fitting parameters was found among different modeling techniques ($p < 0.05$). However, the post hoc test (LSD) revealed that stochastic gradient boosting (SGB), random forest (RF) and support vector machine (SVM) modeling techniques were not significantly different ($p > 0.05$; Table 3). However, the model fitting parameters of these three modeling techniques were significantly different from the least performing partial least square regression (PLSR; $p < 0.05$; Table 3). The relative improvement in terms of MAE, RMSE and R^2 for the SGB, RF and SVM were estimated against PLSR (the lowest performance model) is shown in Table 2. SGB modeling technique showed best performance in terms of relative improvement

of RMSE and R^2 while SVM showed best performance in of relative improvement of MAE (Table 3). A similar trend in performance to RMSE and R^2 (relative improvement) was also found for MAE and RMSE percentage of field-based mean C:N (Table 3).

Table 3. Comparison of model fitting parameters among four modeling methods. Relative improvement of model accuracy, precision and fitness was calculated against partial against partial least squared regression. Columns with the same letter (a or b; group of modeling methods) are not significantly different based on the LSD test ($p < 0.05$).

Models	Model Accuracy			Relative Improvement			Error Percentage	
	MAE	RMSE	R^2	MAE%	RMSE%	$R^2\%$	MAE%	RMSE%
SGB	2.16 ± 0.58^a	2.88 ± 0.85^a	0.51 ± 0.22^a	−11.11	−11.11	37.84	6.15	8.20
RF	2.21 ± 0.60^a	2.99 ± 1.15^a	0.48 ± 0.24^a	−9.05	−7.72	29.73	6.30	8.52
SVM	2.06 ± 0.62^a	3.12 ± 1.20^a	0.46 ± 0.24^a	−15.23	−3.70	24.32	5.86	8.89
PLSR	2.43 ± 0.68^b	3.24 ± 1.15^b	0.37 ± 0.23^b	0.00	0.00	0.00	6.93	9.23

As the three model-evaluating parameters were not varied significantly among SGB, RF and SVM, we also plotted the predicted C:N ratio based on these three models against the field-based C:N ratio for testing its fitness in terms of slope and intercept parameters against the 1:1 line (Figure 2). Observed versus predicted scatter plots showed a strong predictability of the SGB model as the slope was deviated only 0.26 from one, and points were concentrated near the 1:1 line (Figure 2a; Table 4). While the SVM had least predictability (Figure 2c; Table 4).

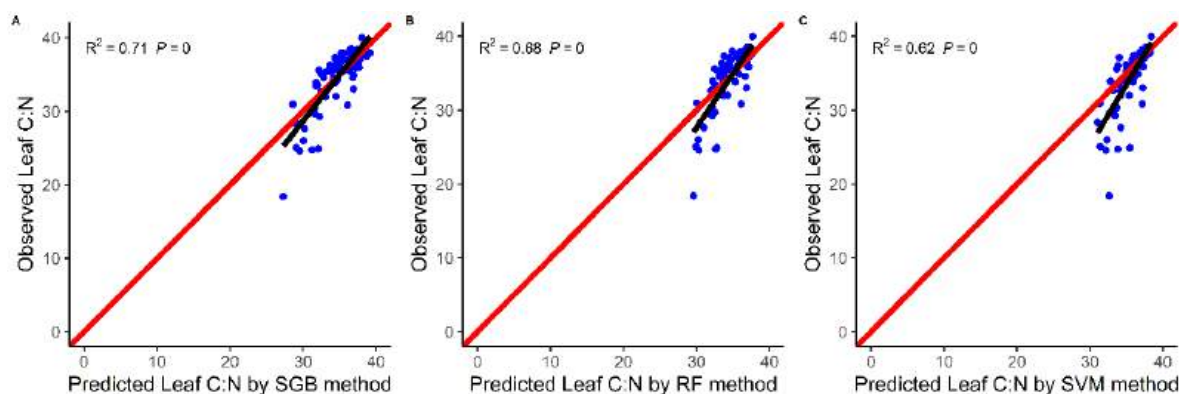


Figure 2. Scatter plot of observed leaf carbon to nitrogen (C:N) ratio versus predicted leaf C:N by of stochastic gradient boosting (a), random forest (b) and support vector machine (c) methods. The bold point (blue) indicates the value of C:N in each plot. Red and black lines indicate the 1:1 reference line and best fit linear, respectively.

Table 4. Internal validation statistics of observed (field) versus predicted C:N ratio of stochastic gradient boosting (SGB), random forest (RF), and support vector machine (SVM). a: intercept and b: slope.

Model Coefficients		Variables		
$y = a + b \cdot x$		Obs_C:N vs. Pre_CN_SGB	Obs_C:N vs. Pre_CN_RF	Obs_C:N vs. Pre_CN_SVM
a		−9.04	−16.15	−24.38
b		1.26	1.46	1.65
MAE		1.58	1.55	1.51
RMSE		2.12	2.23	2.44
MAE%		4.50	4.41	4.30
RMSE%		6.03	6.34	6.94

4.2. Importance of Predictors

In different modeling approaches, the importance of the top ten predictors and the type of predictors were different (Figure 3). For instance, in SGB, RF and SVM models top ten predictors were a mixture of different bands, GLCM texture and vegetation indices while in PLSR model top ten predictors consist of GLCM texture of different bands (Figure 3a–d). Moisture and temperature related bands, GLCM texture and vegetation indices were enlisted in the top 10 important predictors in all the models except PLSR where GLCM texture of different bands were in the top ten predictors (Figure 3a–d).

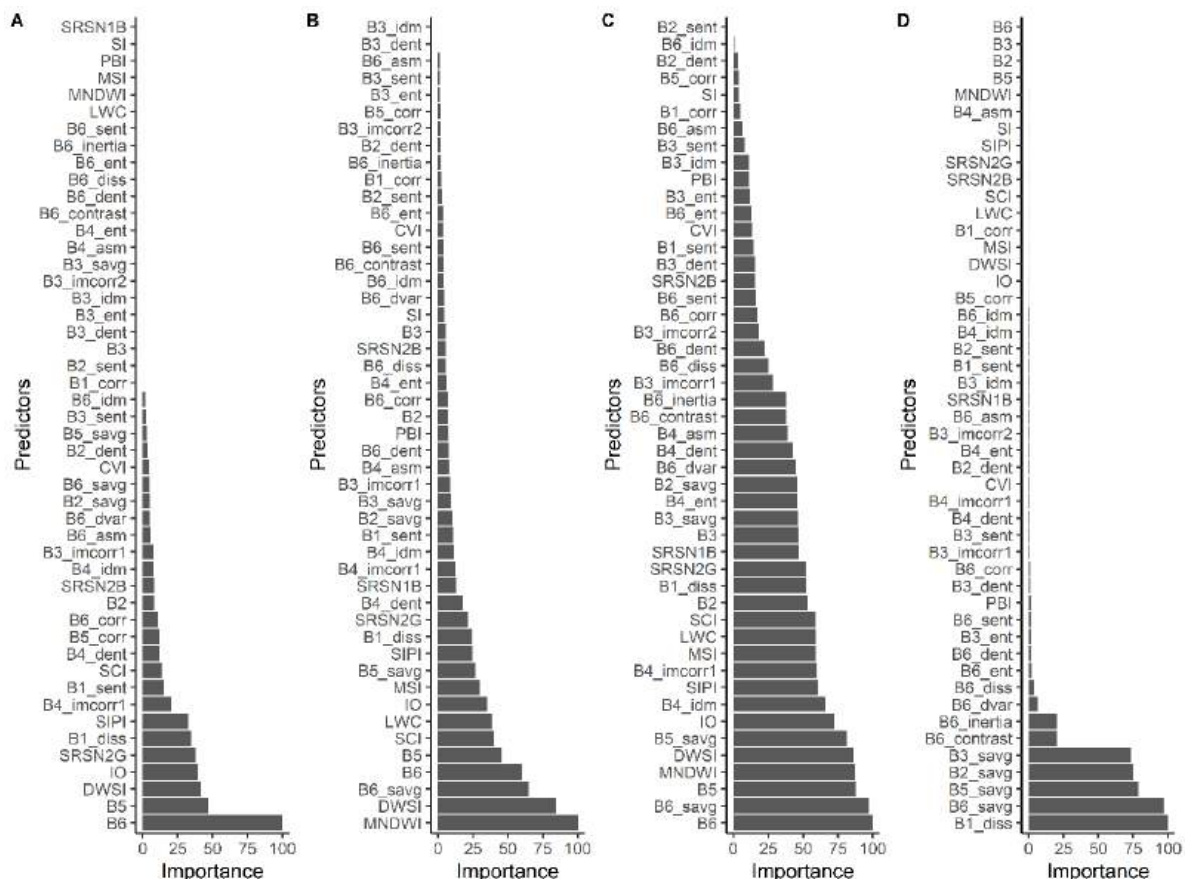


Figure 3. Importance value of predictors in each of the four modeling techniques based on 10-fold cross-validation with 10 times repetition. The codes after band (e.g., B6_savg) and other predictors were explained in Table 2. (a) Stochastic gradient boosting; (b) random forest; (c) support vector machine; (d) partial least square regression.

4.3. Spatial and Temporal Distribution of C:N

The spatial distribution of C:N based on the stochastic gradient boosting (SGB), random forest (RF) and support vector machine (SVM) modeling techniques and its prediction difference between the three-modeling techniques is shown in Figure 4. The difference map and statistics of pixel to pixel association of prediction of C:N ratio by different modeling techniques revealed that SGB and RF predictions were slightly different (Figure 4; Table 4). However, a large difference was observed in production of C:N ratio between SGB and SVM and RF and SVM (Table 4; Figure 4). Although, the prediction difference of C:N ratio was quite low between SGB and RF modeling techniques, the SGB modeling technique captured variation of C:N ratio that was closer to field distribution compared to RF (Table 5).

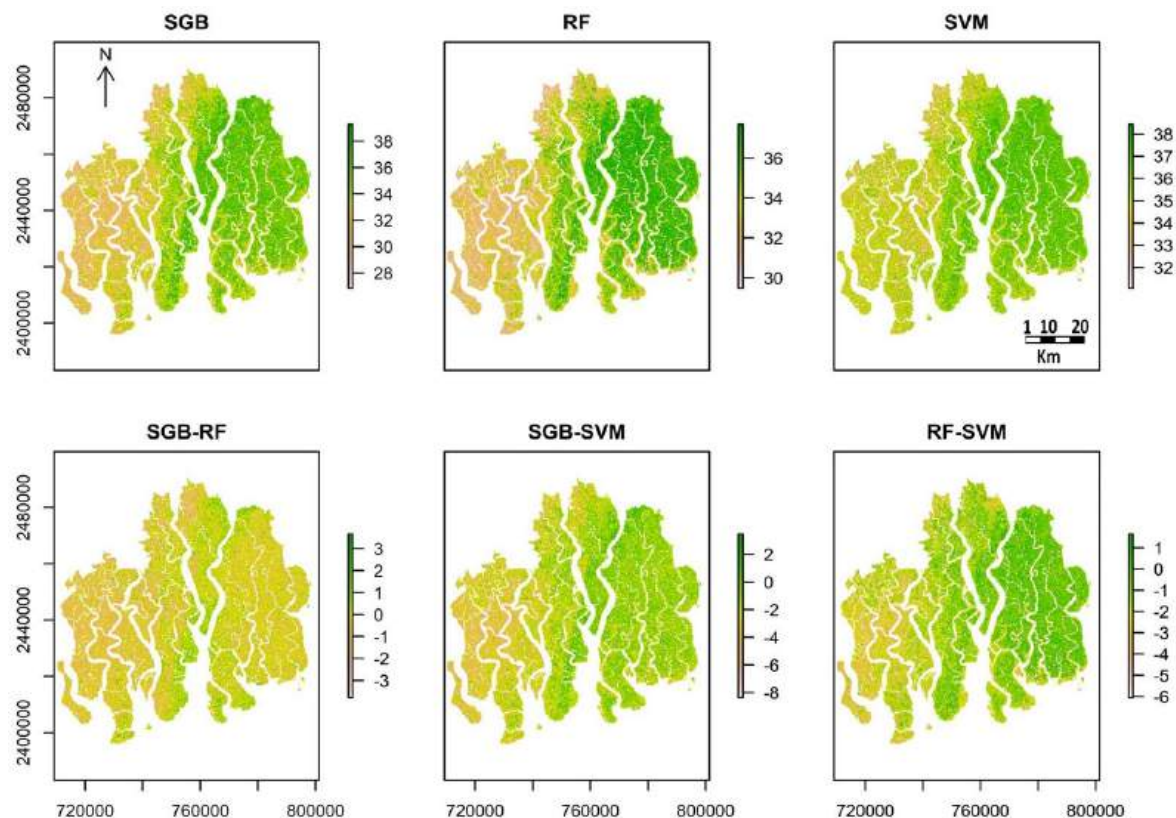


Figure 4. Spatial distribution (upper row) and difference (lower row) of C:N ratio between models across the whole Sundarbans Reserved Forest for 2009–2010. SGB: Stochastic gradient boosting, RF: Random Forest and SVM: Support Vector Machine.

Table 5. Spatial relationship between different modeling methods in predicting C:N ratio across the whole Sundarbans Reserved Forest. SGB: Stochastic gradient boosting, RF: Random Forest and SVM: Support Vector Machine. a: intercept and b: slope.

Model Coefficients		Variables		
$y = a + b \cdot x$	SGB vs. RF	SGB vs. SVM	RF vs. SVM	
a	−3.65	−21.10	−13.86	
b	1.10	1.54	1.35	
R ²	0.92	0.73	0.77	
N (pixel)	4,499,989.00	4,499,989.00	4,499,989.00	
F (F-test significance)	23,384,941.95	5,155,875.87	6,358,556.93	

Higher C:N ratio was associated in the eastern part of Sundarbans Reserved Forest with a clear homogenous patch compared to the western part and the landward side of the central part (Figure 4). The streamline zone (near to river and canal bank) had lower C:N ratio compared to stable stands (forest proper zone; 100 to 200 m away from river and canal bank) (Figure 4).

As the SGB modeling technique covered much spatial variation of C:N and were similar to field C:N, we applied the SGB based on Landsat 8 composite image for 2019 to observing the temporal change of C:N (Table 6; Figure 5). The difference between 2009–10 and 2019 C:N showed that C:N ratio increased mainly at the central and landward parts of Sundarbans Reserved Forest (Figure 5). In the seaward and eastern side, it showed a decreasing trend in C:N ratio (Figure 5).

Table 6. Comparison between spatial statistics (models based) and field C:N ratio. SGB: Stochastic gradient boosting, RF: Random Forest and SVM: Support Vector Machine.

	Mean	Min	Max	SD
Field	35.15	18.4	39.99	3.96
SGB 2010	33.92	26.95	39.43	2.45
RF 2010	34.26	29.36	37.69	2.05
SVM2010	35.68	30.99	38.51	1.16
SGB 2019	32.51	27.04	37.28	1.77

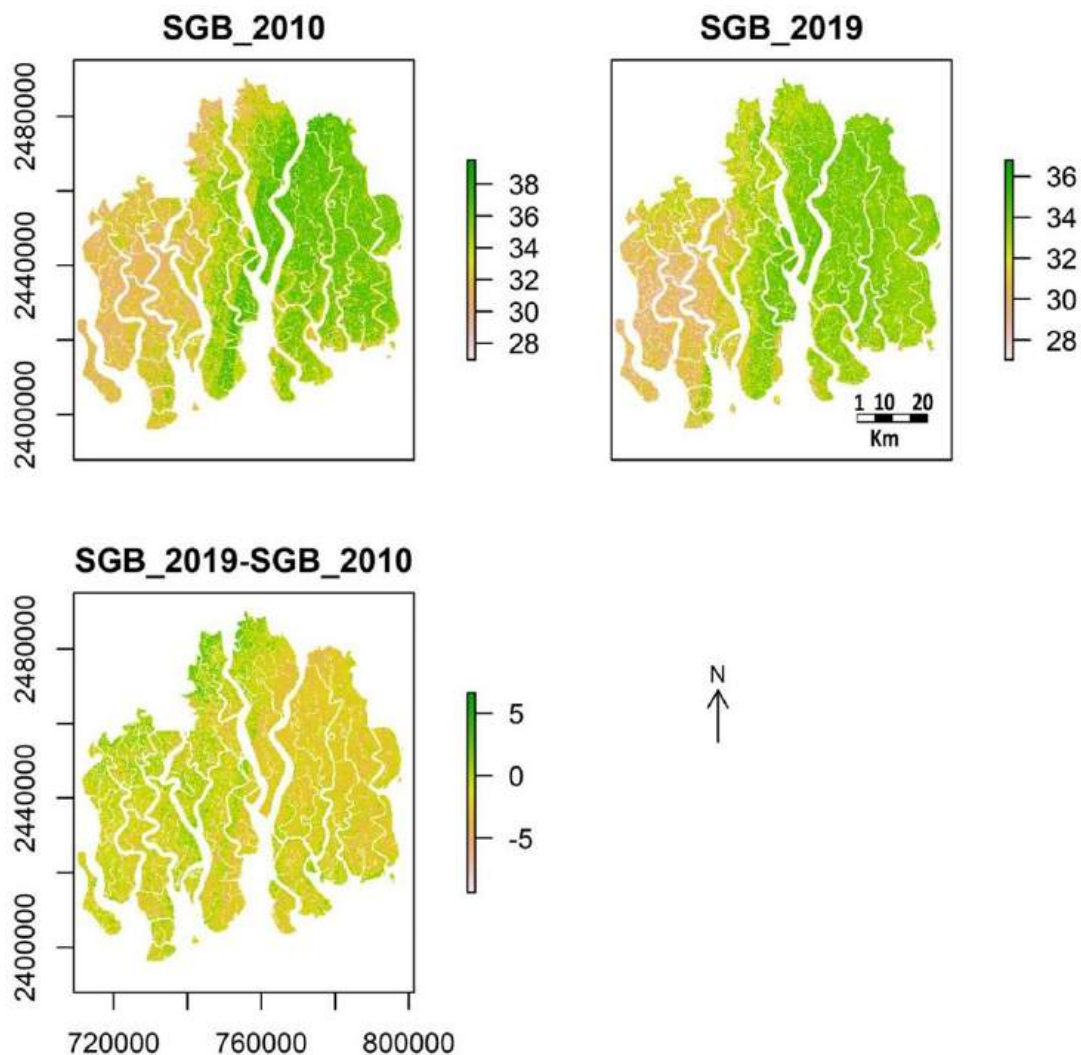


Figure 5. Spatial (upper row) and temporal change (lower row) map of C:N ratio based on stochastic gradient boosting modeling method from 2010 to 2019. SGB: Stochastic gradient boosting, RF: Random Forest and SVM: Support Vector Machine.

5. Discussion

To our current knowledge, this study assessed, for the first time, the spatial distribution and temporal change of the CWM of senescent leaf C:N ratio in Sundarbans Reserved Forest, Bangladesh as a model in mangrove ecosystem with remote sensing. Four machine learning modeling approaches were compared in order to select the best modeling technique for mapping C:N ratio using spectral and textural properties of seven bands and vegetation indices of Landsat TM 5. The modeling approach that combined predictors from the above-mentioned Landsat TM 5 image properties that also related to moisture and temperate, showed better performance compared to the modeling approach that used

only textural properties. The SGB modeling technique captured wide variation of C:N ratio and used a lower number of predictors compared to the other three modeling methods. Thus, this modeling technique was applied to the Landsat 8 image in 2019 for assessing temporal change of C:N ratio.

5.1. Comparison of Model Performance

The nonlinear based modeling technique showed better performance in terms of lower MAE, RMSE and higher R^2 (Table 3). This difference in performance among different modeling approaches could be due to several reasons [47,87]. First, how a specific modeling approach arranges and handles the highly correlated predictors in the top rank of importance and had high correlation with C:N. Although, all the modeling approaches used in this study have the capacity to handle the multicollinearity issue, the SGB showed strong ability to handle multicollinearity over other models [47,87]. In a study of biomass modeling using RapidEye satellite with SGB and RF in Africa terrestrials forest, Dube [50] reported the outperformance of the SGB modeling approach. PLSR showed a weak performance. A similar performance of PLSR modeling technique in comparison with other machine learning modeling approaches has also been reported in biomass modeling studies [88–90]. This finding could reveal that C:N ratio and remote sensing-based predictors are nonlinearly related in the Sundarbans Reserved Forest. This could not be handled well by PLSR modeling technique which is linear based [46].

The second reason is the combination of predictors from different dimensions (e.g., either band spectral properties or GLCM textural properties or vegetation indices or all of them) in modeling. The modeling approaches that aggregate both spectral and textural properties have higher predictability compared to the methods that aggregate only a single type of predictor [23,91]. The top three modeling approaches (SGB, RF and SVM) combined the most important predictors from band spectral and GLCM textural properties and vegetation indices compared to PLSR modeling technique, which mainly aggregated the GLCM texture properties of different bands (Figure 3). This incorporation of homogeneous predictors in modeling resulted in the poor performance of PLSR technique.

Finally, the type of predictors in the top rank of important value in different modeling methods are related to vegetation, canopy moisture, salinity and water content. Among the top rank predictors, Band 6 is related to moisture and temperature while the Disease Water Stress Index and Modified Normalized Difference Water Index are related to water stress and canopy moisture, respectively [92–95]. Previous studies also confirmed that in mangrove, nutrient status and dynamics vary along with water, temperature and salinity gradient [3,7]. Therefore, the modeling technique that had a combination of those predictors in the top rank of importance value had lower errors in prediction and higher capacity of explaining the variance of C:N (Table 2; Figure 3).

5.2. Spatial Distribution and Temporal Change of C:N ratio in SRF

Among the top three modeling approaches, the predicted C:N map stochastic gradient boosting (SGB) technique was closer to the field-based C:N distribution (Figure 4; Table 6). Although, the mean and maximum values of C:N in the predicted map of C:N by the top three models was closer to field value, the SGB technique predicted the lower value of C:N better than RF and SVM (Table 6). The spatial distribution of C:N ratio in Sundarbans Reserved Forest (SRF) was closely related to vegetation types and salinity gradient. For example, the higher C:N value (>36) in the eastern part of SRF is mainly dominated by *Heritiera fomes* (Figure 4) [16,22,96]. The higher C:N ratio in these areas indicates that *Heritiera fomes* dominated stand has higher photosynthetic nitrogen use efficiency or nutrient resorption capacity [3,16,97]. However, most of the western parts of SRF were under low C:N ratio coverage, which indicates species in those areas are in stress conditions with low photosynthetic nitrogen use or have lower nutrient resorption capacity and thereby there may be low productivity or carbon sequestration capacity [16]. These areas are mainly dominated by *Ceriops decandra* and *Excoecaria agallocha* with scattered *Xylocarpus granatum*, *X. mekongensis* [16,96]. A higher level of salinity may cause enzymatic metabolism disorder in the chloroplast; decline in K^+ ion, the key photosynthetic

element and other philological activities, thus decreasing the rate of photosynthesis [97]. For this reason, photosynthetic nitrogen use efficiency could be lower in the high saline zone and streamlined areas in the eastern part of SRF, which could be the cause of low C:N ratio in those zones. The lower range of C:N ratio was also found in the 2010 map in the landward (the northern side of SRF), although this area is associated with freshwater or moderate saline water coverage. The reason of this lower C:N value could be the dominance of *Avicenna officinalis* species, which has a lower C:N ratio (24.47; Table 1) [13].

The temporal change map (difference between 2009–2010 and 2019) indicates that a gradual increasing trend was observed from seaward to landward (central and western part of SRF; Figure 5). Similar trend also observed at streamline to forest proper line (Figure 5). This could be due to the change of species composition over the 10-years period where species with higher C:N may be shifted. In most cases it was *H. fomes* because other species (*A. corniculatum*, *R. mucronata*) are mainly distributed in high saline area (south-western part of SRF) and some also confined in waterlogged areas (*B. sexangular*) in SRF. However, temporal change map also shows that across the seaward and eastern part of SRF the C:N ratio was declined (Figure 5). Two reasons may be caused this declination of C:N: (i) a super cyclone (Bulbul) passed over SRF on 9 November 2019 which severely damaged forest canopy and (ii) may be the change of species composition where species with low C:N ratio may come to canopy layer.

5.3. Implication for Forest Health Monitoring and Policy

In the current climate change regime, monitoring of health conditions is a key forest management task [22,33,98]. Litter quality such as C:N ratio as an essential biodiversity variable or functional trait can be used as a potential indicator of decomposition rate, plant nitrogen retention capacity, primary productivity and soil organic carbon storage [6,9,10,99,100].

We developed advance machine learning models for mapping senescence leaf C:N in SRF using Landsat TM 5 imagery and applied the best model on Landsat 8 imagery for 2019. The C:N ratio map developed in the current study can be used for improving the biogeochemical and ecosystem models in the mangroves. In this way, our developed method will contribute to evaluable ecosystem service policy and forest management at the local and regional levels [53,101]. For example, it can be used to track the endangered Red list species in Mangrove that were enlisted by the International Union for Conservation of Nature. The higher value of C:N ratio in the eastern parts of SRF is dominated by *H. fomes*, the endangered species in SRF. Therefore, by tracking C:N ratio as an indicator of Essential Biodiversity Variable, we can assess the progress of Biodiversity targets that related to IUCN Red List species. For example, two of 20 specific targets (targets 9 and 12) of Aichi Biodiversity Targets-2020 are related to endangered species. The aim of these biodiversity targets is to conserve the threatened species and update their conservation status as to whether they are being improved and sustained by 2020 [102,103]. Furthermore, our developed method directly contributes to monitoring the progress of the biodiversity targets (Aichi Targets 2, 3 and 15), which are related to nutrient dynamic, productivity and ecosystem health [102,104,105]. The C:N ratio is directly associated with key ecosystem functions (decomposition) that have been regarded as biodiversity indicators such as net primary productivity, nutrient retention, disturbance regime [102,104,105], helping in this way to achieve National Biodiversity Target 2 of Bangladesh [106]. While the usage of Landsat imagery in our modeling technique is related to National biodiversity Target 19 of Bangladesh which is aimed at developing a sustainable monitoring technique by means of remote sensing [106]. Ultimately, our finding will be contributed towards achieving sustainable development goal 15 and Aichi Biodiversity target 2, 11 and 15 [51,106–108] for Bangladesh.

6. Conclusions

The current study investigated the potentiality of Landsat imagery for spatiotemporal mapping of CWM of C:N ratio of senescent leaf by comparing four machine learning techniques in SRF. SGB technique showed the best performance by incorporating key predictors from the spectral and textural

properties and vegetation indices group, which had higher relative improvement against the least performing method (PLSR) compared to RF and SVM. The spatial pattern of C:N ratio map based on SGB method captured a wide variation of C:N, which was closer to field C:N ratio. The map of C:N ratio reveals that the spatial distribution of C:N mainly depends on the species type and salinity gradient. Stands close to the streamline are in a more stressed condition than forest proper or inland side. The eastern side of SRF, which is mostly dominated by *H. fomes* and receives more freshwater from the upstream river, had a higher C:N ratio, and thereby is in less stressed condition with higher nitrogen use or retention efficiency compared to the western and landward sides of SRF. The temporal change of C:N ratio could also be influenced by species composition change and cyclone damage. Our developed advance machine learning technique thus can be applied for future prediction of litter quality in SRF and for improving the biogeochemical and ecosystem models related to nutrient dynamic in the mangroves.

Supplementary Materials: The following are available online at <http://www.mdpi.com/2072-4292/12/9/1375/s1>, Pearson correlation between forty-eight predictors and community weighted mean of C:N ratio of senescent leaf used in modeling. The first column represents the type of different predictors (e.g., Surface reflectance). Other column and row represent the different variables (Dependent: CWM C:N, Predictors: all other variables) with a value of either no sign before (positive relation) or negative sign (negative relation between variables). The full names of the variables were given below at the end of Pearson correlation with type (yellow highlight).

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Unraveling the drivers and impacts of leaf phenological diversity in a subtropical forest: A fine-scale analysis using PlanetScope CubeSats

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Summary

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- Leaf phenology variations within plant communities shape community assemblages and influence ecosystem properties and services. However, questions remain regarding quantification, drivers, and productivity impacts of intra-site leaf phenological diversity.
- With a 50-ha subtropical forest plot in China's Heishiding Provincial Nature Reserve (part of the global ForestGEO network) as a testbed, we gathered a unique dataset combining ground-derived abiotic (topography, soil) and biotic (taxonomic diversity, functional diversity, functional traits) factors. We investigated drivers underlying leaf phenological diversity extracted from high-resolution PlanetScope data, and its influence on aboveground biomass (AGB) using structural equation modeling (SEM).
- Our results reveal considerable fine-scale leaf phenological diversity across the subtropical forest landscape. This diversity is directly and indirectly influenced by abiotic and biotic factors (e.g. slope, soil, traits, taxonomic diversity; $r^2 = 0.43$). While a notable bivariate relationship between AGB and leaf phenological diversity was identified ($r = -0.24$, $P < 0.05$), this relationship did not hold in SEM analysis after considering interactions with other biotic and abiotic factors ($P > 0.05$).
- These findings unveil the underlying mechanism regulating intra-site leaf phenological diversity. While leaf phenology is known to be associated with ecosystem properties, our findings confirm that AGB is primarily influenced by functional trait composition and taxonomic diversity rather than leaf phenological diversity.

Introduction

Leaf phenology, the timing of recurring leaf-related biological events such as leaf emergence and senescence, reflects plants' environmental responses throughout their growth and development (Lieth, 1974; Kikuzawa, 1995). Significant variability in leaf phenology exists across large geographic gradients and climate zones (Guan *et al.*, 2015; Gao *et al.*, 2020; Satake *et al.*, 2022), as well as within individual vegetated landscapes, particularly in (sub)tropical forests with diverse woody plant species (Reich, 1995; Lopes *et al.*, 2016; Lee *et al.*, 2023). Intra-site leaf phenological diversity, characterized by variations in leaf phenological events among individuals/species within a community, is increasingly observed in these areas. It indicates intricate and

specific adaptation strategies to environmental conditions, which in turn have various ecological consequences on individual survival, species fitness, and ecosystem stability (Keenan *et al.*, 2014; Forzieri *et al.*, 2020). Despite the ecological importance of understanding and predicting climate change responses, our knowledge of intra-site leaf phenological diversity patterns, underlying drivers, and links to ecosystem properties remains limited.

Various methods have been employed to monitor leaf phenology across a wide range of scales, including ground inventory, proximate remote sensing techniques, and satellite observations. Ground inventory and proximate remote sensing methods, such as tower-mounted phenocam measurements and frequent drone flights, offer accurate characterizations of leaf phenology across tens-to-hundreds of individual tree crowns, but they have limited spatial extents (Klosterman *et al.*, 2018; Park *et al.*, 2019; Del-pierre *et al.*, 2020). Traditional satellite remote sensing, like

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Moderate Resolution Imaging Spectroradiometer (MODIS), Landsat, and Sentinel-2, can monitor ecosystem-scale leaf phenology; however, their coarse resolution makes it challenging to observe intra-site leaf phenological diversity (Blanchard *et al.*, 2016; Panzou *et al.*, 2021). In recent years, PlanetScope CubeSats have emerged as a valuable alternative, offering a 3-m spatial resolution and near-daily global coverage to improve both intra-site and across-ecosystem phenology monitoring. Several recent studies have shown that fine-scale phenology characterizations derived from PlanetScope are comparable to ground and proximate remote sensing observations, with demonstrated accuracy and scalability across both temperate and tropical landscapes (Wang *et al.*, 2020, 2023; Moon *et al.*, 2021; Wu *et al.*, 2021; Zhao *et al.*, 2022). However, the use of PlanetScope Cubesats for characterizing intra-site leaf phenological diversity across spatially continuous landscapes remains under-explored.

Moreover, despite the importance of understanding intra-site leaf phenological diversity, the underlying mechanisms driving this phenomenon remain largely unknown. Past research has primarily concentrated on the ecosystem scale, examining the relationship between macro-climate and phenological variability across vast landscapes and environmental gradients (Kharouba *et al.*, 2018; Gallinat *et al.*, 2021), or delving into organism-scale or single species-scale phenological connections with microclimate and traits (Yang *et al.*, 2017; Su *et al.*, 2023). For example, Sporbert *et al.* (2022) demonstrated that spatial variability of phenostage timing (e.g. leaf emergence and senescence) strongly correlated with functional traits, including plant height, leaf area, leaf dry matter content, and so on using *in situ* records from Germany. Goulart *et al.* (2005) presented the phenological diversity of a Brazilian tree species (*Plathymenia reticulata* Benth.) within and among populations in two biomes (woody savanna and tropical forest) and found that most phenological diversity was attributed to differences of leaf fall among individuals rather than difference of microclimates between biomes. However, the lack of comprehensive characterizations of the underlying mechanisms driving intra-site leaf phenological diversity has indeed limited the scope of similar investigations. We hypothesize that both biotic and abiotic factors may influence intra-site leaf phenological diversity. Specifically, biotic factors could arise from either the abundance of a few dominant species with key functional traits (selection effect hypothesis) or multiple co-existing species with greater trait variation (niche complementarity hypothesis). A thorough understanding of these underlying mechanisms is essential, as it can provide valuable insights into predicting how ecosystems will respond to climate change and other environmental stressors (Tilman *et al.*, 1997; Richardson *et al.*, 2018a; Rahman *et al.*, 2021).

The lack of comprehensive characterizations of intra-site leaf phenological diversity has further implications, particularly given the growing acknowledgment of leaf phenology as a crucial phenotype (Ware *et al.*, 2019; Segrestin *et al.*, 2020), which exhibits connections with ecosystem properties (Zhang *et al.*, 2020; Kim *et al.*, 2021). While the relationship between taxonomic diversity and/or functional trait diversity with essential ecosystem properties, such as aboveground biomass (AGB), has been relatively

well-established (Gross *et al.*, 2017; Rahman *et al.*, 2021; Huxley *et al.*, 2023), the role of leaf phenological diversity in driving ecosystem properties remains less clear. Thus, it is essential to not only expand our understanding of intra-site leaf phenological diversity but also explore its potential impact on ecosystem properties and processes.

In this study, we propose that leaf phenological diversity patterns involve the interplay of both biotic and abiotic factors, and could be intrinsically linked to ecosystem property. Based on this premise, this study aims to enhance the characterization of intra-site leaf phenological diversity, investigate its associated abiotic and biotic drivers, and assess its implications to AGB. To achieve these objectives, we address three specific questions: (1) What are the patterns of leaf phenological diversity over a subtropical landscape in South China detected by using high spatial and temporal resolution PlanetScope data? (2) How and to what extent do abiotic and biotic factors affect intra-site leaf phenological diversity? (3) What is the association between intra-site leaf phenological diversity and AGB variability? To answer these questions, we used a subtropical forest site of the Heishiding Provincial Nature Reserve (part of the global ForestGEO network) in Southern China as a testbed, where all the relevant data for these explorations are available. Specifically, we first assessed the intra-site leaf phenological diversity using PlanetScope time-series images, and then conducted the structural equation modeling (SEM) to explore how abiotic and biotic factors directly and indirectly influence intra-site leaf phenological diversity, as well as how intra-site leaf phenological diversity influences AGB.

Materials and Methods

Study site

The study site was located in the Heishiding Provincial Nature Reserve (23°27'N, 111°52'E), Guangdong, China. It is a 50-ha (1000 m × 500 m) subtropical evergreen broadleaved forest plot with rough terrain, ranging in altitude from 435 to 698 m (Supporting Information Fig. S1). The dominant soil type in this area is ferralsol. The climate of the forest plot is characterized as a subtropical moist monsoon climate. It has a mean annual temperature of 19.6°C and a mean annual precipitation of 1797 mm yr⁻¹. The region experiences a 5-month-long dry season from October to February of the following year, during which monthly precipitation is below 100 mm (Fig. S2). The forest plot exhibits rich plant diversity, comprising 273 000 individual trees and shrubs belonging to 245 species, 160 genera, and 71 families, and includes vast variability in leaf phenology ranging from evergreen to semi-deciduous or fully (but briefly) deciduous (Song *et al.*, 2017). Among the canopy species, the most dominant ones include *Altingia chinensis* Oliv. ex Hance (Hamamelidaceae), *Castanopsis carlesii* (Hemsl.) Hayata (Fagaceae), *Artocarpus styracifolius* Pierre (Moraceae), and *Pinus massoniana* Zucc. (Pinaceae) (He & Deane, 2016). For a comprehensive understanding of the floristic composition and structural characteristics of this forest, please refer to He & Deane (2016).

Two kinds of data are available at the study site, including satellite remote sensing data and ground-based observations as shown below. For clarity purposes, a flowchart summarizing the materials and methods is presented in Fig. 1.

Satellite remote sensing data

PlanetScope surface reflectance products Four-band (blue, green, red, and near-infrared (NIR)) PlanetScope data with a 3-m spatial resolution from Planet Labs PBC (San Francisco, CA, USA) were used in this study. The data were accessed through a research and education license from the Planet website (<https://www.planet.com/>). The specific product used was the level 3B surface reflectance product of PlanetScope, which has undergone orthorectification and preprocessing, including geometric, radiometric, and atmospheric corrections (Planet, 2022). The analysis focused on the full-year data of PlanetScope acquired in 2017, aligning with the year of conducting the field inventory at the study site. A total of 31 daily images with low cloud cover (< 40%; predetermined by cloud filter provided by Planet Labs PBC) were selected for analysis (Fig. S3). It is worth noting that optical satellite data, such as PlanetScope, are prone to contamination by clouds and cloud shadows, leading to data missing in the time-series images. Additionally, given that PlanetScope comprises over 180 CubeSats, there can be radiometric inconsistencies across different CubeSat sensors. To address these challenges, a series of preprocessing steps developed by our previous work were applied before calculating leaf phenological diversity metrics (Fig. 1). The preprocessing steps included: (1) masking out clouds and cloud shadows (Wang *et al.*, 2021), (2) recovering the cloud/cloud shadow-induced missing data (J. Wang *et al.*, 2022), and (3) cross-calibrating PlanetScope to MODIS data (Wang *et al.*, 2020).

MODIS surface reflectance products The daily MODIS BRDF/Albedo model parameter product, MCD43A1 (Schaaf *et al.*, 2002), with a spatial resolution of 500 m was used in this study. The MCD43A1 product was accessed from

<http://ladsweb.nascom.nasa.gov/> and data acquisition from January 2016 to December 2021 was selected to aid the cross-calibration of PlanetScope data by aligning it with corresponding MODIS data from the same day and year to minimize cloud/aerosol contaminations. To address the cross-sensor radiometric inconsistency issue and variable view and illumination geometries of PlanetScope observations, a two-step approach was implemented. First, the MODIS data was adjusted to simulate a fixed sun-sensor geometry (nadir view and solar zenith angle of 45°) to generate the BRDF-adjusted MODIS data (Morton *et al.*, 2014; Saleska *et al.*, 2016). Next, the derived BRDF-adjusted MODIS data was used to cross-calibrate PlanetScope data following the approach of Wang *et al.* (2020).

Shuttle Radar Topography Mission (SRTM) data The digital elevation data, SRTM, with a spatial resolution of 30 m were used in this study. The SRTM data was assessed in the year 2000 at <https://earthexplorer.usgs.gov/>. This data was used for georegistration purposes between the SRTM elevation data and the ground-based elevation data. To achieve this, the ground-based elevation data was resampled to a 30-m resolution to match the spatial resolution of the SRTM data. Subsequently, the SRTM data was co-registered with the resampled ground-based elevation data. To assess the accuracy of the georegistration, the Pearson's correlation coefficient was calculated between the two elevation maps. The obtained high correlation coefficient of 0.94 (Fig. S1) indicates a strong agreement between the satellite-based SRTM data and the ground-based elevation data, demonstrating a successful match between the two datasets.

Ground-based observations

Forest inventory data The forest inventory data for the Heishiding forest plot used in this study was collected in 2017. All free-standing individuals were tagged, mapped, measured, and identified to the species following the ForestGEO (Forest Global Earth Observatory; <https://www.forestgeo.si.edu>) protocol. To encompass a broader range of individuals potentially

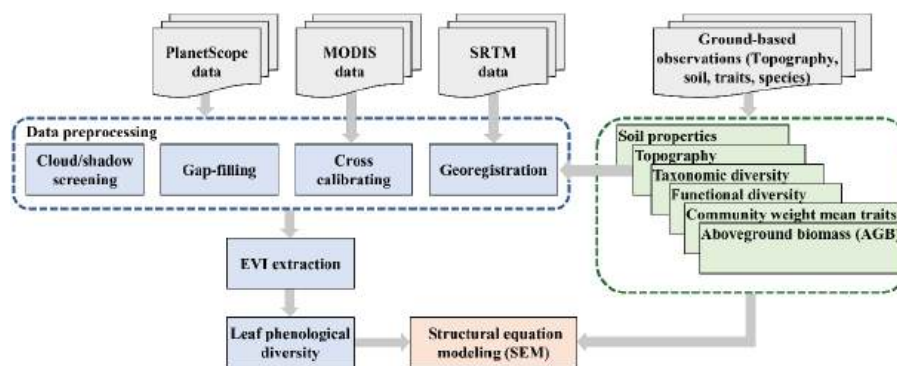


Fig. 1 Flowchart summarizing the materials and methods section. It includes data preprocessing of PlanetScope data, that is cloud and cloud shadow screening, gap-filling, cross-calibrating with MODIS data, and georegistration between Shuttle Radar Topography Mission (SRTM) and ground-based elevation data; extracting soil properties, topography, taxonomic diversity, functional diversity, community weight mean of functional traits, and aboveground biomass (AGB) from ground-based observations; quantifying leaf phenological diversity using the extracted Enhanced Vegetation Index (EVI); and statistical analysis using a structural equation modeling (SEM) approach.

representing canopy tree crowns and aligned with satellite remote sensing, we used a diameter at breast height (DBH) threshold > 5 cm, resulting in a subset of 82 552 individuals for our analysis. With the inventory data, several taxonomic metrics highlighting different aspects of diversity were calculated. These metrics include species richness (the number of species within a given area), Simpson diversity (Simpson, 1949), and species evenness (Pielou, 1966). Additionally, the aboveground biomass (AGB) of individual trees was calculated using a commonly used allometric equation (Table S1; Chave *et al.*, 2014).

Functional trait data Functional traits were measured on a total of 205 randomly distributed species within the Heishiding forest plot. The details regarding how these traits were sampled and measured refer to He & Deane (2016) and He & Biswas (2019) for leaf and stem traits, Luo *et al.* (2021) for root traits, and Feng *et al.* (2018) for photosynthesis traits. Twelve functional traits were selected for analysis based on their recognized importance in vegetation survival and growth (Table S2; Wright *et al.*, 2010; Adler *et al.*, 2014; Ordway *et al.*, 2022). The selected traits are as follows: leaf area (LA), specific leaf area (SLA), leaf dry matter content (LDMC), leaf nitrogen concentration (LNC), leaf phosphorus concentration (LPC), wood density (WD), wood dry matter content (WDMC), specific root length (SRL), average root diameter (ARD), semi-light saturation point of Chl fluorescence (Chl_i), maximum photosynthesis rate (PSN_{Am}), and apparent quantum yield (PSN_b). With these traits, we standardized all the traits and calculated the functional composition of each trait using the community weight mean (CWM) as well as several functional diversity metrics using Rao's quadratic entropy (FRao), richness (Fric), divergence (Fdiv), evenness (Feve), and dispersion (Fdis) (Table S1; Rao, 1982; Cornwell *et al.*, 2006; Lavorel *et al.*, 2008; Villéger *et al.*, 2008; Laliberté & Legendre, 2010).

Soil property data A total of nine soil properties were measured from 625 surface soil samples (0–10 cm depth) that were randomly collected within the Heishiding forest plot. The nine soil properties include soil organic matter (S.SOM), soil acidity (S.pH), electrical conductivity (S.EC), available nitrogen content (S.AN), available phosphorus content (S.AP), available potassium content (S.AK), total nitrogen content (S.TN), total phosphorus content (S.TP), and total potassium content (S.TK). For more details regarding how these soil properties were sampled and measured please refer to He *et al.* (2018) and Y. Li *et al.* (2021). The measured soil properties were interpolated into regular grids using the Ordinary Kriging method (Matheron, 1967).

Topography data The local elevation data across the study area was surveyed in 2011 using a combination of Global Positioning System (GPS) and Real-Time Kinematic (RTK) techniques (Lin, 2017). The elevation data obtained from the survey were interpolated into regular grids using the Ordinary Kriging method. Additionally, two other topographic attributes, namely aspect, and slope were derived from the interpolated elevation data. The calculation of aspect and slope was carried out using

the 'gradientm.m' package in MATLAB R2022a (MathWorks, Natick, MA, USA).

Data analysis

Quantifying leaf phenological diversity In this study, the leaf phenological diversity was assessed using the Enhanced Vegetation Index (EVI) extracted from the time series of PlanetScope images. We followed Sánchez-Ochoa *et al.* (2022), who proposed a Hill numbers-based phenological diversity metric (Table S1), which quantifies the degree of overlap in phenology curves between pairs of individuals or species within a community, as a measure of phenological diversity. Building upon the principles of these metrics, it is possible to first adopt a Hill numbers-like approach to compare any pair of pixels within a spatial grid (e.g. 50 m $\times$ 50 m), and then average the pairwise Hill numbers-like metrics to assess leaf phenological diversity at the grid level. Specifically, we calculated this metric with the following steps: (1) determined the integration areas (i.e. the extent of vegetation presence) of any pair of two EVI time series within each spatial grid and calculated the intersection and union of these integration areas separately; (2) calculated the ratio of the intersection area to the union area for each pair, representing the synchrony rate between the pairwise EVI time series; (3) averaged these ratios across all matched pairs within the spatial grid, providing an assessment of the degree of synchrony in vegetation dynamics across the grid; (4) used the asynchrony rate, namely the one minus synchrony rate, as the metric of leaf phenological diversity to characterize the difference of phenological strategies.

Statistical analysis To examine the relationships of various variables, both abiotic (topographic attributes and soil properties) and biotic factors (taxonomic diversity metrics, community weight mean of functional traits, and functional diversity metrics), with leaf phenological diversity, and the potential impact of leaf phenological diversity on AGB, the structural equation modeling (SEM) approach (Bollen, 1989) was employed. SEM is a statistical method used to analyze complex causal networks and evaluate hypotheses about cause-effect relationships among predictor and response variables (Grace *et al.*, 2010). To assess the linkages between the predictor and response variables in our study, we constructed a conceptual SEM model (Fig. S4) based on several key hypotheses: (1) both abiotic (topography and soil) and biotic factors (taxonomic diversity, functional diversity, and community weight mean of functional traits) can directly influence leaf phenological diversity and AGB. The community weight mean of functional traits were categorized into morphological traits (CWMMT) and physiological traits (CWMPPT) using principal component analysis (PCA) (Fig. S5). Both sets of functional traits may exert an influence on AGB and leaf phenological diversity (Chave *et al.*, 2005, 2014; Blumenthal *et al.*, 2020; González-Rebeles *et al.*, 2021). (2) abiotic factors can indirectly influence leaf phenological diversity and AGB through biotic factors; (3) taxonomic diversity can indirectly influence leaf phenological diversity and AGB through other

biotic factors; (4) topography can indirectly influence leaf phenological diversity and AGB through soil properties; (5) leaf phenological diversity can directly influence AGB (P. Li *et al.*, 2021; Zohner *et al.*, 2021). The fitness of the SEM model was evaluated using four indices: chi-square statistic (χ^2), comparative fit index (CFI), Tucker-Lewis index (TLI), and standardized root mean square residual (SRMR) (Tucker & Lewis, 1973; Bentler, 1990; Steiger, 1990). A good-fitting model typically exhibits a non-significant χ^2 ($P > 0.05$), CFI and TLI > 0.95 , and SRMR < 0.08 (Hu & Bentler, 1999). Notably, the LAVAAN package (Rosseel, 2012) in R v.4.2.2 (R Development Core Team, 2022) was used for implementing and evaluating the SEM models. These calculations were carried out in a grid size of $50 \text{ m} \times 50 \text{ m}$, which aims to minimize geometric errors between remote sensing images and ground-based observations. The results obtained using a smaller spatial grid size of $20 \text{ m} \times 20 \text{ m}$ are comparable and are provided in Fig. S6, but not presented in the manuscript due to space constraints.

Results

PlanetScope-derived intra-site leaf phenological diversity

We analyzed the spatial pattern of leaf phenological diversity in the forest plot using the asynchrony rate metric across four spatial grids of $10 \text{ m} \times 10 \text{ m}$, $20 \text{ m} \times 20 \text{ m}$, $50 \text{ m} \times 50 \text{ m}$, and $100 \text{ m} \times 100 \text{ m}$ (Fig. 2). The results revealed considerable spatial heterogeneity in asynchrony rates throughout the forest plot landscape and across various spatial scales. Generally, the western region exhibited a lower asynchrony rate compared to the eastern region, and this pattern persisted consistently across all spatial scales. However, with an increase in spatial scale, numerous fine-scale phenological differences were inevitably obscured, leading to less pronounced contrast in leaf phenological diversity throughout the forest landscape. Moreover, the upper and lower boundaries of the asynchrony rate elevated, and the histogram distribution of asynchrony rates shifted toward higher values as the spatial scale increased (Fig. S7). This shift in range suggests

an increased variability with the spatial scale. Taken together, our analysis of asynchrony rates unveiled considerable intra-site leaf phenological diversity in the subtropical forest landscape, and the observed intra-site leaf phenological diversity varied with the spatial scale.

To further analyze the intra-site phenological variability across seasons throughout the entire year, we analyzed the coefficient of variation (CV) and mean of monthly EVI at a spatial grid of $50 \text{ m} \times 50 \text{ m}$ (comparable results across other spatial grids shown in Fig. S8). The CV of monthly EVI (variability; Fig. 3a) exhibited a U-shaped curve for the entire landscape, with a decreasing trend from January to July and an increasing trend from July to December, while the mean of monthly EVI (greenness; Fig. 3b) illustrates an overall opposite seasonality to the CV for the entire landscape, with lower values from January to March, reaching the maximum value in July, and then gradually decreasing until December. The results indicate that phenological variability exhibited a distinct seasonality, with greater variability observed during the dry season (October to February) compared to the wet season (March to September), highlighting that the intra-site leaf phenological diversity in subtropical forests is primarily due to large spectral diversity in the early and late growing seasons, corresponding to plants exhibiting substantial variability in leaf abscission and flushing events.

The drivers underlying leaf phenological diversity and its impact on AGB

We next employed structural equation modeling (SEM) to evaluate the causal relationships between intra-site leaf phenological diversity and a variety of abiotic factors (topographic attributes and soil properties) and biotic factors (taxonomic diversity, functional diversity, functional trait composition) as well as AGB at a spatial grid of $50 \text{ m} \times 50 \text{ m}$ (Fig. 4; Table 1). The derived SEM model revealed direct and indirect paths from biotic and abiotic factors to the leaf phenological diversity metric. Functional community weighted mean of morphological traits (CWMMT; WD and SRL) and physiological traits (CWMPT;

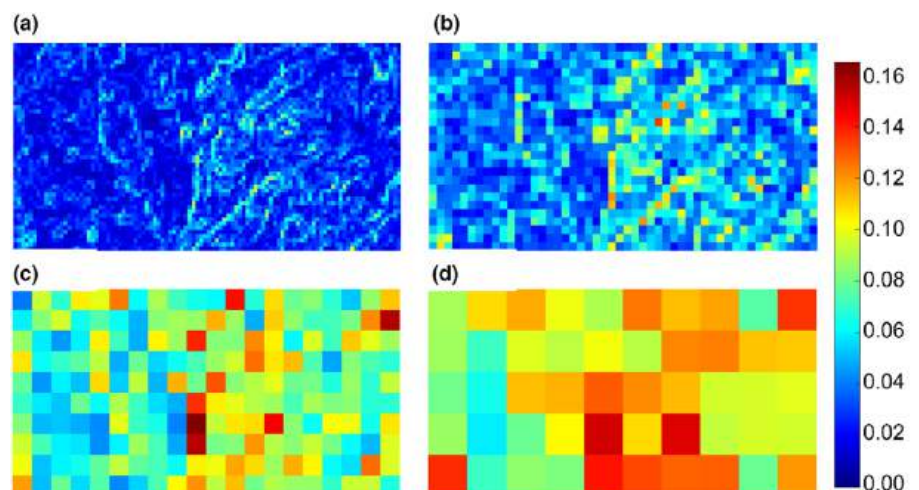


Fig. 2 Spatial pattern of leaf phenological diversity (asynchrony rate) across four different spatial grids of (a) $10 \text{ m} \times 10 \text{ m}$, (b) $20 \text{ m} \times 20 \text{ m}$, (c) $50 \text{ m} \times 50 \text{ m}$, and (d) $100 \text{ m} \times 100 \text{ m}$ in the Heishiding forest plot. The color bar represents the asynchrony rate value within an image pixel.

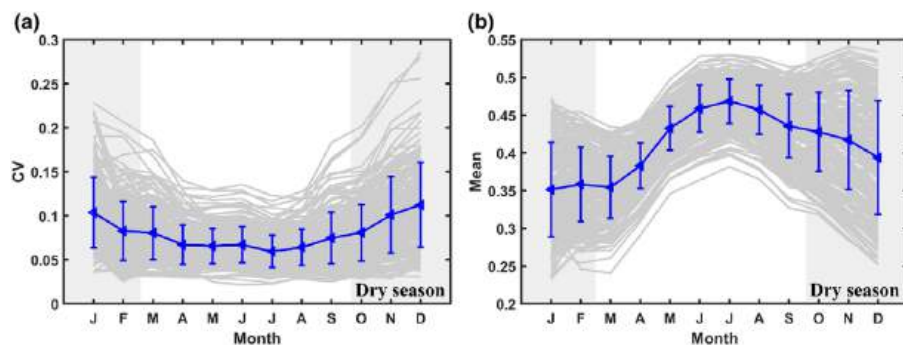


Fig. 3 Seasonality of phenological variability at a spatial grid of 50 m × 50 m in the Heishiding forest plot. (a) Coefficient of variation (CV) and (b) mean of monthly Enhanced Vegetation Index (EVI) were calculated for each grid. Gray lines indicate the seasonality of a total of 200 grids, blue lines indicate the mean value, and error bars indicate one SD. Shading indicates the dry season (< 100 mm of monthly precipitation).

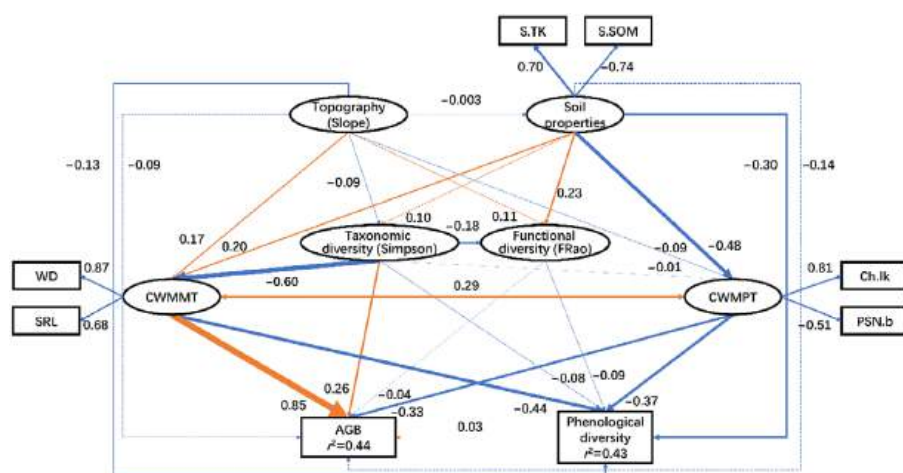


Fig. 4 The SEM model graph for understanding the leaf phenological diversity association with abiotic (topographic attributes and soil properties) and biotic factors (taxonomic diversity, functional diversity, community weighted mean of functional traits (morphological and physiological traits; CWMMT and CWMPT), and aboveground biomass (AGB)) at a spatial grid of 50 m × 50 m. The SEM model had a nonsignificant Chi-square P -value of 0.105, comparative fit index (CFI) value of 0.982, Tucker-Lewis index (TLI) value of 0.955, and standardized root mean square residual (SRMR) of 0.043. The SEM model explains 43% of the variance in leaf phenological diversity ($r^2 = 0.43$) and 44% of the variance in AGB ($r^2 = 0.44$), respectively. Orange and blue lines represent positive and negative influences, respectively. The values besides the paths are the standardized path coefficients (β), with solid and dash lines shown for significant ($P < 0.05$) and non-significant influences ($P > 0.05$). Ch.lk, semi-light saturation point of Chl fluorescence; FRao, functional diversity metric using Rao's quadratic entropy; PSN.b, apparent quantum yield; S.SOM, soil organic matter; S.TK, soil property of total potassium content; Simpson, Simpson diversity; Slope, topographic attribute of slope; SRL, specific root length; WD, wood density. Squares indicate observed variables; circles indicate latent variables.

Ch.lk and PSN.b) showed strong direct influences on leaf phenological diversity ($\beta = -0.44$ and -0.37 ; Fig. 4), while functional diversity (FRao) and taxonomic diversity (Simpson) did not show significant direct influences on leaf phenological diversity ($P > 0.05$). Soil properties (S.TK and S.SOM) and topographic attributes (slope) exhibited considerable direct influences on leaf phenological diversity ($\beta = -0.30$ and -0.13 ; Fig. 4). Apart from the direct influences of different biotic and abiotic factors on leaf phenological diversity, we found some of these factors also indirectly influenced leaf phenological diversity (Fig. 4; Table 1). For instance, soil properties displayed indirect influences mediated through influences on CWMPT ($\beta = 0.180$; Table 1); taxonomic diversity, soil properties, and topographic attributes, respectively, displayed indirect influences mediated through influence on CWMMT ($\beta = 0.261$, -0.086 , and -0.076 ; Table 1). Collectively, the total influences of topography, soil, and taxonomic diversity on leaf phenological diversity

are quite comparable ($\beta = -0.200$, -0.210 , and 0.198 ; $P < 0.05$; Table 1).

We also observed strong direct paths from CWMMT, CWMPT, and taxonomic diversity to AGB with standardized paths of 0.85, -0.33 , and 0.26, respectively, and weak direct paths from functional diversity, soil properties, and slope to AGB ($\beta = -0.04$, -0.14 , and -0.09). In addition, a significant path from leaf phenological diversity to AGB was not found, although there was a significant correlation between AGB and leaf phenological diversity ($r = -0.24$, $P < 0.05$; Fig. S9). These findings highlight the complex interrelationships between biotic and abiotic factors and their collective influence on leaf phenological diversity. Taken together, these results demonstrate that soil properties and functional trait composition directly influence leaf phenological diversity, while topographic attributes, soil properties, and taxonomic diversity indirectly influence leaf phenological diversity via mediation through functional trait composition

Table 1 Indirect and total standardized influence of topography, soil, taxonomic diversity, functional diversity, and community weighted mean of functional traits (morphological and physiological traits; CWMMT and CWMPT) on leaf phenological diversity and AGB.

Indirect and total influence pathways	Leaf phenological diversity		AGB	
	Standard path coefficient	P-value	Standard path coefficient	P-value
Indirect influence of topography through soil	0.001	0.972	0.000	0.972
Indirect influence of topography through taxonomic diversity	0.007	0.476	−0.023	0.267
Indirect influence of topography through soil, taxonomic diversity	0.000	0.972	−0.000	0.972
Indirect influence of topography through functional diversity	−0.010	0.343	−0.004	0.660
Indirect influence of topography through soil, functional diversity	0.000	0.972	0.000	0.972
Indirect influence of topography through taxonomic diversity, functional diversity	−0.001	0.409	−0.001	0.671
Indirect influence of topography through soil, taxonomic diversity, functional diversity	−0.000	0.972	−0.000	0.972
Indirect influence of topography through CWMMT	−0.076	0.041	0.148	0.021
Indirect influence of topography through soil, CWMMT	0.000	0.972	−0.001	0.972
Indirect influence of topography through taxonomic diversity, CWMMT	−0.023	0.243	0.045	0.225
Indirect influence of topography through soil, taxonomic diversity, CWMMT	−0.000	0.972	0.000	0.972
Indirect influence of topography through CWMPT	0.032	0.323	0.028	0.341
Indirect influence of topography through soil, CWMPT	−0.001	0.972	−0.000	0.972
Indirect influence of topography through taxonomic diversity, CWMPT	−0.000	0.931	−0.000	0.931
Indirect influence of topography through soil, taxonomic diversity, CWMPT	−0.000	0.974	−0.000	0.974
Indirect influence of topography through leaf phenological diversity			−0.007	0.752
Indirect influence of soil through taxonomic diversity	−0.008	0.470	0.026	0.309
Indirect influence of soil through functional diversity	−0.021	0.261	−0.009	0.650
Indirect influence of soil through taxonomic diversity and functional diversity	0.002	0.437	0.001	0.676
Indirect influence of soil through CWMMT	−0.086	0.043	0.168	0.035
Indirect influence of soil through taxonomic diversity and CWMMT	0.026	0.285	−0.050	0.271
Indirect influence of soil through CWMPT	0.180	0.024	0.158	0.053
Indirect influence of soil through taxonomic diversity and CWMPT	0.000	0.930	0.000	0.930
Indirect influence of soil through leaf phenological diversity			−0.007	0.751
Indirect influence of taxonomic diversity through functional diversity	0.017	0.276	0.007	0.652
Indirect influence of taxonomic diversity through CWMMT	0.261	0.003	−0.508	0.000
Indirect influence of taxonomic diversity through CWMPT	0.003	0.931	0.002	0.931
Indirect influence of taxonomic diversity through leaf phenological diversity			0.007	0.752
Indirect influence of functional diversity through leaf phenological diversity			−0.003	0.741
Indirect influence of CWMMT through leaf phenological diversity			−0.014	0.762
Indirect influence of CWMPT through leaf phenological diversity			−0.012	0.741
Total influence of topography	−0.200	0.004	0.098	0.164
Total influence of soil	−0.210	0.014	0.143	0.096
Total influence of taxonomic diversity	0.198	0.004	−0.232	0.001
Total influence of functional diversity	−0.093	0.232	−0.043	0.624
Total influence of CWMMT	−0.436	0.002	0.833	0.000
Total influence of CWMPT	−0.371	0.006	−0.339	0.014

Four distinct shades of grey indicate the indirect influences of topography, soil, taxonomic diversity, and the other factors (functional diversity and functional trait composition), respectively.

(Table 1). Besides, leaf phenological diversity could not directly influence AGB.

Discussion

Understanding the patterns, drivers, and implications of intra-site leaf phenological diversity is important for ecological studies that explore ecological scaling processes across a wide range of dynamics from single tree crowns up to forest community and diverse ecosystems and the modelling of forest ecosystem as a whole sensitivity response to climate change which originates from diversity of individual-scale climate sensitivity response. However, this understanding is often limited by the coarse spatial resolution of conventional satellite data. PlanetScope CubeSats, with their 3-m spatial resolution, provide an unprecedented opportunity to monitor intra-site leaf phenological diversity, but their potential has not been fully explored. In this study, we characterized the leaf phenological diversity of a 50-ha subtropical forest plot using PlanetScope data. Our results revealed that intra-site leaf phenological diversity in the subtropical forest is considerable and directly and indirectly influenced by both abiotic and biotic factors. We also found there is no direct pathway from leaf phenological diversity to AGB (Fig. 4). These findings provide valuable insights into the underlying mechanisms that govern intra-site leaf phenological diversity and their implications for ecosystem properties and services.

Considerable intra-site leaf phenological diversity within a subtropical evergreen forest plot

The study revealed substantial intra-site leaf phenological diversity. This intra-site leaf phenological diversity is primarily because of the higher spectral diversity in early and late growing seasons (Fig. 3), suggesting that plants evolve in diverse strategies to adapt to sunlight (early growing season) or water stress (late growing season). This finding is consistent with the literature that shows quite diverse tropical/subtropical tree species belonging to different functional groups coexist in the same plant community (Reich, 1995; Lopes *et al.*, 2016; Lee *et al.*, 2023). Meanwhile, this is also an important addition to the apparent ecosystem greenness as commonly used in the literature previously (Morton *et al.*, 2014), suggesting that the apparent ecosystem-scale greenness is a result of still considerable fine-scale leaf phenological diversity (Lopes *et al.*, 2016; Wu *et al.*, 2016). However, relative to temperate forest landscapes, the variability is still low (remaining dominated by the evergreen trees, showing by asynchrony rate, for example lower than 0.15; Fig. 2), suggesting that subtropical landscape is still more evergreen all year round, especially in late/early growing season compared to temperate forest landscape. Collectively, these suggest that there is indeed considerable intra-site leaf phenological diversity even in the subtropical evergreen forest, and our proposed metric is sufficient to characterize this dimension of diversity. Meanwhile, it also highlights the importance of using the high-resolution remote sensing technique, for example the PlanetScope satellite of 3-m resolution as

used here, for characterizing the intra-site leaf phenological diversity and is able to expand it to even larger forest landscapes.

Underlying mechanism of the relationship between biotic and abiotic factors, leaf phenological diversity, and productivity

Intra-site leaf phenological diversity has long been hypothesized subject to the underlying abiotic and biotic constraints, and here we applied a structural equation modeling approach to confirm this hypothesis. We observed that biotic factors exert a more substantial influence than abiotic factors, and the interaction between biotic and abiotic factors also plays a considerable role (Fig. 4; Table 1). This is because, within a confined subtropical forest landscape with a relatively uniform background climate, the impact of abiotic factors could be minimized, while biotic factors emerge as the dominant regulators of intra-site leaf phenological diversity. Specifically, biotic factors of morphological traits (WD and SRL) and leaf photosynthesis traits (Ch.lk and PSN.b) exhibited direct control on intra-site leaf phenological diversity ($\beta = -0.44$ vs -0.37 ; Fig. 4). This indicated that water and resource use efficiency and photosynthesis reactivity could influence phenology synchronizing, pointing to the paramount importance of life-history strategies and adaptations (Fu *et al.*, 2012; Lasky *et al.*, 2016). Particularly, the dominant species with low values of morphological and physiological traits enhance leaf phenological diversity, demonstrating that the selection effect hypothesis is the main driver of leaf phenological diversity in our studied subtropical forest landscape. Meanwhile, soil properties (S.TK and S.SOM), which are indicators of soil nutrient availability (John *et al.*, 2007; Aguirre-Gutiérrez *et al.*, 2020), exerted a comparable influence on intra-site leaf phenological diversity ($\beta = -0.30$; Fig. 4). The underlying reason is that soil properties are significantly correlated with phenostages (Ramos *et al.*, 2014; Du *et al.*, 2019). Therefore, higher soil nutrients generate a more consistent phenology rhythm. Furthermore, we also observed that biotic factors were induced by the variation in soil properties, topographic attributes, and taxonomic diversity. Soil properties and the topographic attributes of slope contribute to the distribution of solar radiation, water availability, and energy exchange across the landscape (Gao *et al.*, 2019; Lyu & Alexander, 2022). This can directly influence plant photosynthetic capacity, competition, and community assembly (Jucker *et al.*, 2018; Joswig *et al.*, 2022), thereby indirectly affecting leaf phenological diversity. Although no direct path from taxonomic diversity to leaf phenological diversity, different species could influence functional traits (Bongers *et al.*, 2021) thereby influencing leaf phenological diversity. Together, the spatial pattern of leaf phenological diversity is directly and indirectly linked to different ecological strategies related to light and water use efficiency, soil moisture and nutrient absorption, or sensitivity to microclimates (Tian *et al.*, 2018). Despite our abiotic and biotic factors collectively explaining 43% of intra-site phenological variability (Fig. 4), there remains *c.* 57% variance unexplained. This could be attributed to observation uncertainties and errors, or undiscovered processes, such as

interactions among plant species and their interactions with organisms from other trophic levels like herbivores, and soil microbes including mycorrhizal fungi (Kharouba & Wolkovich, 2020; O'Brien *et al.*, 2021).

Previous studies have highlighted the relationships between biotic/abiotic factors and the variability in phenostage timing (Liu *et al.*, 2021; W. Wang *et al.*, 2022). Similar to the existing literature, biotic factors such as functional traits (e.g. PSN.Am and WD; McKown *et al.*, 2013; Liu *et al.*, 2021) and species diversity (e.g. richness; W. Wang *et al.*, 2022), as well as abiotic factors like topography (e.g. slope and aspect; W. Wang *et al.*, 2022) and soil properties (e.g. S.TP; Du *et al.*, 2019), were found to be correlated with variations in phenostages such as leafing, senescing, flowering, fruiting and so on. However, the relationship between functional diversity and phenostages has rarely been explored. Meanwhile, only a few studies have explored the mechanisms of how these biotic and abiotic factors determine leaf phenological diversity (Elmore *et al.*, 2012; Adams *et al.*, 2021), especially in subtropical forests where ground-based observations are lacking. For example, Adams *et al.* (2021) demonstrated that topographic attributes (elevation, slope, and aspect), stem traits and species composition emerged as dominant drivers of the variability in spring phenology using a SEM model in a temperate forest landscape. However, soil properties and their interaction with biotic factors were not accounted for in the SEM model. Therefore, our study provides comprehensive and direct evidence of how biotic and abiotic factors collectively shape intra-site leaf phenological diversity in a subtropical forest landscape and could improve our understanding of the ecophysiological fundamental mechanisms driving spatial patterns of leaf phenological diversity in large-scale forest landscapes in the future.

We also found that the direct influence on AGB is primarily driven by functional trait composition and taxonomic diversity rather than leaf phenological diversity (Fig. 4), although a significant relationship was identified between them with higher leaf phenological diversity simulating lower AGB ($r = -0.24$, $P < 0.05$; Fig. S9). This suggests a mutual regulation of biomass storage by both the selection effect hypothesis and niche complementarity hypothesis (Tilman *et al.*, 1997; Rahman *et al.*, 2021), with the former playing a dominant role. The negative correlation between leaf phenological diversity and AGB may be attributed to a community exhibiting a consistent phenological rhythm, characterized by leaf dropping or exchange during the dry season and retaining leaves in the wet season, is more likely to efficiently utilize essential resources such as solar radiation, water, and nutrients, consequently fostering increased biomass production. Numerous studies have reported that shifts in leaf phenology influence ecosystem productivity across multiple scales from species to community level (Zhang *et al.*, 2020; Kim *et al.*, 2021). However, the specific influence from leaf phenology to productivity comprehensively considering both abiotic and biotic factors remain unclear (P. Li *et al.*, 2021), especially in subtropical forests. We thus highlight that intra-site leaf phenological diversity does not exert direct control over the variations in AGB, but it may serve as an important indicator of diversity metrics that reflect certain aspects of ecosystem properties. In addition to the

examined AGB, other dimensions of ecosystem processes, functioning, and stability also merit future in-depth exploration on a larger spatial scale, incorporating our proposed leaf phenological diversity as well (Bartomeus *et al.*, 2013; Kharouba *et al.*, 2018; Kharouba & Wolkovich, 2020).

Caveats and future directions

Our study identifies three crucial next steps that need to be considered for future advances. First, while PlanetScope data with a 3-m spatial resolution facilitates intra-site leaf phenological diversity monitoring, there are challenges in evaluating the intra-site phenological patterns of specific individual tree crowns or species without integrating finer-scale remote sensing data. This challenge arises in large part from the geolocation discrepancy between satellite- and ground-based data sources (Richardson *et al.*, 2018b). Additionally, in dense subtropical forests, the presence of adjacent and understory vegetation can introduce uncertainty in the PlanetScope-derived phenology for target canopy tree crowns (Wu *et al.*, 2021). Since the real-time kinematic (RTK) positioning, proximate remote sensing, and high-resolution LiDAR approaches can accurately identify the geolocation and vertical structure of each individual tree crown or species, aiding in differentiating target canopy trees from adjacent canopy/understory trees, we recommend integrating PlanetScope data with field or drone surveys for a comprehensive evaluation of individual/species phenology in future attempts. Second, while various biotic factors (taxonomic diversity, functional diversity, and functional trait composition) have been comprehensively assessed, the presence of large tree crowns and canopy-dominant pine species could also influence leaf phenological diversity. To investigate such impacts, we conducted an additional analysis using tree size distributions and the proportion of canopy-dominant pine species (e.g. *Pinus massoniana*) from the plot data to assess their relationships with leaf phenological diversity. Our findings in Fig. S10 reveal a significant correlation between the average DBH and leaf phenological diversity ($r = -0.31$; $P < 0.001$), demonstrating that pixels dominated by large crowns exhibit significantly lower leaf phenological diversity, while relationships with other factors remain nonsignificant. However, when average DBH was included in the SEM model, its influence on leaf phenological diversity was deemed nonsignificant (Fig. S11), suggesting that its influence may be mediated or overshadowed by other factors in the broader context of ecosystem functioning. Further analysis integrating forest structure, such as canopy complexity, or environmental variables like microclimate conditions, might provide a more comprehensive understanding of the influence of tree size distributions on leaf phenological diversity. Third, although the intra-site leaf phenological diversity has been comprehensively assessed in the 50-ha forest plot, how the biotic and abiotic factors would drive the leaf phenological diversity in a border spatial landscape beyond the current plot remains unknown. The integration of high-resolution satellite data, like PlanetScope data, with larger-scale environmental data, functional trait data, and species inventory data could help to understand underlying mechanisms and

implications of leaf phenological diversity in a border spatial landscape. Such integration holds great potential for improving the understanding and modeling of leaf phenological diversity patterns, as well as the associations between leaf phenological diversity and ecosystem function and stability, at broader spatial scales.

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Competing interests

None declared.

Author contributions

JW conceptualized and designed the research. JW performed the data analysis with the help from YL and MMR. BL and YL were responsible for data collection. CC, JW, YL, MMR, ZY, GS, and YZ participated in the result interpretation and rigorousness evaluation of the method. JW drafted the manuscript, CC, JW and all other authors contributed to the manuscript editing and revision. The co-corresponding authors contributed equally to this work.

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Data availability

The PlanetScope product can be accessed via <https://www.planet.com/>, the MODIS product is available at <http://adsweb.nascom.nasa.gov/>, SRTM data can be obtained from <https://earthexplorer.usgs.gov/>. For specific functional trait and environmental data pertaining to the Heishiding forest plot, please refer

to the TRY Plant Trait Database: <https://www.try-db.org/TryWeb/Data.php#68>.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Elevation map derived from ground survey and Shuttle Radar Topography Mission satellite data.

Fig. S2 The mean seasonality of rainfall from 2000 to 2019 at the study site.

Fig. S3 Percentage of data missing of PlanetScope images analyzed in this study in 2017 at the Heishiding site.

Fig. S4 Conceptual structural equation modeling model graph for understanding the leaf phenological diversity association with

abiotic (topographic attributes and soil properties) and biotic factors (taxonomic diversity, functional diversity, community weighted mean of morphological and physiological traits (CWMMT and CWMPT), and aboveground biomass).

Fig. S5 Principal component analyses of 12 traits that were measured in the Heishiding forest plot.

Fig. S6 The structural equation modeling model graph for understanding the leaf phenological diversity association with abiotic (topographic attributes and soil properties) and biotic factors (taxonomic diversity, functional diversity, community weighted mean of morphological and physiological traits (CWMMT and CWMPT), and aboveground biomass) at a spatial grid of 20 m $\times$ 20 m.

Fig. S7 Frequency distribution of leaf phenological diversity (asynchrony rate) across four different spatial grids of 10 m $\times$ 10 m ($n = 5000$), 20 m $\times$ 20 m ($n = 1250$), 50 m $\times$ 50 m ($n = 200$), and 100 m $\times$ 100 m ($n = 50$).

Fig. S8 Seasonality of phenological variability at a spatial grid of 10 m $\times$ 10 m, 20 m $\times$ 20 m, and 100 m $\times$ 100 m in the Heishiding forest plot.

Fig. S9 The linear relationship between aboveground biomass and the leaf phenological diversity (asynchrony rate) at a spatial grid of 50 m $\times$ 50 m in the Heishiding forest plot.

Fig. S10 The linear relationship between the leaf phenological diversity (asynchrony rate) and the mean value of diameter at breast height, the number of large tree crowns, the proportion of canopy-dominant pine species, and the number of canopy-dominant species at a spatial grid of 50 m $\times$ 50 m in the Heishiding forest plot.

Fig. S11 The structural equation modeling model graph for understanding the leaf phenological diversity association with abiotic (topographic attributes and soil properties) and biotic factors (taxonomic diversity, functional diversity, community weighted mean of morphological and physiological traits (CWMMT and CWMPT), diameter at breast height, and aboveground biomass).

Table S1 A list of all the equations used in this study.

Table S2 Plant functional traits and their ecological significance.

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Soil nutrients and leaf area index interact with species and structural diversity to buffer mangrove productivity against salinity

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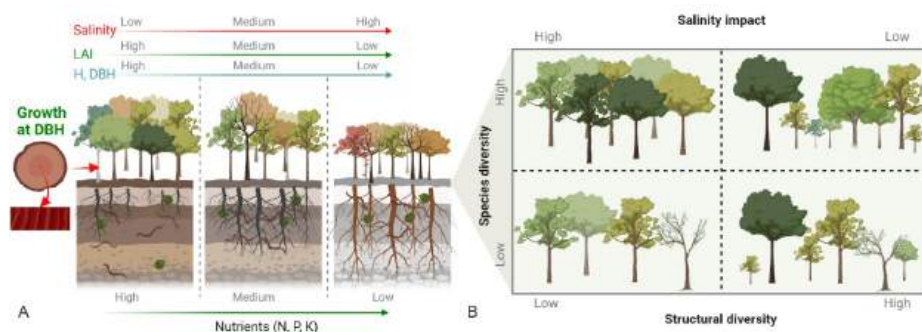
ABSTRACT

• The comparative roles of species and structural diversity in mitigating the impacts of salinity were evaluated.

• Greater diversity contributes to mitigating salinity impacts by interacting with nutrients and leaf functional trait.

• Nutrients and leaf functional trait (leaf area) significantly influenced the effects of salinity on mangrove growth.

• Future growth models should incorporate functional traits and nutrient availability to improve predictions of mangrove growth under saline conditions.



Mangroves show a biogenic response to adjust sea-level rise by accumulating sediment and carbon (vertical soil accretion), reshaping their structure and composition to minimize the effects. Additionally, the often-overlooked factors of soil nutrient availability, functional traits, and stand structure can alter the mangrove diversity-salinity-productivity link. However, how these multiple drivers interplay to maintain growth against salinity still needs to be better understood. Considering all these, we answered two questions: (QI) How do species diversity and structural heterogeneity modulate growth vs. salinity relationships? (QII) To what extent can structural heterogeneity and species diversity create optimal conditions by minimizing the adverse effects of salinity while concurrently maximizing forest growth? To comprehensively understand the interplay between structural and species diversity, nutrient availability, functional traits, and rising salinity, we examined a dataset from 60 permanent plots established in the Sundarbans mangrove forest in Bangladesh. Our results indicated that species diversity less directly contributed to forest growth than structural heterogeneity, nutrient availability (N, P, and K), and leaf area index. While forest structural and species diversity alone is unlikely to optimize growth, incorporating nutrients into the models showed a slight improvement in buffering against salinity. However, when nutrients were combined with the leaf area index, the models indicated a much stronger enhancement in the forest's resilience to salinity through interactions with these factors, allowing continued growth. In conclusion, our study highlights the relative contributions of species and structural diversity to mangrove growth under stress and the potential roles of nutrients and functional traits. These findings are valuable for forest growth modelling, informing conservation and management strategies for mangroves, particularly in coastal plantations facing environmental changes.

Keywords mangrove productivity, structure-functions relation, competition, density, sediment nutrients, functional traits, climate change

1 Introduction

Mangrove forests, situated at the intersection of terrestrial and marine environments, hold a significant role in coastal ecosystems (Alongi, 2008). They offer many ecosystem services, including livelihoods to millions of coastal people, shoreline stabilization, protection of coastal communities from tidal surges and cyclones, habitats for various species, and sequester huge amounts of carbon per year (210 Tg C yr⁻¹) (Donato et al., 2011; Alongi, 2012, 2014; Yoshikai et al., 2021; Ahmed et al., 2023). However, these vital ecosystems are facing unprecedented challenges due to the impacts of climate change, in particular, the rising level of sea level and salinity and shifts in hydrological patterns (Lovelock et al., 2015; Singh, 2020). Global sea level rise, attributed to climate change, has resulted in extensive changes to mangrove habitats and coastal alterations (Sofawi et al., 2017). This phenomenon not only threatens water availability necessary for mangrove survival but also triggers complex ecological responses that intertwine with other intrinsic factors governing the growth of mangrove forests by reducing structure and changing species composition, which is now well recognized in the Indo-Pacific regions (Lovelock et al., 2015; Yoshikai et al., 2021; Perri et al., 2023). Furthermore, the intensity of salinity is projected to increase and is identified as one of the most critical factors that may retard mangrove functional and physiological activities by reducing seed germination (Mittra et al., 2010), leaf size and shape (Mollick et al., 2021), fine root production (Ahmed et al., 2021), tree height (Perri et al., 2023), wood density (Rahman et al., 2021a), and site quality (Ahmed et al., 2022), for example. In response to climate change-induced sea level rise, mangroves adopt a biogenic approach by accumulating organic carbon and sediment (Saintilan et al., 2023). Although few studies explain species diversity through facilitation may help to reduce the salinity impacts (Huxham et al., 2010). However, less is understood regarding the biogenic response of mangroves to salinity, mainly how this response may involve changes in structural and species diversity combined with nutrients and functional traits for maintaining growth.

Biodiversity-productivity relationship is often observed in both tropical and temperate forests, as demonstrated by various studies (Vanelander et al., 2009; Williams et al., 2017; Ali et al., 2019; Park et al., 2019; Zheng et al., 2019). The finding of this relationship plays a critical role in understanding the growth patterns against stress as tree productivity is mainly dependent on the structural and species diversity (Ahmed et al., 2023; Astigarraga et al., 2023; Pretzsch et al., 2023), while other driver effects can be manifested into accumulated tree structure. Forest stands

with higher species and structural diversity facilitating each other through complementarity (Huxham et al., 2010; Jucker et al., 2015). Complementarity occurs when different species occupy different ecological niches, enabling efficient resource utilization by plants and increasing aboveground biomass (Paquette and Messier, 2011; Forrester et al., 2013; Fotis et al., 2018). This is achieved by reducing inter-specific competition between species via stratification in crown and root areas (Riofrío et al., 2017; Condés et al., 2023). Several studies reported that structural diversity is more important than species diversity to maintain aboveground biomass growth (Ali et al., 2016; Dănescu et al., 2016; Park et al., 2019; Astigarraga et al., 2023), as species diversity impact can be mediated by structure (Park et al., 2019). Structural diversity is considered a reliable ecosystem function predictor (LaRue et al., 2019, 2023a, 2023b). In contrast, with the increasing of studies on the association of species diversity with forest growth in different locations and/or biomes, conflicting findings emerged—positive, negative, or even neutral (Cavard et al., 2010; Del Río et al., 2022), which initiates dispute among scientists (Dănescu et al., 2016). This debate is likely to occur due to inappropriate consideration of tree age, nutrients, stand density, and different aspects of biodiversity because these driving factors can define the structure and growth from the individual tree to the stand level under stress (Pretzsch, 2009; Condés et al., 2013; Ali et al., 2016; Schmied et al., 2023). In particular, when stand density is a critical factor that may increase competition and reduce growth (Pretzsch, 2005; Ng et al., 2016). When stand density is so high that light becomes a limiting factor, the significance of the tree's spatial arrangement in defining interspecific competition, relationships, and other competitive events may diminish (Amoroso and Turnblom, 2006). A recent experimental study has reported that constant tree growth in mixed forest stands under low to medium tree density (Thurm and Pretzsch, 2021).

Understanding the intricate dynamics influencing mangrove growth amid changing environmental conditions poses a significant challenge. The delicate interplay between rising salinity and density-induced competition is further complicated by the crucial role of nutrients—nitrogen (N), phosphorus (P), and potassium (K), with nitrogen identified as the primary growth-limiting factor for mangroves (Alongi, 2020). As salinity levels increase, nutrient availability tends to decrease, potentially leading to decreased site conditions and productivity (Ahmed et al., 2022), as productivity is strongly correlated to site conditions (Sun et al., 2017). To adjust the site conditions, plants modify the functional traits (Yaseen et al., 2023) and increase the functional trait variability (Price et al., 2014). Later, the modified functional traits will likely modulate plant dynamics (Pérez-Ramos

et al., 2019). Despite the widespread acknowledgment of the adverse effects of increased salinity, the influence of diversity in associations with nutrients and functional traits on growth in mangroves remains a largely unexplored aspect. Notably, the role of density-modified structure and the buffering potential of nutrient availability among mangrove trees have received limited attention, even though they hold paramount importance in scientific research and management considerations. Adding to the complexity, the relative extent to which structural and species diversity act as buffers against salinity to optimize growth remains largely unexplored. Bridging these gaps in our understanding is crucial for comprehensive insights into mangrove ecosystems under changing conditions.

Several studies on a regional to global scale have demonstrated that species diversity, tree density management, and thinning can increase forest growth by lowering competition for resources like light, water, and nutrients (Miller, 1997; Zhang et al., 2012; LaManna et al., 2017; Güney et al., 2022; Qu et al., 2022). If forest growth needs to be sustained for a long time, management strategies must be modified in response to the effects of climate change (Millar et al., 2007). Long-term growth can be maximized by employing adaptive forest management techniques, including monitoring and modifying forest structure in response to changing conditions (Lindenmayer et al., 2011). By integrating these principles and practices into forest management, forest ecosystems' ecological health and resilience can be preserved while maximizing growth. Consequently, forest regions' future composition and growth will depend on how tree species respond to projected climate change and local environmental conditions. Despite the strong impact of species and structural diversity on forest functions in climate

change scenarios, understanding their comparative role remains contentious.

After considering the above discussion, the overarching objective of this study is to gain a comprehensive understanding of the multifaceted interactions among structural diversity, species diversity, nutrient availability, density-induced competition, and rising salinity and how these interactions collectively or individually shape the growth patterns of mangrove forests (Fig. 1). To do this, we formulated two specific research questions and hypotheses.

(QI) How do species diversity and structural heterogeneity influence growth-salinity relationships? We hypothesized that (HI): Structural and species diversity has a buffering role in mitigating salinity impacts by interacting with nutrient availability and functional traits.

Moreover, (QII): To what extent does the structural heterogeneity and species diversity contribute to buffer forest conditions by minimizing the adverse effects of salinity while concurrently maximizing forest growth potential? To evaluate whether structural and species diversity can optimize growth by reducing salinity impacts, we hypothesized (HII) that the combined influence of structural heterogeneity, species diversity, nutrients, and functional traits would optimize growth. However, we also hypothesized that nutrients and functional characteristics might play a crucial role in enhancing the model's capability and potentially help to reduce salinity impacts.

To address our research questions, we employed a dataset from the Sundarbans Reserve mangrove Forest (SRF) of Bangladesh, one of the largest and most diverse mangrove forests in the world (Rahman et al., 2015; Islam et al., 2016). This dataset includes information on stand structural and compositional diversity, salinity gradients, and

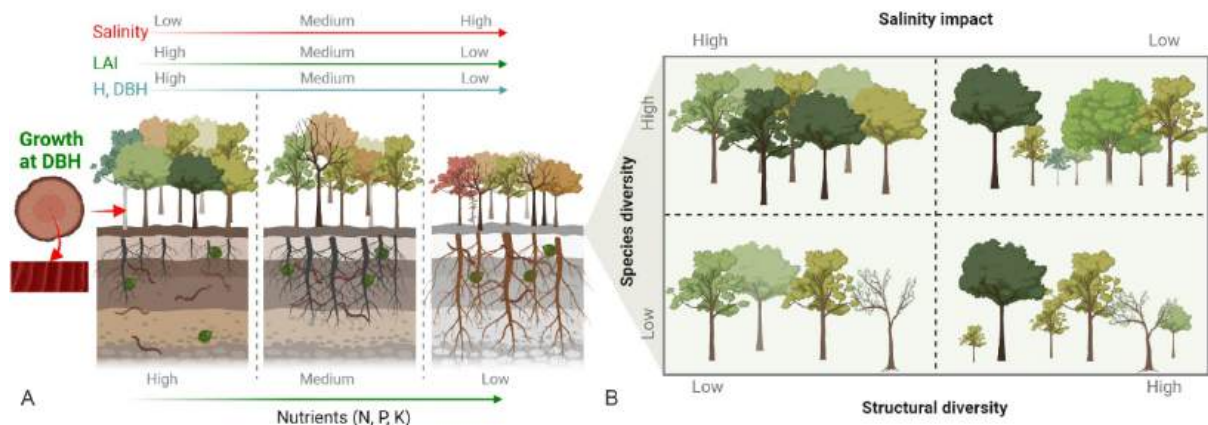


Fig. 1 Conceptual framework illustrating the influence of salinity on forest parameters and growth buffering mechanisms. (A) The variability of forest parameters across salinity levels and (B) the interaction of salinity and structural diversity, species diversity (proportional weight), and nutrients used in this study to describe growth differences and responses. The conceptual diagram explains that salinity stress actively reduces tree growth by limiting structure (tree height, H; diameter at breast height, DBH) and inhibiting resources (nutrients) and leaf functions (leaf area index, LAI). When the forest is structurally diverse in species and structure and resources are abundant, the stress impacts are buffered and promote growth. In panel A, darker soil color indicates higher availability of nutrients (N, P, and K).

nutrients (N, P, and K). We chose this dataset because of the ecological significance of the Sundarbans, which encompasses a large area and is located in an active delta (Bangladesh part: 6017 km²). The SRF exhibits salinity gradients species diversity and plays a significant role in mitigating climate change through sequestering and storing atmospheric carbon and addressing natural calamities like tropical cyclones (Rogers and Goodbred, 2014; Akber et al., 2018; Sarker et al., 2019a, 2019b; Ahmed et al., 2022). This region, in turn, holds the potential to inform evidence-based strategies for conserving and managing these crucial coastal ecosystems in the face of dynamic global changes. By enhancing our understanding of the intricate interactions between ecological variables, this research could contribute to scientific knowledge and offer practical guidance for safeguarding endangered mangrove ecosystems from future threats and degradation.

2 Materials and methods

2.1 Study site description and tree inventory

This research was carried out within the unique ecosystem of the Sundarbans Reserve Mangrove Forest (SRF) in Bangladesh, as illustrated in Fig. S1. SRF is located in an active delta (Rogers and Goodbred, 2014) and faces increasing salinity challenges (Lovelock et al., 2015; Ahmed et al., 2022). Due to increasing salinity, SRF loses productive species or less salinity-tolerant species (Sarker et al., 2019a), which makes SRF less productive in high-salinity areas by restricting tree height growth (Ahmed et al., 2022; Perri et al., 2023). SRF has been classified as three saline zones based on river water salinity levels, namely oligohaline

(<14 ppt), mesohaline (14–25 ppt), and polyhaline (>25 ppt) (Ahmed et al., 2022). To comprehensively assess the forest's structure, species composition, and carbon storage, we established 60 permanent sample plots (PSPs). Each of these plots covered an area of 0.01 hectares. Our sampling strategy involved the creation of 20 PSPs within each of the three salinity eco-regions. This allocation was achieved through a stratified random sampling approach, ensuring representation across the salinity zones. This first fieldwork was conducted in April 2018.

To document the tree diversity, we identified and labeled all trees with a minimum diameter at breast height (DBH) of 4.6 cm, measured at 1.3 meters from the ground. Aluminum tags were affixed to these trees for identification purposes. Concurrently, tree heights were measured utilizing an electrical Dendrometer, specifically the Criterion RD 1000 model from Laser Technology Incorporation in the USA. Adopting the 4.6 cm DBH criterion is in line with historical practice, dating back to the 1900s, and is particularly suited to mangroves given their relatively slow aboveground growth pattern.

In November 2020, we revisited all 60 PSPs to remeasure and assess the biomass growth patterns of the tagged trees—data collection involved measuring the DBH and heights of all the 1378 trees tagged in 2018. However, we excluded all dead trees during the growth (aboveground biomass changes) calculation.

2.2 Stand structure, species composition, nutrients, and biomass estimation

We utilized various tree measurements to assess the characteristics of the forest stand (Table 1). These measurements included stand density (number of stems per hectare), mean

Table 1 List of plot-level variables used in this study.

Variables and metrics names	Abbreviated form	Unit/calculation
Mean diameter at breast height	DBH	cm
Horizontal diversity (coefficient of variation of DBH)	cv of DBH	variation of DBH (cv of DBH=sd DBH/mean DBH)
Mean tree height	H	m
Vertical diversity (coefficient of variation of H)	cv of H	variation of H (cv of H=sd H/mean H)
Initial stand density	Density	stems ha ⁻¹
Aboveground biomass stocks	AGB	mg ha ⁻¹ (biomass equation adopted from Ahmed et al. (2022))
Aboveground biomass changes over time	AGB growth or increment	mg ha ⁻¹ yr ⁻¹
Species diversity	Shannon diversity	Shannon's index
Ammonia (NH ₄ ⁺) termed as nitrogen	N	mg kg ⁻¹
Phosphorus	P	mg kg ⁻¹
Potassium	K	mg kg ⁻¹
Nutrients	Nutrients	Total nutrients=N+P+K
PAR leaf area index	LAI	(m ² m ⁻²)

tree height in meters, and the quadratic mean diameter at breast height (DBH), i.e., 130 cm. We employed Shannon's index to evaluate species diversity (our survey covered 13 species). We calculated the coefficient of variation of H and DBH to analyze structural diversity, specifically the distribution of tree sizes. We also determined the Leaf Area Index (LAI) to check the impacts of functional traits on tree growth. We collected five LAI readings from each plot to determine potential tree variability. Lastly, we estimated tree above-ground biomass non-destructively using allometric equations, which had been developed locally to estimate dry biomass for all tree species (Rahman et al., 2021b) (Table S1). Additionally, to assess the influence of nutrient availability, we examined the presence of nitrogen, phosphorus, and potassium within the top 50 cm of soil depth. The above methodology employed is based on the approach outlined in Ahmed et al. (2022).

2.3 Statistical analyses

In our analyses, we started by checking the normality of our dataset using a Shapiro–Wilk normality test. To address any deviations from the distribution of variables, we made a logarithmic transformation to align them with the assumptions of normality and homogeneity, if necessary. Moving on to exploring the first research question (Q1), we initially looked at how forest growth, salinity, density, structural diversity, and species richness correlate through linear regression using the “lm” function in R.

Furthermore, to elucidate the relative impacts of species diversity and structural diversity (direct vs. indirect) on mitigating salinity effects on AGB growth across varying nutrient levels, we employed a structural equation model (SEM) utilizing the “lavaan” package (Rosseel, 2012) and plotted the model by “tidySEM” package Van Lissa (2020). To determine the most suitable model, we employed various fit indices, including the chi-square (χ^2) test (with the criteria of $0 \leq \chi^2/df \leq 2$), the comparative fit index (CFI) (requiring $CFI > 0.95$), the standardized root mean square residual (SRMR < 0.05), and the significance of paths ($p > 0.05$), as recommended by Schermelleh-Engel et al. (2003). We calculated the indirect and total association of sediment salinity and Shannon diversity with aboveground biomass growth by following Rahman et al. (2021a).

As we delved deeper into our analysis to test H11, we performed several multiple linear models (MLM) using the “lm” function (global model) by choosing variables from the SEM in a progressive way. We employed an MLM with standardized predictor variables. Standardization was performed using the formula $f(x) = (x_i - x_{\min}) / (x_{\max} - x_{\min})$, where x_i represented the variables with their respective maximum ($x_{\max}$) and minimum ($x_{\min}$) values. This approach facilitated a direct comparison of effect sizes, thereby enhancing the inter-

pretability of the model (Schielzeth, 2010). Finally, we optimized all models to maximize the growth by reducing salinity impacts. We used “optim” functions with advanced optimization methods “L-BFGS-B” in base R. Similar to model fitting, we first tried to optimize the model without considering nutrients and leaf area index, as forest structure and species composition management are easily attainable through management (Pretzsch, 2009). We expected that the impact of nutrients might manifest in structure and diversity. In the second and third optimization steps, we included nutrients and leaf area index in the model. We compared the model's performance (i.e., with vs. without nutrients and leaf area index based on AIC values by using the “performance” functions from the “performance” package (Lüdecke et al., 2021). Besides, all the visualizations were done by using the “ggplot2” package (Wickham, 2011).

All statistical analyses and visualizations were performed in R (version 4.3.1) (R Core Team, 2023).

3 Results

3.1 Patterns and distribution of species and structural diversity across salinity, nutrient availability, and density gradients

Stand structure, species diversity, and growth varied across salinity zones and levels as well as stand density (Fig. 2). Most structural diversity, such as H, DBH, cv of H, and DBH, were higher, and distributions were skewed towards the oligohaline and mesohaline zones. In contrast, species diversity tends to be higher in the polyhaline salinity zones (Fig. 2A–2E). Stand structural properties and forest growth (above-ground biomass changes over time) varied across salinity and density gradients (Fig. 2). Forest growth was reduced with salinity. In contrast, growth increased with nutrient availability and density (see the trend in Fig. 2F–2H).

3.2 Role of structural and species diversity in modulating salinity impacts on growth via nutrient availability and functional traits (Q1)

Species diversity enhanced growth across salinity levels but declined under varying nutrient regimes and density gradients (Fig. 3, right column). Structural diversity (measured as the coefficient of variation of DBH) positively influenced growth across nutrient regimes and density gradients but decreased with increasing salinity levels (compare the trends, Fig. 3, middle column). Additionally, AGB growth increased with AGB stocks (Fig. S2). Growth responses to structure and species density across salinity zones are presented in Fig. S3.

Figure 4 shows that growth response varies by species

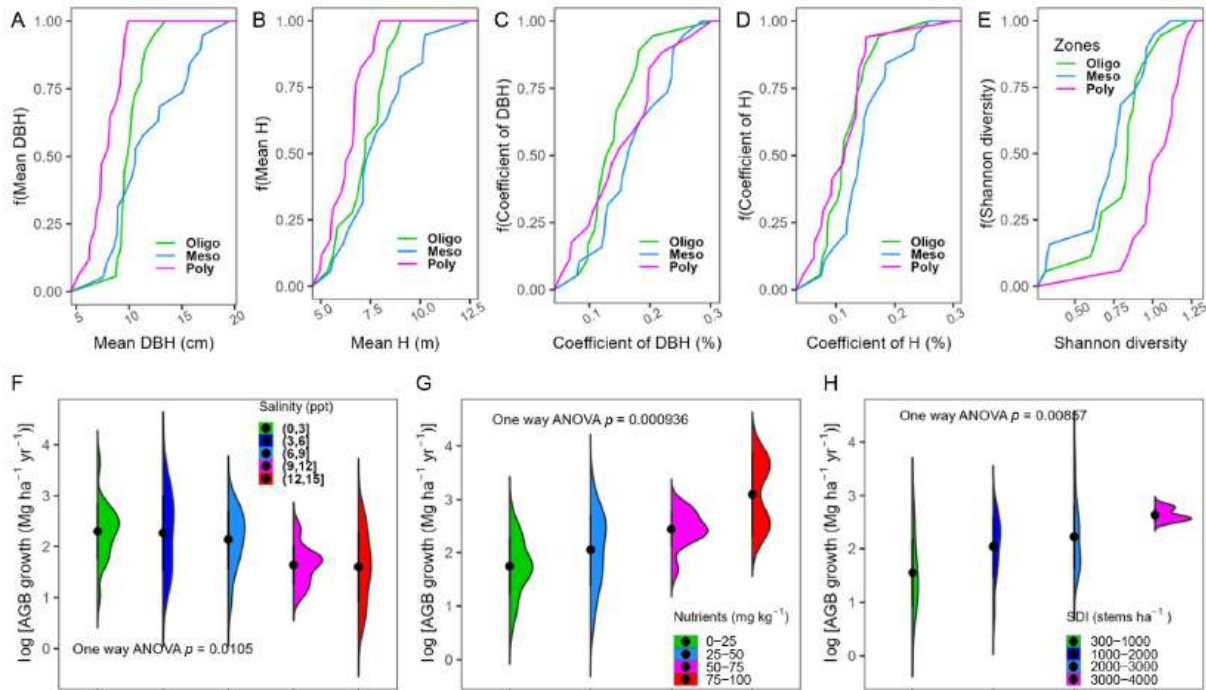


Fig. 2 The descriptive representations of patterns of structure and AGB growth. Top row: (A–E) shows cumulative density distributions of structural properties (A) mean DBH, (B) mean height, (C) horizontal diversity or coefficient of DBH, (D) mean coefficient of height, (E) Shannon species diversity. In the Bottom row, figures (F–H) show growth response. (F) growth response to different salinity levels, (G) growth response to nutrient availability, and (H) growth variability with different stand density levels. In Fig. F–H, vertical solid lines indicate a 95% confidence interval and black circles indicate median values.

and is influenced by functional traits. Growth was positively correlated with mean tree height (H), diameter at breast height (DBH), and LAI, with species-specific variations observed across nutrient availability and salinity levels.

Our Structural Equation Model (SEM) demonstrated a good fit ($\chi^2=12.889$, $p=0.301$), providing insights into how species diversity and structural diversity influence the relationship between salinity and growth via nutrient availability and leaf area index (see Fig. 5). The SEM accounted for 84% of the variation in Aboveground Biomass (AGB) growth. Specifically, species diversity had a significant direct effect on growth ($\beta=0.16$, $p<0.01$) and an indirect impact by improving leaf trait levels ($\beta=0.10$, $p>0.05$). In contrast, structural diversity showed a positive and statistically significant direct effect on growth ($\beta=0.23$, $p<0.01$).

As expected, salinity was found to be negatively correlated with nutrient availability ($\beta=-0.34$, $p<0.001$) and showed a weak, non-significant direct effect on AGB growth ($\beta=-0.02$, ns). This suggests that salinity indirectly affects AGB growth by reducing nutrient availability and LAI rather than having a pronounced direct impact. The LAI had a strong, positive, and statistically significant direct effect on AGB growth ($\beta=0.71$, $p<0.001$), while LAI elevated with nutrients ($\beta=0.40$, $p<0.001$).

The total effect of sediment salinity was negative ($\beta=-0.309$, $p<0.01$), while species diversity ($\beta=0.07$, $p>0.05$)

and nutrients ($\beta=0.156$, $p>0.05$) had positive but not significant effects (see Table 2).

3.3 Role of structural heterogeneity and species diversity to create optimal growth conditions (Q2)

The optimized model's results, as shown in Fig. 6, highlight how—species diversity, structural diversity, nutrients, and LAI—influence forest growth in the presence of salinity stress. Fig. 6A shows that predictions fall below zero when only species diversity and cv of DBH are included in the model. Fig. 6B indicates a reduction in the negative trend when nutrients are added, with predictions hovering near the zero line. Figure 6C demonstrates that including LAI results in most predictions exceeding zero.

4 Discussion

Our results indicate that structural and species diversity reduces salinity impacts by associating with nutrient availability and leaf area index. More precisely, structural diversity is more effective in low-salinity areas where competition for light is intense. Species diversity matters more in high-salinity areas where usable water for trees is scarce. Besides, nutrient availability improves forest condi-

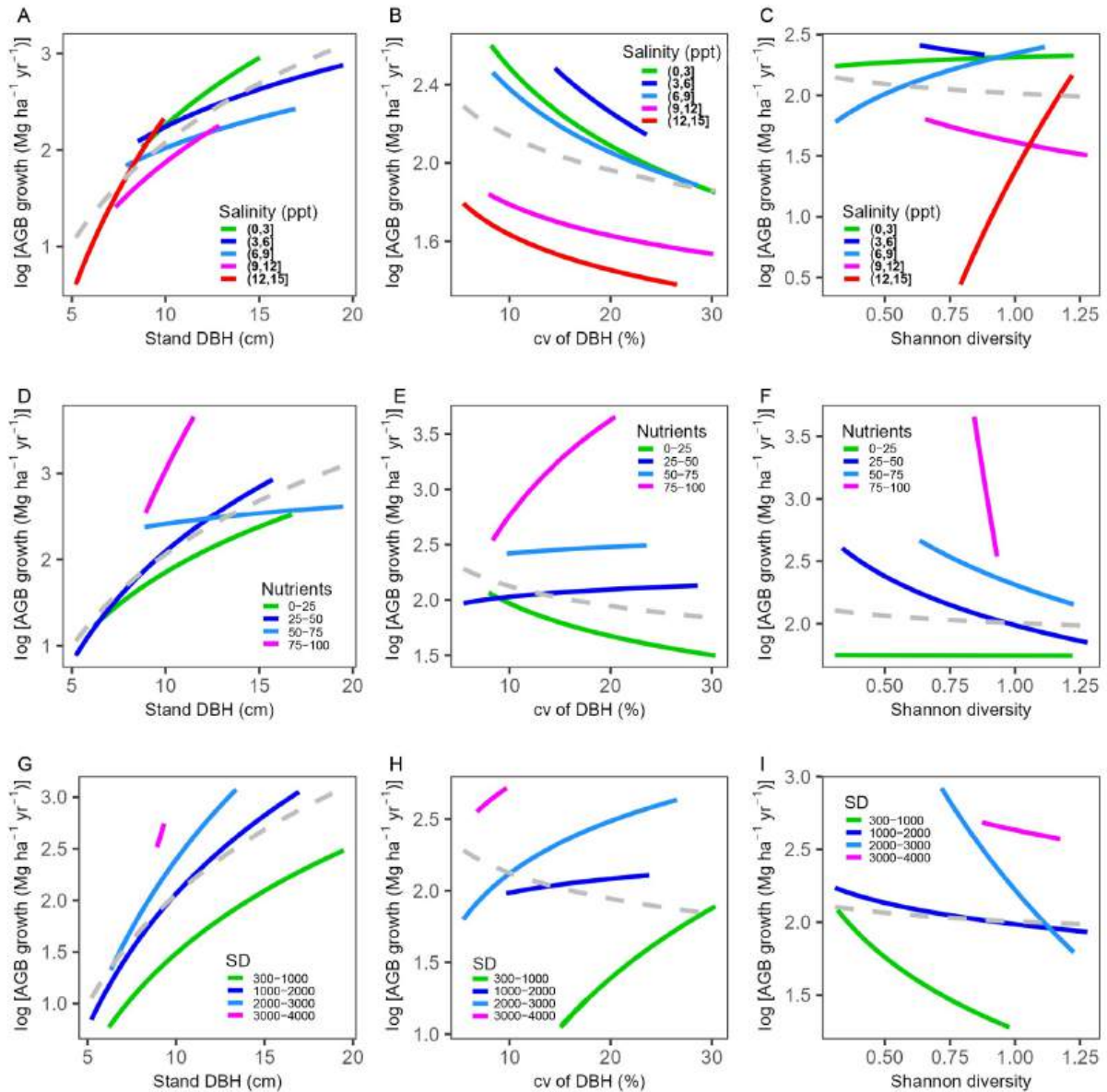


Fig. 3 Growth response to stand structure and species diversity across different salinity, nutrients, and stand density levels. SD indicates stand density (stems ha^{-1}). The Grey dashed line indicates the mean trend line.

tions by helping leaf function traits, further helps the forest reduce salinity stress and competition for light by modifying structure and species.

4.1 Role of structural and species diversity in modulating salinity impacts on growth via nutrients, functional traits, and stand density (Q1)

Our results indicate that forest growth predominantly occurred in lower salinity areas (Fig. 2), driven by salinity's impact on stand structural diversification. High salinity areas showed lower overall growth, likely due to reduced hydraulic conductivity and water availability (Lovelock et al., 2006). Besides, it could also be possibly due to low stand dominant

height and stocking degrees (Condés et al., 2013) in high salinity areas.

Mangrove growth's multidimensional response to salinity was observed through tree structure (DBH, cv of DBH), species diversity, nutrient availability, and stand density (Figs. 3–4). Bivariate relationships showed species diversity had a negative and insignificant impact across salinity gradients, but SEM indicated a positive impact on AGB growth when interacting with nutrients and leaf area index. High salinity may increase mortality and create gaps, potentially filled by saline-tolerant species like *C. decandra* (Ahmed et al., 2023), increasing species diversity without enhancing growth. Nutrient levels did not increase with species diversity as expected, possibly due to flash flooding or distinct pheno-

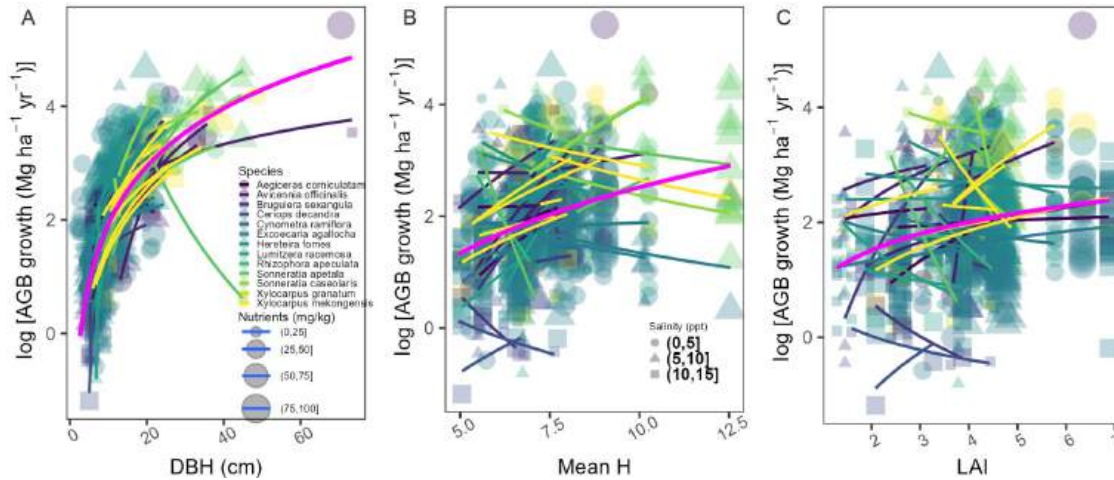


Fig. 4 The changes in AGB growth concerning tree and species level structure (H, DBH), leaf area index (LAI), and varying levels of salinity and nutrient availability. The pink color indicates the mean trend line. Legends are applicable for all sub-figures.

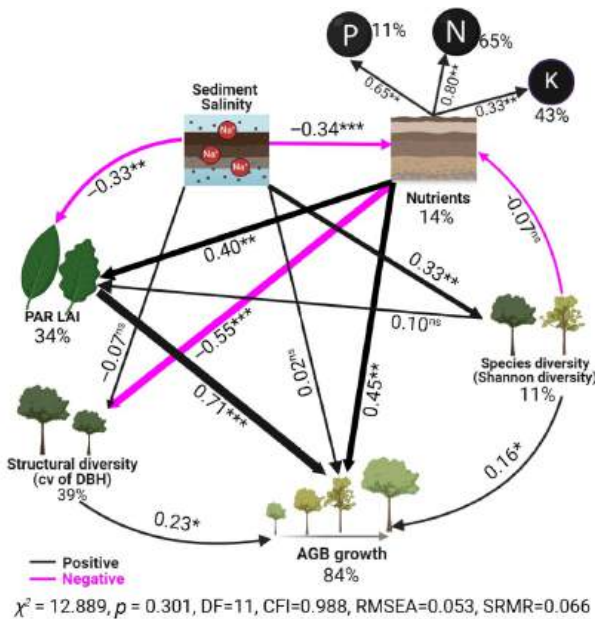


Fig. 5 The Structural Equation Model (SEM) illustrating the nuanced contributions, both direct and indirect, of species diversity and structural diversity in mitigating the impact of salinity on growth along varying nutrient gradients. The goodness-of-fit tests for the model produced the following outcomes: $\chi^2=12.889$, $p=0.301$, indicative of a commendable fit with the data. Moreover, the Comparative Fit Index (CFI) registered at CFIc 0.988, while the Root Mean Square Error of Approximation (RMSEA) and Standardized Root Mean Square Residual (SRMR) were 0.053 and 0.066, respectively, signifying no significant deviations from the model's underlying dataset. These assessments were conducted with a degree of freedom of two. The numerical values on the arrows denote the standardized strength of the relationships between the predictors and the dependent variables. The figures adjacent to the variables signify the proportion of variance accounted for, quantified as the coefficient of determination (R^2) is denoted by %, by all the predictors. The thickness of the lines signifies the magnitude or robustness of the relationships. Moreover, the proximate path values indicate standardized path coefficients and incorporate significance levels denoted by asterisks (ns>0.05; * $p<0.05$; ** $p<0.01$; and *** $p<0.001$). ns, not significant.

logical patterns or terrestrial inputs. We expected continuous litterfall from various species most likely enriched the soil with nutrients (Ahmed and Kamruzzaman, 2021), and diverse root depths allowed better nutrient utilization (Soethe et al., 2006). This variation could be attributed to differences in nutrient availability across salinity gradients. A previous study highlighted that highly saline, less-diverse Sundarbans mangroves are characterized by elevated phosphorus (P) and potassium (K) levels, whereas less saline, more diverse mangrove communities exhibit higher nitrogen (N) levels. Notably, nitrogen appears to have minimal influence on community diversity (Sarker, 2017). These patterns are likely associated with the location of these areas—lower salinity zones are typically situated near estuaries that receive terrestrial nutrient inputs, while higher salinity zones are found along open coasts with limited or no terrestrial nutrient inputs. Additionally, litterfall decomposition rates and microbial activity tend to be higher in lower salinity areas (Khan et al., 2009; Alongi, 2011), contributing to enhanced nutrient availability.

Besides, these contrasting results indicate that forest functions might species specific and their response to with disturbance or stress type (i.e., competition for nutrients, light or salinity) (see Figs. 3, 4 and Sarker et al. (2021)).

Structural diversity positively correlated with AGB growth but negatively with nutrient levels, suggesting smaller trees benefit more from nutrients. Structural diversity and functional traits had a more substantial influence on biomass growth than species diversity (Fig. 5), possibly due to synchrony or disturbances like cyclones. Species diversity's weak correlation with biomass growth may explain species loss with stand maturity in high saline areas (Schall et al., 2018; Sarker et al., 2019a; Ahmed et al., 2022). LAI strongly influenced AGB growth, enhanced by nutrient availability (Pérez-Ramos et al., 2019; Yaseen et al., 2023), which is consistent with our results.

Table 2 Indirect and total standardized association of sediment salinity, nutrients, functional traits, species, and structural diversity on forest increment.

Indirect and total association pathways	Increment	
	std. as	p-value
Indirect association of sediment salinity through Shannon diversity	0.052	0.105
Indirect association of sediment salinity through LAI	-0.232	0.016
Indirect association of sediment salinity through nutrients	-0.153	0.062
Indirect association of sediment salinity through cv of DBH	-0.015	0.623
Indirect association of sediment salinity through Shannon diversity and LAI	0.022	0.432
Indirect association of sediment salinity through Shannon diversity and cv of DBH	0.007	0.454
Indirect association of sediment salinity through Shannon diversity and nutrient	-0.01	0.646
Indirect association of sediment salinity through Shannon diversity, nutrient, and LAI	-0.007	0.647
Indirect association of sediment salinity through Shannon diversity, nutrient, and cv of DBH	0.003	0.654
Indirect association of Shannon diversity through cv of DBH	-0.095	0.045
Indirect association of Shannon diversity through LAI	0.068	0.411
Indirect association of Shannon diversity through nutrients	-0.05	0.639
Indirect association of Shannon diversity through nutrients and LAI	-0.02	0.642
Indirect association of Shannon diversity through nutrients and cv of DBH	0.009	0.65
Indirect association of nutrients through cv of DBH	-0.127	0.084
Indirect association of nutrients through LAI	0.283	0.004
Total association of sediment salinity	-0.309	0.008
Total association of Shannon's diversity	0.071	0.552
Total association of nutrients	0.156	0.17

The indirect and total associations of sediment salinity, nutrients, and Shannon diversity were based on structural equation models. The significant standardized associations (std. as) have a p -value < 0.05.

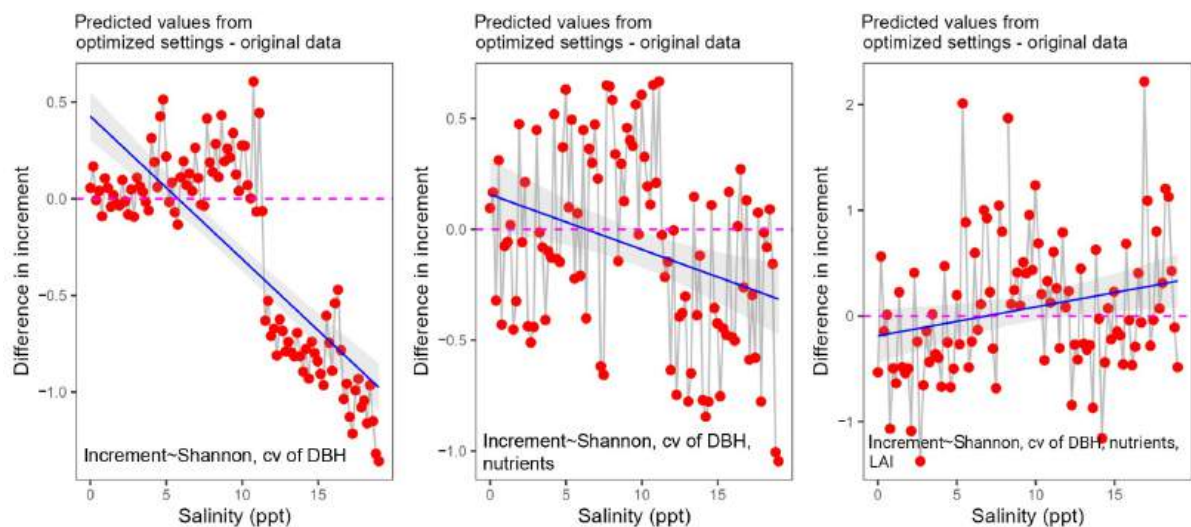


Fig. 6 Model optimization to maximize forest increment or AGB growth. In A–C, the x-axis represents the salinity values, and the y-axis represents the difference in increment values (the predicted growth values obtained from the optimized settings for a specific set of predictor variables—the actual AGB growth values or the original data). The blue line shows how the contrast changes as salinity changes. The magenta dashed line at $y=0$ indicates where the optimized and actual increment values are equal. If the blue line exceeds the red dashed line, optimized settings reduce salinity's impact on AGB growth; minimal difference suggests predictor variables have limited influence on the outcome. In the subset, respective models optimize the settings, and predictions are presented.

Growth increased with forest stand density, despite SEM not including density in the final model. Higher density fostered greater height growth in young trees, especially in high salinity areas (Fig. 2). Heightened competition for light among densely packed trees led to increased aboveground biomass (López-Hoffman et al., 2006; Peng et al., 2022). Biodiversity and nutrient richness in densely populated stands also contributed to growth (Vanellander et al., 2009; Cleland, 2011).

However, structural diversity, nutrients and leaf trait positively buffered salinity impacts, while they are interacting together.

4.2 Role of structural heterogeneity and species diversity to create optimal growth conditions (Q2)

Our model optimization settings explain that combined structural diversity and species proportions may not maximize forest growth. However, adding nutrients to optimization settings enabled the forest to buffer a bit (until a certain salinity level was reached). Still, when added functional trait optimization showed the forest to sustain its growth against salinity (see Fig. 6). These results indicate that a nutrient-free setting may not mitigate salinity impacts and nutrient availability most likely trees to capture light and accumulate biomass efficiently through functional traits. However, continuously increasing salinity led to a reduction in growth, eventually reaching the zero line. This supports our second hypothesis (HII). These results additionally indicate that species diversity and structural diversity, coupled with canopy complementarity, assist trees in effectively utilizing the available nutrients and mitigating the impacts of salinity. This could be related to more plastic and extensive crown space occupation by species-diverse stands compared to less diverse stands, which could increase aboveground biomass in higher salinity stands. It could be attributed to species diversity and density, which increased canopy packing, reduced crown shyness, and enhanced light use efficiency, leading to increased growth (Pretzsch, 2014; Jucker et al., 2015). However, the mean line suggests that better outcomes (growth) may be attainable by adjusting predictor variables, i.e., increasing canopy packing through mixing. Several studies have demonstrated that tree density management and thinning can increase forest growth by lowering competition for resources like light, water, and nutrients (Miller, 1997; Güney et al., 2022; Qu et al., 2022).

Nevertheless, sustainable forest growth needs modified management strategies in response to the effects of climate change (Millar et al., 2007; Lindenmayer et al., 2011). Our model optimizations also indicate that structure and the growth of stands can be enhanced under stress by applying suitable silvicultural strategies. By integrating these principles

and practices into forest management, forest ecosystems' ecological health and resilience can be preserved while maximizing growth. However, large-scale datasets or more studies are required to conclude.

5 Research implications and future directions

The findings of our study shed light on the crucial role of species and structural diversity in mitigating the impact of salinity, particularly in regions with higher salinity levels. We observe a buffering effect in the areas where species diversity is naturally more abundant. This suggests that the functional differences between species, specifically their varying levels of salt tolerance, work synergistically to support continued forest growth.

Conversely, in regions with lower salinity levels but higher structural diversity, we see a different dynamic at play. Here, the growth persists, indicating that species diversity might not be the primary driver. Instead, less salty areas and salt-tolerant species, while maintaining structural diversity, tend to utilize crown space more extensively through denser canopy packing. This might lead to an increase in aboveground biomass growth.

Besides, in the case of the natural mangrove forest, managing both the structure and species composition becomes critical due to its vast expanse. However, this insight is particularly relevant in specific natural forest parts where natural mortality rates tend to be higher (could be due to salinity, biological age, or natural calamities). More precisely, stands-dominated pioneer species like *Sonneratia apetala* and *Avicennia officinalis* often experience frequent mortality due to their biological age, or diseases like dying back increase mortality (Giri et al., 2007; Rahman et al., 2010) and form significant gaps. Furthermore, the Sundarbans region is prone to tropical cyclones, which lead to tree mortality by causing severe damage, further creating gaps, especially in areas with higher salinity levels, which are more exposed to the coasts with little or no terrestrial nutrient inputs. Introducing different species in these gaps could be beneficial in expediting forest recovery and enhancing resilience against future climatic events (natural calamities and salinity).

This research holds strong applicability in coastal single species-based plantation efforts to mixing plantations, especially when the objective is to safeguard coastal areas while simultaneously harnessing the co-benefits of carbon sequestration (growth under salinity stress) in the face of cyclones. It underscores the potential of nature-based solutions in fortifying our coastal belts against environmental challenges.

However, our study was conducted in a single mangrove forest (Sundarbans), which may limit generalizability to other

regions. Examining multiple sites across varying salinity gradients could strengthen conclusions. Besides, we expect some interplay between water depth and seasonal rainfall patterns due to storms effect on salinity level and impacts, which could be interesting to explore. However, our dataset could not capture any disturbance (e.g., cyclones) effect on growth across species within study periods (~2.5 years). Overall, the study takes an essential first step in highlighting biodiversity's role in increasing mangrove resilience to salinity stress. However, ongoing research across broader spatial scales and including additional functional measures is needed. For example, the mechanisms underlying diversity-productivity relationships still need to be fully elucidated.

Further research into functional traits and niche differentiation could provide more mechanistic explanations. Translation to the application requires examining feasibility constraints. A systems perspective incorporating socio-ecological factors could maximize conservation impact. While promising, the findings represent an early stage in fully elucidating diversity's multifaceted influence.

6 Conclusion

This study evaluated the comparative role of structural and species diversity in modulating salinity impacts on growth. We also assessed how some growth-modifying factors, for example, nutrient availability, and leaf functional traits, modulate the salinity-growth relationships. Our analyses revealed that the growth of mangrove forests is strongly controlled by salinity, with its effects mediated by structural heterogeneity, nutrient dynamics, and leaf functional traits. Conversely, the leaf area index emerged as the primary facilitator of forest growth in areas marked by elevated salinity and limited water access. While species diversity may influence growth indirectly through its interaction with forest structure, structural diversity, including variations in stand size classes and horizontal heterogeneity. These insights are vital for improving mangrove growth modelling, enhancing ecological understanding, and informing coastal forest conservation and management strategies.

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Authors contributions

Conceptualization: SA; Data collection: SA, SKS (lead) and MK; Data analysis: SA (lead), MMR; Manuscript writing: SA (lead), RH, NT, SP; Manuscript revision: MMR, HP, SKS, SP, TH, TTA, EU, CSSN; Supervision: HP (lead), SKS

Competing interest

The authors declare no competing interest.

Data and code availability

Data is available in Ahmed et al. (2022). All R code is available from the corresponding author upon request.

Electronic supplementary material

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Article

Assessing Tree Coverage and the Direct and Mediation Effect of Tree Diversity on Carbon Storage through Stand Structure in Homegardens of Southwestern Bangladesh

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Abstract: Dealing with two major challenges, climate change mitigation and biodiversity loss, under the same management program, is more noteworthy than addressing these two separately. Homegardens, a sustainable agroforestry system and a home of diverse species, can be a possible choice to address these two issues. In this study, we assessed tree coverage, and the direct and indirect effects of tree diversity on carbon storage in different carbon pools through stand structure in homegardens of southwestern Bangladesh, using Sentinel 2 and field inventory data from 40 homesteads in eight villages. An unsupervised classification method was followed to assess homegardens' tree coverage. We found a high tree coverage (24.34% of total area of Dighalia) in homesteads, with a high overall accuracy of 96.52%. The biomass and soil organic carbon ($p < 0.05$) varied significantly among the eight villages, while total carbon stock did not vary significantly ($p > 0.05$). Shannon diversity had both direct and indirect effects on biomass carbon, upper layer soil organic carbon and total carbon storage, while basal area mediated the indirect effect. Both basal area and tree height had positive effects on biomass carbon and total carbon storage, with basal area having the strongest effect. These findings suggest that we must maintain higher diversity and tree height in order to maximize and sustain carbon storage, where tree diversity increases stand basal area and improves total carbon storage (including soil organic) in homegardens. Therefore, privately managed homegardens could be a potential nature-based solution for biodiversity conservation and climate change mitigation in Bangladesh.

Keywords: traditional homegardens; agroforestry system; biodiversity and carbon; optical remote sensing

1. Introduction

Nature-based solutions are one of the best strategies for adapting to global climate change and biodiversity loss [1]. To combat global warming, world leaders committed in the Paris Agreement to keep atmospheric temperature below 2 °C by end of the century [2]. Each signatory country to the Paris Agreement has pledged to cut a part of its greenhouse gas emissions by 2030 [2]. Under the framework of Nationally Determined Contributions (NDCs), each country is meeting their obligation by lowering either low carbon sectors (transport, industry, power, etc.) or by increasing forestry activities (afforestation or restoration) [3]. Homegardens, a sustainable and integrated agroforestry system in tropical and subtropical countries, has huge carbon sequestration potential [4,5]. This climate regulatory role of homegardens can be a potential NDC component for many countries in meeting its carbon emission commitment [3].

Sustainably managed lands that have the ability to supply food and other domestic products to conserve biodiversity and regulate climate are of high demand in the current changing climate [6]. Homegardens are a tiny piece of land just in and around the homestead in rural areas of the tropical and subtropical regions. This unique type of agroforestry system provides multiple ecosystem services over the year to members of the household [7]. It is an intensive cropping system of domesticated annual and perennial plants and/or animals, which are the primary source of household demand, such as food, fruits, fodder, fuel wood and furniture [7,8]. Different coexisting species of trees, shrubs and herbs in homegardens with vertical differentiation make this traditional agroforestry system a forest-like structure [7]. Within human-dominated landscapes, homegardens are considered as highly biodiverse agroforestry systems which play an important role in biodiversity conservation and carbon sequestration [4].

Studies on homegardens' carbon storage have shown a high capacity for minimizing atmospheric carbon through photosynthesis compared to other agroforestry systems [4,9,10]. Findings on homegardens carbon stocks have revealed that aboveground carbon is mainly contributed by tree and palm species in homegardens, and that soil carbon occupies the largest percentage of the total carbon stored in homegardens [9]. Homegardens with high species diversity and stem density lead to a more basal area and biomass (above and below ground), and therefore, they store more biomass and soil carbon compared to homegardens with less species density and stem density [5,11,12]. In the forest ecosystem, the positive role of species diversity in carbon storage is underpinned by the niche complementarity hypothesis, which believes that higher numbers of species in a stand with greater trait variation have used more resources, and therefore, store more carbon [13]. Some studies have, however, documented no particular relationship between species diversity and carbon stock, which can be explained by selection effect. The selection effect believes that among the coexisting species, the most dominant species with its key traits such as size vigor (height and diameter) regulates carbon storage [12]. Both of these hypotheses can be a driver for homegarden agroforestry carbon regulation, because it is a multi-storied-species landscape. Furthermore, inconsistent results of species diversity and carbon storage relationships have also been reported in different studies in agroforestry systems [5,14]. Thus, for understanding the species diversity and carbon storage and their relationship in a specific study area, we need to conduct direct field investigations.

Bangladesh committed to the Paris Agreement to reduce its greenhouse gas emission from power, transport and industry sectors with an equivalent to 12 Mt CO_{2e} by 2030, which is 5% below the total business as usual emission level from those sectors. However, Bangladesh is a country of villages where almost every household has a homegarden, which are well-established land use systems [7]. A recent study reported that the total tree coverage in non-forest area is about 4.5 million ha, where homegardens have dominant contributions [15]. The homegardens in Bangladesh are biologically diverse and most of the studies on homegardens in Bangladesh remained descriptive on the floristic characteristics, structure, uses and the relation between household and homegarden characters [7,16]. Homegardens in Bangladesh can be a potential component of NDCs, similar to other developing countries. However, few studies focused on the total carbon storage capacity and carbon and biodiversity relationship in Bangladesh [17,18]. This knowledge gap in homegardens agroforestry system in Bangladesh might be one of the most important reasons for excluding agroforestry as a key NDCs component by policy makers, despite its huge coverage and high potential for climate change mitigation.

In this study, we assessed homegarden tree coverage, stand structure, species diversity and carbon storage in southwestern Bangladesh. We set three specific objectives in this study. First, to assess the total tree coverage in homesteads of Dighalia upazila (administrative unit) in the Khulna district, Bangladesh by using Sentinel 2 imagery. Second, to quantify the direct and indirect effects of species diversity on biomass and soil organic carbon stocks through different stand structures in homegardens. Third, to ex-

amine how results in line with carbon emission targets have contributed to homegardens' management system.

2. Materials and Methods

2.1. Study Site

The study was conducted in Dighalia Upazilla (administrative unit) in the Khulna district of Bangladesh (Figure 1). Dighalia Upazilla is located in the southwestern part of Bangladesh. The study area is primarily a flood plain land mass between 22.50° to 22.60° N latitude and 89.31° to 89.37° E longitude. The deltaic landscape of this region is mainly low (<10 m above average sea level), flat and fertile [19]. The average area of the homegarden over the study site is 0.05 ha and rich in biodiversity [19]. It generally enjoys a tropical to subtropical monsoon climate with an average annual temperature of 26 °C. January is the coolest month (average temperature: 12.4 °C), while April is the hottest month (average temperature: 34.6 °C) in this region. The average annual rainfall for this region is 1986 mm (range: 1400–2600 mm). This upazila is enclosed by three main rivers, i.e., the Bhairab, Citra and Naboganga, and is adjacent to Khulna City, the largest city in southwestern Bangladesh [20].

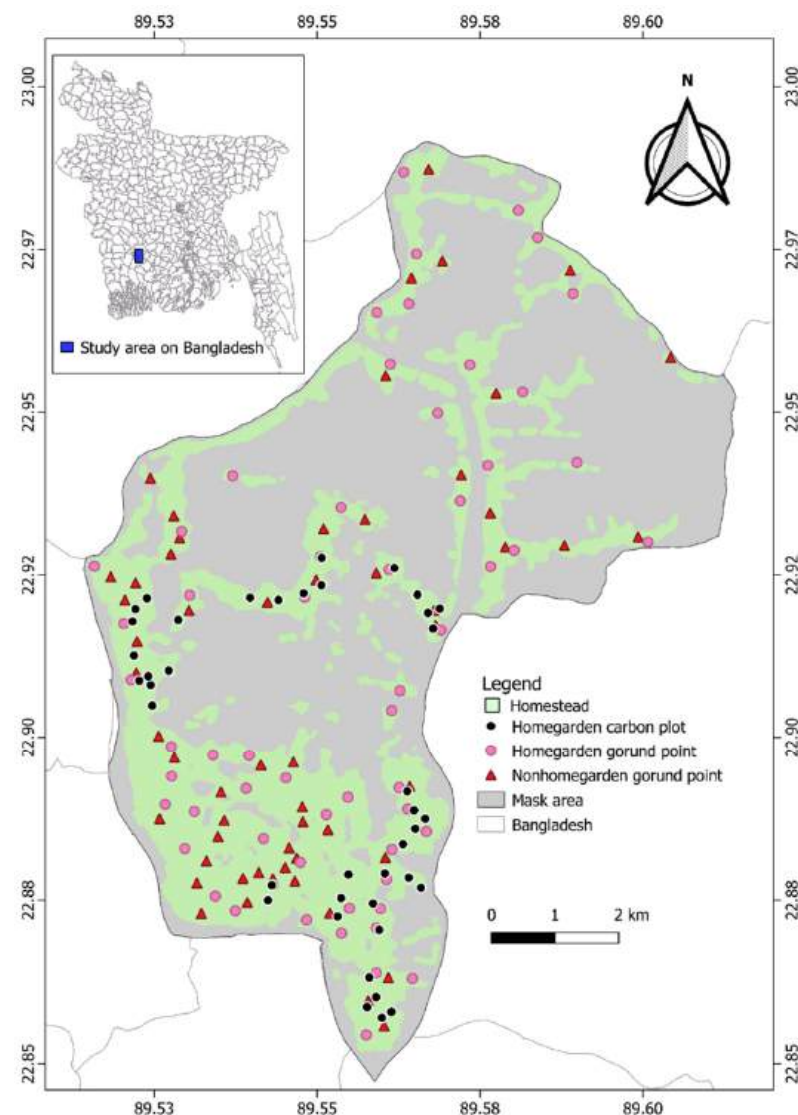


Figure 1. Map showing carbon sampling plots and ground truthing points in Dighalia Upazilla.

2.2. Sampling Design and Field Inventory

A multi-stage random sampling procedure was followed to select homegardens. First, two unions (smallest administrative unit), namely Senhati and Barakpur, from the total of six unions of Dighalia Upazila were selected randomly. Then, eight villages were selected randomly from the two randomly selected unions (four villages from each union). A total of 40 plots of 10×10 m (five from each randomly selected village) were taken purposively (homestead with homegarden), as shown in Figure 1. We measured the diameter at breast height (DBH) and canopy height of all species with a $\text{DBH} \geq 3$ cm. All species in each sampled plot were identified and recorded to species level or by local name and were later confirmed from an authentic source [19]. We collected soil samples for soil organic carbon assessment from the center of each plot. We pulled out soil cores using an open face peat augur. From the soil core, we took a total of 80 ($40 \text{ homegardens} \times 1\text{-plot} \times 2 \text{ depths}$) soil samples; two from each plot using a five-centimeter-long steel core at the midpoint of 0–15 cm depth and 15–30 cm depth. All samples were air-dried and analyzed for estimating bulk density and soil organic carbon. We analyzed all the soil samples at the Nutrient Dynamics Laboratory of Khulna University, Bangladesh.

2.3. Remote Sensing Data and Processing

Sentinel-2 satellite image that was acquired on 13 December 2016 was used in this study for assessing the vegetation coverage. The Level-1C product of Sentinel-2 is a version of Top-Of-Atmosphere reflectance image, which is radiometrically and geometrically corrected. Sentinel-2 is a multispectral imager launched on 23 June 2015 by the European Space Agency, which has 13 bands of different resolutions (10–60 m) [21]. In this study, we used the 10 m resolution bands (Red and Near Infrared) for the Normalized Difference Vegetation Index (NDVI). We masked non-homestead areas (water body, crop field and fishpond) in order to distinguish rural settlement with homesteads. Then, we applied an ISODATA algorithm, an iteration method that collapses clusters of similar groups into one by measuring the Euclidean distance between the cluster centers [22,23]. We set the number of classes in the classification to 10, the total class size to 10 and the sample interval to 2. Using Google Earth visual interpretation and field data, we assigned homegardens and non-homegardens classes (houses, paddy fields, fish ponds, waterbodies and built-up areas) to the resultant one band image of ISODATA classification with 10 clusters [24]. We collected 115 points randomly from homegardens (59) and non-homegardens (56) within homestead areas (Figure 1) by using Google Earth visual interpretation for accuracy assessment of our classification result [23–25]. A confusion matrix was used for assessing the accuracy of our classification [25]. We used ENVI 5.00 and QGIS 3.14 for classification and mapping.

2.4. Data Analysis

2.4.1. Biomass Carbon

Common allometric equations were used (Table 1) to calculate the biomass [26,27], as it was developed from wide graphical (tropical countries) data and has diameter range (Table 1). The wood density data were obtained from the Global Wood Density Database [28]. For species with missing wood density in this database, we used the average wood density. Biomass carbon was calculated by multiplying 0.5, as it is assumed that wood biomass contains 50% carbon [28].

Table 1. Allometric equations used for biomass estimation in this study.

Equation(s)	Notes	Reference
$\text{AGB} = \rho \times \exp(-1.499 + 2.148 \times \ln(\text{DBH}) + 0.207 \times (\ln(\text{DBH}))^2 - 0.0281(\ln(\text{DBH}))^3)$	For all dicot trees	Chave et al. [26]
$\text{AGB} = 6.666 + 12.826 \times \text{ht}^{0.5} \times \ln(\text{ht})$	For palm, coconut and date trees	Pearson et al. [29]
$\text{BGB} = \exp(-1.0587 + 0.8836 \times \ln(\text{AGB}))$	For root biomass	Cairns et al. [27]

AGB = Aboveground biomass, ρ = Wood density (gcm^{-3}), DBH = Diameter at breast height, ht = Height, BGB = Belowground Biomass.

2.4.2. Soil Carbon Calculation

Soil carbon storage was estimated by the product of soil organic carbon concentration, soil bulk density and soil depth range. Soil bulk density was determined for each soil layer by dividing the oven-dried soil sample mass by the volume of the sample (Equation (1)) [30]:

$$\text{Bulk Density} = \frac{Wt_{105^{\circ}\text{C}}}{V_{\text{core}}} \quad (1)$$

$$V_{\text{core}} = \frac{(\pi D_{\text{core}}^2 L_{\text{core}})}{4}$$

where $Wt_{105^{\circ}\text{C}}$ is the weight of oven dried soil, V_{core} is the volume of the core, D_{core} is the inner diameter of the core and L_{core} is the length of the core.

The Loss of Ignition method was used for calculating soil organic matter [30]. Generally, soil organic matter contains 58% organic carbon [31]. Thus, we divided the organic matter by 1.724 (a universal conversion factor (called the van Bemmelen factor) for converting the soil organic carbon percentage [31].

After getting the soil bulk density and organic carbon percentage, we calculated the soil carbon stock for each layer using Equation (2):

$$\text{Soil C (Mg ha}^{-1}\text{)} = \text{bulk density (g m}^{-3}\text{)} \times \text{soil depth interval (m)} \times \% \text{OC} \times 0.01 \quad (2)$$

where Soil depth interval = 0.15 for 0–15 cm depth and 15–30 cm depth, %OC is expressed as a decimal fraction (e.g., 5% is expressed as 0.05) and 0.01 is a conversion factor to convert units to Mg ha^{-1} .

2.4.3. Total Carbon Stock

We calculated total carbon stocks (Equation (3)) per hectare by summing the biomass carbon stocks (aboveground and belowground biomass carbon) and soil organic carbon stocks (soil carbon, 0–15 cm and soil carbon, 15–30 cm):

$$\text{Total C stock (Mg ha}^{-1}\text{)} = \text{Biomass carbon}_{(\text{AGC} + \text{BGC})} + \text{TSOC}_{\text{SOC (0–15cm)}} + \text{SOC (15–30)} \quad (3)$$

2.4.4. Woody Species Diversity Calculation

The Shannon–Wiener index (H ; Equation (4)) was used for species diversity [32], while for species richness, the Margalef index (D ; Equation (5)) was used [33]:

$$H = \sum_{i=1}^S (P_i \times \ln(P_i)) \quad (4)$$

$$D = \frac{(S - 1)}{\ln(N)} \quad (5)$$

where H is the Shannon diversity for a plot, S is the number of species (species richness), P_i is the proportional of individuals of species i in the plot, N is the total number of individuals in a plot and “ln” is the natural logarithm.

One-way analysis of variance (ANOVA) was used to test the significant difference of carbon pools across the eight villages and Duncan’s Multiple Range Test was used for multiple comparisons. Pearson’s correlation tests were performed using the “rstatix” package in R environment (Version 3.6.3) [34], to explore the correlation between stand structure, woody species diversity and carbon pools. Structural equation model (SEM) was used to quantify the direct and indirect effects of tree diversity on carbon pools through stand structure [35]. We used the Lavaan package for employing SEM in R environment (Version 3.6.3) [34]. As recommended, we log transformed and standardized all the covariables before applying the structural equation model. Multiple linear regression was used to quantify the variance inflation factors (VIF) with a threshold value of 2 and assuming that covariables having a value below this threshold have no multicollinearity [36]. In the final

steps of the variance inflation factors test, mean tree height, basal area, stem density and Shannon diversity were retained (MDBH had a VIF value that exceeded the threshold value, Supplementary Table S1). Then, we employed SEM for quantifying the direct, indirect and total effect of the Shannon diversity on different carbon pools through stem density, canopy height and basal area. A total 14 SEMs were tested for seven carbon pools, where highly insignificant covariables were removed until the model satisfied fit statistics [37] (Supplementary Table S2). The indirect effects of Shannon diversity were assessed by the path value through its mediator to different carbon pools (response variable) [35]. We assessed the total effect of the Shannon diversity on a carbon pool by adding the direct and indirect effect through different mediators [35,37].

3. Results

3.1. Homegardens Cover

The spatial distribution of homegardens and non-homegardens based on the unsupervised classification of Sentinel 2 images in Dighalia Upazilla showed a good match with the Google Earth map (same year image as of Sentinel 2; Figure 2). Our classification separated homegardens and non-homegardens in homestead areas in Dighalia with a higher overall accuracy and kappa coefficient (overall accuracy = 96.52% and Kappa = 0.93; Table 2). The overall coverage of the homegardens was 17.32 km² and the total percentage of homegardens in Dighalia was 24.34 (the total area of Dighalia is 71.16 km²).

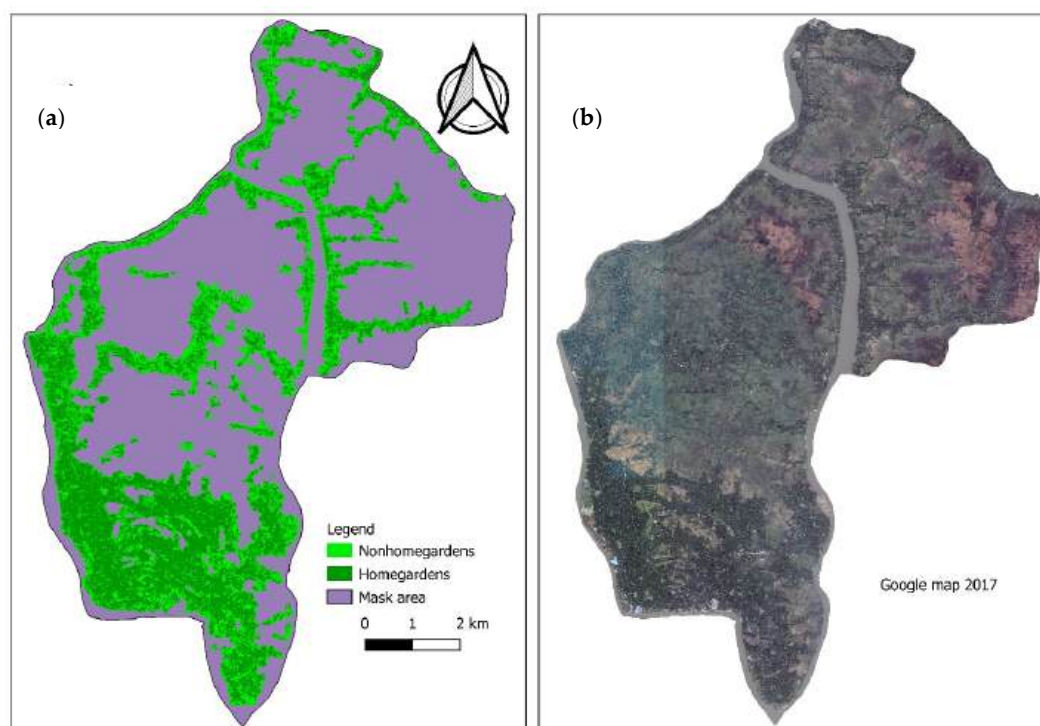


Figure 2. Extent of homegardens and non-homegardens in Dighalia Upazila. (a) Based on unsupervised classification of Sentinel 2 imagery; (b) Google Earth satellite image[®] shows homegardens covered by dark area.

Table 2. Result of the classification accuracy based on the Confusion Matrix. Column represents the Google Earth vegetation class, while the row represents the Sentinel 2-based class.

Pixel Based	Google Earth Based				
	Homegardens	Non-Homegardens	Total	Users (%)	Commission (%)
Homegardens	56	1	57	98.25	1.75
Non-homegardens	3	55	58	95.16	5.17
Total	59	56	115		
Producers (%)	94.92	98.21		Overall	96.52
Commission (%)	5.08	1.79		Kappa	0.93

3.2. Diversity and Stand Structure

The mean Shannon diversity, Margalef diversity and stand structures exhibited a spatial variation across the 40 plots (Table 3). The range of Shannon diversity was 1.91 to 3.01 with a mean of 2.56 ± 0.04 in the 40 plots (Table 3). The Margalef diversity index had a mean value of 2.31 ± 0.06 (range: 1.44–3.03). The estimated mean stem density, DBH, tree height and basal area were 1055.00 ± 35.62 trees ha^{-1} , 13.30 ± 0.49 cm, 7.88 ± 0.17 m and 13.76 ± 0.96 m^2 ha^{-1} , respectively, across the 40 sample plots (Table 3).

Table 3. Descriptive statistics of stand structures and diversity indices across the 40 plots in Dighalia.

Variables	Minimum	Maximum	Mean	SE
Shannon–Wiener index (H')	1.91	3.01	2.56	0.04
Margalef index (R)	1.44	3.03	2.31	0.06
Stem density (trees ha^{-1})	600	1600	1055	35.61
Mean DBH (cm)	8.77	24.45	14.49	0.49
Tree height (m)	5.84	11.00	7.88	0.17
Basal area (m^2 ha^{-1})	4.04	29.31	13.76	0.96

3.3. Carbon Stocks

High variations of above- and belowground biomass carbon were observed (aboveground: 45.01 ± 6.95 to 88.00 ± 6.39 Mg ha^{-1} ; belowground: 9.98 ± 1.36 to 18.11 ± 2.6 Mg ha^{-1}). We found similar biomass carbon stocks in Senhati, Baracpur and Lakhoati villages, but these three villages have significantly lower biomass carbon than in the Chandani mahal village ($p < 0.05$; Table 4). However, the biomass carbon stocks in these four villages were not significantly different from the other four villages ($p > 0.05$). Overall, above- and belowground carbon stocks across the 40 plots in Dighalia were 60.42 ± 3.93 and 12.89 ± 0.75 Mg ha^{-1} , respectively. Although the upper layer (0–15 cm) soil organic carbon stocks were insignificant across the different villages ($p > 0.05$; Table 4), the lower layer (15–30 cm) soil organic carbon stocks were found to be significantly different ($p < 0.05$; Table 4). In the Chandani mahal and Hagigram villages, soil organic carbon stocks at lower layers were significantly lower than at the Bativita village ($p < 0.05$; Table 4). Although we found a significant difference in biomass and lower layer soil organic carbon stocks, the differences in overall carbon stocks were insignificant across the villages ($p > 0.05$). The total carbon stock (biomass and soil carbon pools) in the eight villages ranged from 109.10 ± 10.89 Mg ha^{-1} (in Senhati) to 152.22 ± 10.15 Mg ha^{-1} (in Chandani mahal) with an overall average total carbon stock of 129.47 ± 5.10 Mg ha^{-1} (Table 4). The overall biomass carbon stock (73.31 ± 4.67 Mg ha^{-1}) contributes 56.62 percent of the total carbon stock, while soil carbon (56.16 ± 1.71 Mg ha^{-1}) contributes 43.38 percent of the total carbon stock. As we mapped the total area of homegardens and mean total carbon per hectare in Dighalia, we also assessed the total carbon stock in Dighalia by multiplying the mean carbon stocks by the total area. The total carbon stocks in Dighalia homegardens was 224,242.2 Mg C , which is equivalent to 822,968.29 Mg CO_2 .

Table 4. Mean ($\pm$ SE) difference of different carbon pools between villages. According to the Duncan's Multiple Range Test, the columns with a similar letter suggest a non-significant difference ($p > 0.05$).

Village	Aboveground Biomass Carbon (Mg ha^{-1})	Belowground Biomass Carbon (Mg ha^{-1})	Soil Organic Carbon (0–15 cm) (Mg ha^{-1})	Soil Organic Carbon (15–30 cm) (Mg ha^{-1})	Total Biomass Carbon (Mg ha^{-1})	Total Soil Organic Carbon (Mg ha^{-1})	Total Carbon (Mg ha^{-1})
Senhati	45.01 ± 6.95 b	9.98 ± 1.36 b	26.18 ± 2.25 a	27.94 ± 3.38 ab	54.99 ± 8.30 b	54.12 ± 3.99 ab	109.10 ± 10.89 a
Chandani mahal	88.00 ± 6.39 a	18.11 ± 1.16 a	26.04 ± 2.18 a	20.08 ± 3.29 b	106.11 ± 7.56 a	46.12 ± 4.63 b	152.22 ± 10.15 a
Bativita	55.05 ± 10.07 ab	11.89 ± 1.94 ab	29.44 ± 3.02 a	34.79 ± 2.00 a	66.94 ± 12.01 ab	64.23 ± 3.85 a	131.17 ± 12.14 a
Hagigram	61.47 ± 10.47 ab	13.12 ± 1.99 ab	27.48 ± 3.63 a	22.21 ± 3.35 b	74.59 ± 12.46 ab	49.70 ± 6.28 ab	124.30 ± 16.22 a
Baracpur	49.96 ± 14.79 b	10.82 ± 2.81 b	31.28 ± 2.67 a	27.04 ± 3.85 ab	60.78 ± 17.59 b	58.32 ± 5.66 ab	119.10 ± 19.06 a
Ghosgati	66.83 ± 7.28 ab	14.18 ± 1.38 ab	30.80 ± 2.20 a	29.23 ± 3.27 ab	81.01 ± 8.67 ab	60.03 ± 4.21 ab	141.03 ± 10.83 a
Kamargati	67.82 ± 10.86 ab	14.34 ± 2.00 ab	30.03 ± 6.46 a	29.32 ± 4.39 ab	82.16 ± 12.86 ab	59.35 ± 3.74 ab	141.51 ± 14.14 a
Lakhoati	49.20 ± 13.18 b	10.70 ± 2.51 b	30.30 ± 2.27 a	27.10 ± 2.06 ab	59.91 ± 15.69 b	57.40 ± 3.38 ab	117.31 ± 18.40 a
Average	60.42 ± 3.93	12.89 ± 0.75	28.94 ± 1.12	27.21 ± 1.25	73.31 ± 4.67	56.16 ± 1.71	129.47 ± 5.10

3.4. Bivariate Relationship between Stand Structure, Diversity and Carbon Pools

All four stand structure parameters were positively related to biomass carbon (above-ground and belowground), where basal area had a stronger relation compared to stem density, mean tree height and mean diameter ($p < 0.05$; Figure 3). However, these four-stand structure parameters had no significant relation with any of the soil organic carbon layers ($p > 0.05$; Figure 3). When we considered all the carbon pools together as total mean carbon stocks, all four stand structure parameters except mean tree height had a significant relation and basal area also had a stronger relation ($p < 0.05$, Figure 3). Shannon diversity index also had positive effects on all carbon pools (except lower layer soil organic carbon; $p > 0.05$; Figure 3) and total carbon stocks as well as stem density and basal area ($p < 0.05$; Figure 3).

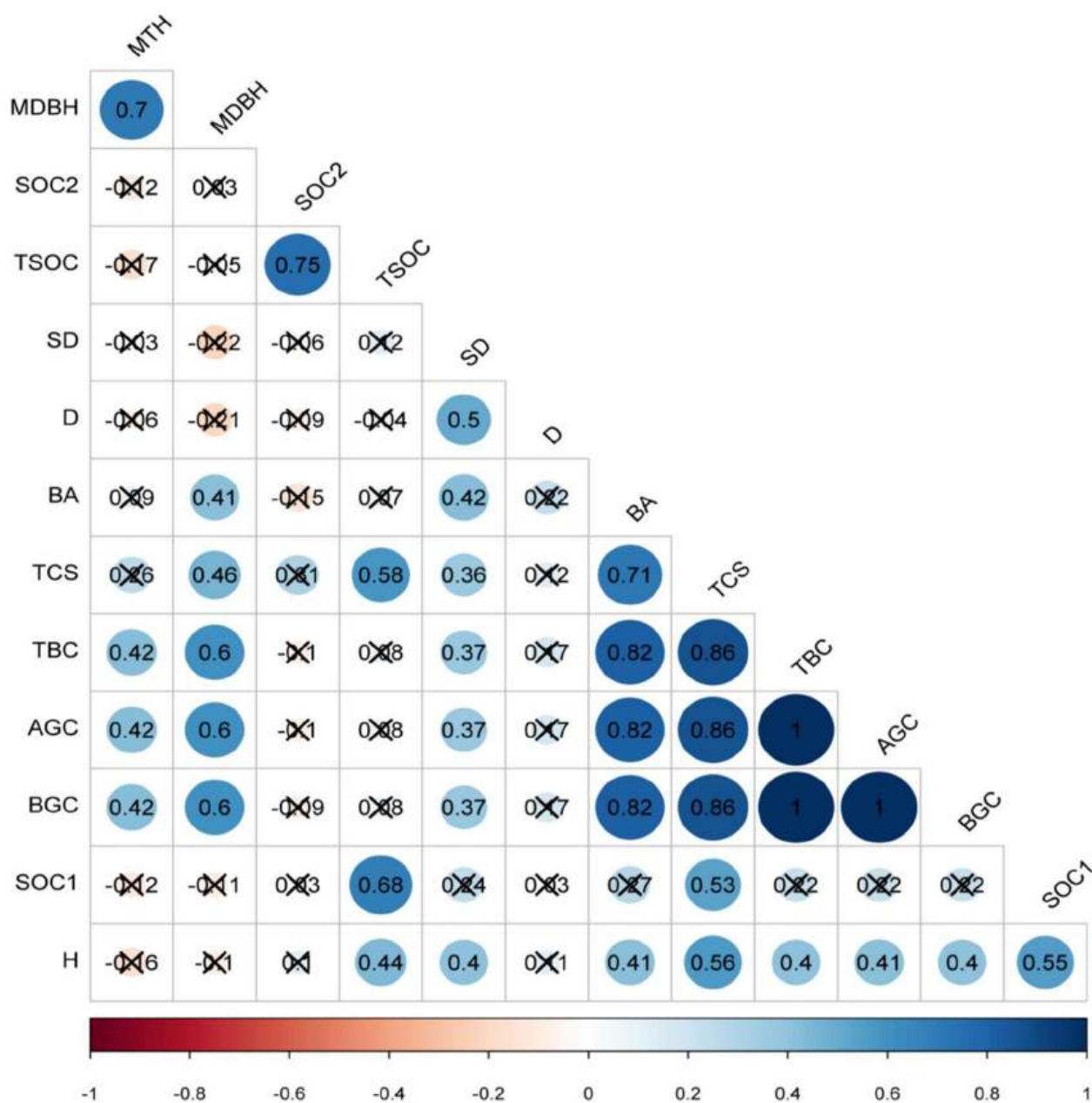


Figure 3. Correlation matrix between stand structure, diversity and different carbon pools. Block with cross sign indicates insignificant relation ($p > 0.05$). MDBH: Mean Diameter at Breast Height (1.37 m), MTH: Mean tree height, SD: Stem Density, BA: Basal Area, D = Margalef Diversity index, H = Shannon Diversity Index, AGC: Aboveground Carbon, BGC: Belowground Carbon, SOC1 = Soil Organic Carbon at 0–15 cm depth, SOC2 = Soil Organic Carbon at 15–30 cm, TBC = Total Biomass Carbon (AGC + BGC), TSOC = Total Soil Organic Carbon (0–30 cm).

3.5. Direct, Indirect and Total Effects of Biodiversity and Stand Structure on Carbon Pools

Shannon diversity had significant direct and indirect effects on aboveground biomass carbon, belowground biomass carbon, total biomass carbon, and total carbon storage, but had only direct effects on the upper layer soil organic carbon and total soil organic carbon (Figure 4a–g). The basal area only significantly mediated the Shannon diversity effect on biomass carbon (above- and belowground) and total carbon stocks (Figure 4a,b,e,g). Basal area had the strongest direct effect on biomass and total carbon storage (Figure 4a,b,e,g), while Shannon diversity had the strongest effect on upper layer and total soil organic carbon storage (Figure 4c,f). Tree height also had positive effects on biomass carbon and total carbon storage (Figure 4a,b,e,g). As the Shannon diversity has indirect effects on the biomass and total carbon storage and, which were mediated by the basal area, the total effect of the Shannon diversity on the aboveground biomass carbon, belowground biomass carbon, total biomass carbon and total carbon storage were 0.393, 0.393, 0.513, 0.393, 0.411 and 0.544, respectively (Table 5). The Shannon diversity, basal area and mean tree height together explained 80%, 80%, 27%, 04%, 80%, 19% and 61% variations in aboveground biomass carbon, belowground biomass carbon, upper layer soil organic carbon, total biomass carbon, total soil organic carbon and total carbon storage in homegardens, respectively (Figure 4a–g).

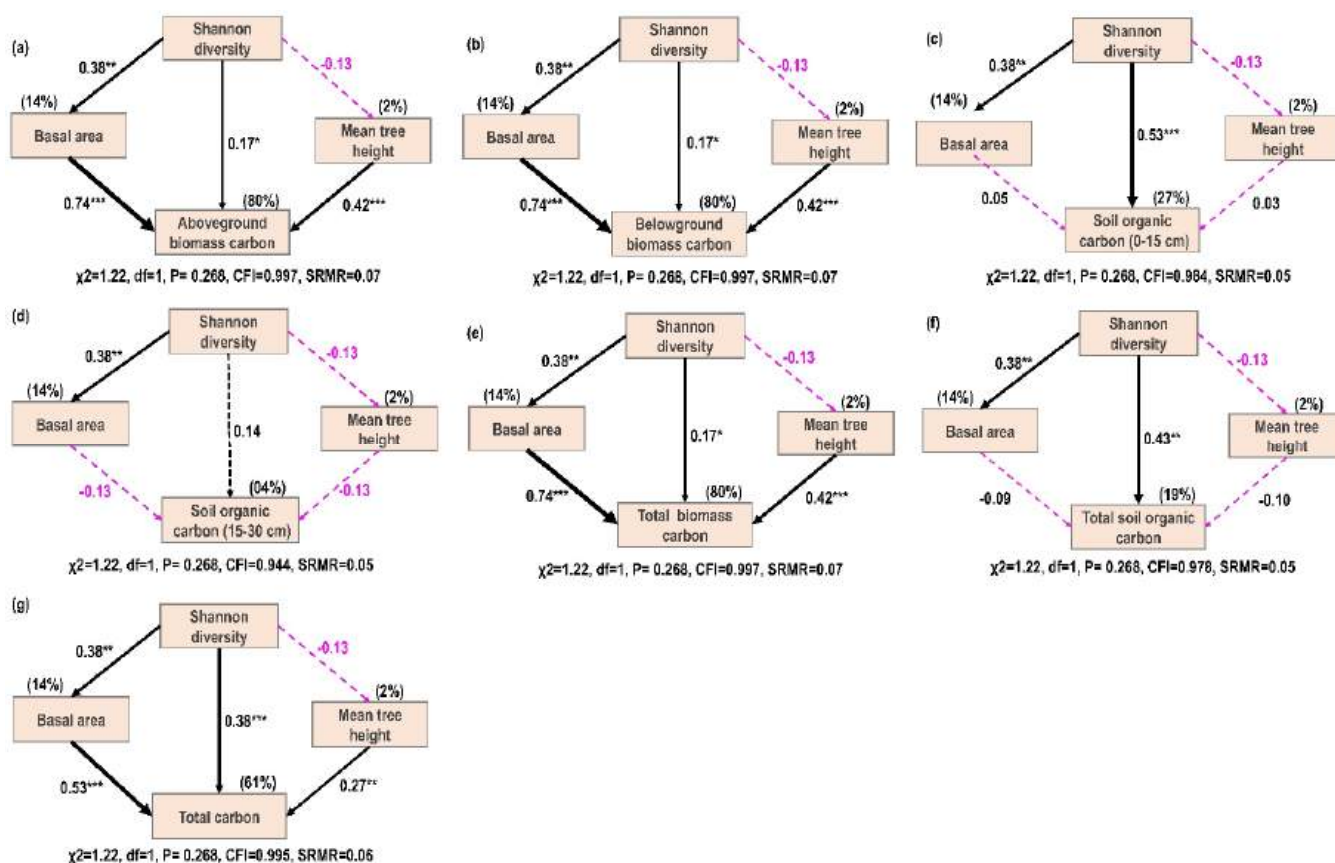


Figure 4. Structural equation models. (a–g) Aboveground biomass carbon (a), belowground biomass carbon (b), soil organic carbon (0–15 cm) (c), soil organic carbon (15–30 cm) (d), total biomass carbon (e), total soil organic carbon (f) and total carbon (g). All SEMs had a similar insignificant χ^2 (Chi-Square) of 1.22 ($p = 0.268$), with a comparative fit index close to one (CFI; 0.98–0.99) and a standardized root mean square residual close to zero (0.08), indicating no significant deviation from model and datasets at one degree of freedom. The lines with pink and black indicate a negative and positive association between the two covariables. Arrows with numbers indicate the standardized association of predictors with dependent variables. Numbers with percentages above boxes independent variables indicate their explained variance (Coefficient of determinant: R squared) by all the predictors. The path values with asterisks indicate their significance level (** $p < 0.001$, * $p < 0.01$, * $p < 0.05$), while the insignificant paths were indicated as dotted lines ($p > 0.05$).

Table 5. Indirect and total standardized effect of the Shannon diversity on different carbon pools. The path values with asterisks indicate their significance level (** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$).

Indirect and Total Effects	AGC	BGC	SOC1	SOC2	TBC	TSC	TCS
	Std. eff.	Std. eff.	Std. eff.	Std. eff.	Std. eff.	Std. eff.	Std. eff.
Indirect effect of the Shannon diversity through basal area	0.282 *	0.282 *	−0.017	−0.049	0.282 *	−0.035	0.201 *
Indirect effect of the Shannon diversity through tree canopy height	−0.056	−0.056	0.004	0.017	−0.056	0.014	−0.035
Total effect of the Shannon diversity	0.393 **	0.393 **	0.513 ***	0.106	0.393 **	0.411 **	0.544 ***

4. Discussion

Homegardens' agroforestry system is rich in biodiversity in the tropics and provides a variety of ecosystem services, including carbon sequestration. We quantified homegardens' agroforestry vegetation coverage and the direct and mediation effects of tree diversity on carbon storage via stand structure in Dighalia, Khulna, Bangladesh. We found that homegardens had high vegetation coverage and carbon stocks where tree diversity promotes carbon storage directly and indirectly through basal area. The stand basal area had a stronger effect on carbon storage than other stand structural parameters.

4.1. Homegardens Vegetation Coverage

We found a high vegetation coverage (as homegardens) in homesteads, which accounted for 24% of the total area of Dighalia Upazila. Two recent studies reported that the percentage of tree coverage outside forests (woodlots, homegardens and other plantations) in Bangladesh is 13.01 (1,920,700 ha/country area) by Landsat Satellite with 30 m × 30 m pixel [38] and 15.1% by Sentinel 1 and 2 with 10 m × 10 m pixel [15]). While the other two studies focus on all types of trees outside forests, we focused only on the homegardens that were found in villages, which play a key role in biodiversity conservation and carbon sequestration. The difference in percentage of tree coverage in our study could be due to the spatial resolution of satellite data and the specific types of trees outside forests. The forest area percentage within the total land area is one of the key indicators of the Sustainable Development Goals (15.1.1), Aichi biodiversity targets of the convention on biological diversity (Target 5; related to forest coverage or extent) and a key activity data for estimating countries' emission factor [6]. Higher accuracy in the estimation of homegardens' coverage with high-resolution imagery (Sentinel 2 10 m band), thus, contributes to assessing the accurate uncertainty in the carbon emission factor.

4.2. Stand Structure

Forest stand structures (tree height, DBH, stand density and basal area) are the key indicators of growth and productivity modeling, and recently, the stand level means of these parameters have been used in stand level biomass carbon modeling [39]. Higher amounts of those parameters at the stand level reveals standing resources and, thereby, growth and biomass carbon stocks [18,39]. The mean stem density in our study (1055.00 ± 35.62 trees ha^{−1}) was very high compared with the natural forests in the Chittagong Hill Tracts (South) Forest Division (381 trees ha^{−1}; Nath et al., 1998) in the Chunati Wildlife Sanctuary, Cox's Bazar (459 trees ha^{−1}) [40]. Higher stem density is also reported in homegardens in Ethiopia (1125.23 ± 334.6) [41]. Homegardens are an intensive land use system, where low planting space is being maintained, which may be the cause of the higher stem density in homegardens compared to natural and secondary forests in Bangladesh. These traditional homegardens with high stem density, thus, play a major role in meeting the timber demand in a sustainable manner while reducing pressure on natural forests [7,18].

4.3. Carbon Stocks

We found a high carbon stock in the studied homegardens, which indicates a huge potential for homegardens to mitigate climate change through sequestering atmospheric CO₂.

The total average carbon stock in our study ($129.47 \pm 5.10 \text{ Mg ha}^{-1}$; biomass + soil 0–30 cm) was within the range of total carbon stocks in Ethiopian homegardens (91.75 ± 4.31 to $156.17 \pm 13.78 \text{ Mg ha}^{-1}$) [14], but higher than Indonesian homegardens (107 Mg ha^{-1}) [9]. Our result was also higher than that of other agroforestry system's carbon stocks, such as agroforest (120 Mg ha^{-1} ; soil carbon at 40 cm only) in Panama and traditional agroforestry systems ($14\text{--}70.08 \text{ Mg ha}^{-1}$) in Mali [42]. The variations in total carbon stocks in different studies are mainly due to the difference in biomass and soil carbon pools that depend on higher stem density, basal area and tree size (height; Table 4 and Figure 3) as well as homestead age [9,11]. For example, in our study, the stem density (1055 stem/ha) in homegardens was higher than that of Indonesian homegardens (624.4 stem/ha). Our findings also showed that across villages, some had higher biomass carbon and lower soil carbon stocks than other villages and vice versa, where the biomass carbon varies mainly due to the difference in mean stem density, basal area and canopy height (Table 3; Figure 3). The positive correlation between these stand structures and different carbon pools indicates that high carbon storage is present in homegardens, and thus, villages are associated with higher stand structure (Figure 3).

4.4. Effects of Tree Diversity and Stand Structure on Carbon Pools

Dealing with two major challenges, climate change mitigation and biodiversity loss under the same management program is more noteworthy than addressing these two separately [43]. In order to practice this type of management, we need to show the evidence that species diversity and carbon storage are positively linked. However, conflicting results (positive, negative or neutral) in different ecosystems (or forest types), sparked debate among scientists and posed a barrier to the adoption of a unified management plan [5,11,18,44]. In our study, we found that tree diversity (Shannon) promotes all carbon pools (except for lower layer soil organic carbon) including total carbon stock (Figures 3 and 4). Tree diversity also had a mediation effect on all carbon pools (except soil organic carbon) through increasing stand basal area. These findings indicate that homegardens associated with high diversity had higher carbon stocks. Our result, therefore, supports the complementary niche hypothesis which believes that higher niche differences between coexisting species can promote more efficient resource acquisition, leading to higher productivity and higher carbon stocks. The basal area and tree height are both good indicators of biomass and productivity, and hence, carbon storage, at the stand level [18,39,45]. Both of these stand structural features had a positive influence on biomass and total carbon storage in homegardens, according to bivariate relationships and subsequently verified by structural equation models, with the basal area having the strongest effect (Figures 3 and 4a,b,e,g). As a result, in order to maximize and sustain both above- and belowground carbon storage (including soil organic) in homegardens, we must maintain higher diversity and stand tree height, where tree diversity increases stand basal area and improves total carbon storage in homegardens. Recent studies in tropical forests have shown that both selection and complementarity directed the biomass carbon stock together [13]. Potential work should consider different components of diversity, e.g., species richness, functional trait diversity and functional trait composition, to investigate the mechanism between diversity and carbon relationship in homegardens.

4.5. Implication for Nationally Determined Contributions

Agricultural activities are among the main sources of carbon emissions in developing countries. Integrated agroforestry systems, however, are considered to be sustainable land-use systems, such as traditional homegardens, which can be a crucial tool for achieving carbon mitigation goals because it has high potential for sequestering atmospheric carbon. Nevertheless, some countries have not included homegardens as possible NDCs, such as Bangladesh, whose homegardens coverage is so rich. Here, we showed that in the Digholia sub-district, each village had high tree coverage (Figure 2) with high diversity (Table 3) and carbon storage (Table 4). In its initial forest reference level, homegardens have not been

listed as forests by the FAO (Food and Agricultural Organization) definition in Bangladesh. Nevertheless, our study shows that both tree coverage (24%) and mean tree canopy height (7.87 m) in homegardens within homesteads satisfied the forest definition by the UNFCCC (United Nations Framework Convention on Climate Change; tree cover: 10–30%; Canopy height: 2–5 m) [46]. Through considering many homegardens as a unit, we can view it as continuous forests (Figure 2). Therefore, homegardens also follow the concept of minimum area of land (0.05–1.0 ha) [42]. Thus, districts with homegardens in Bangladesh satisfy the requirements of REDD+ or the IPCC (Intergovernmental Panel on Climate Change) definition of forests, which has been used in other forest-based mitigation alternatives such as the clean development mechanism (CDM). Nonetheless, since homegardens are not currently included in the REDD+ scheme, we should explicitly enlist homegardens as a choice for prospective NDCs (although a large-scale study is needed in other districts of Bangladesh). Therefore, by setting up a Village Forestry Committee or a group (composed of several homegardens owner), we can manage it like community-based forestry program [18], e.g., social forestry (managed by the Social Forestry Committee with Gender Equity under direct oversight of the Forestry Department of Bangladesh). In this way, it can retain some of the contributions to the Government's climate change adaptation and mitigation measures promised by the Paris Agreement (NDCs; 5% reduction in greenhouse gas emissions by 2030).

5. Conclusions

In this study, we assessed the tree coverage and direct and indirect effects of tree diversity on carbon storage through stand structure in homestead agroforestry systems (homegardens) in Dighalia, using both remote-sensing and field data. Remote sensing-based tree coverage and consistency of extent of homegardens and field-based canopy height met the forest definition as per IPCC or UNFCCC protocol. High tree diversity with a higher stand structure and carbon storage both above- and belowground indicates that homegardens in Bangladesh play a key role in the conservation of biodiversity and climate change mitigation. We also found that tree diversity promotes carbon storage directly and indirectly via basal area. Therefore, we suggest that for maximizing and sustaining carbon storage, we need to maintain diverse tree species which would lead to a higher stand basal area and, hence, high carbon storage in homegardens. In this way, districts with high homegardens in Bangladesh could contribute to mitigating carbon emissions, and thereby, be a prospective choice (as an agroforestry system) for the NDC.

Supplementary Materials: The following are available online at <https://www.mdpi.com/article/10.3390/f12121661/s1>, Table S1: Result of multicollinearity test of different covariates using variance inflation factors analysis. Table S2: Fit statistics of the rejected and accepted structural equation models for predicting different carbon pools in combination of plant diversity and stand structure.

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RESEARCH ARTICLE

Functional composition outweighs taxonomic and functional diversity in maintaining ecosystem properties and processes of mangrove forests

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Abstract

Biodiversity loss can have significant consequences for human well-being, as it can affect multiple ecosystem properties and processes (MEPP) that drive ecosystem services. However, a comprehensive understanding of the link between environmental factors, biodiversity, and MEPP remains elusive, especially in mangrove ecosystems that millions of people along tropical coastlines worldwide depend upon. Here, we collated a comprehensive dataset on forest inventory, plant traits, and environmental factors across 93 plots in the Sundarbans Reserved Forests, Bangladesh. The functional composition (FC) of leaf area showed a stronger positive association with MEPP, being determined by total biomass and productivity of the mangroves, sediment organic carbon, and ammonium, phosphorus, and potassium contents of the sediment, than species richness (SR) or functional diversity (FD). Further, FC mediated a strong negative association of sediment salinity, and a positive association of SR, with MEPP. The similar but opposite total associations of SR and sediment salinity with MEPP suggest that species-rich mangroves could offset the negative impacts of rising salinity on MEPP. When focusing on a single aspect of MEPP, both FD and FC mattered, with the FD of leaf area showing a strong association with mangrove productivity and sediment potassium content, while the FC of leaf litter nitrogen showed the strongest associations with sediment ammonium and phosphorus contents. Therefore, to sustain mangrove ecosystems as a reliable nature-based solution for climate change mitigation, conservation and (re-) establishment projects should prioritize regionally dominant species with high leaf area and nitrogen content, plus functionally different species to support the ecosystem processes and services provided by mangroves.

KEYWORDS

biodiversity, functional traits, multiple ecosystem properties and processes, nature-based

1 | INTRODUCTION

Anthropogenic disturbances have increasingly threatened biodiversity and its capacity to provide and maintain multiple ecosystem processes (Cardinale et al., 2012; Hector & Bagchi, 2007). The simultaneous consideration of a variety of ecosystem properties and processes, coined as ecosystem multifunctionality and defined herein as "multiple ecosystem properties and processes" (MEPP), is a multidimensional concept in ecology that encompasses both ecosystem services and their underlying ecosystem properties and processes (Cardinale et al., 2012; Hector & Bagchi, 2007). For example, productivity (a property), biomass (a property), and sediment organic carbon stocks (a property) are related to carbon storage, a key ecosystem process that underlies the service of climate change mitigation. On the other hand, soil nitrogen (N), phosphorus (P), and potassium (K) contents of the leaf litter are ecosystem properties related to nutrient cycling (Cardinale et al., 2012; Hector & Bagchi, 2007; Jing et al., 2015; Schuldt et al., 2018). Despite the fact that human disturbances influence a variety of ecosystem processes simultaneously, most research remains focused on a single dimension (Adame et al., 2014; Conti & Díaz, 2013; Islam et al., 2015; Kauffman et al., 2020; MacKenzie et al., 2016; McKee, 1995; Mensah et al., 2016; Poorter et al., 2015; Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015; Saintilan, 1997; Tamooch et al., 2008; Tinh et al., 2020), particularly in mangrove forests (Rahman et al., 2021; Sarker et al., 2021). These tropical forests of the upper intertidal zone of soft sediment coastlines are considered important blue carbon ecosystems for climate change mitigation, particularly along large environmental gradients.

To investigate the mechanisms underlying biodiversity–ecosystem process relationships, two hypotheses have been widely proposed (Tilman et al., 1997): niche complementarity and selection effect. Commonly used diversity measures for testing these hypotheses (Le Bagousse-Pinguet et al., 2019; Schuldt et al., 2018) are taxonomic diversity (SR: species richness of a community) and diversity related to plant traits, entailing resource-use strategies within a community, such as functional diversity (FD: variation of species traits within a community) and functional composition (FC: community-wide weighted mean traits). A positive association of taxonomic and functional diversity in regulating ecosystem processes would lend support for the niche complementary hypothesis which suggests that a higher diversity of species or traits ensures optimal utilization of available resources and enhances ecosystem processes and, thus, services. A positive association of functional composition in regulating ecosystem processes, on the other hand, would lend support for the selection effect hypothesis (Tilman et al., 1997), which highlights that the dominant species, represented by their key functional traits, most importantly drive one or multiple key ecosystem processes. These different measures of biodiversity are ultimately linked to different environmental factors of an ecosystem (characterized by various physical and geochemical properties, such as salinity, pH, sediment texture, or elevation) as they

directly or indirectly influence the taxonomic and functional composition of a community (Joswig et al., 2021; Rahman et al., 2021; Sarker et al., 2021).

Studies from terrestrial ecosystems demonstrate that the functional composition and functional diversity of plant traits are more important than the taxonomic diversity in regulating ecosystem processes, as plant traits directly reflect a species' contribution to these ecosystem processes (Hector & Bagchi, 2007; Schuldt et al., 2018). Yet hardly any study on this topic has been conducted in mangrove forests. Studies of mangrove ecosystems primarily focus on the taxonomic aspect of biodiversity, and often only examine a single dimension of ecosystem processes (e.g., ecosystem carbon storage; Rahman et al., 2021) or productivity (Sarker et al., 2021). While a recent study (Rahman et al., 2021) showed how different measures of biodiversity regulate carbon storage in a mangrove ecosystem in Bangladesh, again only a single ecosystem process was considered. However, other ecosystem processes, such as nutrient cycling and primary production, have also been shown to effectively indicate key ecosystem processes and critically determine the vulnerability of mangrove ecosystems through climate change (Alongi, 2018; Jing et al., 2015). Therefore, a holistic assessment of biodiversity–ecosystem process relationships, especially considering the interactions among environmental factors, different biodiversity measures, and MEPP, remains critically needed. The closure of this knowledge gap would also provide pivotal information to test the hypotheses of community and ecosystem ecology as well as to aid managers and policymakers in adopting nature-based solutions to climate change challenges.

In this study, we aim to provide a holistic assessment of the relationships between biodiversity, environmental factors and key ecosystem processes in the Sundarbans Reserved Forest (SRF) in Bangladesh, one of the largest, most diverse, and intact mangrove forests in the world (Rahman et al., 2019; Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). We quantify these relationships using structural equation modeling (SEM), an advanced modeling approach based on causal relationships (Grace et al., 2014). SEM is capable of handling multidimensional interactions among biotic and abiotic factors, as well as ecosystem properties and processes found in natural ecosystems (Grace et al., 2014). Employing this sophisticated modeling approach, we can explore key mechanisms underlying multivariate associations (Rahman et al., 2021). This can be achieved by calculating the direct, indirect, and total associations of an exogenous variable (independent) through its mediators (here: different biodiversity components) with an endogenous variable (dependent). Such results cannot be obtained through conventional bivariate relationships (Grace et al., 2016). Utilizing SEM, we explore the interactions among biodiversity, environment, and individual and MEPPs by testing the following three hypotheses: (H1) Environmental variables (e.g., sediment salinity, pH, texture, and elevation) exhibit direct and indirect (through mediators—taxonomic diversity, functional diversity, and functional composition) associations with individual and MEPPs; (H2) Taxonomic diversity produces direct and indirect associations

(through its alternative pathways of functional composition and functional diversity of different traits); (H3) Functional composition, as opposed to taxonomic diversity and functional trait diversity, exhibits strong direct associations with individual and MEPPs as found in terrestrial ecosystems. To test these hypotheses, we compiled a unique, comprehensive dataset across extensive environmental gradients and forest compositions in SRF, Bangladesh, covering various dimensions of taxonomic and functional diversity, various dimensions of ecosystem properties and processes, and key environmental variables known to drive mangrove ecosystem processes.

2 | MATERIALS AND METHODS

We conducted this study in the SRF in Bangladesh. SRF locates between 21°30'N and 22°20'N and 89°00'E and 89°55'E. The SRF, encompassing an area of 6017 square kilometers, is renowned for being one of the world's most diverse mangrove forests (Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). Mangroves dominate approximately 69% of the total area, while the remaining 31% is covered by water bodies. This forest is home to a remarkable variety of plant and animal species, including 528 species of vascular plants, with 24 being true mangroves and 70 being mangrove associates (Islam et al., 2014; Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). The dominant tree species include *Heritiera fomes*, *Excoecaria agallocha*, and *Ceriops decandra* (Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). The faunal biodiversity is equally impressive, with a recorded total of 1135 species, including endangered creatures like the Royal Bengal Tiger, Ganges and Irrawaddy dolphins, and saltwater crocodiles (Aziz & Paul, 2015).

The SRF plays a vital role in supporting the livelihoods of local communities and holds significant socioeconomic and ecological value (Rahman et al., 2017; Uddin et al., 2013). Recognizing its importance, it was designated as a Ramsar Wetlands Site in 1992 and a UNESCO World Heritage Site in 1997. With its diverse ecosystem services and biodiversity, the SRF is poised to have regional and global environmental implications, especially in the face of climate change and the various anthropogenic and natural disturbances it experiences (Aziz & Paul, 2015; Rahman et al., 2017; Uddin et al., 2013).

To test our hypotheses, we first assembled an extensive dataset of forest carbon inventory data from 93 sampling sites nearly equally distributed across the entire SRF, each comprising five 10-m radius subplots. From this data, we derived biomass (living tree aboveground biomass and belowground root biomass) and sediment organic carbon content at the sampling site level. At each sites, we calculated diversity metrics, including taxonomic diversity (species richness), functional diversity (functional dispersion of specific traits), and functional composition (community-weighted mean of specific traits). Next, we derived multiple ecosystem metrics at each site by averaging the Z-score (Jing et al., 2015; Maestre et al., 2012; a mean of zero and a standard deviation of one) of six field/satellite-derived

ecosystem properties and processes: (1) total biomass carbon (both aboveground biomass and belowground root biomass), (2) sediment carbon content (total sediment organic carbon content down to 1-m depth), (3) forest productivity (approximated by satellite-based Enhance Vegetation Index; Huete et al., 1999), and (4–6) sediment nutrient concentrations (Sarker et al., 2021) of ammonium-nitrogen (NH_4), phosphorus (P), and potassium (K). We used sediment salinity, pH, texture (sand, silt, and clay), and elevation as environmental factors, generating spatial layers for the entire SRF using data from 110 permanent sample plots (Sarker et al., 2021). We then estimated the average value for these environmental factors for each plot using their five subplot values from these spatial layers. Further details of data and analyses involved in this study are provided below.

2.1 | Biomass and sediment organic carbon stock

For our study, we utilized live tree species with a DBH (diameter at breast height) ≥ 10 cm from the SRF inventory data (2009–2010) for biomass calculation and biodiversity measures. The field inventory conducted in 2009–2010 encompassed 150 plots, each containing five 10-m radius circular subplots (Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). Out of these 150 plots, we selected 93 plots, as one to four subplots in the excluded plots were either completely underwater or partially occupied by canals and rivers. To calculate aboveground biomass, we employed a species-specific allometric equation for *E. agallocha* (Hossain et al., 2015) and a common allometric equation for all other species (Chave et al., 2005). Additionally, we used a universal allometric equation for estimating belowground biomass of all trees (Komiya et al., 2005). The sediment carbon included in this study was estimated at a depth of 1 m (Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015).

2.2 | Functional traits and diversity metrics

We used eight plant functional traits related to biomass, productivity, and leaf litter quality in our study. These traits included wood density, maximum canopy height, leaf photosynthesis rate, leaf area, specific leaf area, leaf dry matter content, and leaf litter carbon and nitrogen content. Maximum canopy height for each tree was extracted from the 1997 forest inventory data in the SRF (Rahman et al., 2021), while wood density data were sourced from the Bangladesh Forest Research Institute (Yakub et al., 1972). In this study, we used plot specific trait values for each mangrove forest plot (Chanda et al., 2016; Nandy (Datta & Ghose, 2001). The values for leaf litter traits (Chanda et al., 2016) were averaged across monsoon, pre-monsoon, and post-monsoon periods. For species with missing traits (relative abundance species $< 2\%$), we employed the community average trait value (bold value; Table S1). We used the average value for leaf area and specific leaf from a recent published study in SRF (Sarker et al., 2021). To

determine the species richness of each plot, we counted the number of different species present.

We calculated the functional composition of traits using the community-weighted mean (Equation 1). For computing functional diversity, we employed the functional dispersion (Equation 2) metric using the FD package in R (R Core Team, 2020). As the contribution of a species to forest structure is best represented by its basal area (Finegan et al., 2015), we used species-specific total basal area in each plot as a weighted factor for calculating both metrics. Before calculating the community-weighted mean and functional dispersion, all traits were log transformed and standardized to a zero mean and SD of 1.

$$\text{CWM}_{ij} = \sum_{i=1}^S P_{ij} t_i, \quad (1)$$

where CWM_{ij} represents the community-weighted mean of trait t for plot j , S the species richness, P_{ij} refers to the proportion of relative basal area of species i for plot j , and t_i signifies the mean value of the trait of species i .

$$\text{FDis} = \frac{\sum a_j Z_j}{\sum a_j}, \quad (2)$$

where a_j represents the abundance of species j in terms of basal area, while Z_j denotes the distance of species j to the weighted centroid c , which signifies the centroid of the n species within the trait space.

2.3 | Environmental factors and ecosystem properties related to nutrients

Here, we examined various environmental factors, including sediment salinity, pH, texture (sand, silt, and clay) and elevation, and three key nutrients essential for plant growth: NH_4 , total phosphorus, and total potassium. These variables were measured for surface sediment (0–15 cm depth) at 110 permanent sample plots distributed across the SRF (Sarker et al., 2021). We used Inverse Distance Weighted Interpolation (Basso et al., 1999) in Google Earth Engine to generate spatial maps for these environmental factors and nutrients. Subsequently, we extracted the environmental factors and nutrients from these spatial maps using the shapefile for each subplot (five subplots within each plot). Finally, we calculated the mean values of environmental factors and nutrients for each plot by averaging the values obtained from the five subplots.

2.4 | Individual and MEPPs

Several methods have been proposed to generate metrics that indicate MEPPs, often referred to as ecosystem multifunctionality. These methods include the single-process approach, the turnover approach, the averaging approach, and the threshold approach (Jing et al., 2015; Maestre et al., 2012). Each method has its own advantages and limitations; therefore, in this study, we computed the

metrics using two straightforward approaches: the single-process approach and the average approach (Jing et al., 2015; Maestre et al., 2012). For the single-process approach, we first log transformed each ecosystem characteristic/process, then standardized each of them with a zero mean and a SD of 1. For the average approach, we used the standardized metric of each ecosystem characteristic/process obtained in the previous step and computed their average to derive MEPP (Jing et al., 2015; Maestre et al., 2012).

2.5 | Statistical analysis

To better satisfy linearity assumptions (Grace et al., 2016), we log transformed and normalized all metrics of environment, biodiversity measures, and individual and MEPPs. We found that functional diversity and functional composition of various traits are highly correlated (Figures S1 and S2). To eliminate highly correlated variables from functional diversity and functional composition of different traits, we used multiple linear regressions to quantify variance inflation factors (a method for measuring multicollinearity) with a threshold value of 10 (Graham, 2003). We began with all covariates and then gradually removed variables with the highest variance inflation factor until the threshold value was reached (all covariates with a variance inflation factor below 10). Following this multicollinearity test, we found that the variance inflation factor for the functional diversity of leaf area, leaf litter nitrogen, and dry matter content, as well as the functional composition of maximum canopy height, leaf area, leaf litter carbon, and nitrogen content, was less than 10 (Table S2). Finally, using the Lavaan Package (Rosseel, 2012) in R, we assessed the direct, indirect, and total associations of environmental factors and various biodiversity measures with MEPP and individual ecosystem properties or processes by constructing the associated structural equation models (Grace et al., 2016; Grace & Bollen, 2005).

We constructed structural equation models based on published theories relating diversity to ecosystem properties and processes, which have been tested in terrestrial ecosystems (Mensah et al., 2020). We focused on niche complementarity and selection effect hypotheses, with species abundance and functional diversity of traits supporting the former hypothesis while functional composition of traits supporting the latter hypothesis. Three hypothesized pathways (as our H1–H3 in the Introduction), connecting different environment and diversity measures to individual and MEPPs, guided SEM. Based on these hypotheses, the structural equation model was tested in various combinations of species diversity, functional composition, functional diversity of different traits (variance inflation factor value less than 10; Table S2), and environmental factors (Table S2), respectively, for MEPP and each of the six ecosystem properties and processes. For each SEM, we began with all the covariates for single or MEPPs, then gradually eliminated the hypothesized path with the highest p -value until the model satisfied the model-fitting measures (Rahman et al., 2021). Using MEPP as an example, the combination of environmental factors (sediment salinity and pH, silt content, and elevation), species richness, functional diversity of leaf area, and functional composition of leaf area and leaf litter nitrogen content

provided the best fit (Table S3). Then, with the same predictors, we performed the SEM respectively for each of the six individual ecosystem properties and processes, that is, total biomass, sediment carbon storage, productivity, and sediment NH₄, P, and K contents. By multiplying the standardized associations of each path from environmental factors to species richness to functional composition and functional diversity with individual/MEPPs (Grace & Bollen, 2005), we assessed the indirect association of environmental factors on driving single/multiple ecosystem properties or processes. We multiplied the standardized associations of all paths from environmental factors and species richness with individual/MEPPs via various mediators (Grace & Bollen, 2005) to estimate the total indirect associations of sediment salinity and species richness, respectively. By adding the standardized direct and indirect associations (Grace & Bollen, 2005), we estimated the total association of environmental factors and species richness with individual/MEPPs (Grace & Bollen, 2005).

We last assessed the performance of the structural equation model using several standard fit indices. Typically, a nonsignificant chi-squared test ($p > .05$; Equation 3), a comparative fit index (Equation 4) greater than 0.90, and a standardized root mean square residual (Equation 5) close to 1 indicate that the structural equation model does not significantly deviate from the data (Ali et al., 2019; Grace & Bollen, 2005; Laliberté & Legendre, 2010). To calculate all the fit indices required for evaluating SEM (Laliberté & Legendre, 2010), we employed the FD package in R. Furthermore, we conducted a variance inflation factor test to assess the multicollinearity among all variables represented in the final, fitted structure equation models (Table S4).

$$\chi^2 = (N - 1)f \quad (3)$$

$$CFI = \frac{\max(\chi_0^2 - df_0) - \max(\chi_k^2 - df_k)}{\max(\chi_0^2 - df_0)} \quad (4)$$

$$SRMR = \sqrt{[\rho^{*-1}(e'W_s e)]} \quad (5)$$

where, χ^2 chi-squared test statistic; CFI, comparative fit index; SRMR, standardized root mean square residual; N , sample size; f , minimized discrepancy function; 0, baseline model; df , degrees of freedom; k , tested or hypothesized model, ρ^* , number of non-duplicated elements in the covariance matrix; e , a vector of residuals from a covariance matrix; s , a vector of the ρ^* nonredundant elements in the observed covariance matrix; W , a weight matrix; W_s , a diagonal weight matrix used to standardize the elements of covariance matrix.

3 | RESULTS

3.1 | Interactive associations among biodiversity, environment, and MEPP

To test our hypotheses, we employed SEM and investigated the interactive associations among different biodiversity measures, environmental factors, and MEPP (Figure 1). The final model fits our dataset

well, with no significant deviations observed (chi square = 0.698, $p = .705$, CFI = 1.000, and SRMR = 0.010), explaining 48% of the variation in MEPP. Even after accounting for both direct and indirect associations of environmental factors, we found that species richness exhibited a positive direct association with MEPP (Figure 1; Table 1). Additionally, species richness had an indirect positive association with MEPP through its influence on the functional composition of leaf area (Figure 1; Table 1).

Functional composition of leaf area showed the strongest association with MEPP, while neither functional diversity of leaf area nor functional composition of leaf litter nitrogen content showed any significant association with MEPP (Figure 1). The associations between these biodiversity variables and MEPP in the SEM were also found to be consistent with bivariate analysis (Figure S3a–d).

Silt content (a sediment texture property) was directly and negatively associated with MEPP (Figure 1), although it was not significantly correlated with MEPP according to the bivariate relationship (Figure S4c). In contrast, bivariate analyses showed a significant negative correlation between sediment salinity and MEPP (Figure S4a), but this negative association was found to be mediated by the strong decrease in functional composition of leaf area with increasing sediment salinity, as revealed by SEM analysis (Figure 1; Table 1). A similar pattern was observed for the indirect association of sediment pH with MEPP, mediated by the functional composition of leaf area (Figure S4; Table 1). Sediment pH did not show a significant correlation with MEPP ($p > .05$; see Figure S4b), but its indirect effect was evident in the SEM analysis through the declining functional composition of leaf area (Figure 1). The total association (combining direct and indirect associations) of sediment salinity ($\beta = -0.383$) with MEPP was negative, while it was positive for species richness ($\beta = -0.385$; Table 1).

3.2 | Interactive associations among biodiversity, environment, and single ecosystem properties and processes

We applied structural equation models to individual ecosystem properties and processes, which also demonstrated a good fit with the data, as indicated by similar goodness-of-fit indices (chi-square = 0.698, $p = .705$, CFI = 1.000, and SRMR = 0.010; Figure 2). These models explained 55%, 20%, 41%, 25%, 25%, and 19% of the variation in total biomass, sediment organic carbon, productivity, sediment ammonium, sediment phosphorus, and sediment potassium, respectively (Figure 2a–g). The individual ecosystem property or process showed varying associations with different diversity measures and environmental factors ($p < .05$; Figure 2). For instance, the functional composition of leaf area had the strongest positive association with total biomass and significantly mediated a positive association of species richness, as well as negative associations of sediment pH and salinity, with total biomass ($p < .05$; Figure 2a). For sediment organic carbon stock, SEM revealed that it was significantly enhanced by the functional composition of leaf

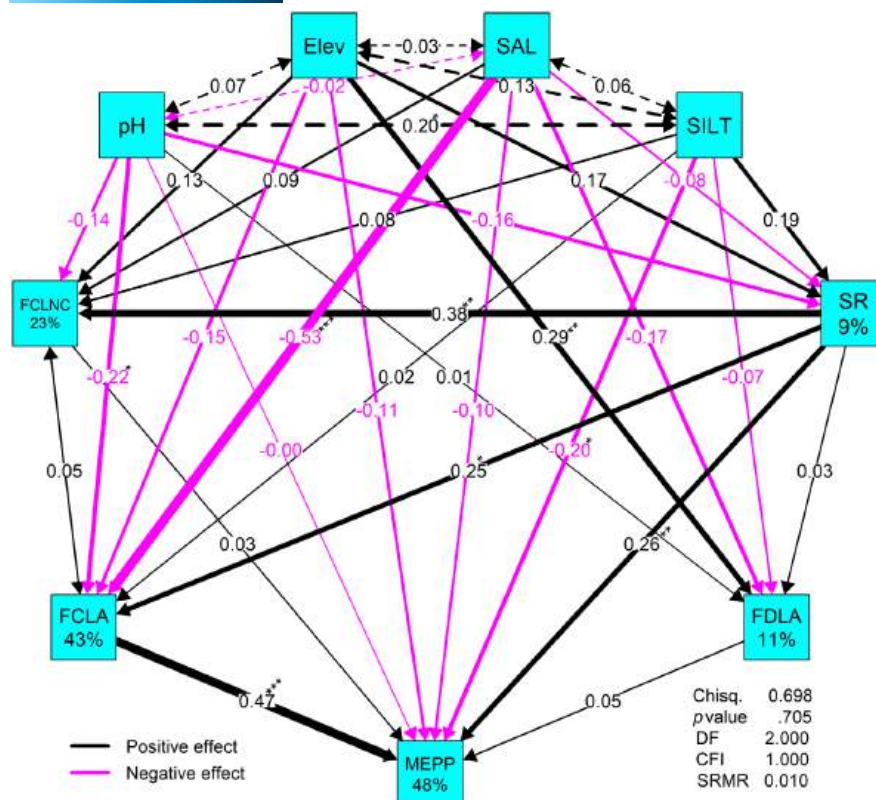


FIGURE 1 Structural equation model (SEM) illustrating the mechanisms underlying the relationships among environmental variables, biodiversity measures, and multiple ecosystem properties and processes (MEPP). In SEM, the environmental variables were sediment salinity (SAL), elevation (Elev), sediment pH (pH), and sediment silt content (SILT), while the biodiversity variables were species richness (SR), functional composition leaf area (FCLA), functional composition of leaf litter nitrogen concentration (FCLNC), and functional diversity leaf area (FDLA). The model demonstrates a non-significant χ^2 (chi-square) of 0.698 ($p = .705$), a comparative fit index close to one (CFI; 1.000), and a standardized root mean square residual close to zero (0.010), indicating no significant deviation between the model and datasets at 2 degrees of freedom. Magenta and black lines indicate negative and positive associations between covariables, respectively. Arrows with numbers indicate the standardized association of predictors with dependent variables. Numbers with percentages inside boxes indicate the explained variance of dependent variables (coefficient of determinant: R^2) by all predictors. Path values with asterisks indicate their significance level (** $p < .001$, ** $p < .01$, * $p < .05$) while paths without asterisk represent insignificant associations ($p > .05$).

area, the dominant biodiversity variable, while it significantly declined with increasing silt content in sediment (Figure 2b). Species richness had an indirect association with sediment organic carbon content via increasing functional composition of leaf area (Figure 2b, Table 1).

In contrast, both species diversity and functional diversity of leaf area displayed a positive association with productivity, with the latter having a stronger association than the former ($p < .05$; Figure 2c). However, in bivariate analysis, the functional composition of leaf area was the strongest predictor ($R^2 = .18$) for productivity among all biodiversity variables (Figure S3m–p). SEM analysis also confirmed that productivity decreased with sediment salinity ($p < .05$; Figure 2c). Interestingly, elevation showed both a direct negative association and an indirect positive association with productivity, mediated by an increase in functional diversity of leaf area ($p < .05$; Figure 2c), despite not exhibiting a significant correlation with productivity in the bivariate relationship ($p > .05$; Figure S4p).

SEM analysis with sediment ammonium confirmed that it directly decreased with sediment salinity but increased with the functional

composition of leaf litter nitrogen content ($p < .05$; Figure 2d). However, the functional composition of leaf area did not show a significant association with sediment ammonium in the SEM analysis (Figure 2d), despite being the only biodiversity variable that exhibited a significant correlation with sediment ammonium in the bivariate analysis (Figure S3t). Species richness indirectly increased sediment phosphorus through an increase in the functional composition of leaf nitrogen litter content (Figure 2e and Table 1). SEM analysis also indicated that sediment phosphorus decreased with increasing sediment pH (Figure 2e), although sediment pH did not show a significant correlation with sediment phosphorus (Figure S2v).

Sediment salinity, elevation, and silt content did not show any significant association with sediment phosphorus (Figure 2e), and neither the functional composition nor functional diversity of leaf area exhibited any significant association, despite both biodiversity variables showing a significant correlation with sediment phosphorus (Figure S3w,x). SEM analysis revealed that sediment potassium significantly increased with elevation and the functional diversity of leaf area ($p < .05$; Figure 2f). Elevation also had an indirect positive

TABLE 1 Indirect and total associations of environmental variables and species richness with single/multiple ecosystem properties and processes through mediators.

Indirect and total association pathways	EMF		TB		PDT		SOC		NH ₄		P		K	
	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value
Indirect association of sediment salinity through species richness	−0.020	.455	−0.006	.542	−0.019	.459	−0.015	.479	−0.005	.617	0.000	.963	−0.004	.667
Indirect association of sediment salinity through FD leaf area	−0.008	.579	0.036	.135	−0.055	.113	0.025	.256	−0.013	.483	0.024	.254	−0.039	.164
Indirect association of sediment salinity through FC leaf litter nitrogen content	0.003	.738	0.002	.802	−0.001	.904	0.002	.864	0.018	.388	−0.022	.373	0.009	.508
Indirect association of sediment salinity through FC leaf area	−0.249	.000	−0.327	.000	−0.083	.148	−0.192	.007	−0.071	.263	0.088	.167	−0.044	.500
Indirect association of sediment salinity through species richness and FD leaf area	0.000	.824	0.000	.811	−0.001	.811	0.000	.813	0.000	.819	0.000	.813	0.000	.811
Indirect association of sediment salinity through species richness FC leaf litter nitrogen content	−0.001	.746	−0.001	.806	0.000	.904	−0.001	.866	−0.006	.477	0.008	.468	−0.003	.558
Indirect association of sediment salinity through species richness and FC leaf area	−0.009	.460	−0.012	.458	−0.003	.505	−0.007	.469	−0.003	.532	0.003	.509	−0.002	.615
Indirect association of sediment pH through species richness	−0.040	.175	−0.013	.404	−0.038	.186	−0.030	.247	−0.011	.547	−0.001	.963	−0.009	.623
Indirect association of sediment pH through FD leaf area	0.001	.905	−0.003	.903	0.004	.903	−0.002	.903	0.001	.904	−0.002	.903	0.003	.903
Indirect association of sediment pH through FC leaf litter nitrogen content	−0.004	.728	−0.003	.798	0.001	.904	−0.003	.863	−0.028	.239	0.035	.210	−0.013	.437
Indirect association of sediment pH through FC leaf area	−0.103	.019	−0.135	.013	−0.034	.195	−0.079	.047	−0.029	.296	0.036	.211	−0.018	.511
Indirect association of sediment pH through species richness and FD leaf area	0.000	.819	0.001	.804	−0.001	.804	0.001	.806	0.000	.813	0.001	.806	−0.001	.804
Indirect association of sediment pH through species richness FC leaf litter nitrogen content	−0.002	.728	−0.001	.798	0.001	.904	−0.001	.863	−0.012	.241	0.015	.212	−0.006	.439
Indirect association of sediment pH through species richness and FC leaf area	−0.018	.190	−0.024	.181	−0.006	.316	−0.014	.216	−0.005	.383	0.006	.327	−0.003	.544
Indirect association of silt through species richness	0.050	.111	0.016	.378	0.048	.124	0.038	.195	0.013	.536	0.001	.963	0.011	.617
Indirect association of silt through FD leaf area	−0.003	.660	0.014	.517	−0.022	.512	0.010	.543	−0.005	.615	0.010	.543	−0.016	.522
Indirect association of silt through FC leaf litter nitrogen content	0.003	.740	0.002	.804	−0.001	.904	0.002	.865	0.017	.421	−0.022	.408	0.008	.526
Indirect association of silt through FC leaf area	0.011	.775	0.015	.775	0.004	.779	0.009	.776	0.003	.782	−0.004	.779	0.002	.792
Indirect association of silt through species richness and FD leaf area	0.000	.818	−0.001	.803	0.002	.803	−0.001	.805	0.000	.812	−0.001	.805	0.001	.803
Indirect association of silt through species richness FC leaf litter nitrogen content	0.002	.726	0.002	.798	−0.001	.904	0.001	.863	0.016	.189	−0.019	.154	0.007	.417

(Continues)

TABLE 1 (Continued)

Indirect and total association pathways	EMF		TB		PDT		SOC		NH ₄		P		K	
	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value	Std.as	p value
Indirect association of silt through species richness and FC leaf area	0.022	.128	0.029	.118	0.007	.276	0.017	.158	0.006	.354	-0.008	.289	0.004	.532
Indirect association of elevation through species richness	0.043	.144	0.014	.391	0.041	.156	0.033	.222	0.012	.541	0.001	.963	0.010	.620
Indirect association of elevation through FD leaf area	0.013	.566	-0.061	.041	0.092	.022	-0.043	.184	0.021	.459	-0.041	.182	0.066	.071
Indirect association of elevation through FC leaf litter nitrogen content	0.004	.729	0.003	.799	-0.001	.904	0.002	.863	0.028	.245	-0.034	.217	0.013	.440
Indirect association of elevation through FC leaf litter nitrogen content	0.013	.566	-0.061	.041	0.092	.022	-0.043	.184	0.021	.459	-0.041	.182	0.066	.071
Indirect association of elevation through species richness and FD leaf area	0.000	.818	-0.001	.803	0.001	.803	-0.001	.805	0.000	.812	-0.001	.805	0.001	.804
Indirect association of elevation through species richness FC leaf litter nitrogen content	0.002	.727	0.001	.798	-0.001	.904	0.001	.863	0.014	.216	-0.017	.185	0.006	.428
Indirect association of elevation through species richness and FC leaf area	0.000	.818	-0.001	.803	0.001	.803	-0.001	.805	0.000	.812	-0.001	.805	0.001	.804
Indirect association of species richness through FD leaf area	0.001	.817	-0.005	.801	0.008	.801	-0.004	.803	0.002	.810	-0.004	.803	0.006	.802
Indirect association of species richness through FC leaf litter nitrogen content	0.012	.721	0.008	.796	-0.004	.903	0.007	.862	0.080	.069	-0.099	.032	0.038	.369
Indirect association of species richness through FC leaf area	0.116	.011	0.152	.006	0.038	.184	0.089	.036	0.033	.289	-0.041	.201	0.021	.508
Sum of indirect association of sediment salinity	-0.284	.000	-0.306	.000	-0.161	.029	-0.187	.021	-0.080	.272	0.101	.174	-0.084	.257
Sum of indirect association of sediment pH	-0.166	.006	-0.176	.006	-0.073	.171	-0.128	.020	-0.085	.054	0.091	.053	-0.048	.280
Sum of indirect association of silt	0.085	.140	0.076	.215	0.037	.456	0.076	.124	0.051	.167	-0.043	.293	0.018	.619
Sum of indirect association of elevation	0.011	.864	-0.112	.086	0.116	.041	-0.047	.428	0.059	.230	-0.073	.157	0.087	.080
Sum of indirect association of species richness	0.129	.021	0.154	.020	0.042	.442	0.092	.123	0.115	.032	-0.143	.012	0.064	.254
Total association of sediment salinity	-0.383	.000	-0.375	.000	-0.461	.000	-0.071	.491	-0.313	.001	0.295	.002	-0.044	.649
Total association of sediment pH	-0.168	.074	-0.159	.084	0.033	.719	0.026	.805	-0.197	.041	-0.115	.236	-0.013	.900
Total association of silt	-0.115	.225	-0.091	.325	0.095	.310	-0.139	.187	-0.129	.186	-0.077	.432	0.051	.608
Total association of elevation	-0.103	.269	-0.224	.014	-0.105	.253	-0.016	.876	-0.032	.737	-0.216	.025	0.334	.001
Total association of species richness	0.385	.000	0.235	.010	0.287	.001	0.286	.006	0.184	.058	-0.139	.159	0.121	.231

Note: The indirect and total associations of sediment salinity, sediment pH, silt, elevation and species richness were derived based on structural equation models. The significant standardized associations (std.as) have $p < .05$.

Abbreviations: FC, functional composition; FD, functional diversity.

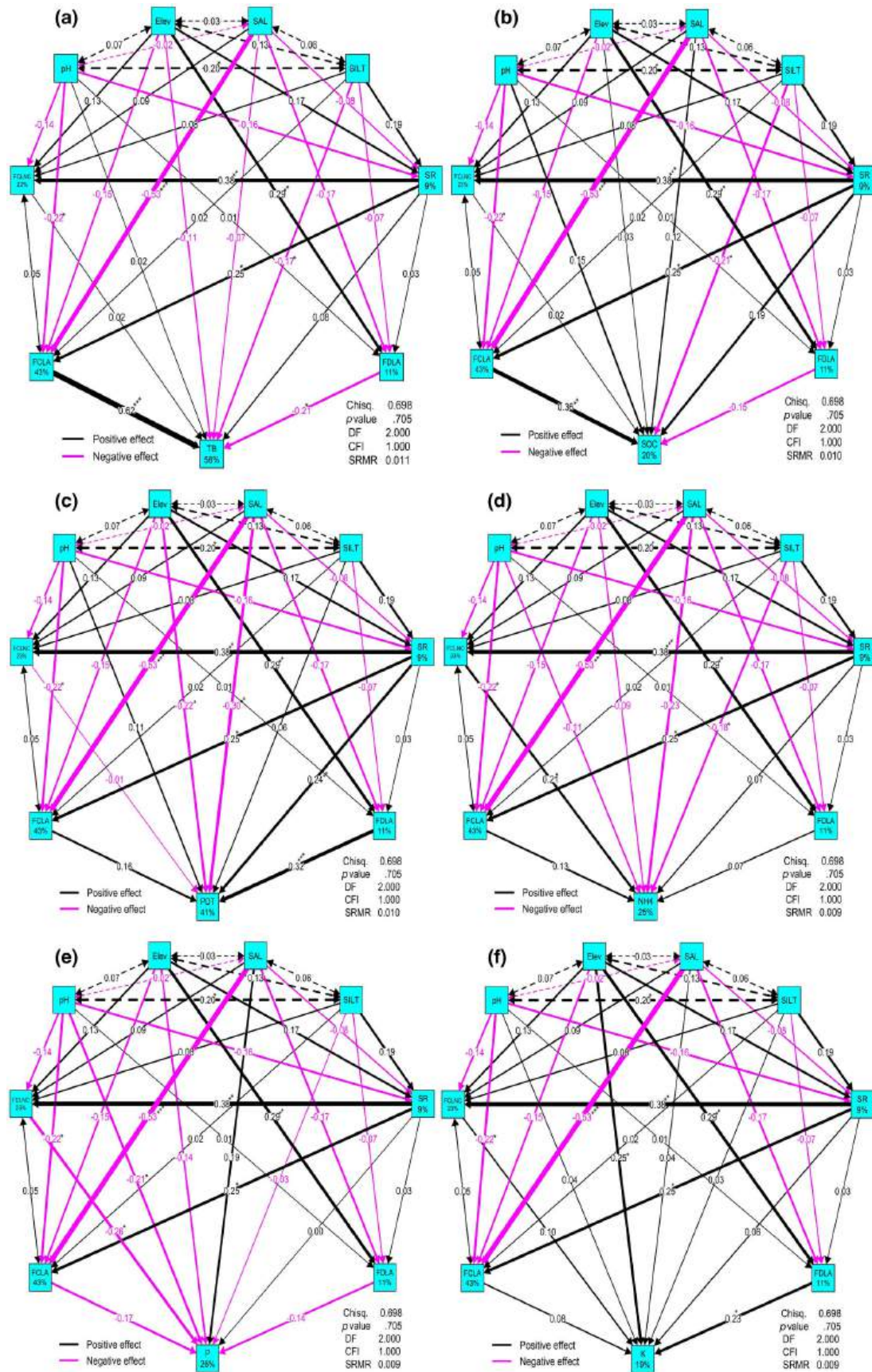


FIGURE 2 Structural equation models (SEMs) illustrating the mechanisms underlying the relationships among environmental variables, biodiversity measures, and individual ecosystem properties and processes (panels a–f). The six SEMs represent the models for (a) total biomass (TB), (b) sediment organic carbon content (SOC), (c) productivity (PDT), (d) sediment ammonium (NH_4), (e) total sediment phosphorus content (P), (f) total sediment potassium content (K). In each SEM, the environmental variables were sediment salinity (SAL), elevation (Elev), sediment pH (pH), and sediment silt content (SILT), while the biodiversity variables were species richness (SR), functional composition leaf area (FCLA), functional composition of leaf litter nitrogen concentration (FCLNC), and functional diversity leaf area (FDLA). SEMs had non-significant χ^2 (chi-square) with a comparative fit index close to one (CFI; 1.000) and standardized root mean square residual close to zero (0.010), indicating no significant deviation between models and datasets at 2 degrees of freedom. Magenta and black lines indicate negative and positive associations between covariables, respectively. Arrows with numbers indicate the standardized association of predictors with dependent variables. Numbers with percentages inside boxes indicate the explained variance of dependent variables (coefficient of determinant: R^2) by all predictors. Path values with asterisks indicate their significance level ($***p < .001$, $**p < .01$, $*p < .05$), while insignificant paths are indicated without asterisk ($p > .05$).

association with sediment potassium through an increase in the functional diversity of leaf area (Figure 2f; Table 1). None of the other environmental or biodiversity variables showed a significant association with potassium (Figure 2f; Table 1).

4 | DISCUSSION

Over the past few decades, biodiversity research has evolved from focusing solely on taxonomic diversity to considering functional diversity and, more recently, the functional composition of communities (Conti & Díaz, 2013; Finegan et al., 2015; Rahman et al., 2021; Tilman et al., 1997). In this study, we investigated how different diversity measures (species richness, functional diversity, and functional composition) and environmental factors interact to determine six major individual ecosystem properties or processes and their combined entity, coined as MEPP. We used mangrove ecosystems in the SRF as a case study. Our analysis revealed that the properties and processes of the studied mangrove ecosystems were partly driven by environmental conditions (supporting H1). Species richness had both positive direct and indirect associations with MEPP (supporting H2). However, the functional composition of leaf area, along with the environmental factor of sediment salinity, showed the strongest association with MEPP (supporting H3), while functional diversity per se did not exhibit any significant association with MEPP. The functional diversity of leaf area, however, demonstrated strong positive associations with productivity (a single ecosystem process) and sediment K content (a single ecosystem property related to nutrient cycling, which is a multifaceted ecosystem process). Moreover, species richness indirectly enhanced sediment ammonium through increasing the functional composition of leaf litter nitrogen content (Figure 2d) which, in turn, promotes microbial decay and results in elevated sediment nitrogen levels (Hossain et al., 2013; Quadros et al., 2021). Mangrove leaves are usually rich in potassium but poor in nitrogen and phosphorus as the plant roots are often inundated by seawater providing most of the potassium to the ecosystem (Alongi, 2018; Reef et al., 2010). Some salt-tolerant species, such as *E. agallocha*, *Avicennia officinalis*, and *Xylocarpus mekongensis*, can also take up more potassium than other less salt-tolerant species (e.g., *H. fomes*) from saline porewater, making the plant tissues rich in potassium (Alam et al., 2021; Hossain et al., 2013; Reef et al., 2010).

The decay of the potassium-rich tissues, in turn, results in an enrichment of potassium in the sediment (Alam et al., 2021; Hossain et al., 2013). Our correlation analysis confirmed such a relationship between functional diversity of leaf area and sediment potassium contents (Figure S3z) and our SEM model (Figure 2) also reconfirmed the relationship.

Incorporating abiotic environmental factors into our understanding of biodiversity and its effects on MEPP improves the predictability of ecosystem service-provisioning. This is because these factors influence both species and their functional traits, which are related to services and the underlying properties and processes, such as productivity (Komiya et al., 2008), carbon storage (MacKenzie et al., 2016), and nutrient cycling (Quadros et al., 2021). Our SEM analysis revealed that the combination of different biodiversity measures (primarily species richness and functional composition) and environmental factors (sediment salinity, pH and silt content, and elevation) raised the predictability of MEPP (48%) compared to the simple linear analysis of their individual contributions (0%–45%). The remaining unexplained variation (52%) could be attributed to a lack of representation of phylogenetic diversity (Le Bagousse-Pinguet et al., 2019) or belowground microbial diversity (Shi et al., 2021), or other environmental factors not considered in this study. Although taxonomic diversity (species richness) is highly correlated with phylogenetic diversity (Yuan et al., 2018), and the environmental factors used in this study could also display tight correlations with microbial diversity (Liang et al., 2016), incorporating these additional diversity measures in future analyses could potentially improve the explanatory power of the model for MEPP.

Understanding the mechanisms behind the links between biodiversity and MEPP is critical for both fundamental science and ecosystem management (Hector & Bagchi, 2007; Le Bagousse-Pinguet et al., 2019; Schuldt et al., 2018; Soliveres et al., 2016). The association of species richness and functional composition with MEPP has strong implications for the management and sustainable use of mangrove forests in the Sundarbans and beyond. Our finding of a positive association of species richness with MEPP lends our support to the niche complementarity hypothesis (Mensah et al., 2020; Tilman et al., 2009), which predicts that forest stands with diverse coexisting species utilize available resources more efficiently than monospecific stands. This finding suggests that mangrove forests experiencing high species loss rates due

to deforestation, illegal logging, or overexploitation (Goldberg et al., 2020; Polidoro et al., 2010; Richards & Friess, 2016) may struggle to maintain MEPPs simultaneously. Conversely, our findings imply that current coastal mangrove restoration programs with monospecific plantations are unlikely to provide MEPPs altogether, whereas implementing mangrove stands with only a few, but functionally diverse mangrove species might be more successful. Furthermore, the SEM with individual ecosystem properties and processes suggests that, in addition to increasing productivity, total stand biomass, and sediment organic carbon storage, species richness promotes sediment nutrient (NH_4) availability through increasing litter quality, particularly leaf litter N content (Figure 2d). All these ecosystem properties and processes, which are enhanced by increased species richness, are essential for maintaining a healthy and stable ecosystem capable of coping with climate change. Therefore, depending on environmental conditions and the functional composition of the regionally available species pool, mangrove (re-)establishment of (functionally) diverse communities might better support MEPPs. This is particularly relevant when comparing species-rich mangrove forests of Asia to those in the Americas, which are often naturally species-poor or even monospecific but exhibit large trait differences among mangrove species (Quadros et al., 2021).

Our findings also provide critical insights into the mechanisms that underlie the maintenance of MEPP in mangrove forests, particularly concerning the relative importance of various biodiversity measures (corresponding to different hypotheses) in accounting for the complex interplay between environmental factors, diversity measures and MEPP. In addition to the positive association of species richness with MEPP, which supports the niche complementarity hypothesis, our results reveal that functional composition exhibits an even stronger positive association with MEPP. This indicates that functional composition, and thus the selection effect hypothesis, plays a more significant role in MEPP in mangrove forests. Consequently, a few dominant species with larger leaf areas in a species-rich mangrove ecosystem drive its MEPP. It is likely that forest stands with functionally diverse tree species utilize sunlight and other resources more efficiently (Williams et al., 2021) than monocultures or communities of highly similar species, and the few dominant species with key traits (e.g., larger leaf area and higher leaf litter) can further enhance resource-use efficiency in these mixed forest stands, resulting in higher productivity (Lange et al., 2015; Pettoirelli et al., 2021). This increased productivity leads to the production of leaf litter, which in turn promotes faster nutrient cycling and higher sediment carbon storage (Hossain et al., 2013; Lange et al., 2015; Quadros et al., 2021; Rahman et al., 2021).

Nature-based solutions for climate change mitigation and adaptation inherently address both the biodiversity crisis and the negative ecosystem consequences associated with climate change. Knowledge regarding the associations among biodiversity, environment, and MEPP (Pettoirelli et al., 2021) remains limited but our study is among the first to elucidate these associations. Our

findings, on one hand, pledge for protecting species-rich mangrove forests that support MEPP without human intervention, such as in the old-grown and highly diverse SRF (Rahman et al., 2019; Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). On the other hand, our findings imply that well-designed mangrove (re-)establishment strategies (Zimmer, 2018; Zimmer et al., 2022) that focus on a few regionally dominant species, for example, with large leaf area and high leaf litter nitrogen content, along with adding some functionally different species to support a distinct functional composition of the community would help to enhance ecosystem service-provisioning. This ecosystem codesign approach (Zimmer, 2018) aims to implement simple communities with traits that drive those ecosystem properties and processes essential to key ecosystem services that are most demanded by local societies (Zimmer et al., 2022).

The negative associations of sediment salinity, pH (indirectly through different biodiversity measures), and silt content (directly with individual and MEPPs (Figures S4 and S5) indicate that changes in these environmental factors could have adverse impacts on service provisioning by mangrove ecosystems. While sediment salinity had only negligible effects on species richness in the Sundarbans mangrove forests, it was negatively related to two crucial plant functional traits (maximum canopy height and wood density), which in turn were strongly associated with carbon storage (Rahman et al., 2021).

Reductions in water discharge upstream of the mangrove forest, such as those caused by dams and polders in the hinterland, could increase sediment salinity and siltation (Alongi, 2018; Rahman et al., 2021; Rahman & Rahaman, 2017). Similarly, rising sea levels may increase inundation frequencies in currently less saline mangrove stands, dominated by tall species with large leaf areas and plant-accessible nutrients (e.g., NH_4) but low salt tolerance (Rahman, Hossain, et al., 2015; Rahman, Kabir, et al., 2015; Rahman, Khan, et al., 2015). Although the relations between tidal regimes, inundation frequency and duration, and sediment salinity are complex, sea level rise is expected to not only shift species composition (Alongi, 2018; Rahman et al., 2020) toward salt-tolerant species with smaller leaves and other functional traits (e.g., wood density, canopy height) relevant to carbon storage (Rahman et al., 2021) but also impair the functional distinctiveness of leaf area and nutrient availability, negatively impacting MEPP. Similarly, increased siltation (Alongi, 2018) is expected to exert direct impacts on both individual and MEPPs (Figures 1 and 2a,b,d). As a result, a regional treaty on upstream freshwater sharing is urgently needed for India, Bangladesh, Nepal, and China, as the major rivers (Ganges, Teesta, Bramaputra, Padma, and Megna) that carry freshwater into the SRF and Indian Sundarbans are distributed among these countries (Rahman & Rahaman, 2017).

Finally, our findings on the associations between species richness, sediment salinity, and MEPP provide a basis for optimism for the sustainability of mangrove ecosystems in the face of climate change. The comparable, yet contrasting, total associations of species richness and sediment salinity with MEPP (Table 1)

suggest that mangrove stands with a diverse array of species can effectively counteract the negative impact of rising salinity on MEPP. Consequently, to maintain MEPP and ensure the resilience of mangrove ecosystems against unfavorable environmental conditions, it is necessary to focus not only on regionally dominant species and their functional traits (such as large leaf area and high leaf litter nitrogen content) but also on incorporating a select few functionally distinct species that contribute to a unique functional composition.

AUTHOR CONTRIBUTIONS

Md Mizanur Rahman: Conceptualization; data curation; formal analysis; investigation; methodology; software; visualization; writing – original draft. **Martin Zimmer:** Writing – review and editing. **Daniel Donato:** Formal analysis; writing – review and editing. **Imran Ahmed:** Investigation; writing – review and editing. **Ming Xu:** Supervision; writing – review and editing. **Jin Wu:** Resources; supervision; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from Rahman et al. (2024) in FigShare at <https://doi.org/10.6084/m9.figshare.24948975.v1>.

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Article

Opposing Ecological Strategies Together Promote Biomass Carbon Storage in Homegardens Agroforestry of Southern Bangladesh

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Abstract: Exploration of the biodiversity–environmental factors–carbon storage relationships have been a central research question in the changing global climate over the last few decades. However, in comparison to other forest ecosystems, very few studies have been conducted in homegarden agroforestry plantations, which have a tremendous capacity to battle global climate change sustainably. We hypothesized that (i) soil organic matter content has both a direct and indirect effect on aboveground carbon storage through species richness, structural diversity, functional diversity (FD) and functional composition (FC); (ii) some facets of diversity (structural diversity, FD and FC) would be more important in linking species richness to aboveground carbon; (iii) species richness, FC, structural diversity and FD would have a positive impact on aboveground carbon storage (AGC) after considering the effect of soil fertility; and (iv) FC would have a greater effect on AGC than the other three components of biodiversity. These hypotheses were tested using structural equation modeling with field data obtained from 40 homesteads in southwestern Bangladesh. We observed that species richness, FC of maximum canopy height and structural diversity had significant effects on AGC, while soil organic matter and FD of wood density had an insignificant effect. Among the four biodiversity components, the structural diversity had a greater influence on AGC. Contrary to our hypothesis, soil fertility and species richness did not have a significant indirect effect on AGC through their mediators. These four components of biodiversity, along with soil organic matter together explained 49% of the variance in AGC. Our findings indicate that both niche complementarity and selection effects regulate AGC in homegardens, where the former theory had stronger control of AGC in homegardens. Therefore, we need to maintain not only the species diversity but also structural diversity (DBH) and functional composition (canopy height) for enhancing aboveground carbon storage on a sustainable basis in homegardens and other restoration programs under nature-based solution.

Keywords: soil organic matter; mechanism of carbon storage; homegardens; functional traits diversity; nature-based solution



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1. Introduction

Understanding the relationship between biodiversity and ecosystem services, in particular carbon storage, has been a central research theme in ecology over the recent decades [1–3]. This is due to the ongoing loss of biodiversity and its impact on the carbon storage capacity of forest ecosystems [4]. However, in comparison to tropical, subtropical and temperate forests [1,2,5,6], few studies have investigated homegardens agroforestry systems with an emphasis on multiple aspects of biodiversity (species, functional trait diversity and structural diversity) [7,8]. Homegardens in the tropical and subtropical

countries are rich in biodiversity, with uneven ages and a stand multilayer structure that resembles a forest. They play an important role in climate change mitigation and adaptation by sequestering and storing atmospheric CO₂ as biomass and soil organic matter [9,10]. In addition to this regulatory service (carbon sequestration), it also provides provisional services, such as food, fodder, fuel wood, timber and other domestic products [11–13]. Thus, homegardens can be viewed as one of the key nature-based solutions for climate change mitigation and adaptation.

Multiple abiotic and biotic factors interact with each other in the forest ecosystem and affect carbon storage [5,14]. Among the abiotic factors, some crucial factors which are interlinked to other physical, chemical and biological factors related to soil can be viewed as a key indicator of soil productivity [15,16]. For example, soil organic matter enhances common soil properties, such as bulk density, composition, texture and water holding capacity, cation exchange capacity, nutrient cycling, nitrogen mineralization bacteria, dinitrogen fixation, mycorrhizae fungi and microbial biomass productivity [15–17]. As a result, organic matter plays a key role in plant productivity and hence in aboveground carbon storage. In the case of biotic factors, the multiple aspects of biodiversity—species diversity, structural diversity, functional diversity and composition—are commonly being used for assessing their effects on carbon storage in recent years [2,14]. These multiple aspects of diversity can also mediate the effect of organic matter content on aboveground carbon storage, as they are also influenced by organic matter content. Thus, by considering these abiotic and biotic influences in an integrated modeling framework (structural equation model, a robust approach that considers both direct, indirect and total effects of predictors on response), this enriches our understanding of the biodiversity–environmental factor–carbon relationship in the homegarden agroforestry systems.

Several theories have been developed to explain the relationship between biodiversity and carbon, of which niche complementarity and selection theories are mostly supported by the results of biodiversity–carbon studies in different habitats [2,18,19]. In a study on whether the species richness (count of species in a plot or community), functional diversity (variation of different traits between species, e.g., maximum canopy height, wood density) or structural diversity (variation of DBH in a plot or community) has a positive effect on carbon storage, the niche complementarity effect is assumed to be effective in carbon storage [2,7,20]. The explanation behind this assumption is that niche complementary theory assumes that the ecosystem carbon capacity is mostly determined by the diverse coexisting species, with their greater trait variation in a community [2,21]. For example, balancing light demanding species in the top canopy and shade tolerant species in the lower canopy to optimize light use efficiency will lead to overall community carbon gain [2,18,19,22]. On the other hand, if the functional composition of plant traits—the community weighted mean of traits—in a community or stand has a positive effect on carbon storage [2,22–24], the selection effect is assumed to be efficient. The selection effect assumes that among the coexisting species, the dominant species with key characteristics determine the carbon storage [2,22,24]. For instance, maximum canopy height, a key indicator of light interception [25] and of structural change reflecting the higher resource use capacity of a species than its neighbor in a community, determines community scale biomass allocation and carbon storage [2,22–24]. While dominant species with their high tissue quality (e.g., high wood density indicates higher investment of biomass in per unit structure), has a positive effect on aboveground biomass and therefore carbon storage [22].

Bangladesh is a downstream country of major rivers, with a vast flat plain of fertile soil favorable for plant growth, making it one of the world's richest homegardening countries [10,26]. There are about 4.5 million ha of tree cover area outside the designated forest area of Bangladesh, in which homegarden's contribution is predominant [26,27]. Every household in each village has a homegarden, which is a well-established land use system and is biologically diverse [28,29]. Most of the studies on homegardens in Bangladesh remain descriptive on the floristic, structure, uses and the relation between household and home garden characters, with more recent studies on the taxonomic diver-

sity and carbon storage [30,31]. In this study, we are focused on the understanding of the biodiversity–carbon relationship after accounting for the effect of soil organic matter as a proxy of soil fertility. We hypothesized that (i) soil organic matter content has both a direct and indirect effect on aboveground carbon storage through species richness, structural diversity, functional diversity and functional composition; (ii) some facets of diversity (structural diversity, functional diversity and functional composition) would be more important in linking species richness to aboveground carbon; (iii) all the four biodiversity components have a positive effect on aboveground carbon storage after considering the effect of soil fertility; and (iv) functional compositions have a greater effect on aboveground carbon storage compared to other biodiversity components.

2. Materials and Methods

2.1. Study Area

The study was conducted in the Dighalia Upazila, Khulna district of Southwestern Bangladesh (Figure 1). The study area is primarily a flood plain land mass between 22.50° to 22.60° N latitude and 89.3° to 89.37° E longitude. The deltaic landscape of this region is mainly low (<10 m above average sea level), flat and fertile [32]. The average area of the homestead over the study site is 0.05 ha [10]. It generally enjoys a tropical to subtropical monsoon climate with an average annual temperature of 26 °C. January is the coolest month, and April is the hottest month in this region. The average annual rainfall for this region is 1986 mm (range: 1400–2600 mm) [32].

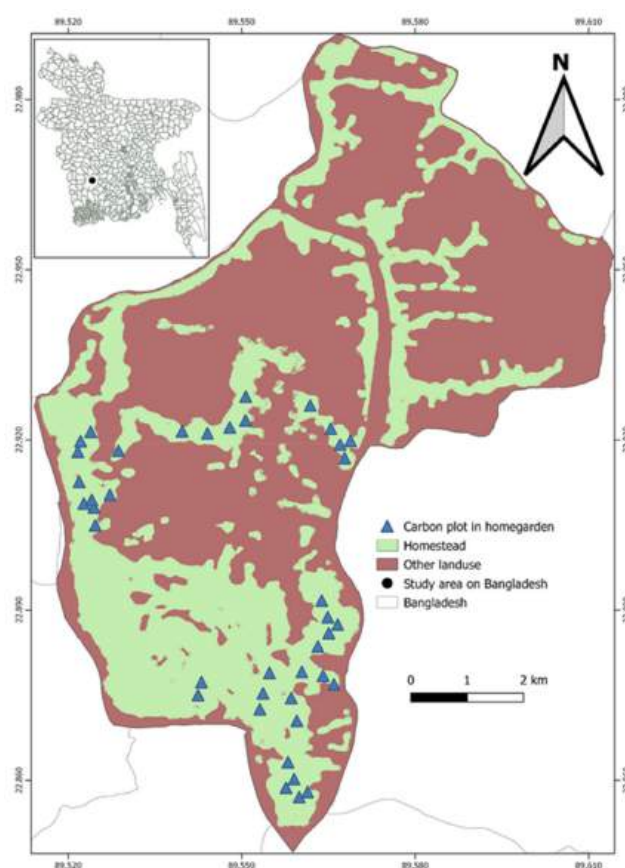


Figure 1. Location carbon plots within homesteads in Dighalia Upazila, Khulna, Bangladesh.

2.2. Field Inventory

We laid out 40 plots (10 m × 10 m) at 40 household with homegardens in eight villages of Dighalia, Khulna, Bangladesh (Figure 1). The diameter and height of all the species having DBH (diameter at breast height) ≥ 3 cm were measured using diameter tape and a

range finder, respectively. All species in each sampled plot were identified and recorded to species level or by local name, and later were confirmed from an authentic source [10]. At each plot from the center, a 5 cm sample soil core was collected at the middle point of 0–15 cm soil depth that was pulled out using an open face peat augur. All samples were air-dried and then analyzed for soil organic content measurement at the Nutrient Dynamics Laboratory of Khulna University, Khulna, Bangladesh.

2.3. Data Analysis

Biomass Carbon

We used a common allometric equation for aboveground biomass estimation for trees because site-specific allometric equations were not available for all the species (38 species; Equation (1)) [33]. The wood density data were obtained from the Bangladesh Forest Research Institute [34] and Global Wood Density Database [35]. For species with missing wood density in these databases, we used community average wood density. The biomass of palm, coconut, betel nut and date species was estimated by using a height based allometric model (Equation (2)) [36]. We calculated biomass carbon by multiplying biomass with 0.5, as it is assumed that wood biomass contains 50% carbon [37].

$$\text{AGB (Mg/ha)} = \rho \times \exp(-1.499 + 2.148 \times \ln(\text{DBH}) + 0.207 \times (\ln(\text{DBH}))^2 - 0.0281(\ln(\text{DBH}))^3) \quad (1)$$

$$\text{AGB (Mg/ha)} = 6.666 + 12.826 \times \text{ht} \times 0.5 \times \ln(\text{ht}) \quad (2)$$

where AGB = aboveground biomass, ρ = wood density (g cm^{-3}), DBH = diameter at breast height, ht = H= height.

2.4. Functional Traits and Diversity Metrics

We used wood density and maximum canopy height, the two widely used functional traits in biodiversity–carbon relationship studies because they are highly related to aboveground biomass carbon [2,38]. Maximum canopy height for each tree species was extracted from field inventory data. We assessed species richness by counting the number of species in each plot. Plot level structural diversity was estimated by calculating standard deviation of DBH. Functional composition and functional diversity of traits were calculated using community weight mean (Equation (3)) and functional dispersion (Equation (4)) metric, respectively, in R by using FD package [39]. Both metrics were abundance weighted and the species wise total basal area in each plot was used as abundance because species contribution in an ecosystem is best represented by basal area [24,40]. As recommended, we standardized all the traits before calculating community weighted mean and functional dispersion.

$$\text{CWM}_{tj} = \sum_{i=1}^S P_{ij} t_i \quad (3)$$

where CWM_{tj} is the community weighted mean of trait t for j plot, S is the species richness, P_{ij} is the proportion of relative basal area of species i for j plot and t_i is the mean value of trait of species i .

$$\text{FDis} = \frac{\sum a_j Z_j}{\sum a_j} \quad (4)$$

where a_j is the abundance of species j in terms of basal area and z_j is the distance of species j to the weighted centroid c which represents the centroid of the n species in trait space.

For calculating soil organic matter, we used the Loss of Ignition method [41]. In this method, one gram of soil was taken in a pre-weighted porcelain cup and oven-dried at 105 °C for 24 h [41]. The oven-dried sample was ignited by using a digital muffle furnace

(WiseTherm F, Wids, Korea) at 450 °C for four hours. After ignition and cooling at room temperature, we weighed the ashes to calculate the loss of ignition (Equation (5)).

$$\text{Loss of Ignition (\%)} = \frac{(W_{t_{105\text{ }^{\circ}\text{C}}} - W_{t_{450\text{ }^{\circ}\text{C}}})}{W_{t_{105\text{ }^{\circ}\text{C}}}} \times 100 \quad (5)$$

Here, $W_{t_{105\text{ }^{\circ}\text{C}}}$ is the weight of soil at 105 °C and $W_{t_{450\text{ }^{\circ}\text{C}}}$ is the weight of soil at 450 °C.

2.5. Statistical Analysis

The structural equation model was used for assessing the direct, indirect and total effects of predictors on aboveground carbon storage [14,42–44]. The carbon taken up by plants and accumulated in plant biomass and soil organic matter over time is influenced by integrated mechanisms involving both abiotic and biotic influences [14,42,44]. For understanding the underlying mechanisms, simple linear or multiple linear regression is insufficient because they only provide insight into direct effects [43]. However, biotic and abiotic factors interact with each other and regulate ecosystem function and services jointly [14,42–44]. Thus, for addressing this complex system in an integrated way, we need to use complex modeling approaches, such as structural equation models [14,42–44]. Structural equation models are a robust approach (compared to multiple linear or bivariate relationships) that consider both the direct and indirect (through mediators) as well as the total effects (direct + indirect) of predictors [14,42–44]. All the biodiversity components, soil organic matter and aboveground carbon storage were log transformed and standardized for attaining linearity assumption before applying the structural equation model [43]. Analyses were performed using in R environment (Version 3.6.3) [45]. The structural equation models were employed using the Lavaan Package in [46] R for assessing the direct, indirect and total effects of soil nutrient, species richness, structural diversity and functional composition and functional diversity of maximum canopy height and wood density on aboveground carbon storage [43,47]. Based on our hypotheses, we started the structural equation model by including all the predictors and gradually removed the predictor which had high a p -value. We followed this process because multicollinearity among covariates may lead to the model dissatisfying the goodness of fit indices [42]. In the second structural equation model, we removed functional composition of wood density and in the third structural equation model we removed functional diversity of maximum canopy height (Supplementary Materials; Table S1). In the third structural equation model for assessing the effect of the predictors on aboveground carbon storage, the combination of soil organic matter, species richness, functional composition of maximum canopy height and functional diversity of wood density attained all the goodness of fit indices (Supplementary Materials; Table S1). The indirect effect of soil nutrient and species richness were calculated by multiplying the standardized effects of all the paths from them to aboveground carbon through different mediators [47]. The total effect of soil nutrient and species richness on aboveground carbon was estimated by summing the standardized direct and the standardized indirect effects [47]. We evaluated structural equation models by using different fit indices that are widely applied for structural equation models. For instance, we can assume that the structural equation model shows no significant deviation from the dataset if we found an insignificant chi-square test ($p > 0.05$), a comparative fit and Tucker–Lewis index with >0.90 and a standardized root mean square residual close to zero [47,48]. The standardized effect of each path of fitted SEMs was validated using a bootstrap (1000 times) resampling method using Lavaan package in R [46]. Furthermore, we applied multiple linear regression for quantifying the variance inflation factors for checking measure multicollinearity of the covariate in the final SEM, with a threshold value of two (Supplementary Materials; Table S2) [49]. We also applied a simple linear relationship for all the hypothesized paths of the final structural equation model for interpreting the result of the structural equation model [43].

3. Results

3.1. Bivariate Relationship between Different Predictors and Carbon Storage

Soil organic matter had an insignificant positive effect on aboveground carbon storage (Figure 2a; $p > 0.05$). Aboveground carbon was increased with all the biodiversity components except functional diversity of wood density (Figure 2b–e). However, among the four diversity components, structural diversity had the only significant effect on aboveground carbon storage (Figure 2b–e; $p < 0.05$). All the biodiversity components except functional composition of maximum canopy height showed an increasing trend with organic matter content, but none of the relationships were significant (Figure 2f,i, $p > 0.05$). Functional composition of maximum canopy height declined with species richness, while functional diversity of wood density increased with species richness (Figure 2j,l). Species richness also had an insignificant positive effect on structural diversity (variation of DBH; Figure 2k; $p > 0.05$).

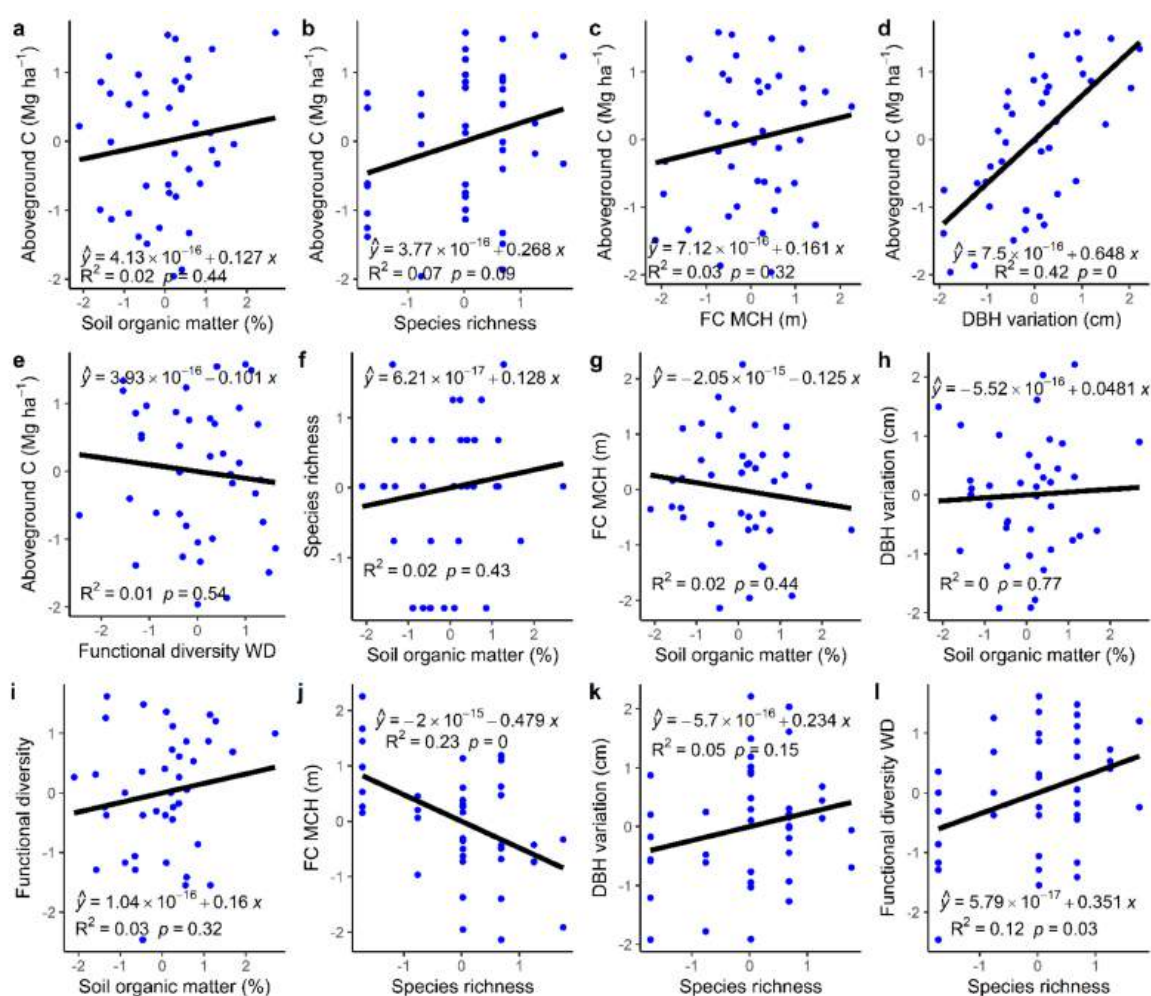


Figure 2. Bivariate relationship between predictors and response variables (all the direct paths of structural equation model in Figure 3). All the data were log transformed and standardized mean zero and one standard deviation. (a–e) Aboveground biomass carbon vs. soil organic matter (a), species richness (b), functional composition of maximum canopy height (FC MCH; (c)), variation of diameter at breast height (DBH at 1.37 m; (d)) and functional diversity of wood density (FC WD), respectively. (f–i) The effects of soil organic matter on species richness, FC MCH, variation of DBH and FC WD, respectively. (j–l) The effects of species richness on FC MCH, variation of DBH and FC WD, respectively.

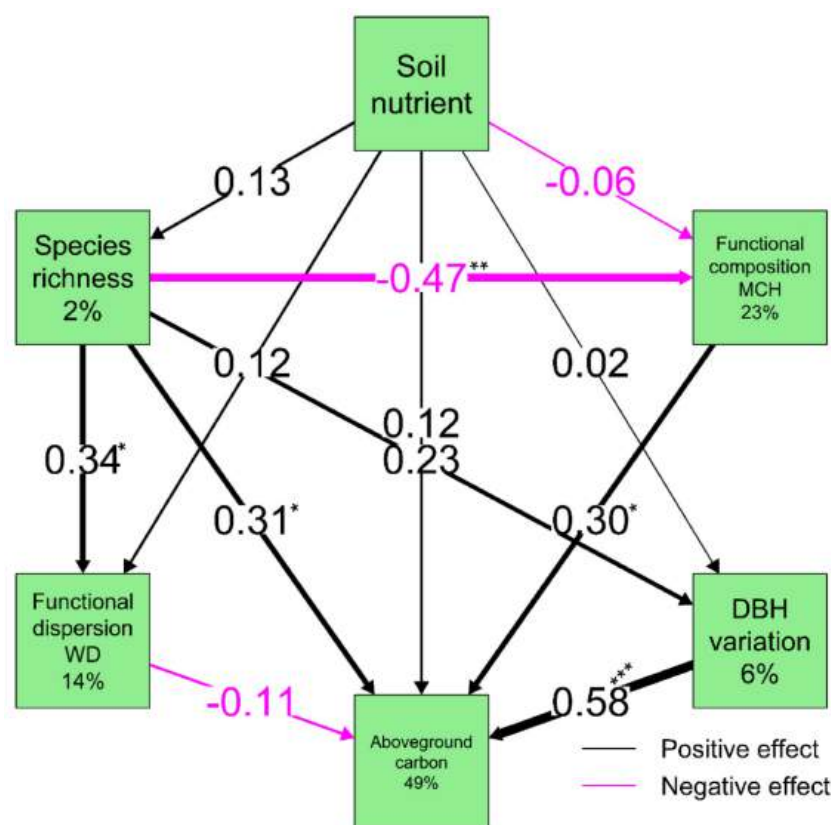


Figure 3. Underlying mechanism of the relationship between different components of biodiversity and aboveground carbon with structural equation modeling. The best fitted model with insignificant lower chi-square value ($\chi^2 = 2.198$, $p = 0.532$), along with comparative fit index (1) and Tucker–Lewis index (1) close to one and standardized root mean square residual close to zero (0.057), indicates no deviation of model from observed dataset and thus indicates good fit. The percentage value in the box of endogenous variables shows their explained variance (coefficient of determinant: R squared). The path values with a star mark indicates their significant level (***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$), while the insignificant paths were indicated without a star mark ($p > 0.05$). MCH: maximum canopy height, WD: wood density and DBH: diameter at breast height (at 1.30 m).

3.2. Underlying Mechanism of Relationship between Soil Nutrient, Biodiversity and Carbon Storage

Soil organic matter had no direct and indirect effect on aboveground carbon storage (Figure 3; $p > 0.05$; Table 1). The effect of all biodiversity components (except functional diversity of wood density) on aboveground carbon stock were significant and positive (Figure 3; $p < 0.05$). Although functional composition of maximum canopy height partly mediated species richness effect on aboveground carbon storage (Figure 3), it was not retained when we validated it with bootstrapping (Table 1). Structural diversity had a strong effect on aboveground carbon, although it was not affected by soil organic matter and species richness (Figure 3; Table 1). Functional diversity of wood density had no effect on aboveground carbon storage, although it was positively influenced by species richness (Figure 3; Table 1).

Both soil organic matter and species richness had no total effect on aboveground carbon, we also did not find any significant effect ($p > 0.05$; Table 1).

The four biodiversity components along with the effect of soil organic matter in combination explained 49% of the variation of aboveground carbon stock ($R^2 = 0.49$). This is while the soil organic matter and species richness together explained 23% ($R^2 = 0.23$), 6% ($R^2 = 0.06$) and 14% ($R^2 = 0.06$) of functional composition of maximum canopy height, structural diversity and functional diversity of wood density, respectively (Figure 3). Soil nutrient also explained 2% of the variation of species richness (Figure 3).

Table 1. Direct, indirect and total standardized effect of soil nutrient and different diversity component on aboveground carbon. *p*-value in bold indicates significant effect ($p < 0.05$), based on 1000 times bootstrapping. DBH: diameter at breast height, FC: functional composition, FD: functional diversity, MCH: maximum canopy height, WD: wood density.

Predictors	Path Ways	Response	Std. Effect	95% CI		<i>p</i> -Value
				Lower	Upper	
Soil nutrient	Direct	Aboveground carbon	0.120	−0.100	0.330	0.310
Soil nutrient	Indirect effect via species richness	Aboveground carbon	0.040	−0.060	0.130	0.440
Soil nutrient	Indirect effect via FC MCH	Aboveground carbon	−0.020	−0.100	0.060	0.650
Soil nutrient	Indirect effect via DBH variation	Aboveground carbon	0.010	−0.160	0.180	0.900
Soil nutrient	Indirect effect via Functional diversity WD	Aboveground carbon	−0.010	−0.050	0.030	0.560
Soil nutrient	Indirect effect via species richness and FC MCH	Aboveground carbon	−0.020	−0.060	0.030	0.450
Soil nutrient	Indirect effect via species richness and DBH variation	Aboveground carbon	0.020	−0.030	0.060	0.480
Soil nutrient	Indirect effect via species richness and functional diversity WD	Aboveground carbon	0.000	−0.020	0.010	0.560
Soil nutrient	Sum of indirect effect	Aboveground carbon	0.010	−0.200	0.220	0.910
Soil nutrient	Total effect	Aboveground carbon	0.130	−0.170	0.420	0.400
Soil nutrient	Direct	Species richness	0.130	−0.180	0.440	0.410
Soil nutrient	Direct	FC MCH	−0.060	−0.340	0.210	0.640
Soil nutrient	Direct	DBH variation	0.020	−0.280	0.320	0.900
Soil nutrient	Direct	Functional diversity WD	0.120	−0.180	0.410	0.440
Species richness	Direct	Aboveground carbon	0.300	0.030	0.560	0.030
Species richness	Indirect effect via FC MCH	Aboveground carbon	−0.140	−0.270	0.000	0.060
Species richness	Indirect effect via DBH variation	Aboveground carbon	0.140	−0.050	0.310	0.150
Species richness	Indirect effect via functional diversity WD	Aboveground carbon	−0.040	−0.120	0.050	0.400
Species richness	Sum of indirect effect	Aboveground carbon	−0.040	−0.280	0.200	0.750
Species richness	Total effect	Aboveground carbon	0.260	−0.030	0.550	0.080
Species richness	Direct	FC MCH	−0.470	−0.740	−0.200	0.000
Species richness	Direct	DBH variation	0.230	−0.070	0.540	0.130
Species richness	Direct	Functional diversity WD	0.330	0.040	0.620	0.030
FC MCH	Direct	Aboveground carbon	0.300	0.040	0.530	0.020
DBH variation	Direct	Aboveground carbon	0.580	0.340	0.790	0.000
Functional diversity WD	Direct	Aboveground carbon	−0.110	−0.340	0.120	0.360

4. Discussion

We explored the mechanism between biodiversity (species richness, functional composition of maximum canopy height, structural diversity and functional composition of wood density) and carbon storage in the managed landscape after considering soil organic matter as a proxy for soil fertility. We found that species richness and functional composition of maximum canopy height had a positive effect on aboveground carbon storage in the structural equation model. However, both soil organic matter and species richness had no significant mediation effect on aboveground carbon storage.

4.1. Relative Importance of Biodiversity Components to Maintain Aboveground Carbon

Multiple aspects of biodiversity have been widely used in the field of ecology in recent years, to understand their relative importance in maintaining carbon storage in different forest ecosystems [1,2,14,42]. The carbon stored in forests is the result of a long-term

accumulation of plant biomass, which is generally linked to biodiversity (e.g., species, functional trait and structural) [3,7,8,14,30,40]. In our study, we included four components of biodiversity (species richness, functional diversity, functional composition and structural diversity) and hypothesized that all of them will have a positive effect on aboveground carbon storage (Hypothesis iii). We intended to explore their relative importance in Hypothesis iv. In line with our hypothesis, we found all the biodiversity components except functional diversity had a positive effect on aboveground carbon storage where structural diversity (opposing to our fourth hypothesis) posed the dominant effect compared to other three diversity components. Previous studies in different forest types have reported varying importance of these biodiversity components in the context of the study area [1–3,7,8,14,30,40]. For example, in tropical forests (the most diverse ecosystems on the earth), functional diversity plays the dominant role compared to functional composition [2]. However, in temperate and other ecosystems, such as mangrove forest and agroforestry systems with relatively lower species diversity than tropical forests, the dominant role is exerted by functional composition of plant traits or structural diversity [2,3,8,22,40,42]. Our findings in homegardens, thus revealing that in order to maximize aboveground carbon storage we need to manage stand with diverse stem size and in combination with dominant species and few other species (as species richness and functional composition of maximum canopy height also had a positive effect on aboveground carbon).

4.2. Underlying Theory of Aboveground Carbon Storage in Homegardens

Although we did not find any significant mediation effect of soil organic matter and species richness on aboveground carbon storage with hypothesis one and two (both are rejected), the findings of hypothesis three and four disclosed the underpinning mechanism of biodiversity–carbon relationship in our study area. As we found a positive effect of species richness, structural diversity (variation of DBH), functional composition of maximum canopy height on aboveground carbon storage, our study confirmed that both niche complementarity and selection effect theories are mutually inclusive in deriving aboveground carbon storage in studied homegardens. Our study also emphasized that between these two ecological theories, the niche complementarity effect theory had strong control on aboveground carbon storage in our studied homegardens. Structural diversity and species richness represent the niche complementary hypothesis, which believes that a community with diverse species ensures optimum utilization of limited resources through niche differentiation or facilitation among them [7]. While functional composition represents the selection effect theory, which assumes that carbon storage is driven by the key functional traits of the most dominant species among the diverse species in a forest stand [1,2]. Our finding is in line with a recent study on the relationship between biodiversity and aboveground carbon storage in a Sri Lankan homegarden agroforestry system, where the authors reported that structural diversity plays the dominant role in shaping the aboveground biomass [7]. However, we oppose the hypothesis that structural diversity (tree size variation), rather than species diversity determined the aboveground biomass or carbon storage in a homegarden [7,50], as we also found that species richness had a significant positive effect on aboveground carbon storage. The mutual existence of both niche complementarity effect and selection effect theories in deriving aboveground carbon storage has also been reported in other forest types. For example, a recent study in Sundarbans Reserved Forest, Bangladesh, one of the largest mangrove forests in the world, reported that the niche complementarity effect (represented by species richness) and selection effect (represented by functional composition of maximum canopy height and leaf litter nitrogen content) together enhanced carbon storage [42]. However, the dominant role was posed by the selection effect theory [42]. While in African tropical forests, Mensha et al. [2] found the dominant role of the niche complementarity effect theory over the selection effect theory in determining aboveground biomass carbon. Another study in Australian temperate forests reported that structural diversity, representing niche complementarity, suppressed the selection effect theory (represented by functional composition

of maximum canopy height and wood density) for regulating plant biomass carbon [51]. Although the above-mentioned studies (including our current study in homegardens) support the mutual existence of both theories in controlling carbon storage, the differences in the dominant role and representation of different diversity components highlight the importance of exploring biodiversity–carbon relationships in any specific study area.

Individual tree DBH variation is a forest health predictor which indicates that forest stand with high DBH variation is demographically rich, more resilient and has more niche variations compared to less structural diversity [1,7]. Diverse species in a forest stand lead to a higher stem density with varying growth forms and also provide opportunities for plant–plant mutual benefits through enhancing soil nutrients through nitrogen fixing species and thereby enhance carbon storage [2]. On the other hand, the tallest tree (representation of maximum canopy height) with a high stem DBH contains more biomass in forest and leads to a higher aboveground carbon storage at stand level [2,33]. As our findings revealed that these drivers had positive effects on aboveground carbon storage, we can conclude that homegardens with diverse species have a structurally diverse stand, favor of some dominant species and enhance carbon storage through both niche complementarity and selection effects in homegardens in a sustainable manner, similar to other types of forest ecosystems [44,52].

4.3. Limitations of This Study

We did not find any significant indirect and total effect of both soil organic matter and species richness (Figure 3; Table 1). Soil organic matter was used as a proxy of soil fertility, while for testing the mediation effect of species richness we used structural diversity, functional diversity and functional composition of wood density and maximum canopy height. However, soil nitrogen and phosphorus as a measure of soil nutrient could have direct as well as indirect effects on aboveground carbon storage in homegardens, similar to other forested ecosystems [53]. For example, it has been reported that soil phosphorus content can explain 45% of variation of aboveground carbon storage in a tropical forest in southern Costa Rica [53]. Similarly, foliar functional traits, such as specific leaf area, leaf dry matter content, leaf nitrogen and phosphorus contents, and their ratios may mediate the indirect effect of species richness on aboveground carbon storage in homegardens via functional diversity and functional composition [3,24,40,54,55]. Future research in homegarden agroforestry systems focusing on the aforementioned soil nutrients and plant foliar traits would help us better understand the biodiversity–environmental factor–carbon storage relationship.

5. Conclusions

We assessed the interaction effect of various aspects of biodiversity and soil organic matter on aboveground carbon storage in homegarden agroforestry plantations of southwestern Bangladesh. Species richness, functional composition of maximum tree height and structural diversity (variation in stand DBH) promote aboveground carbon storage. However, variation in stand DBH had a stronger influence on aboveground carbon storage than the other two components of diversity. Thus, in homegardens, both niche complementarity and selection effect in combination determine aboveground carbon storage. Our findings also suggest that for maximizing carbon storage in homegardens, future plantations should carry on with site-specific dominant species, and few other species having a different growth form rather than simple monocultural practice. In this way, homegarden agroforestry systems can contribute to carbon emission reduction and adaptation targets as one of the nature-based solutions in tropical and subtropical countries.

Supplementary Materials: The following are available online at <https://www.mdpi.com/article/10.3390/f12121669/s1>, Table S1: Model fit statistics of all tested structural equation models in exploring biodiversity–carbon relationship, Table S2: Result of variance inflation factor test for checking multicollinearity using multiple linear regression among the covariates in the accepted structural equation model.

Author Contributions: Conceptualization, M.M.R.; methodology, M.M.R. and G.K.K.; software, M.M.R.; formal analysis, M.M.R. and G.K.K.; investigation, G.K.K. and H.A.; writing—original draft preparation, M.M.R.; writing—review and editing, M.M.R. and M.E.K.; visualization, M.M.R.; supervision, M.E.K. and M.X.; funding acquisition, G.K.K. and M.X. All authors have read and agreed to the published version of the manuscript.

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Carbon stock in the Sundarbans mangrove forest: spatial variations in vegetation types and salinity zones

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Abstract The Sundarbans (21°30′–22°30′ N and 89°00′–89°55′ E) is the largest mangrove forest in the world. Forests are very important for sequestering atmospheric carbon and mangroves are amongst the most efficient in carbon sequestration. This study presents the estimation of ecosystem carbon (above- and belowground) stock in the Sundarbans using a large scale data sets collected from systematic grid samples throughout the forest. The variation of carbon stock in different vegetation types and in different salinity zones in Sundarbans was investigated. The relationships between carbon stock and different vegetation functional attributes (basal area, mean tree height, crown coverage etc.) were also investigated. The amount of carbon stored varied significantly among vegetation types, salinity zones and vegetation functional attributes ($P < 0.05$). Sundri (*Heritiera fomes*) dominated forest types store more ecosystem carbon ($360.1 \pm 22.71 \text{ Mg C ha}^{-1}$) than other vegetation types. The fresh water zone shows the highest ecosystem carbon stock ($336.09 \pm 14.74 \text{ Mg C ha}^{-1}$) followed by moderate and strong salinity zones.

Salinity was found to enhance belowground carbon stock as revealed by the lowest proportion of belowground carbon stock (57.2 %) with respect to ecosystem carbon in fresh water zone and by the highest (71.9 %) in strong salinity zone. The results also reveal that no matter whether the mangroves are tall or dwarf, a significant amount of carbon is stored into the sediment. The vegetation attributes (basal area and mean tree height) of the dominant mangrove species in each vegetation type were identified as the key indicator of ecosystem carbon stock. We recommended some generalized regression equations to predict ecosystem carbon stock from basal area or mean tree height.

Keywords Carbon sequestration · Aboveground carbon · Belowground carbon · REDD+ · Vegetation functional attributes · Ecosystem carbon

Introduction

The World today is facing major challenges caused by atmospheric CO₂ causing global warming (IPCC 2013). Global warming is mostly due to man-made emissions of greenhouse gases (mainly CO₂) (IPCC 2013). Combustion of fossil fuel and deforestation are the two main sources of CO₂ emission to the atmosphere (Detwiler and Hall 1988; Woodwell et al. 1983). It has been predicted that atmospheric

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CO₂ will range between 467 and 555 ppm by the year 2050, which ranged between 278 ppm in 1750 and 390 ppm in 2011 and that the average global temperature will increase by 2–42 °C by the year 2050 (IPCC 2007; IPCC 2013; Anderson and Bows 2011). The global warming of this magnitude could significantly alter the earth's climate, land use, and major vegetation zones and perhaps more importantly melting of polar ice to raise sea level by 5 m causing severe loss of life and property (Detwiler and Hall 1988; IUFRO 2009), especially in the developing countries. Bangladesh is among the most vulnerable countries affected by global warming.

Forests play an important role in mitigating global climate change through sequestering atmospheric carbon (Adame et al. 2013). Mangroves are particularly efficient sinks, sequestering four times carbon per unit area compared with terrestrial forests in the tropics (Khan et al. 2007; Donato et al. 2011). However, deforestation of mangroves, which was very widespread in the last few decades (Giri et al. 2011), may render them significant sources of atmospheric carbon (Donato et al. 2011) and policy makers are looking at new ways to save this unique ecosystem through different mitigation approaches such as 'Reducing emissions from deforestation and forest degradation (REDD+)'. So, in order to participate in UNFCCC's mitigation programs (e.g., REDD+) and thereby generate economic benefit for the country, it is imperative to make a baseline assessment of the ecosystem carbon stock (Adame et al. 2013).

In mangrove forests carbon sequestration (e.g., Khan et al. 2007; Bouillon et al. 2008) and organic carbon dynamics (Machiwa and Hallberg 2002; Kristensen et al. 2008) have been studied much. Carbon stock in mangrove ecosystem varies with species (Laffoley and Grimsditch 2009), vegetation type (Adame et al. 2013; Cerón-Bretón et al. 2011; Mitra et al. 2011; Sapit et al. 2011; Laffoley and Grimsditch 2009) and salinity (Adame et al. 2013). However, less attention has been paid on the spatial variation of carbon stock among different vegetation types and variation in the above- and belowground carbon.

The Sundarbans is the largest single tract of mangrove forest in the world (6,017 sq km in Bangladesh and 4,000 sq km in India). It is a RAMSAR SITE having three wildlife sanctuaries which are designated as World Heritage by UNESCO during 1997. The forest is, nationally and

internationally, of great conservation significance for its environmental services and biodiversity (Seidens-ticker and Hai 1983; Iftexhar and Saenger 2008). To aid in conservation of the forest and to benefit from various global initiatives (e.g., carbon trading) an assessment of the carbon sequestration (above- and belowground) in Sundarbans is immensely important. Moreover, the heterogeneity of the mangrove forest in terms of large area of forest cover, salinity zone (Wahid et al. 2007), dominant mangrove types and vegetation functional attributes (basal area, mean tree height, canopy cover etc.), which might influence the aboveground and belowground carbon stock would be of great interest to mangrove ecologists. Therefore, the objectives of the present study were: (i) to estimate the carbon above- and belowground carbon stock in Sundarbans, (ii) to investigate the variation of carbon stock in different vegetation types and salinity zones in the forest and (iii) to establish a generalized method for assessing the ecosystem carbon stock in Sundarbans.

Materials and methods

Description of study area

The study was carried out in the Sundarbans, which lies between 21°30' N and 22°30' N and 89° 00' E and 89°55' E, in the southwest of Bangladesh. The forest covers an area of 6,017 km² (Bangladesh part) of which 4,120 km² are covered by the forests and the remaining 1,897 km² are in rivers, canals and creeks of varying width and depth (Islam 2011). The soil of the Sundarbans is silty clay loam with alternate layers of clay, silt and sand (Gopal and Chauhan 2006). Trees in the Sundarbans include 22 families representing 30 genera. Important tree species of the Sundarbans along with their life form are presented in the Table 1 (Iftexhar and Islam 2004). In Sundarbans there are ten prominent vegetation types (Table 2) (Chaffey et al. 1985; Iftexhar and Saenger 2008).

Sampling design

Through systematic grid sampling method, 155 plots (1,570.79 m² each) were selected at 4 min intervals of latitude and 2 min intervals of longitude. Of the 155 plots, five plots were fallen over water channels,

Table 1 Life form characteristics of main flora in the Sundarbans

Local name	Scientific name	Life form
Amoor	<i>Amoora cucullata</i>	A small tree
Baen	<i>Avicennia officinalis</i>	A large tree
Bola	<i>Hibiscus tiliaceous</i>	A small tree and semi climbing liana
Dhundul	<i>Xylocarpus granatum</i>	A small tree
Gewa	<i>Excoecaria agallocha</i>	A fair sized tree
Golpata	<i>Nypa fruticans</i>	A palm with soboliferous stem
Goran	<i>Ceriops decandra</i>	A shrub with coppice like growth
Hantal	<i>Phoenix paludosa</i>	A small gregarious palm
Hargoza	<i>Acanthus ilicifolius</i>	A small prickly leaved shrub
Hodo(Tiger fern)	<i>Acrostichum aureum</i>	A rigid tufted fern under growth
Kankra	<i>Bruguiera sexangula</i>	A medium sized tree
Keora	<i>Sonneratia apetala</i>	A large tree generally with spreading habit
Kewa kanta	<i>Pandanus odoratissimus</i>	A gregarious screwpin under-growth
Passur	<i>Xylocarpus mekongensis</i>	A fair sized tree
Sundri	<i>Heritiera fomes</i>	A fair sized tree

Source (Siddiqi 2001)

Table 2 Major vegetation types in the Sundarbans with distribution and area

Code	Vegetation type	Distribution in Saline zone	Area %
VT1	Sundri	FR	21.0
VT2	Sundri–Gewa	FR, MO and ST	25.8
VT3	Sundri–Passur–Kankra	FR,ST	1.7
VT4	Gewa Mathal, Passur–Kankra–Baen, Sundri–Passur–Keora,	FR, MO and ST	2.9
VT5	Gewa	MO	5.2
VT6	Gewa–Sundri	MO,ST	18.4
VT7	Non tree vegetation (NTV)	FR,ST	1.2
VT8	Goran–Gewa	MO, ST	13.7
VT9	Goran	ST	1.6
VT10	Gewa–Goran	MO, ST	8.4

FR fresh water zone, MO moderate saline zone, and ST strong saline zone

Source (Chaffey et al. 1985; Iftekhhar and Saenger 2008)

and the remaining 150 sample plots (total sample area of 23.56 ha) were considered for this study (Fig. 1). In each sample plot, five circular plots of 10 m radius were used for data collection (Fig. 3). These five subplots were nested and arranged in a cluster manner (Fig. 2) in Sundarbans carbon inventory. The cluster plot designs tend to capture more microsite variation in vegetation, soils, etc., thereby reducing among-plot variation (Pearson et al. 2005).

From the centre subplot, the other four subplots (Fig. 3) were arranged towards the four cardinal directions with a distance of 50 m (Ahmed and Iqbal 2011). The vegetation type and salinity zone for each plot were determined from the Sundarbans vegetation map and salinity map developed by Chaffey et al. (1985). The formation of salinity zones in Sundarbans is a recognized phenomenon (Wahid et al. 2007).

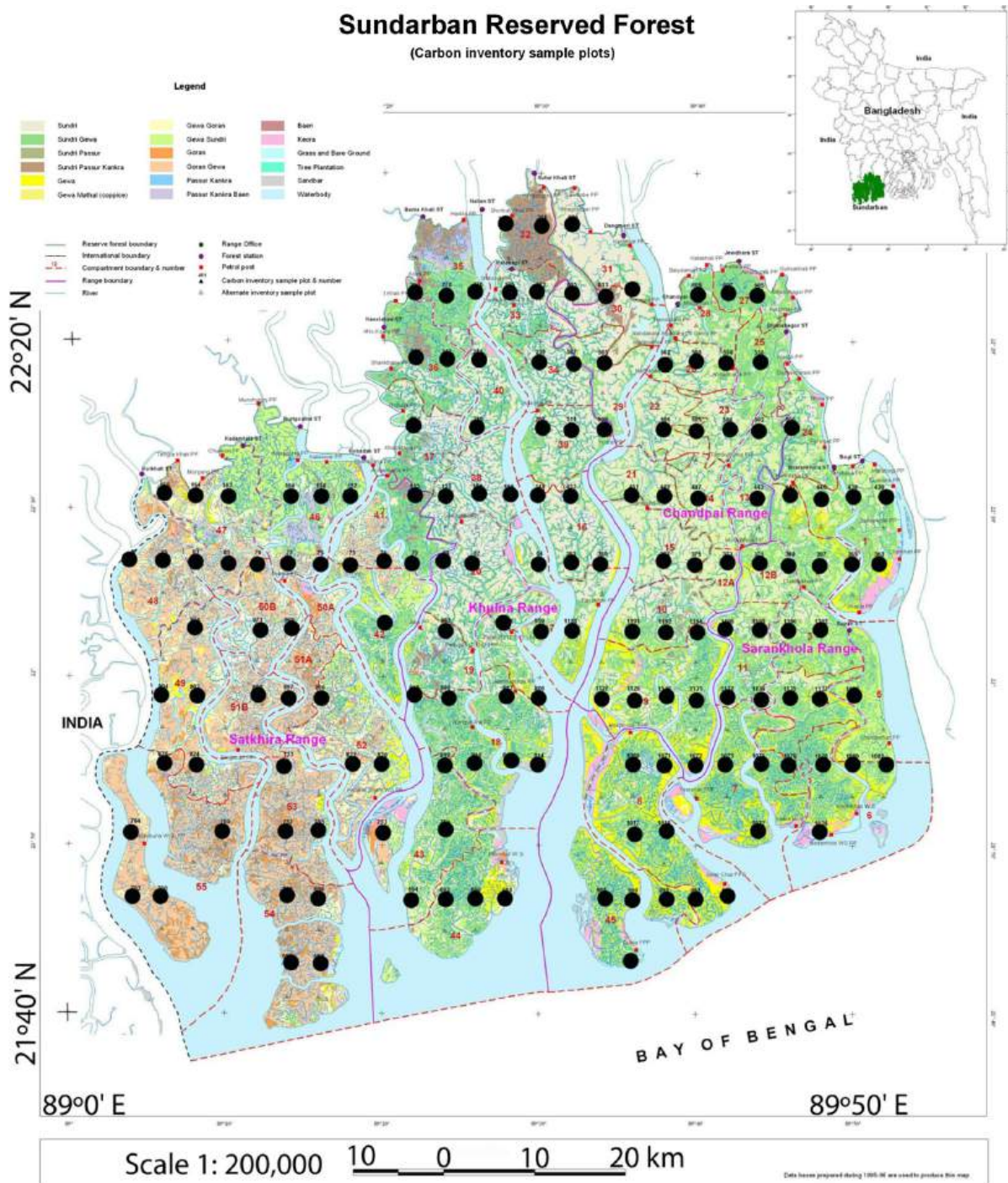


Fig. 1 Location of sample plots (black circles) in Sundarbans mangrove forest, Bangladesh

Tree inventory

Within 10 m radius from the centre of each subplot, diameter of all trees (live or dead with

diameter ≥ 10 cm at breast height (bh) was measured. The height of three co-dominant trees within each of the five subplots (Fig. 2) was measured with digital range finder. Species name, diameter at breast height

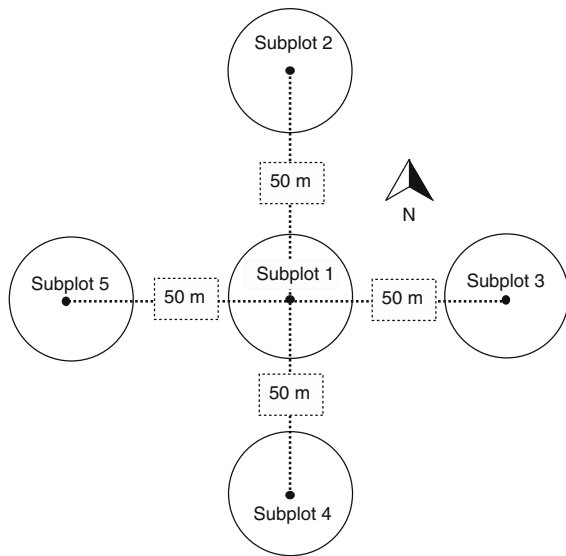


Fig. 2 Cluster plot composed of five nested subplots in a main plot

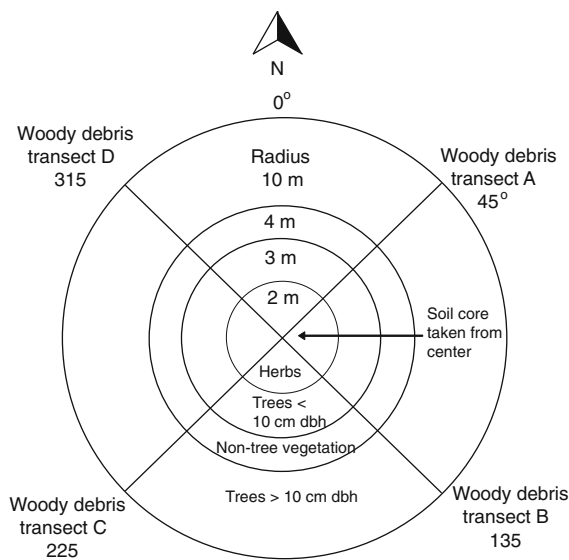


Fig. 3 Layout of a nested subplot

(dbh), living status and height in case of dead and broken trees was recorded. For saplings (diameter < 10 cm at bh) data were recorded as it was done for trees but within 3 m radius from the center of each of the five subplots in a sample plot (Fig. 3). However, for seedlings (trees < 1.3 m high), the number of individuals and the dominant species were recorded.

Table 3 Average biomass of species developed by destructive process

Species name	Average biomass (kg)
Golpatta (per frond)	2.00
Tiger fern (Per clump)	0.30
Hargoza (Per stem)	0.05
Pandanus	0.29
Goran (Small)	0.0048
Goran (Medium)	0.37
Goran (Large)	3.64
Goran (Extra large)	10.55

Palms with woody trunks, but not reaching breast height were counted as seedling.

Woody debris survey

Mass of dead tree (that lean at an angle > 45° from true vertical) and downed wood materials (twigs, branches or stems of trees or shrubs) that have fallen and lies up to a height of 2 m above the forest floor, was estimated by using planar intersect technique (Van Wagner 1968; Harmon and Sexton 1996). In each subplot (10 m radius), four 10 m long transects were laid out; of which first one was directed to 45° from the subplot centre and the other three transects were established at clock wise of 90° off from the previous transect. Woody debris was categorized into four size classes: small (0–0.6 cm), medium (0.6–2.5 cm) large (2.5–7.6 cm), and extra-large (≥7.6 cm) (Brown 1971). Again, the extra-large class was divided into two subclasses: sound (machete bounces off or only sinks slightly when struck) and rotten (machete sinks deeply and wood is crumbly as it decomposed).

An aluminum downed-wood gauge (Fuel Gauge) was used to determine the size class of each piece encountered. Small, medium, and large pieces were tallied as the number of pieces that crossed the transect tape. For extra-large pieces the actual diameter over which the transect line was crossed was measured. Small pieces were only tallied for 2 m of transect (from meter 10 to meter 8), while medium pieces were only tallied for 5 m of transect (from meter 10 to meter 5) and the large and extra-large pieces were measured along the 10 m transect.

In order to quantifying specific gravity and quadratic mean piece diameter (small, medium and large),

21, 22, 20, 19, and 20 pieces of debris were collected respectively for small, medium, large, extra-large rotten, and extra-large sound size classes. The quadratic mean diameter of small, medium and large sized classes of woody debris was used to calculate the volume of these three classes (Brown 1971; Kauffman and Cole 2010).

Non-tree vegetation

Among the non-tree vegetation, goran (*Ceriops decandra*) and harbacious plants were measured within 2 m radius from the center of each of the five subplots, while other non-woody palms (e.g., *Nypa*), *Pandanus*, Tiger fern and woody shrubs were measured within 4 m radius (Table 1). Goran was tallied by 4 size classes (0–0.6 cm, 0.6–2.5 cm, 2.5–7.6 cm, >7.6 cm). For lianas dbh measurement was taken like as tree. Herbaceous vegetation was visually estimated and recorded as percent ground cover of herbs and grasses separately (USDA 2008). In case of *Nypa*, the number of stems rooted in the subplot was counted (not individuals or clumps, but separate stems), whereas for *Pandanus* and Tiger fern the number of clumps in the subplot was recorded.

Canopy cover

Canopy cover % was estimated by a spherical crown densitometer standing at the subplot center. To take readings, it was assumed that there are four equispaced dots in each square of the grid and systematically count the dots equivalent to quarter-square openings. Next the total count was multiplied by 1.04 to obtain percent of overhead area not occupied by canopy. The difference between this and 100 is an estimation of over story density in percent.

Soil sampling

Two soil samples (5 cm length) were collected at the midpoint of 0–30 and 30–100 cm depth intervals from each of the five subplots for determining soil bulk density and organic carbon concentration. Soil core of 1 m length was pulled out near the subplot centre by using a one meter long open face peat auger. Soil samples were air-dried in the field, oven-dried to constant mass at 60 °C (to stop microbial decomposition) at the Khulna IPAC cluster office for

determining bulk density, and then sent to Bangladesh Forest Research Institute (BFRI), Chittagong for carbon analysis. Soil carbon analyses were conducted in the laboratory of the soil sciences division of the BFRI. Soil sample were oven dried at 105 °C before homogenizing and organic carbon concentration was determined by following Walkley–Black's wet oxidizing method (Nelson and Sommers 1996).

Biomass and carbon computation

Aboveground biomass of live trees, poles, saplings and dead ones (decay status 1: having stem, branch and twigs) was estimated by the following general equation (Eq. 1) for mangrove tree species (Chave et al. 2005)

$$AGB = \rho \cdot e^{\{-1.349 + 1.98 \cdot \ln(D) + 0.207 \cdot \ln(D^2) - 0.0281 \cdot \ln(D^3)\}} \quad (1)$$

where AGB = aboveground biomass, ρ = wood density, D = dbh. The wood density data were obtained from destructive samples supplemented with local literatures, World Agroforestry Database (Carsan et al. 2012) and the Global Wood Density Database (Chave et al. 2009; Zanne et al. 2009).

Belowground biomass of trees was computed by using the general mangrove equation (Eq. 2) of Komiyama et al. (2005), while for palms, it was conservatively took 15 % of aboveground biomass (because general mangrove equation for belowground biomass does not apply to palms) (Macdicken 1997).

$$BGB = 0.199 \cdot \rho^{0.899} D^{2.22} \quad (2)$$

where BGB = belowground biomass, ρ = wood density, D = dbh.

Pearson et al. (2005) equation was applied for computing the biomass of woody palms that reaching breast height. For dead trees having decay status-2 (trees with no twigs/small branches but had large branches or stem only) whose base diameter was smaller than DBH (due to decay at base), the standard calculation would result in artificially large top diameter. Therefore, it was adjusted base diameter for these records (relatively few) based on average ratio of dbh: base-diameter, which was 0.82. It also followed for heavily buttressed trees, for which standard calculations yielded artificially low or even negative top diameters; it was defined heavily

buttressed trees as those with a dbh: base-diameter ratio that was two standard deviations below the mean ratio. For small palms not reaching dbh, these were measured as non-tree understory and a destructive harvest was carried out to estimate average biomass per understory palm. The short stumps, those not reaching breast height, were simply modeled as a cylinder shape to obtain volume, and then multiplied by species-specific wood density. For belowground biomass of these individuals, the base diameter was used to estimate the projected dbh based on the average ratio of dbh to base diameter (0.82), and then entered this into the equation.

In case of some non-tree vegetation including seedling, subsamples were collected from each destructively harvested individual to determine moisture content. The wet: dry ratios were averaged over the whole sample, then this average value applied to each individual (Table 3). Lianas biomass was quantified by using Schnitzer et al.'s (2006) allometric equation.

To calculate the mass of downed wood, it is necessary to know the mean specific gravity of the downed wood in the mangroves as well as quadratic mean diameters for the wood size classes <7.6 cm diameter (debris diameter greater than 7.6 cm were measured in the field) (Kauffman and Cole 2010). The quadratic mean diameter for small, medium and large was 0.45, 1.21 and 3.17, respectively. The mean specific gravity was 0.59, 0.55, 0.49, 0.25 and 0.40 g cm⁻³, respectively for small, medium, large, extra-large rotten and extra-large sound. By using these specific gravity and mean quadratic data, woody debris biomass was estimated via standard volumetric equations (Brown 1971).

Conversion of dry biomass of trees, understory, and downed wood to carbon mass was done by multiplying 0.5 as forest biomass contains half carbon by mass (Gifford 2000). Soil C stock was determined as the product of soil carbon concentration, bulk density, and depth intervals (Donato et al. 2011). The total carbon stock per plot was calculated by adding each of the carbon pool of the five subplots as below:

$$\begin{aligned} \text{Total C stock} = & C_{\text{treeAG}} + C_{\text{treeBG}} + C_{\text{deadtree}} \\ & + C_{\text{sapling seedlingAG}} + C_{\text{sapling seedlingBG}} + C_{\text{deadsapling seedling}} \\ & + C_{\text{nontreevegetation}} + C_{\text{woodydebris}} \\ & + C_{\text{soil}} \end{aligned} \quad (3)$$

Statistical analysis

At first a one sample Kolmogorov–Smirnov test (K–S test) was performed to check whether the carbon stock data under the subgroups of vegetation types and salinity zones are normally distributed. A one-way ANOVA was used to test the significance of the differences among the carbon stock in different partitions, vegetation types and salinity zones. In addition, a two-way-ANOVA was performed to investigate the interaction effect with vegetation types nested within salinity zones. For multiple comparisons among the variables Duncan Multiple Range Test (DMRT) was used. Finally, correlation analysis was performed to obtain some generalized regression equations in order to predict above- and belowground carbon stock from vegetation attributes, such as stand basal area or mean tree height. All the statistical and graphical analyses were performed using SPSS version 16 and R version 3.1.0 (R Core Team 2014).

Result

Vegetation types and carbon stock

Aboveground carbon

The carbon stock data under the subgroups of vegetation types and salinity zones showed normal distribution as tested with Kolmogorov–Smirnov test (K–S test). Among the vegetation types, VT1 showed significantly higher ($P < 0.05$) tree aboveground carbon (TAGC) stock (152.57 Mg ha⁻¹) than other vegetation types, whereas VT7 showed significantly higher ($P < 0.05$) shrubs and herbs carbon (SHC) stock (61.35 Mg ha⁻¹) than other vegetation types (Fig. 4). There was no significant difference ($P > 0.05$) in down wood carbon (DWC) among the vegetation types (Fig. 4). If the total aboveground carbon stock were considered, the VT1 again showed significantly higher ($P < 0.05$) carbon stock than other vegetation types. The lowest carbon stock of 45.24 Mg ha⁻¹ was found in VT8 (Fig. 4). The aboveground carbon stock showed no significant interaction effect ($P > 0.05$) between the vegetation types nested within salinity zones as tested with two-way-ANOVA.

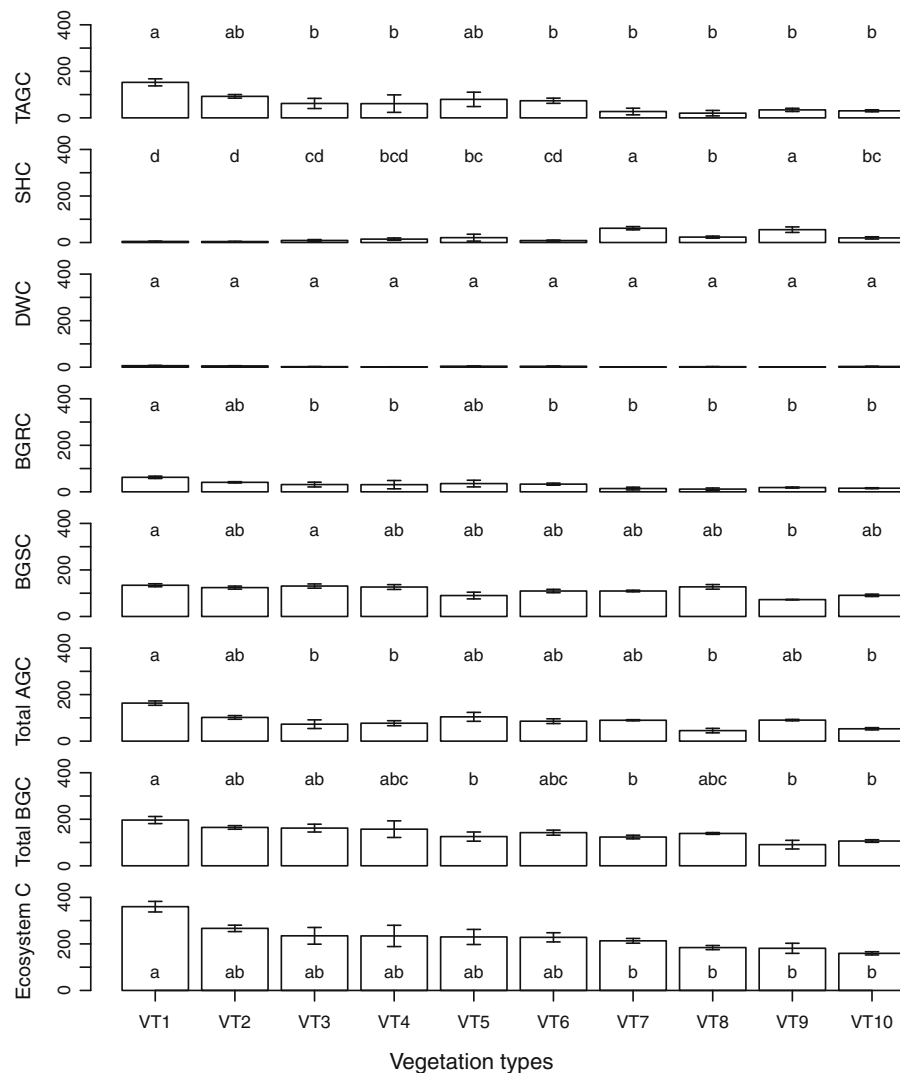


Fig. 4 Carbon stock (Mg ha^{-1} , Megagram per hectare $\pm$ SE) in different partitions (TAGC tree aboveground carbon, AGC aboveground carbon, SHC shrubs and herbs carbon, DWC downed wood carbon, BGRC belowground root carbon, BGSC belowground soil carbon), in vegetation types (VT1 = Sundri, VT2 = Sundri-Gewa, VT3 = Sundri-Passure-Kankra,

VT4 = others (Gewa Mathal, Passur-Kankra-Baen, Sundri-Passur-Keora), VT5 = Gewa, VT6 = Gewa-Sundri, VT7 = Non tree vegetation, VT8 = Goran-Gewa, VT9 = Goran, VT10 = Gewa-Goran). The same letter(s) indicated on bars are not significantly different ($P > 0.05$) as tested with Dancun Multiple Range Test

Belowground carbon

As observed in the aboveground carbon, the belowground root carbon (BGRC) stock also showed significant differences among the vegetation types and the VT1 showed significantly higher ($P < 0.05$) carbon stock (62.37 Mg ha^{-1}) than other vegetation types (Fig. 4). The minimum amount (11.72 Mg ha^{-1}) of BGRC was observed in VT8. Although the

belowground soil carbon (BGSC) stock showed significant differences ($P < 0.05$) among the vegetation types, the BGSC stock showed a comparatively high range ($90.03\text{--}134.17 \text{ Mg ha}^{-1}$). If the belowground carbon stock were considered, the vegetation type VT1 showed significantly higher ($P < 0.05$) carbon stock ($196.54 \text{ Mg ha}^{-1}$) than other vegetation types and the minimum value (90.83 Mg ha^{-1}) was observed in VT9 (Fig. 4). The belowground carbon stock showed no

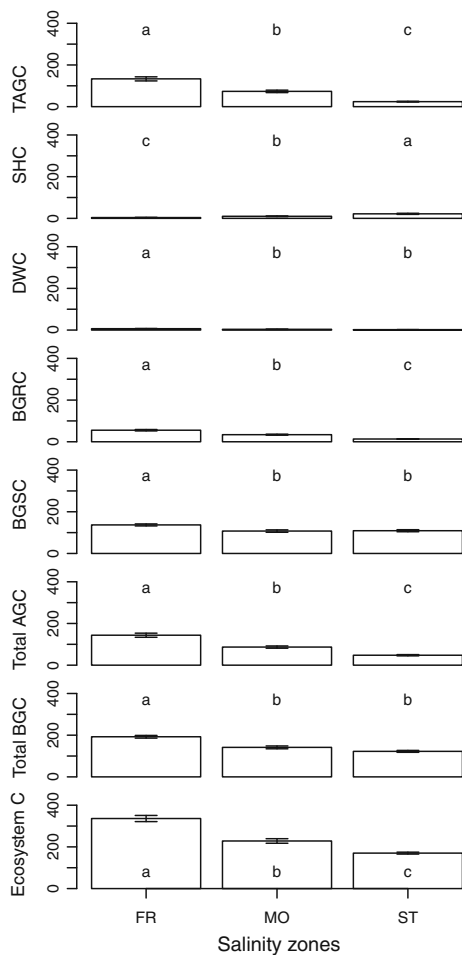


Fig. 5 Carbon stock ($\text{Mg ha}^{-1} \pm \text{SE}$) in different partitions (codes are same as Fig. 2) in salinity zones (FR fresh water zone, MO moderate saline zone and ST strong saline zone). The same letter(s) indicated on bars are not significantly different ($P > 0.05$) as tested with Duncan Multiple Range Test

significant interaction effect ($P > 0.05$) between the vegetation types nested within salinity zones as tested with two-way-ANOVA.

Salinity zone and carbon stock

The fresh water zone (FR) showed significantly higher ($P < 0.05$) carbon stock among the three major salinity zones (Fig. 5). The ecosystem carbon stock increased from strong salinity zone (ST) to moderate salinity zone (MO) to FR ($P < 0.05$; Fig. 5). This trend was observed in all the partitions (tree species, aboveground, BGRC, BGSC, downed wood, ecosystem) except in SHC stock ($P < 0.05$; Fig. 5).

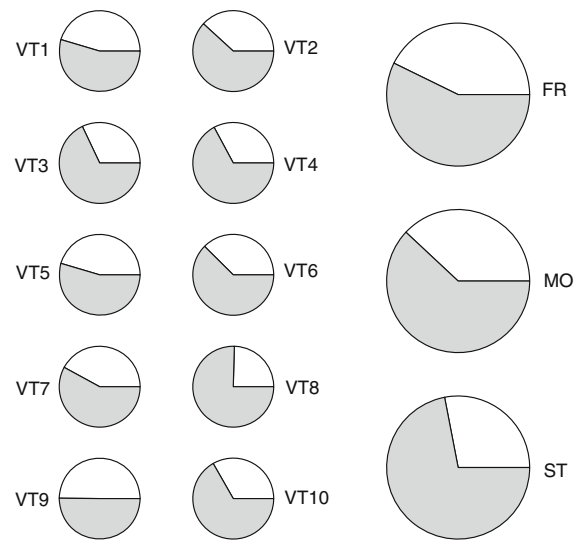


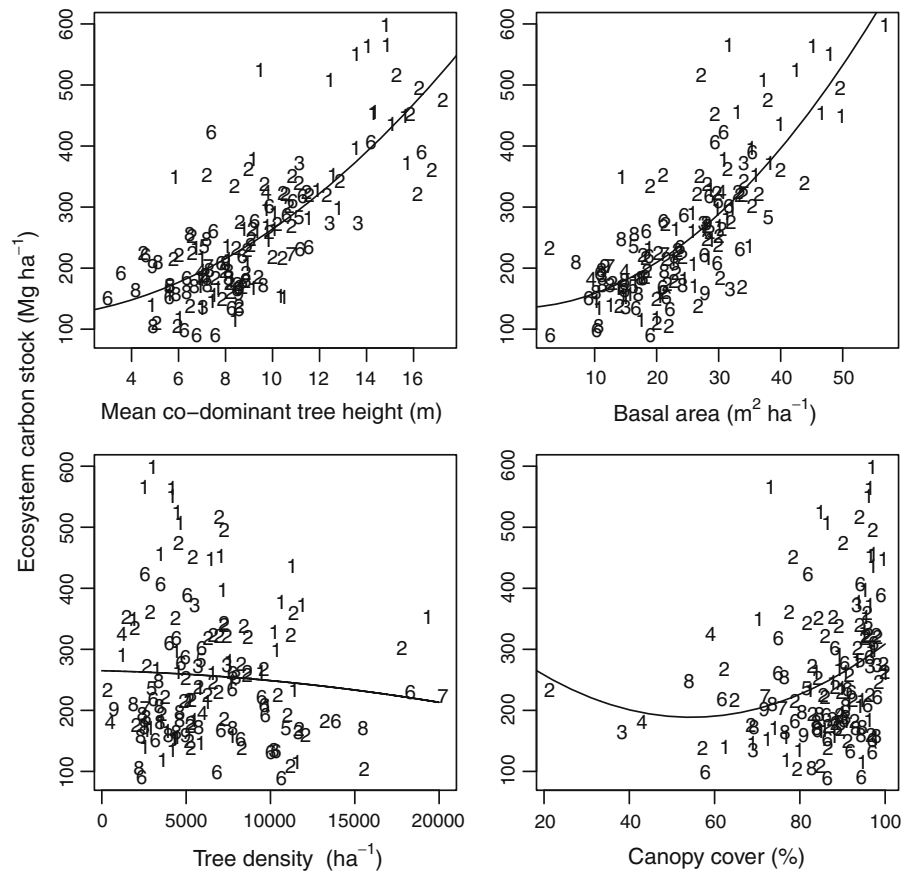
Fig. 6 Proportions of above- (white colored) and belowground (grey colored; including soil) carbon stock in relation to vegetation types (codes are same as Fig. 4) and salinity zones (codes are same as Fig. 5)

Ecosystem carbon stock

The ecosystem carbon stock ranged from $159.49 \pm 6.86 \text{ Mg ha}^{-1}$ in Gewa-Goran dominated vegetation type (VT10) to $360.01 \pm 22.71 \text{ Mg ha}^{-1}$ Sundri dominated VT1 (Fig. 4). VT1 showed significantly higher carbon stock than any other vegetation types (one-way-ANOVA, $P < 0.05$). As tested with DMRT, there was no significant difference in ecosystem carbon stock among the vegetation types VT2, VT3, VT4, VT5 and VT6 ($P > 0.05$; Fig. 4), followed by the next homogenous subset of comprising the vegetation types VT7, VT8, VT9 and VT10 ($P > 0.05$; Fig. 4).

If vegetation type is taken into account, it was found that the minimum proportion of belowground carbon stock constitute 50.15 % of the ecosystem carbon stock in VT9, which reaches up to 75.44 % in VT8 (Fig. 6). If ST is taken into account, the salt showed a positive influence with belowground carbon partitioning and it was found that the minimum proportion of belowground carbon stock constitute 57.24 % of the ecosystem carbon stock in FR, which reaches up to 71.95 % in strong salinity zone (ST) (Fig. 6). The ecosystem carbon stock showed no significant interaction effect ($P > 0.05$) between the vegetation types nested within STs as tested with two-way-ANOVA.

Fig. 7 Relationships of ecosystem carbon stock (Mg ha^{-1}) to mean co-dominant tree height (m) ($R^2 = 0.58$, $P < 0.05$), basal area ($\text{m}^2 \text{ha}^{-1}$) ($R^2 = 0.61$, $P < 0.05$), tree density (ha^{-1}) ($R^2 = 0.01$, $P > 0.05$) and canopy cover (%) ($R^2 = 0.09$, $P > 0.05$). Data points represents vegetation types (VT1 = 1, VT2 = 2, VT3 = 3, VT4 = 4, VT5 = 5, VT6 = 6, VT7 = 7, VT8 = 8, VT9 = 9, VT10 = 10; codes are same as Fig. 4). The curves were fitted using quadratic equation (Eqs. 4, 5)



Vegetation functional attributes and carbon stock

The ecosystem carbon stock were plotted against several vegetation functional attributes such as basal area (BA), mean co-dominate tree height (CDTH), tree density and canopy cover percent (CCP) considering all the vegetation types (Fig. 7) and all the STs (Fig. 8). It was found that BA holds a strong positive relationship ($R^2 = 0.61$, $P < 0.05$), which can be expressed by the following equation:

$$\text{EcosystemC} = 255.2 + 1016.6 \times \text{BA} + 239.3 \times \text{BA}^2 \quad (4)$$

The CDTH also showed a strong positive relationship ($R^2 = 0.58$, $P < 0.05$), which can be expressed by the following equation:

$$\text{EcosystemC} = 255.2 + 1007.4 \times \text{CDTH} + 171.2 \times \text{CDTH}^2 \quad (5)$$

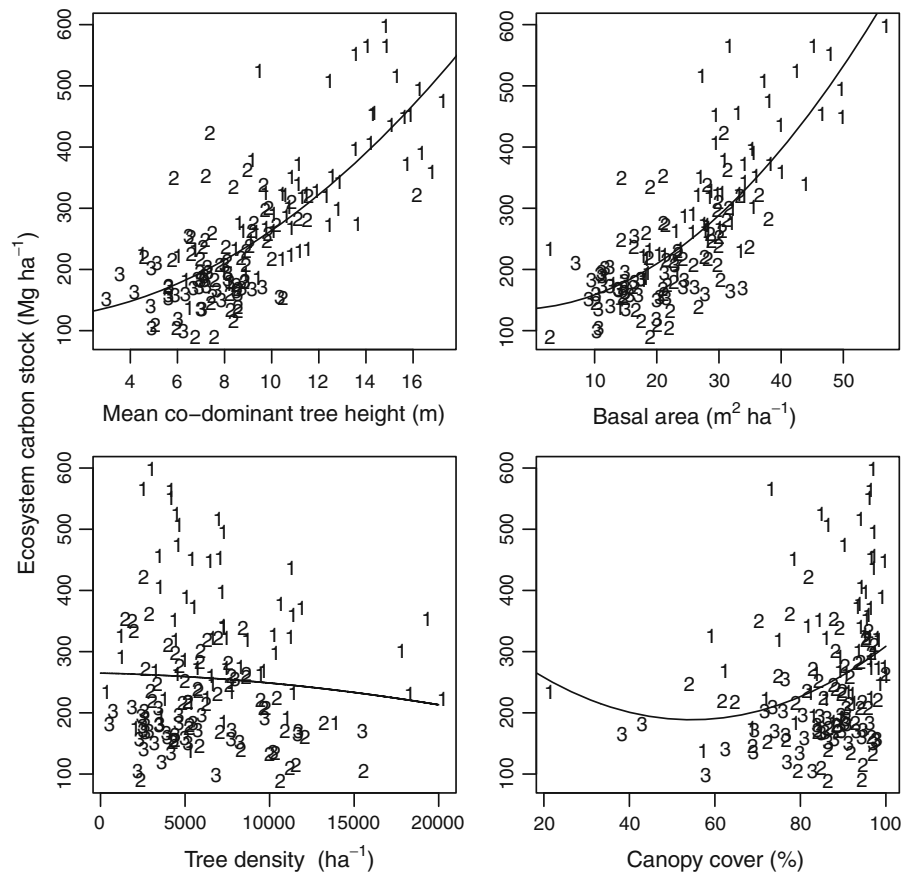
However, tree density and CCP showed weak relationships with ecosystem carbon stock.

Discussion

The richness of mangrove tree species in Sundarbans restricts the use of species-wise allometric equations for biomass estimation. Therefore, in this study we used universal allometric equations (Chave et al. 2005; Komiyama et al. 2005) for estimating the above- and below-ground biomass of tree species using the variables tree dbh and wood density, in order to avoid destructive sampling of trees.

The results of this study suggest significant differences ($P < 0.05$) in carbon stock among different mangrove vegetation types. However, no significant interaction effect ($P > 0.05$) of the vegetation types and nested within salinity zones on carbon stock was detected, which indicates that the variations of carbon

Fig. 8 Ecosystem carbon stock in relation to salinity zone. Data points represents salinity zone (1 = Fresh water zone, 2 = Moderate Saline zone and 3 = Strong Saline Zone). The curves were fitted using quadratic equation (Eqs. 4, 5)



stock are caused independently by vegetation types or salinity. Among the vegetation types, Sundri dominated forest contained highest amount of ecosystem (above- and belowground) carbon (360.0 Mg ha^{-1}) per unit area followed by Sundri-Gewa, Sundri-Passure-Kankra, others (Gewa Coppice, Passur-Kankra-Baen, Sundri-Passur), Gewa, Gewa-Sundri, non-tree vegetation, Goran-Gewa, Goran, and Gewa-Goran (Fig. 4). The reason behind this variation could be aboveground vegetation stature as the carbon stock was found strongly correlated with the size of the trees (height and diameter) and basal area. Klimešová et al. (2008), Westoby et al. (2002), Westoby (1998) also observed that trees attaining greater height outcompeting their neighbors accumulate more carbon. In a natural ecosystem, the diameter of the tree is a determinant of aboveground biomass (Chave et al. 2004). The dominant species in each vegetation type may have an effect on aboveground biomass depending on their basal area (Ruiz-Jaen and Potvin 2010) as observed in this study (Figs. 7 and 8). While canopy

cover is a weak indicator of carbon stock as in natural forest the canopy cover is formed by almost all the size classes of trees. Similarly, tree density forms a weak indicator of ecosystem carbon stock (Figs. 7 and 8), which is a general phenomenon, because a young forest with high density may show a lower biomass than in a low density mature forest. However, tree density may form a strong indicator of ecosystem carbon stock for cases such as Gross et al. (2014), when seedlings and saplings are few or absent. So, vegetation types with higher canopy height, trees having larger diameter and thus more basal area, contain more ecosystem carbon. It is important to note that Sundri is among the tallest tree species in the Sundarbans (height 15–21 m; Karim 1988) and comparatively large diameter (except Baen and Keora). Therefore, vegetation types with fewer or no Sundri have lower carbon stock.

Significant differences in ecosystem carbon stock were also observed in different STs in the Sundarbans. The FR contained the highest amount of carbon followed by moderately and strongly saline zones

(Fig. 5). In the Sundarbans as salinity increases plants become dwarfed. Here, salinity is highly dependent on the fluctuating volume of freshwater coming from the upstream (Wahid et al. 2007) and literally the absence of fresh water flow makes the strong saline zones. There is also spatial variation of nutrients in tidal water (Wahid et al. 2007). Therefore, the variation of carbon stock in different STs in Sundarbans could be due to adverse impact of increased salinity on the biomass productivity and due to spatial variation of fresh water nutrients (Wahid et al. 2007). Generally, in mangrove ecosystem especially in riverine mangrove forest, which experience of the incursion of larger amount of freshwater with fluvial nutrients (Kathiresan and Bingham 2001), there is a prominent tradeoff between salinity and the distribution of species, their productivity and growth of mangrove forests (Twilley and Chen 1998) and thus on carbon stock (Crooks et al. 2011). In mangrove ecosystem, vegetation is more abundant in lower STs where their productivity is higher and higher productivity means higher biomass and higher carbon stock (Crooks et al. 2011; Kathiresan and Bingham 2001; Ball 1998, 2002).

In this study it was found that, the contribution of Soil carbon (up to 1 m) to total ecosystem carbon stock was almost similar to total aboveground carbon stock both among the vegetation types and STs which reveals that mangroves can store larger amount of organic carbon into sediment (Bouillon et al. 2003; Khan et al. 2007; Donato et al. 2011). When soil carbon and root carbon are considered together, like other mangrove forests the contribution of belowground carbon to ecosystem carbon stock, was more than that of aboveground carbon (Kauffman et al.

2011; Donato et al. 2011; Adame et al. 2013). This is due to high root-shoot ratios in mangrove forest (Fujimoto 2004; Fujimoto et al. 1999; Page et al. 2002; Komiyama et al. 2000, 2008; Khan et al. 2007), which means that mangroves store a larger amount of carbon in soil at several meters depth (Bouillon et al. 2003).

The estimated carbon stock of vegetation types and STs, were similar to Indian Sundarbans and mangrove ecosystems in Northwestern Madagascar (Table 4). However, comparing with Indo Pacific region and Mexican Caribbean mangroves (soil carbon up to 1 m), our results are much lower, such as the *Sonneratia alba* dominated Mangrove in Yab, Micronesia, the *Rhizophora apiculata* dominated Mangrove in Babeldoab, Republic of Palau, the *Rhizophora apiculata* and *Bruguiera gymnorhiza* dominated Mangrove in Kalimantan, Indonesia and the *R. mangle* mangroves of the Sian Ka'an Biosphere Reserve, Mexico (Table 4). This differences in mangrove carbon stock would occur due to the amount of peat soil (Smith 1983a, b; Crooks et al. 2011; Siikamäki et al. 2012), mineral sediment (Crooks et al. 2011; Siikamäki et al. 2012), stature of aboveground vegetation (Murdiyarso et al. 2010), wood density (Baker et al. 2004), forest age (Kridiborworn et al. 2012; Cerón-Bretón et al. 2011), disturbance history (Goodale et al. 2002), dominant species (Ksawani et al. 2007), the depth of the organic soil, salinity, available soil phosphorus (Adame et al. 2013).

In conclusion, this study presents estimates of ecosystem carbon stock of a large mangrove forest, the Sundarbans using data sets covering widely distributed samples. The results suggest the existence of significant variations of carbon stock in different

Table 4 Comparison of ecosystem carbon stock (Above- and belowground; soil carbon up to 1 m depth) of current study to other mangroves of the world

Source	Site	Ecosystem carbon stocks (Mg C ha ⁻¹)
Present study	Vegetation types, Sundarbans	159.5–360.0
Present study	Salinity zones, Sundarbans	170.1–336.1
Donato et al. 2011	Indian Sundarbans	~212.5–312.5
Kauffman et al. 2011	Mangrove in Yab, Micronesia	897.8
Kauffman et al. 2011	Mangrove in Babeldoab, Republic of Palau	618.3
Murdiyarso et al. 2010	Mangrove in Kalimantan, Indonesia	488
Adame et al. 2013	Sian Ka'an biosphere reserve, Mexico	631.33
Jones et al. 2014	Mangrove ecosystems in northwestern Madagascar	443.2

mangrove vegetation types and different zones of the forest having variable salinity regimes. The vegetation attributes (basal area and height) of the dominant mangrove species in each vegetation type are the key indicator of determining the nature of ecosystem carbon stock. The results also reveal that no matter whether the mangroves are tall or dwarf, a significant amount of carbon is stored into the sediment, which is a characteristic feature of mangroves. For a gross estimation of ecosystem carbon stock, the generalized regression equations (Eqs. 4, 5) with variables, such as ‘basal area’ or ‘mean tree height’ could be very useful not only for Sundarbans mangrove but also for other mangroves as the results were obtained from a wide variety of mangrove vegetation types and from large scale spatial distribution of samples. The results of this study might be of use for the policy makers to develop suitable adaptive measures to cope with the trends of sea level rise, that is projected to upturn (IPCC 2013). Future studies should focus to find out the driving forces other than salinity of the so called ‘salinity zones’ in the Sundarbans, which might influence the differences in vegetation patterns as well as carbon sequestration in these zones.

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